Therapeutic Decision-Making

TDM Style Case Questions

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IMPORTANT NOTICE

Before You Begin

> These more than 300 cases are based on prior clinical notes and previous exam scenarios. In current MCQ formats, some questions may have more than one correct answer and may require selecting multiple options. All appropriate answers should be chosen based on the clinical context.

1. Each question presents a **clinical scenario** followed by multiple options. Read the stem carefully before selecting your answer.

2. **Some questions require selecting more than one answer.** These are clearly indicated with a red notice specifying the number of correct answers to select.

3. Questions are presented **without answers** in the main section. All correct answers with detailed clinical reasoning are provided in the **Answer Key** section at the end of the book.

4. This collection covers multiple therapeutic areas including neurology, cardiology, pulmonology, dermatology, anticoagulation, and more.

5. Always verify clinical decisions with **current guidelines** and evidence-based references. This material is intended for educational purposes and exam preparation only.

SECTION ONE

Clinical Case Questions

1 A patient presents with cogwheel rigidity, unsteady gait, and resting tremor affecting daily activities. What is the appropriate first-line pharmacological treatment?

- **A.** Amantadine 100 mg PO BID
- **B.** Carbidopa/Levodopa 25/100 mg half-tablet PO TID
- **C.** Pramipexole 0.125 mg PO TID
- **D.** Benztropine 1 mg PO BID
- **E.** Bromocriptine 1.25 mg PO BID
- **F.** Selegiline 5 mg PO BID
- **G.** Entacapone 200 mg PO TID

2 Which of the following are EARLY side effects of Carbidopa/Levodopa? (Select 3)

SELECT 3 CORRECT ANSWERS

- **A.** Nausea
- **B.** Dyskinesia
- **C.** Somnolence
- **D.** Wearing-off phenomenon
- **E.** Dizziness
- **F.** Impulse control disorders
- **G.** End-of-dose akinesia

3 Which of the following are LATE side effects of Carbidopa/Levodopa? (Select 3)

SELECT 3 CORRECT ANSWERS

- **A.** Nausea
- **B.** Wearing-off phenomenon
- **C.** Headache
- **D.** Severe dyskinesia
- **E.** Somnolence
- **F.** Impulse control disorders
- **G.** Orthostatic hypotension

4 How can you minimize the side effects of Carbidopa/Levodopa when initiating therapy? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Start with a high dose for rapid effect

B. Start with a low dose and titrate slowly

C. Use divided dosing

D. Combine with an opioid

E. Give the full dose at bedtime

F. Administer with grapefruit juice

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5 A 24-year-old male presents with 4 patches of non-scarring, non-scaly alopecia in the occipital and parietal areas. His father became bald at age 40. What is the first-line treatment for androgenic alopecia?

A. Finasteride 1 mg PO daily

B. Minoxidil 5% foam applied BID

C. Dutasteride 0.5 mg PO daily

D. Intralesional triamcinolone

E. Spironolactone 100 mg PO daily

F. Baricitinib 2 mg PO daily

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6 What is the minimum duration of Minoxidil 5% therapy before assessing benefit in androgenic alopecia?

A. 2 weeks

B. 1 month

C. 3 months

D. 6 months

E. 12 months

F. 4 months

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7 A patient with androgenic alopecia has used Minoxidil 5% for 5 months with no improvement and wants aggressive treatment. What medication should you prescribe?

A. Baricitinib

B. Intralesional triamcinolone

C. Finasteride 1 mg PO daily

D. Dutasteride 0.5 mg PO daily

E. Spironolactone 100 mg PO daily

8

A patient presents with patchy hair loss that developed after a stressful job. This is consistent with alopecia areata. What is the first-line treatment?

A. Minoxidil 5% foam BID

B. Finasteride 1 mg PO daily

C. Intralesional triamcinolone acetonide

D. Oral prednisone 40 mg daily

E. Baricitinib 2 mg PO daily

F. Topical clobetasol 0.05%

9

After intralesional triamcinolone for alopecia areata, when should you expect to see a response?

A. 1-2 weeks

B. 3-4 weeks

C. 6-8 weeks

D. 3 months

E. 6 months

10

A patient with alopecia areata did not respond to intralesional corticosteroids after 6 months. What is the next medication to consider?

A. Finasteride

B. Minoxidil 5%

C. Baricitinib (JAK inhibitor)

D. Methotrexate

E. Cyclosporine

F. Azathioprine

11

An 85-year-old patient on warfarin for atrial fibrillation has an INR of 7. Which of the following can increase INR? (Select 5)

SELECT 7 CORRECT ANSWERS

A. Fever

B. Hyperthyroidism

C. Heart failure

D. Hypothyroidism

E. Liver disease

F. Diarrhea

G. Wrong dose of warfarin

H. TMP-SMX use

I. Vitamin K supplementation

J. Hypoalbuminemia

12

An elderly patient on warfarin has poor vision and cannot differentiate between 1 mg and 5 mg tablets. What is the best solution?

A. Switch to a DOAC

B. Use a pill organizer/blister pack

C. Have a family member administer the medication

D. Increase the font size on the label

E. Switch to injectable anticoagulant

F. Discontinue anticoagulation

13

A 10-year-old child develops central chest tightness after playing basketball, relieved by rest. What medication should you prescribe?

A. Montelukast 5 mg PO daily

B. Fluticasone inhaler 100 mcg BID

C. Albuterol 100 mcg/inhalation, 2 puffs

D. Ipratropium bromide inhaler

E. Prednisone 1 mg/kg PO daily

F. Theophylline 200 mg PO BID

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14 When should the medication in Question 13 be administered relative to exercise?

- **A.** Immediately before exercise
- **B.** 5-20 minutes before exercise
- **C.** 30-60 minutes before exercise
- **D.** 1-2 hours before exercise
- **E.** After exercise when symptoms develop

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15 A 17-year-old female on phenytoin for seizures since age 10 (seizure-free for 18 months) is now pregnant. What advice should you give? (Select 3)

SELECT 3 CORRECT ANSWERS

- **A.** Immediately discontinue phenytoin
- **B.** Folic acid supplementation 4 mg PO daily throughout pregnancy
- **C.** Assess phenytoin serum level every 4 weeks
- **D.** Switch to valproic acid
- **E.** Counseling about teratogenicity and seizure trigger reduction
- **F.** Double the phenytoin dose
- **G.** No changes needed during pregnancy

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16 What are the serious side effects of phenytoin EXCEPT teratogenicity? (Select 3)

SELECT 3 CORRECT ANSWERS

- **A.** Hepatotoxicity
- **B.** Agranulocytosis
- **C.** Weight gain
- **D.** Stevens-Johnson Syndrome
- **E.** Hair growth
- **F.** Cardiac arrhythmias
- **G.** Acne

17

A 60-year-old patient on high-dose atorvastatin presents with muscle aches, myalgia, and fatigue. What investigation should you order to confirm the diagnosis?

- **A.** ESR
- **B.** Serum CK (Creatine Kinase)
- **C.** Liver function tests
- **D.** Serum lactate
- **E.** Troponin
- **F.** Myoglobin
- **G.** ANA

18

The CK level in the patient from Question 17 is 20 times the upper limit of normal. What is this condition called?

- **A.** Statin-induced myalgia
- **B.** Statin-induced myositis
- **C.** Statin-induced rhabdomyolysis
- **D.** Polymyositis
- **E.** Fibromyalgia
- **F.** Dermatomyositis

19

An 8-month-old boy has frequent regurgitation on formula. He is healthy with a normal examination. What medication class would you prescribe?

- **A.** H2 receptor antagonists
- **B.** Proton pump inhibitors
- **C.** Prokinetics
- **D.** Antacids
- **E.** Sucralfate

20

What is the duration of PPI treatment before follow-up in an infant with GERD?

- **A.** 2 weeks
- **B.** 4 weeks
- **C.** 8 weeks
- **D.** 12 weeks
- **E.** 6 months

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21 Which of the following clinical findings in an infant with regurgitation would make you change your management? (Select 4)

SELECT 5 CORRECT ANSWERS

A. Bilious vomiting
B. Occasional spit-up after feeds
C. Weight loss or faltering growth
D. Normal stool pattern
E. Projectile vomiting
F. Hepatosplenomegaly
G. Smiling after feeds
H. Bulging fontanelle

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22 A child is brought to the ER unconscious after ingesting unknown medication. GCS 11, HR 60, RR 8, BP 90/60. After ABC management, what are your first priorities? (Select 3)

SELECT 3 CORRECT ANSWERS

A. Naloxone IV
B. Dextrose IV
C. Thiamine IV
D. Activated charcoal immediately
E. Gastric lavage
F. IV fluid
G. Ipecac syrup

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23 What advice would you give to prevent accidental pediatric poisoning? (Select 4)

SELECT 5 CORRECT ANSWERS

A. Keep harmful products locked up and out of reach
B. Do not take medicine in view of small children
C. Do not solely rely on child-resistant containers
D. Call medicine "candy" to encourage compliance
E. Properly dispose of unwanted medication
F. Leave medications on the counter for easy access
G. Always re-secure childproof caps after use

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24 A 20-year-old female asks for emergency contraception after a broken condom 6 days ago. What are the first-line options? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Combined oral contraceptive pills

B. Copper IUD

C. Levonorgestrel 1.5 mg PO

D. Ulipristal 30 mg PO

E. Mifepristone 200 mg PO

F. Medroxyprogesterone 150 mg IM

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25 If the patient in Question 24 has a BMI of 45, what is the best emergency contraception option?

A. Levonorgestrel 1.5 mg PO

B. Ulipristal 30 mg PO

C. Copper IUD

D. Combined oral contraceptive pills

E. Depo-Provera injection

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26 A nurse in a long-term care facility brings you a medication list for a resident who hasn't followed up in 4 years. Medications include: Acetaminophen 1g QID, Allopurinol, Levothyroxine, Benzodiazepam, Rabeprazole. Which medications should be stopped or adjusted? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Levothyroxine

B. Benzodiazepine

C. Allopurinol

D. Rabeprazole

E. Acetaminophen

---

27 What is the rationale for stopping/tapering benzodiazepine in an elderly patient? (Select 2)

SELECT 2 CORRECT ANSWERS

A. It causes hepatotoxicity

B. It is addictive

C. It increases the risk of falls

D. It causes renal failure

E. It causes hypertension

---

28 What are the long-term risks of PPI use? (Select 4)

SELECT 4 CORRECT ANSWERS

A. Osteoporosis and fracture risk

B. C. difficile infection

C. Decreased iron and Vitamin B12 absorption

D. Increased HDL cholesterol

E. Hypertension

F. Microscopic colitis

G. Weight loss

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29 A 23-year-old female with moderate acne has failed topical treatments. She is sexually active and on OCP. What oral treatment would you give?

A. Doxycycline 100 mg PO BID

B. Isotretinoin 0.5 mg/kg/day

C. Spironolactone 25 mg PO BID

D. Prednisone 10 mg PO daily

E. Azithromycin 500 mg PO daily

F. Minocycline 100 mg PO BID

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30 What is the most important preventable side effect of isotretinoin?

A. Hepatotoxicity

B. Hyperlipidemia

C. Teratogenicity

D. Depression

E. Dry skin

F. Photosensitivity

---

31 What monitoring is required for a patient on isotretinoin? (Select 4)

SELECT 4 CORRECT ANSWERS

A. CBC with differential

B. Liver function tests

C. Lipid profile

D. Pregnancy test

E. Serum calcium

F. Thyroid function tests

G. Pulmonary function tests

32

A nurse had a needle stick injury. The source patient is Hepatitis B positive, HCV negative, and HIV positive. How many antiretroviral medications should be given for HIV PEP?

A. One

B. Two

C. Three

D. Four

E. Five

33

For how long should HIV post-exposure prophylaxis (PEP) be given?

A. 1 week

B. 2 weeks

C. 4 weeks

D. 6 weeks

E. 3 months

F. 6 months

P. be given?

34

What advice should you give to a healthcare worker after a needle stick injury to prevent infection transmission? (Select 3)

SELECT 3 CORRECT ANSWERS

A. Practice safe sex and use condoms

B. Do not donate blood or organs

C. Do not share personal objects like toothbrushes and razors

D. Avoid all physical contact with others

E. Quarantine for 6 months

F. Stop working immediately

35

An elderly lady with wrist and mid-hand pain has X-ray changes consistent with RA. She has a rheumatology appointment in 3 weeks. You decide to give glucocorticoid. What is the appropriate prescription?

A. Prednisone 10 mg PO daily

B. Dexamethasone 4 mg IV daily

C. Methylprednisolone 125 mg IM

D. Hydrocortisone 100 mg IV BID

E. Betamethasone 6 mg IM

F. Triamcinolone 40 mg intra-articular

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36 What are the systemic side effects of glucocorticoids? (Select 4)

SELECT 6 CORRECT ANSWERS

A. Hyperglycemia

B. Immunosuppression and increased risk of infection

C. Fluid retention and increased blood pressure

D. Hypoglycemia

E. Gastritis/peptic ulcer

F. Weight loss

G. Insomnia and mood changes

H. Decreased bone density

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37 The RA patient from Question 35 could not reach her rheumatologist. You decide to start maintenance treatment. What is the drug class and name?

A. TNF inhibitor - Adalimumab

B. Dihydrofolate reductase inhibitor - Methotrexate

C. JAK inhibitor - Tofacitinib

D. IL-6 inhibitor - Tocilizumab

E. Antimalarial - Hydroxychloroquine

F. Calcineurin inhibitor - Cyclosporine

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38 A patient in his 50s has BP readings of 145/88 and 136/92 on two visits. What non-pharmacological measures should be recommended? (Select 3)

SELECT 3 CORRECT ANSWERS

A. Weight reduction

B. Regular exercise

C. DASH diet

D. Increase caffeine intake

E. Bed rest

F. High sodium diet

G. Stress management

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39 If you start ramipril for hypertension, what investigations should you order? (Select 3)

SELECT 3 CORRECT ANSWERS

A. Serum potassium

B. Serum creatinine

C. TSH

D. Liver function tests

E. eGFR

F. Serum calcium

G. Uric acid

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40 When should the investigations in Question 39 be ordered after initiating ramipril?

A. Immediately

B. 1-2 weeks after initiation

C. 1 month after initiation

D. 3 months after initiation

E. 6 months after initiation

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41 A 60-year-old patient with BPH symptoms. What physical examination finding would make you urgently refer to a urologist? (Select 3)

SELECT 3 CORRECT ANSWERS

A. Abnormal rectal exam with irregular/nodular prostate

B. Palpable urinary bladder

C. Significant weight loss

D. Mild urinary frequency

E. Normal digital rectal exam

F. Bilateral pedal edema

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42 What is the first-line treatment for BPH?

A. 5-alpha reductase inhibitor - Finasteride

B. Alpha-1 adrenergic receptor antagonist - Tamsulosin

C. PDE5 inhibitor - Tadalafil

D. Anticholinergic - Oxybutynin

E. Beta-3 agonist - Mirabegron

F. Muscarinic antagonist - Tolterodine

43

A 24-year-old female has episodes of binge eating and food restriction. Her dentist found multiple tooth decay and enamel wearing. What is the first-line treatment?

A. SSRI - Fluoxetine

B. CBT (Cognitive Behavioral Therapy)

C. Olanzapine

D. Topiramate

E. Naltrexone

F. Bupropion

44

What factors can cause CBT to fail in bulimia nervosa? (Select 5)

SELECT 5 CORRECT ANSWERS

A. Non-adherence

B. Lack of motivation

C. Cognitive instability

D. Substance use disorder

E. Comorbid anxiety or depression

F. High intelligence

G. Young age

45

After 6 months of failed CBT for bulimia nervosa, what is the best pharmacological treatment?

A. Olanzapine 5 mg PO daily

B. Fluoxetine 20 mg PO daily

C. Bupropion 150 mg PO daily

D. Topiramate 25 mg PO daily

E. Lithium 300 mg PO BID

F. Valproic acid 250 mg PO BID

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46 A young couple has 18 months of infertility. What important history questions should you ask? (Select 5)

SELECT 6 CORRECT ANSWERS

A. Fertility history in previous relationships
B. Sexual history and frequency
C. Menstrual history
D. Cigarette smoking, marijuana, alcohol use
E. Medical and surgical history
F. Favorite hobbies
G. History of chemotherapy or radiation
H. Blood type

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47 What medication can be used to increase the chance of pregnancy in anovulatory patients?

A. Metformin 500 mg PO BID
B. Clomiphene citrate 50 mg PO daily
C. Progesterone 100 mg PO daily
D. Estradiol 2 mg PO daily
E. Goserelin 3.6 mg SC monthly
F. Letrozole 2.5 mg PO daily

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48 A man with oligospermia reports going to the gym daily and taking "some pills." What medication might cause his oligospermia?

A. Creatine supplements
B. Anabolic-androgenic steroids
C. Whey protein
D. Multivitamins
E. Caffeine pills
F. Fish oil supplements

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49 What are the side effects of anabolic-androgenic steroids? (Select 3)

SELECT 3 CORRECT ANSWERS

A. Gynecomastia
B. Erythrocytosis
C. Improved fertility
D. Hepatotoxicity
E. Decreased aggression
F. Testicular hypertrophy

50

A patient born female wants a deeper voice and more masculine features. What is the first treatment option?

- **A.** Estradiol cypionate IM
- **B.** Testosterone enanthate 50 mg IM weekly
- **C.** Spironolactone 100 mg PO daily
- **D.** Finasteride 5 mg PO daily
- **E.** GnRH agonist
- **F.** Progesterone 200 mg PO daily

51

What investigations should be monitored while a patient is on testosterone therapy? (Select 4)

SELECT 4 CORRECT ANSWERS

- **A.** Serum testosterone
- **B.** Hematocrit
- **C.** Lipid profile
- **D.** Liver function
- **E.** Serum calcium
- **F.** Thyroid function
- **G.** Pulmonary function

52

A patient presents with chest pain worse when lying down and improved with sitting/leaning forward. Recent respiratory infection. ECG shows diffuse ST elevation. What is the first-line treatment?

- **A.** Aspirin 81 mg PO daily + Clopidogrel 75 mg PO daily
- **B.** Ibuprofen 600-800 mg PO TID + Colchicine 0.6 mg PO BID
- **C.** Prednisone 40 mg PO daily
- **D.** Nitroglycerin 0.4 mg sublingual
- **E.** Metoprolol 25 mg PO BID
- **F.** Heparin IV infusion

53

What is the duration of colchicine therapy in acute pericarditis?

- **A.** 1 week
- **B.** 2 weeks
- **C.** 1 month
- **D.** 3 months
- **E.** 6 months
- **F.** 12 months

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54 The pericarditis patient from Question 52 also has a history of peptic ulcer. What medication should be added to prevent peptic ulcer while on NSAID therapy?

A. Sucralfate

B. Pantoprazole

C. Misoprostol

D. Bismuth subsalicylate

E. Famotidine

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55 After one week, the pericarditis patient still has pain and ECG changes. What is the next medication?

A. Morphine 5 mg IV

B. Prednisone 0.5 mg/kg daily for 2 weeks then taper

C. Azathioprine 50 mg PO daily

D. Methotrexate 7.5 mg PO weekly

E. Anakinra 100 mg SC daily

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56 A female patient has panic attacks and bruises from self-harm behavior. What conditions might be associated? (Select 4)

SELECT 5 CORRECT ANSWERS

A. Borderline personality disorder

B. Substance use disorders

C. Bipolar disorder

D. Major depression

E. Diabetes mellitus

F. Hypothyroidism

G. Schizophrenia

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57 What classes of medication should be given for panic attacks with self-harm? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Antipsychotics

B. SSRI antidepressants

C. Benzodiazepines

D. Mood stabilizers

E. Opioids

F. Stimulants

58

A 66-year-old male presents with SOB, orthopnea, and PND. He is on digoxin and furosemide 80 mg BID. BP 90/60, HR 125, O2 84%. What blood investigation should be done before treatment?

A. Troponin

B. BNP (B-type natriuretic peptide)

C. D-dimer

D. Serum lactate

E. Procalcitonin

F. CRP

59

What is the initial therapy for the patient in Question 58? (Select 5)

SELECT 5 CORRECT ANSWERS

A. Position patient at 45 degrees

B. High-flow O2 via non-rebreather mask

C. IV Furosemide 160 mg bolus

D. Morphine

E. Dobutamine

F. Oral metoprolol

G. IV normal saline bolus

H. Sublingual nitroglycerin

60

What medications should the heart failure patient be discharged on? (Select 4)

SELECT 4 CORRECT ANSWERS

A. ACE inhibitor (Ramipril)

B. Beta-blocker (Metoprolol)

C. Potassium-sparing diuretic (Spironolactone)

D. Statin (Atorvastatin)

E. Calcium channel blocker (Amlodipine)

F. Digoxin loading dose

G. Aspirin 325 mg daily

61

A COPD patient presents with fever 39°C, increased sputum production with color change, and dyspnea. What 3 classes of medication will you give? (Select 3)

SELECT 3 CORRECT ANSWERS

A. Short-acting beta agonist (Albuterol)

B. Short-acting muscarinic antagonist (Ipratropium)

C. Systemic glucocorticoids (Prednisone)

D. Long-acting beta agonist (Salmeterol)

E. Leukotriene receptor antagonist (Montelukast)

F. Theophylline

62

What maintenance medications should be given to a COPD patient after an exacerbation? (Select 2)

SELECT 2 CORRECT ANSWERS

A. ICS (Budesonide)

B. LABA (Formoterol)

C. Short-acting beta agonist PRN only

D. Oral prednisone daily

E. Daily azithromycin

F. Theophylline

63

What non-pharmacological advice should you give a COPD patient? (Select 4)

SELECT 4 CORRECT ANSWERS

A. Smoking cessation

B. Influenza and pneumococcal vaccination

C. Pulmonary rehabilitation

D. Education about proper inhaler technique

E. Avoid all physical activity

F. High-sodium diet

G. Bed rest

64

A 40-year-old with perilimbic congestion, severe eye pain, chronic back/hip problems, slit lamp showing +3 cells, and normal IOP. What medication relieves symptoms?

A. Prednisolone acetate 1% drops

B. Cyclopentolate 1% drops TID

C. Timolol 0.5% drops BID

D. Pilocarpine 2% drops QID

E. Erythromycin ointment

F. Ciprofloxacin 0.3% drops

65

What will the ophthalmologist give the patient in Question 64?

A. Cyclopentolate 1% drops

B. Prednisolone acetate 1% drops BID-QID

C. Timolol 0.5% drops

D. Latanoprost 0.005% drops

E. Ofloxacin 0.3% drops

66

A man with intentional amphetamine overdose presents with BP 240/120. What medications should be given for agitation? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Diazepam 10 mg IV

B. Lorazepam 2 mg IV

C. Haloperidol 5 mg IM

D. Propranolol 10 mg IV

E. Phenytoin 100 mg IV

F. Ketamine 1 mg/kg IV

67

What medications should be given for the severe hypertension in the amphetamine overdose patient? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Nitroprusside 0.25-0.5 mcg/kg/min

B. Phentolamine 2-5 mg IV

C. Propranolol 1 mg IV

D. Enalaprilat 1.25 mg IV

E. Clonidine 0.1 mg PO

F. Atenolol 50 mg PO

68

A 57-year-old man presents with typical pneumonia symptoms. He has a penicillin allergy (to cloxacillin). Fever 39.5, productive cough with yellowish-green mucus. What antibiotic should you prescribe?

- **A.** Amoxicillin 1g PO TID
- **B.** **
- **C.** Cephalexin 500 mg PO QID
- **D.** Ciprofloxacin 500 mg PO BID
- **E.** Doxycycline 100 mg PO BID
- **F.** Azithromycin 500 mg PO daily

69

The pneumonia patient returns with a white lesion in his mouth. He admits to using his wife's puffer. What caused the lesion?

- **A.** Oral candidiasis from inhaled corticosteroid use
- **B.** Herpes simplex virus
- **C.** Aphthous ulcer
- **D.** Leukoplakia
- **E.** Lichen planus
- **F.** Squamous cell carcinoma

70

A patient on nifedipine develops ankle swelling. His daughter thinks he has organ failure. What is the cause?

- **A.** Heart failure
- **B.** Nifedipine-induced peripheral edema
- **C.** Deep vein thrombosis
- **D.** Nephrotic syndrome
- **E.** Liver cirrhosis
- **F.** Lymphedema

71

The patient from Question 70 was given furosemide at a walk-in clinic. Now Na is 125 and K is low. What is the best next step?

- **A.** Increase furosemide dose
- **B.** Add spironolactone
- **C.** Stop furosemide
- **D.** Give hypertonic saline
- **E.** Add potassium supplements only
- **F.** Switch to hydrochlorothiazide

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72 How should nifedipine-induced peripheral edema be managed? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Add a loop diuretic

B. Combine with ACEI or ARB

C. Switch to a non-dihydropyridine CCB (diltiazem or verapamil)

D. Double the nifedipine dose

E. Add compression stockings only

F. Discontinue all antihypertensives

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73 A patient with menorrhagia bleeds 8 days in a 30-day cycle, uses 3 tampons/day for the first 3 days, uterus is 14-week size, feels dizzy and fatigued. What pharmacological treatment should be given along with standard management?

A. Vitamin B12 supplementation

B. Ferrous sulfate 325 mg (65 mg elemental iron) 1-2 tabs daily

C. Folic acid 5 mg daily

D. Erythropoietin injection

E. IV iron infusion

F. Blood transfusion

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74 What may interfere with oral iron absorption? (Select 4)

SELECT 5 CORRECT ANSWERS

A. Celiac disease

B. H. pylori infection

C. PPI use

D. Calcium supplementation

E. Vitamin C supplementation

F. High-protein diet

G. Atrophic gastritis

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75 A 60-year-old male with erectile dysfunction has CVD, HTN, and DM. What medication will you give?

A. Testosterone gel 1%

B. Alprostadil injection

C. Sildenafil 50 mg

D. Tadalafil 20 mg

E. Vardenafil 10 mg

F. Yohimbine 5 mg

76 What medication is absolutely contraindicated with PDE5 inhibitors (sildenafil)?

- **A.** Beta-blockers
- **B.** ACE inhibitors
- **C.** Nitrates (nitroglycerin)
- **D.** Calcium channel blockers
- **E.** Thiazide diuretics
- **F.** Statins

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77 A patient with panic disorder on benzodiazepine (lorazepam) presents with BP 170/110 and hyperreflexia. He stopped his medication 2 days ago. What should you give now?

- **A.** Lorazepam 1 mg PO
- **B.** Diazepam 10-20 mg PO/IV
- **C.** Alprazolam 0.5 mg PO
- **D.** Phenobarbital 60 mg IV
- **E.** Carbamazepine 200 mg PO
- **F.** Gabapentin 300 mg PO

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78 What is the plan for weaning benzodiazepines? (Select 2)

SELECT 2 CORRECT ANSWERS

- **A.** Stop abruptly
- **B.** Avoid sudden discontinuation
- **C.** Switch to long-acting BDZ (diazepam) and slowly taper
- **D.** Switch to an opioid
- **E.** Double the dose then stop
- **F.** Replace with stimulant medication

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79 What is the maximum daily dose of acetaminophen for an adult?

- **A.** 2 grams
- **B.** 3 grams
- **C.** 4 grams
- **D.** 5 grams
- **E.** 6 grams

---

80 What monitoring investigations should be done for a patient taking high-dose acetaminophen regularly? (Select 3)

SELECT 3 CORRECT ANSWERS

A. Liver function tests

B. CBC

C. Kidney function tests

D. Thyroid function

E. Lipid profile

F. Serum acetaminophen level

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81 A patient on ACE inhibitor needs monitoring. What should be checked? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Serum potassium

B. Serum creatinine

C. Serum calcium

D. TSH

E. Uric acid

F. Serum magnesium

---

82 An 8-month-old has not been vaccinated because the mother worries vaccines cause autism. What is your response?

A. Agree with the mother and delay vaccination

B. Educate that epidemiologic evidence does NOT support an association between immunization and autism

C. Tell her she is wrong and insist on vaccination

D. Refer to a specialist

E. Report to child protective services

F. Suggest alternative medicine instead

---

83 What are the common side effects of DTaP vaccine? (Select 3)

SELECT 4 CORRECT ANSWERS

A. Erythema and pain at injection site

B. Excessive crying and irritability

C. Fever

D. Permanent neurological damage

E. Autism

F. Anaphylaxis (common)

G. Drowsiness

---

84 A pregnant woman is on ramipril, isotretinoin, and minocycline. Which medications must be stopped? (Select 3)

SELECT 3 CORRECT ANSWERS

A. Ramipril only

B. Isotretinoin only

C. Minocycline only

D. All three must be stopped

E. None need to be stopped

---

85 What is the rationale for stopping ramipril in pregnancy?

A. Causes neural tube defects

B. Causes renal tubular dysgenesis and oligohydramnios

C. Causes fetal cardiac defects

D. Causes gestational diabetes

E. Causes preterm labor

---

86 What antihypertensive medications CAN be given in pregnancy? (Select 3)

SELECT 3 CORRECT ANSWERS

A. Labetalol

B. Nifedipine

C. Hydralazine

D. Ramipril

E. Candesartan

F. Lisinopril

G. Atenolol

87

A 7-year-old came from camping with a small pimple that changed to yellow on his chest (impetigo). What is the treatment?

A. Oral amoxicillin 500 mg TID

B. Topical mupirocin TID for 5 days

C. Doxycycline 100 mg BID

D. IV cephalexin

E. Topical hydrocortisone 1%

F. Oral clindamycin 300 mg TID

88

Why can you NOT prescribe doxycycline to a 7-year-old?

A. It causes hepatotoxicity in children

B. It is traditionally avoided in children under 8 due to teeth discoloration

C. It causes renal failure in children

D. It is not effective against skin infections

E. It causes growth retardation

F. It is only available in IV form

89

A female patient presents after a restaurant visit with difficulty breathing, wheezing, urticaria, vomiting, and nausea. She is on a beta-blocker. BP 80/50, HR 40. What medication should you give first?

A. Diphenhydramine 50 mg IV

B. Methylprednisolone 125 mg IV

C. Epinephrine 0.3-0.5 mg IM

D. Albuterol nebulizer

E. Normal saline 1L IV bolus

F. Atropine 0.5 mg IV

90

What are the initial treatments for the anaphylaxis patient on a beta-blocker? (Select 3)

SELECT 3 CORRECT ANSWERS

A. Oxygen

B. IV normal saline bolus

C. Glucagon

D. Oral antihistamine

E. Inhaled corticosteroid

F. Oral prednisone

---

91 What should be given on discharge after an anaphylaxis episode?

- **A.** Oral prednisone taper
- **B.** Epinephrine auto-injector (EpiPen)
- **C.** Daily antihistamine
- **D.** Inhaled corticosteroid
- **E.** Prophylactic antibiotics

---

92 What are the discharge instructions after anaphylaxis? (Select 4 - AFEW)

SELECT 4 CORRECT ANSWERS

- **B.** Further evaluation by an allergist
- **C.** Epinephrine auto-injector prescription
- **D.** Written information about anaphylaxis
- **E.** Avoid all restaurants permanently
- **F.** Daily steroid use
- **G.** Weekly allergy shots
- **W.** A) Anaphylaxis emergency action plan

---

93 A traveler returns from Mexico with 6 episodes of watery diarrhea, no blood, no fever. What treatment should you give? (Select 2)

SELECT 2 CORRECT ANSWERS

- **A.** Ciprofloxacin 500 mg PO BID
- **B.** Loperamide 4 mg then 2 mg after each stool
- **C.** Oral rehydration solution (ORS)
- **D.** IV antibiotics
- **E.** Metronidazole 500 mg PO TID
- **F.** Bismuth subsalicylate

---

94 What prevention advice should you give before returning to Mexico? (Select 3)

SELECT 3 CORRECT ANSWERS

- **A.** Drink bottled beverages
- **B.** Eat fruits that you peel just prior to eating
- **C.** Eat thoroughly cooked food served hot
- **D.** Drink tap water freely
- **E.** Eat street food without concern
- **F.** Take prophylactic antibiotics daily

95

A palliative care patient with terminal lung cancer has intractable hiccups. What medications can treat hiccups? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Baclofen

B. Gabapentin

C. Morphine

D. Diazepam

E. Ondansetron

F. **

96

What medication classes treat dyspnea in palliative care? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Opioids

B. Benzodiazepines

C. NSAIDs

D. Antibiotics

E. Antihistamines

F. Stimulants

97

A female patient has headaches 3 times per week, band-like around the head, no photophobia or vomiting. She has asthma and GERD with weak insurance. What affordable medications can she take? (Select 4)

SELECT 4 CORRECT ANSWERS

A. Acetaminophen

B. Sumatriptan

C. Mirtazapine

D. Amitriptyline

E. Gabapentin

F. Topiramate

G. Botox injections

H. Ergotamine

98

An 8-year-old female has unilateral throbbing headache with nausea, worsened by noise, relieved by sleep, once weekly. What is the first-line treatment?

A. Ibuprofen 10 mg/kg

B. Acetaminophen 15 mg/kg

C. Sumatriptan 6 mg SC

D. Propranolol 10 mg PO BID

E. Amitriptyline 10 mg PO qhs

F. Topiramate 25 mg PO daily

99

The child from Question 98 returns with no improvement and has developed a phobia of tablets. What formulations can you give? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Ibuprofen syrup 10 mg/kg

B. Sumatriptan nasal spray 5 mg

C. Acetaminophen suppository

D. Morphine elixir

E. Diazepam rectal gel

F. Prednisone liquid

100

A 75-year-old female is on metformin, metoprolol, and ramipril. She fell once and feels dizzy. K=5.7, GFR=29, BP 110/70, HR 40, blood glucose normal. What medications should be stopped or adjusted? (Select 3)

SELECT 3 CORRECT ANSWERS

A. Stop metformin (GFR <30, risk of lactic acidosis)

B. Hold ramipril (hyperkalemia)

C. Hold metoprolol (bradycardia causing falls)

D. Increase metformin dose

E. Add furosemide

F. Add digoxin

101

A child with otalgia and discharge. What is the drug and dose?

A. Amoxicillin 45 mg/kg PO divided BID for 5 days

B. Amoxicillin 90 mg/kg PO divided BID for 5 days

C. Azithromycin 10 mg/kg PO day 1 then 5 mg/kg for 4 days

D. Cephalexin 25 mg/kg PO BID for 10 days

E. Ciprofloxacin otic drops BID for 7 days

---

102 The child from Question 101 returns after 3 days with high-grade fever, otalgia, and increased discharge. What risky condition are you worried about?

A. Meningitis
B. Acute mastoiditis
C. Brain abscess
D. Cholesteatoma
E. Labyrinthitis
F. Facial nerve palsy

---

103 What is the management for acute mastoiditis? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Oral amoxicillin-clavulanate
B. Piperacillin-tazobactam IV
C. Urgent referral to otolaryngologist
D. Watchful waiting
E. Topical antibiotic drops only
F. Oral prednisone

---

104 A 65-year-old lady has a DEXA scan T-score of -2.5 at the vertebra and -1.8 at the femoral neck. Family history of osteoporosis. What is the most important non-pharmacological advice?

A. Calcium supplementation
B. Vitamin D supplementation
C. Fall prevention
D. Weight-bearing exercise
E. Smoking cessation
F. Alcohol avoidance

---

105 What is the first-line pharmacological treatment for osteoporosis?

A. Raloxifene 60 mg PO daily
B. Alendronate 70 mg PO weekly
C. Teriparatide 20 mcg SC daily
D. Denosumab 60 mg SC every 6 months
E. Calcitonin 200 IU intranasal daily
F. Zoledronic acid 5 mg IV yearly

---

106 A woman at 36 weeks gestation asks about breastfeeding. What are the benefits to the baby? (Select 3)

SELECT 3 CORRECT ANSWERS

A. Meets nutritional needs until 6 months

B. Enhances GI tract growth, motility and maturity

C. Decreases risk of necrotizing enterocolitis and respiratory infections

D. Guarantees higher IQ

E. Prevents all childhood illnesses

F. Eliminates need for any supplementation

---

107 What are the maternal benefits of breastfeeding? (Select 3)

SELECT 3 CORRECT ANSWERS

A. Money savings

B. Enhanced weight loss after pregnancy

C. Decreased risk of breast, ovarian, and endometrial cancers

D. Complete prevention of postpartum depression

E. Guaranteed contraception

F. Prevents all future cancers

---

108 What is the most important step for successful breastfeeding in the delivery room?

A. Immediate formula supplementation

B. Immediate and uninterrupted skin-to-skin contact

C. Separation of mother and baby for assessment

D. Bottle feeding first then breastfeeding

E. Delayed feeding until 6 hours postpartum

---

109 What supplement should be given to a breastfed baby?

A. Iron drops

B. Vitamin D 400 IU drops

C. Vitamin C drops

D. Calcium drops

E. Zinc drops

F. Multivitamin drops

---

110 A patient with hepatic encephalopathy whose consciousness deteriorated. Ethanol zero, agitated, chronic liver disease. What medication should you give?

- **A.** Rifaximin 550 mg PO BID
- **B.** Lactulose 30 mL (20g) PO QID
- **C.** Neomycin 500 mg PO QID
- **D.** Metronidazole 500 mg PO TID
- **E.** Flumazenil 0.2 mg IV
- **F.** Naloxone 0.4 mg IV

---

111 How do you titrate lactulose in hepatic encephalopathy?

- **A.** Titrate to achieve one stool per day
- **B.** Titrate to achieve two to three soft stools per day
- **C.** Titrate to achieve five stools per day
- **D.** Give maximum dose regardless of stool output
- **E.** Titrate based on ammonia levels only

---

112 The hepatic encephalopathy patient does not improve on lactulose. What should you give next?

- **A.** Neomycin 500 mg PO QID
- **B.** Rifaximin 550 mg PO BID
- **C.** Metronidazole 500 mg PO TID
- **D.** Vancomycin 125 mg PO QID
- **E.** Lactulose enema
- **F.** Liver transplant referral only

---

113 A 17-year-old male has recurrent urethritis after treatment. What causes recurrence? (Select 3)

SELECT 3 CORRECT ANSWERS

- **A.** Nonadherence to treatment
- **B.** Atypical pathogen or resistant organism
- **C.** Infected untreated sexual partners
- **D.** Excessive water intake
- **E.** Wearing tight underwear
- **F.** Frequent urination

114 When can a patient with urethritis resume sexual activity after treatment?

- **A.** Immediately after starting treatment
- **B.** 3 days after treatment
- **C.** 7 days after treatment initiation
- **D.** 14 days after treatment
- **E.** 30 days after treatment
- **F.** After negative test of cure

---

115 A woman presents with right-side hemiparesis in upper and lower limbs started 3 hours ago. BP 170/90. What should you give for her hypertension?

- **A.** Labetalol 10 mg IV
- **B.** Nicardipine IV infusion
- **C.** No treatment (treat only if ≥185/110)
- **D.** Amlodipine 10 mg PO
- **E.** Hydralazine 10 mg IV
- **F.** Nitroprusside infusion

---

116 The stroke patient is referred to the specialist team. What medication should they give?

- **A.** Aspirin 325 mg PO
- **B.** Heparin IV infusion
- **C.** Tenecteplase (fibrinolytic)
- **D.** Clopidogrel 300 mg PO
- **E.** Warfarin 5 mg PO
- **F.** Nimodipine 60 mg PO

---

117 What are absolute contraindications for fibrinolytic therapy in stroke? (Select 3)

SELECT 3 CORRECT ANSWERS

- **A.** Intracranial hemorrhage
- **B.** Previous stroke in the last 3 months
- **C.** Severe closed head/facial trauma in last 3 months
- **D.** Mild headache
- **E.** Age over 60
- **F.** Hypertension 150/90
- **G.** Diabetes mellitus

118 A patient on typical antipsychotic (fluphenazine) presents pacing and moving around the room (akathisia). What medications should you give? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Propranolol 10 mg PO BID

B. Benztropine 1 mg PO BID

C. Haloperidol 5 mg IM

D. Chlorpromazine 25 mg PO

E. Olanzapine 5 mg PO

F. Lithium 300 mg PO

---

119 The patient from Question 118 later returns with cogwheel rigidity and unsteady gait (pseudo-parkinsonism). What is the best treatment?

A. Propranolol 20 mg PO BID

B. Benztropine 1-2 mg PO BID

C. Levodopa/Carbidopa

D. Haloperidol 5 mg IM

E. Diazepam 5 mg PO TID

F. Amantadine 100 mg PO BID

---

120 After multiple extrapyramidal side effects, what is the best approach for the patient on typical antipsychotics?

A. Increase the dose of typical antipsychotic

B. Add another typical antipsychotic

C. Change to an atypical antipsychotic

D. Add lithium

E. Discontinue all medications

F. Add benzodiazepine permanently

---

121 Which antipsychotics cause the MOST weight gain? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Aripiprazole

B. Olanzapine

C. Ziprasidone

D. Clozapine

E. Haloperidol

F. Lurasidone

---

122 A 73-year-old man with AF on warfarin is annoyed by regular INR checking. His friend takes a medication that doesn't need regular follow-up. What alternative can you prescribe?

- **A.** **
- **B.** Apixaban
- **C.** Clopidogrel
- **D.** Aspirin
- **E.** Heparin
- **F.** Fondaparinux

---

123 What are contraindications to DOACs? (Select 3)

SELECT 4 CORRECT ANSWERS

- **A.** Acute bleeding
- **B.** Moderate to severe hepatic impairment
- **C.** Prosthetic valve disease
- **D.** Mild hypertension
- **E.** Age over 65
- **F.** Diabetes mellitus
- **G.** Pregnancy

---

124 A patient's renal clearance is 10 mL/1.73m². Why is this concerning for DOAC use?

- **A.** DOACs are hepatotoxic
- **B.** DOACs cannot be given with GFR <30
- **C.** DOACs cause hyperkalemia
- **D.** DOACs require weekly monitoring with low GFR
- **E.** DOACs are only available IV with low GFR

---

125 What is the non-pharmacological management of Type 2 Diabetes? (Select 3)

SELECT 3 CORRECT ANSWERS

- **A.** Weight reduction through diet and exercise
- **B.** Regular exercise
- **C.** Psychological intervention
- **D.** Unlimited carbohydrate intake
- **E.** Bed rest
- **F.** Avoid all physical activity

---

126 What chronic follow-up investigations are needed for a diabetic patient? (Select 5)

SELECT 5 CORRECT ANSWERS

A. HbA1C every 3-6 months

B. Serum creatinine annually

C. Urinary albumin-to-creatinine ratio annually

D. Dilated eye examination annually

E. Comprehensive foot examination annually

F. Weekly blood glucose only

G. Chest X-ray annually

---

127 A patient with signs and symptoms of hyperthyroidism (tachycardia, palpitations, heat intolerance, weight loss, fine tremors). What medication should you give to relieve symptoms?

A. Methimazole 10 mg PO daily

B. Propylthiouracil 100 mg PO TID

C. Atenolol 50 mg PO daily

D. Radioactive iodine

E. Levothyroxine 100 mcg PO daily

F. Prednisone 40 mg PO daily

---

128 What is the long-term medication for hyperthyroidism?

A. Propranolol 40 mg PO BID

B. Methimazole 10-40 mg PO daily

C. Levothyroxine 100 mcg PO daily

D. Hydrocortisone 20 mg PO daily

E. Lithium 300 mg PO BID

---

129 What monitoring is needed for a patient on methimazole? (Select 3)

SELECT 3 CORRECT ANSWERS

A. TFT (TSH, free T4)

B. CBC with differential

C. Liver function tests

D. Serum calcium

E. Chest X-ray

F. ECG weekly

---

130 A patient with hypothyroidism has TSH of 10, weight gain, and constipation. What medication should you give?

- **A.** Methimazole 10 mg PO daily
- **B.** Levothyroxine 1.6 mcg/kg/day PO daily
- **C.** Liothyronine 25 mcg PO daily
- **D.** Propylthiouracil 100 mg PO TID
- **E.** Prednisone 10 mg PO daily

---

131 When should you check TSH after starting levothyroxine?

- **A.** 1 week
- **B.** 2 weeks
- **C.** 4-6 weeks
- **D.** 3 months
- **E.** 6 months
- **F.** 1 year

---

132 A 35-year-old woman with depression. Her husband left her with 5 children. She weeps, can't sleep, low energy, no self-harm ideation. What 2 medications from different classes can you prescribe?

SELECT 2 CORRECT ANSWERS

- **A.** SSRI: Paroxetine 20 mg/day
- **B.** SNRI: Venlafaxine 37.5 mg/day
- **C.** TCA: Amitriptyline 25 mg/day
- **D.** MAOI: Phenelzine 15 mg/day
- **E.** Benzodiazepine: Lorazepam 1 mg/day
- **F.** Stimulant: Methylphenidate 10 mg/day

---

133 How long does it take to see improvement with antidepressants?

- **A.** 1-3 days
- **B.** 1 week
- **C.** 2-4 weeks
- **D.** 2-3 months
- **E.** 6 months

---

134 How long should antidepressant treatment continue after remission of a first episode?

- **A.** 2 weeks
- **B.** 1 month
- **C.** At least 6 months
- **D.** 1 year
- **E.** Lifelong

---

135 A 25-year-old smoker wants to quit. His father died of lung cancer. What are the first-line treatment options for smoking cessation? (Select 3)

SELECT 3 CORRECT ANSWERS

- **A.** Nicotine replacement therapy (patch, gum, lozenges)
- **B.** Varenicline 0.5 mg
- **C.** Bupropion 150 mg
- **D.** Clonidine 0.1 mg
- **E.** Nortriptyline 25 mg
- **F.** Fluoxetine 20 mg

---

136 A patient wanting to quit smoking has a history of head trauma and seizures. What medication is contraindicated?

- **A.** Nicotine patch
- **B.** Varenicline
- **C.** Bupropion
- **D.** Nicotine gum
- **E.** Nicotine lozenges

---

137 Why is bupropion contraindicated in patients with seizure history?

- **A.** It causes hepatotoxicity
- **B.** It reduces seizure threshold and increases risk of seizures
- **C.** It causes cardiac arrhythmias
- **D.** It causes renal failure
- **E.** It interacts with antiepileptic drugs

---

138 A patient with Crohn's disease presents with vomiting, diarrhea, glucose 2.1, BP 80/50, pulse 125. What immediate resuscitative measures should you implement? (Select 3)

SELECT 3 CORRECT ANSWERS

A. Isotonic IV fluid (NS or RL)
B. Correct hypoglycemia with D50W IV
C. Correct electrolyte disturbances
D. Start oral feeding immediately
E. Give IV antibiotics empirically
F. Perform colonoscopy urgently

---

139 A 46-year-old presents with acute swelling of the right big toe (acute gout). What 3 medications can be used? (Select 3)

SELECT 3 CORRECT ANSWERS

A. Prednisone 30 mg PO daily
B. Naproxen 500 mg PO BID
C. Colchicine 0.6 mg PO daily
D. Allopurinol 100 mg PO daily
E. Febuxostat 40 mg PO daily
F. Probenecid 250 mg PO BID

---

140 What advice should you give to prevent gout recurrence? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Maintain healthy weight and regular exercise
B. Dietary moderation in purine intake
C. Increase red meat consumption
D. Increase alcohol intake
E. Take aspirin daily
F. Avoid all exercise

---

141 An 8-year-old with ADHD has failed non-pharmacological management. What is the first-line medication?

A. Atomoxetine 10 mg PO daily
B. Methylphenidate 5 mg PO BID
C. Dextroamphetamine 5 mg PO daily
D. Clonidine 0.1 mg PO qhs
E. Guanfacine 1 mg PO daily
F. Bupropion 75 mg PO daily

142 What should be monitored in a child on methylphenidate? (Select 4)

SELECT 4 CORRECT ANSWERS

A. Weight
B. Height
C. Heart rate
D. Blood pressure
E. Serum drug levels
F. Liver enzymes monthly
G. Chest X-ray

---

143 What are the side effects of methylphenidate? (Select 3)

SELECT 3 CORRECT ANSWERS

A. Cardiovascular events and increased blood pressure
B. Growth retardation
C. Weight gain
D. New onset psychosis or mania
E. Improved appetite
F. Excessive sleepiness

---

144 After 2 months of minimal response to methylphenidate at maximum dose, what should you give?

A. Double the methylphenidate dose
B. Amphetamine or dextroamphetamine
C. Add an SSRI
D. Switch to benzodiazepine
E. Discontinue all medications
F. Add lithium

---

145 A 16-year-old boy with schizophrenia thinks the government is spying on him, covers windows with aluminum, has flat affect. What medications can be used? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Risperidone 0.25-8 mg PO daily
B. Olanzapine 2.5-30 mg PO daily
C. Fluoxetine 20 mg PO daily
D. Lithium 300 mg PO BID
E. Methylphenidate 10 mg PO BID
F. Diazepam 5 mg PO TID

146 The schizophrenia patient returns with neck spasm and involuntary leg movements (acute dystonia). What medications should you give? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Benztropine 1-2 mg PO/IV
B. Diphenhydramine 50 mg PO/IM
C. Haloperidol 5 mg IM
D. Increase antipsychotic dose
E. Diazepam 10 mg IV
F. Phenytoin 100 mg IV

---

147 A patient with hyperkalemia K=7.2 on diuretics. What medication should be given FIRST to prevent cardiac complications?

A. Insulin with glucose
B. Calcium gluconate 1g IV over 2-3 minutes
C. Albuterol nebulization
D. Furosemide 40 mg IV
E. Sodium bicarbonate
F. Kayexalate PO

---

148 What medications can shift potassium into cells? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Insulin with glucose
B. Beta-2 agonist (Albuterol nebulization)
C. Calcium gluconate
D. Furosemide
E. Patiromer
F. Sodium polystyrene sulfonate

---

149 A woman in her 50s with restless leg syndrome and low serum ferritin. What medication should you give first?

A. Gabapentin 300 mg PO qhs
B. Pramipexole 0.125 mg PO qhs
C. Ferrous sulfate 325 mg PO TID
D. Ropinirole 0.25 mg PO qhs
E. Clonazepam 0.5 mg PO qhs
F. Pregabalin 75 mg PO qhs

---

150 What other medications are used for restless leg syndrome? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Gabapentin 300 mg PO qhs

B. Pramipexole 0.125 mg PO qhs

C. Methylphenidate 10 mg PO BID

D. Fluoxetine 20 mg PO daily

E. Haloperidol 2 mg PO qhs

F. Propranolol 20 mg PO BID

---

151 A 40-year-old woman with plantar fasciitis trains for a marathon. Pain 15 minutes after running and in the morning. What investigation should you do?

A. X-ray of the foot

B. MRI of the foot

C. Ultrasound of the foot

D. None

E. CT scan

F. Bone scan

---

152 What is the non-pharmacological treatment for plantar fasciitis? (Select 4)

SELECT 4 CORRECT ANSWERS

A. Stretching exercises for plantar fascia and calf muscles

B. Avoiding flat shoes and barefoot walking

C. Using silicone heel shoe inserts

D. Decreasing aggravating physical activities

E. Increasing running distance

F. Wearing high heels

G. Complete bed rest

---

153 An old man presents with headache and visual disturbance. Temporal tenderness on exam (giant cell arteritis). What medication should you give immediately?

A. Prednisone 1 mg/kg PO daily

B. Methylprednisolone 1000 mg IV x 3 days

C. Ibuprofen 800 mg PO TID

D. Acetaminophen 1g PO QID

E. Methotrexate 15 mg PO weekly

F. Tocilizumab 162 mg SC weekly

154 What are the long-term side effects of steroids that you should tell the giant cell arteritis patient about? (Select 4)

SELECT 5 CORRECT ANSWERS

A. Blood glucose impairment/DM

B. Immunosuppression/infection

C. Osteoporosis

D. Blood pressure elevation

E. Weight loss

F. Improved wound healing

G. Glaucoma/cataracts

---

155 A 32-year-old female wants tips to help her sleep but doesn't want medication. She drinks coffee 5 times a day. What non-pharmacological advice should you give? (Select 5)

SELECT 6 CORRECT ANSWERS

A. Avoid caffeinated drinks especially before sleep

B. Avoid daytime sleep

C. Schedule fixed sleep and wake times

D. Avoid eating or exercise 3 hours before bedtime

E. Keep bedroom for sleep and sex only

F. Use phone/laptop in bed to relax

G. Drink alcohol before bed to relax

H. Avoid using screens before bedtime

---

156 A 55-year-old lady with insomnia wants a non-addictive medication. What can you prescribe?

A. Lorazepam 1 mg PO qhs

B. Zopiclone 3.75 mg PO qhs

C. Ramelteon 8 mg PO qhs

D. Diazepam 5 mg PO qhs

E. Alprazolam 0.25 mg PO qhs

F. Zolpidem 5 mg PO qhs

---

157 A 7-year-old boy has nocturnal enuresis. What non-pharmacological options should you try? (Select 5)

SELECT 6 CORRECT ANSWERS

A. Void 4-7 times per day including before bed

B. Avoid high-sugar and caffeine drinks in the evening

C. Provide smaller fluid intake after 5 PM

D. Use a bed-wetting alarm

E. Punish the child for bedwetting

F. Restrict all fluids after noon

G. Educate parents this is not uncommon and not the child's fault

H. Wake child at night to use toilet

---

158 What medication can be used for nocturnal enuresis if non-pharmacological measures fail?

A. Oxybutynin 5 mg PO qhs

B. Desmopressin 0.2-0.6 mg PO qhs

C. Imipramine 25 mg PO qhs

D. Furosemide 20 mg PO daily

E. Tolterodine 2 mg PO BID

---

159 A patient with a kidney transplant on cyclosporine develops URTI. You give clarithromycin. What drug interaction are you concerned about?

A. Clarithromycin and acetaminophen

B. Clarithromycin and cyclosporine

C. Clarithromycin and ibuprofen

D. Clarithromycin and vitamin D

E. Clarithromycin and calcium

---

160 What is the most common symptom of cyclosporine toxicity from this interaction?

A. Headache

B. Tremor

C. Rash

D. Diarrhea

E. Cough

F. Blurred vision

---

161 How can you avoid cyclosporine toxicity when giving clarithromycin?

A. Increase cyclosporine dose

B. Decrease cyclosporine dose by 50%

C. Add another antibiotic

D. Give both medications at the same time

E. Switch to IV cyclosporine

---

162 A patient with Colles fracture received high-dose ketamine and midazolam. RR suddenly drops to 6. What should you give? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Naloxone

B. Flumazenil

C. Non-invasive ventilation (NIV)

D. Epinephrine

E. Atropine

F. Dantrolene

---

163 How can you prevent respiratory depression from sedation in the future? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Don't combine sedative medications

B. Start at lower doses with slow titration

C. Use higher doses for faster effect

D. Avoid monitoring during procedures

E. Always use general anesthesia instead

---

164 A patient on St. John's Wort is also taking warfarin. What is the concern?

A. St. John's Wort increases warfarin effect

B. St. John's Wort induces CYP enzymes causing treatment failure with warfarin

C. St. John's Wort has no interaction with warfarin

D. St. John's Wort causes warfarin allergy

E. St. John's Wort converts warfarin to a toxic metabolite

---

165 Which medications can have treatment failure when combined with St. John's Wort? (Select 4)

SELECT 4 CORRECT ANSWERS

A. Warfarin

B. Hormonal contraceptives

C. Cyclosporine

D. Digoxin

E. Acetaminophen

F. Vitamin D

G. Calcium supplements

---

166 A 28-year-old female with multiple somatic symptoms, many normal investigations, and has changed many doctors. After reassuring and counseling, what is the most important next step?

A. Order more investigations

B. Refer to a specialist immediately

C. Schedule regular outpatient visits not contingent on symptoms

D. Prescribe opioids for pain

E. Discharge from care

F. Admit to hospital

---

167 What is the first-line pharmacological treatment for somatic symptom disorder?

A. Benzodiazepine

B. Opioid

C. Antidepressant (SSRI)

D. Antipsychotic

E. Mood stabilizer

F. Stimulant

---

168 A 24-year-old male, weight 98 kg, height 172 cm, worried about obesity. Normal BP, lipids, and blood sugar. What should you ask in history to predict cardiovascular risk? (Select 4)

SELECT 5 CORRECT ANSWERS

A. Smoking

B. Family history of premature CHD

C. Sedentary lifestyle

D. Favorite color

E. Shoe size

F. Dyslipidemia history

G. Obstructive sleep apnea

---

169 What single clinical assessment predicts cardiac risk in an obese patient?

A. BMI calculation

B. Central obesity (waist circumference or waist-to-hip ratio)

C. Skin fold thickness

D. Body fat percentage

E. Grip strength

---

170 A patient with bacterial vaginosis presents with fishy odor vaginal discharge, thin green, no fever. What is the treatment?

A. Metronidazole 500 mg PO BID x 7 days

B. Fluconazole 150 mg PO single dose

C. Azithromycin 1g PO single dose

D. Doxycycline 100 mg PO BID x 7 days

E. Clindamycin 300 mg PO BID x 7 days

F. Ciprofloxacin 500 mg PO BID x 7 days

---

171 What precautions should a patient on metronidazole take? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Avoid alcohol during treatment and 24 hours after

B. Complete the full course of antibiotics

C. Take with grapefruit juice

D. Avoid sunlight completely

E. Take on an empty stomach only

---

172 A 22-year-old female wants to start contraception. What are the absolute contraindications to combined OCP? (Select 4)

SELECT 5 CORRECT ANSWERS

A. Age ≥35 and smoking ≥15 cigarettes/day

B. History of VTE

C. Known ischemic heart disease

D. Migraine with aura

E. Mild acne

F. Family history of diabetes

G. BMI 26

H. Current breast cancer

---

173 A woman on the phone says she missed 1 day of her COCP. She hasn't had sex in the last month. What advice should you give?

A. Stop the pack and start a new one

B. Take the missed pill as soon as noticed (OK to take 2 pills same day)

C. Use emergency contraception

D. Take 3 pills the next day

E. Discard the pack and use condoms only

---

174 A woman missed 2 pills in the FIRST week of her OCP pack. What should she do? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Take remaining pills at usual time

B. Use backup contraception

C. Stop the pack entirely

D. Take all missed pills at once

E. No action needed

---

175 A patient with essential tremor that improves with alcohol. His father has a similar condition. What is the first-line medication?

A. Levodopa/Carbidopa

B. Propranolol 60 mg ER PO daily

C. Gabapentin 300 mg PO TID

D. Topiramate 50 mg PO BID

E. Clonazepam 0.5 mg PO BID

F. Primidone 25 mg PO qhs

---

176 What aggravates essential tremor symptoms? (Select 4)

SELECT 4 CORRECT ANSWERS

A. Caffeine

B. Stress

C. Fatigue

D. Cold

E. Sleep

F. Relaxation

G. Warm environment

---

177 A young woman presents with gonorrhea after unprotected sex. What medications should you give? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Ceftriaxone 500 mg IM single dose

B. Doxycycline 100 mg PO BID x 7 days

C. Metronidazole 500 mg PO BID x 7 days

D. Fluconazole 150 mg PO single dose

E. Ciprofloxacin 500 mg PO single dose

F. Penicillin G 2.4 million units IM

---

178 When can the gonorrhea patient resume unprotected sex?

A. Immediately after treatment

B. 3 days after treatment

C. 7 days after treatment initiation

D. 14 days after treatment

E. After negative test of cure

F. 30 days after treatment

---

179 What advice should you give the gonorrhea patient? (Select 4)

SELECT 4 CORRECT ANSWERS

A. All sexual partners should be evaluated and treated

B. Consistent condom use

C. HIV counseling and testing

D. Return at 3 months for retesting

E. No follow-up needed

F. Avoid all sexual contact permanently

G. No need to notify partners

---

180 A mother brings her boy with mosquito bites. She wants to know the safe percentage of DEET in insect repellent for children >2 months.

A. 5%

B. 10-30%

C. 50%

D. 75%

E. 100%

F. DEET should never be used in children

---

181 What non-pharmacological measures protect from mosquito bites? (Select 3)

SELECT 3 CORRECT ANSWERS

A. Wear light-colored, long-sleeved shirts and pants

B. Avoid outdoor exposure between dusk and dawn

C. Sleep within insecticide-treated bed nets

D. Wear dark-colored clothing

E. Use perfumed products to repel mosquitoes

F. Keep windows open at night

---

182 A patient on gentamicin for infective endocarditis. What are the side effects? (Select 3)

SELECT 3 CORRECT ANSWERS

A. Nephrotoxicity

B. Ototoxicity (hearing loss)

C. Neurotoxicity

D. Hepatotoxicity

E. Cardiotoxicity

F. Pulmonary fibrosis

---

183 How should a patient on gentamicin be monitored? (Select 3)

SELECT 3 CORRECT ANSWERS

A. Renal function (BUN/creatinine)

B. Serum drug concentration

C. Hearing test

D. Liver function tests

E. Chest X-ray

F. ECG weekly

---

184 There is a measles outbreak. An unimmunized person comes to you. What vaccine should she get?

A. Single dose of MMR

B. Two doses of MMR separated by at least 28 days

C. Three doses of MMR over 6 months

D. Measles-only vaccine single dose

E. Immunoglobulin only

F. No vaccine available for adults

---

185 What are contraindications to MMR vaccine? (Select 3)

SELECT 3 CORRECT ANSWERS

A. Severe allergic reaction/anaphylaxis to previous dose or component

B. Pregnancy or attempting to become pregnant

C. Immunodeficiency

D. Mild upper respiratory infection

E. History of febrile seizure

F. Age over 50

G. Breastfeeding

---

186 A patient with paroxysmal positional vertigo (BPPV). What maneuver is used to diagnose?

A. Epley maneuver

B. Dix-Hallpike maneuver

C. Semont maneuver

D. Brandt-Daroff exercises

E. Romberg test

F. Rinne test

V. . What maneuver is used to diagnose?

---

187 What is the non-pharmacological treatment for BPPV? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Epley maneuver
B. Semont maneuver
C. Bed rest for 2 weeks
D. Cervical collar
E. Traction
F. Surgery immediately

---

188 What medications can be used for BPPV symptoms? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Betahistine 8-16 mg PO daily
B. Dimenhydrinate 50-100 mg PO q4-6h
C. Propranolol 40 mg PO BID
D. Amitriptyline 25 mg PO qhs
E. Carbamazepine 200 mg PO BID
F. Phenytoin 100 mg PO TID

---

189 A 5-year-old child on steroids for kidney disease with recurrent URTI. Which vaccines are CONTRAINDICATED? (Select 2)

SELECT 2 CORRECT ANSWERS

A. MMR
B. Varicella
C. Inactivated influenza
D. Pneumococcal conjugate
E. Hepatitis B
F. DTaP

---

190 What vaccine SHOULD be given to the immunocompromised child?

A. MMR
B. Varicella
C. Live attenuated influenza
D. Inactivated influenza vaccine
E. BCG
F. Oral polio vaccine

191 A patient with Meniere's disease on diuretics. What 2 medications from different classes can be used? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Betahistine 8 mg PO TID
B. Hydrochlorothiazide 25 mg PO daily
C. Propranolol 40 mg PO BID
D. Amitriptyline 25 mg PO qhs
E. Carbamazepine 200 mg PO BID
F. Gabapentin 300 mg PO TID

---

192 What non-pharmacological management should be advised for Meniere's disease? (Select 3)

SELECT 3 CORRECT ANSWERS

A. Vestibular rehabilitation therapy
B. Salt restriction
C. Limiting caffeine and alcohol
D. High-sodium diet
E. Increased caffeine intake
F. Bed rest for 6 months

---

193 An old man with dementia, MMSE 20/30, his wife wants something for his memory. What class of medication will you give?

A. SSRI
B. Cholinesterase inhibitor (Donepezil)
C. Benzodiazepine
D. Typical antipsychotic
E. Mood stabilizer
F. Stimulant

---

194 The dementia patient becomes aggressive and scatters things. What medication should you prescribe?

A. Haloperidol 5 mg PO BID
B. Olanzapine 2.5 mg PO BID
C. Diazepam 5 mg PO TID
D. Methylphenidate 10 mg PO BID
E. Lithium 300 mg PO BID
F. Carbamazepine 200 mg PO BID

---

195 What non-pharmacological strategies help manage agitation in dementia? (Select 4)

SELECT 5 CORRECT ANSWERS

A. Distraction and redirection

B. Structured routines

C. Providing calm, reassuring responses

D. Establishing consistent sleep and wake times

E. Frequent changes in environment

F. Physical restraints as first line

G. Loud music therapy

H. Exposure to bright light during early part of day

---

196 A patient presents with alcohol withdrawal. He has not had alcohol in 24 hours. What conditions is he at risk for? (Select 4)

SELECT 4 CORRECT ANSWERS

A. Alcohol tremulousness (6-36h)

B. Withdrawal seizures (6-48h)

C. Alcoholic hallucinosis (12-48h)

D. Delirium tremens (48-96h)

E. Alcohol-induced hepatitis (immediate)

F. Wernicke encephalopathy (immediate)

---

197 What class of medication prevents alcohol withdrawal complications?

A. Antipsychotics

B. Benzodiazepines

C. SSRIs

D. Anticonvulsants

E. Opioids

F. Stimulants

---

198 What other medications should be given to an alcohol withdrawal patient? (Select 3)

SELECT 4 CORRECT ANSWERS

A. Isotonic IV fluid

B. Thiamine 100 mg IM/IV

C. Glucose (D50W) IV

D. Naloxone IV

E. Flumazenil IV

F. Phenytoin IV

G. Multivitamins with folate

---

199 A patient with an old digoxin overdose. K=6, Na=128, creatinine=170, ECG shows arrhythmia. What IV medications should be given to correct hyperkalemia? (Select 2 with different mechanisms)

SELECT 3 CORRECT ANSWERS

A. Insulin with glucose

B. Loop diuretics with normal saline

C. **

D. Sodium bicarbonate

E. Oral kayexalate

F. IV potassium chloride

---

200 What is the antidote for digoxin toxicity?

A. Naloxone

B. Flumazenil

C. N-acetylcysteine

D. Digoxin-specific antibody fragments (Fab)

E. Protamine sulfate

F. Vitamin K

G. Activated charcoal

H. Deferoxamine

---

201 A female patient with low BMI and distorted body image. Nutritional therapy and CBT done without good result (Anorexia Nervosa). What medication should you give?

A. Fluoxetine 20 mg PO daily
B. Olanzapine 2.5 mg PO daily
C. Amitriptyline 25 mg PO qhs
D. Bupropion 150 mg PO daily
E. Risperidone 1 mg PO daily
F. Mirtazapine 15 mg PO qhs

---

202 In anorexia nervosa, which medication is contraindicated and why? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Bupropion - higher incidence of seizures in eating disorders
B. TCAs - risk of cardiotoxicity in malnourished patients
C. SSRIs - cause weight gain
D. Olanzapine - causes hypoglycemia
E. Benzodiazepines - cause respiratory depression
F. Metformin - causes lactic acidosis

---

203 A male aged 20s presents with unilateral burning chest pain and skin rash along the 11th rib with vesicles (Herpes Zoster). What medications should you give? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Valacyclovir 1000 mg PO TID x 7 days
B. Ibuprofen 400 mg PO BID
C. Acyclovir 200 mg PO 5 times daily
D. Amoxicillin 500 mg PO TID
E. Prednisone 60 mg PO daily
F. Metronidazole 500 mg PO TID

---

204 What non-pharmacological advice should you give a herpes zoster patient to prevent spread? (Select 3)

SELECT 4 CORRECT ANSWERS

A. Keep the rash covered
B. Wash hands often
C. Avoid contact with pregnant women who never had chickenpox
D. Share towels to build immunity in others
E. Expose the rash to sunlight for healing
F. Avoid contact with immunocompromised individuals

---

205 A patient develops post-herpetic neuralgia 8 weeks after herpes zoster. What medications can be used? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Pregabalin 75 mg PO BID

B. Amitriptyline 10 mg PO qhs

C. Ibuprofen 400 mg PO TID

D. Metformin 500 mg PO BID

E. Omeprazole 20 mg PO daily

F. Atenolol 50 mg PO daily

---

206 What vaccine could have prevented herpes zoster in an elderly patient?

A. Varicella vaccine (Varivax)

B. Recombinant zoster vaccine (Shingrix)

C. Pneumococcal vaccine

D. Influenza vaccine

E. MMR vaccine

F. Hepatitis B vaccine

---

207 A 25-year-old university student from Asia is a chronic carrier of Hepatitis B. What advice should you give to prevent spread? (Select 4)

SELECT 5 CORRECT ANSWERS

A. Have sexual and household contacts vaccinated

B. Use barrier protection during sexual intercourse

C. Do not share toothbrushes or razors

D. Do not donate blood, organs, or sperm

E. Avoid all social contact

F. Isolate at home for 6 months

G. Clean blood spills with detergent or bleach

---

208 What vaccine should be given to a chronic Hepatitis B carrier?

A. Hepatitis B vaccine booster

B. Hepatitis A vaccine

C. Pneumococcal vaccine

D. Influenza vaccine

E. HPV vaccine

---

209 A patient presents with unilateral facial weakness and inability to close the eye (Bell's Palsy). What first-line treatment should you give? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Prednisolone 60 mg PO daily x 7 days

B. Valacyclovir 1000 mg PO TID x 7 days

C. Amoxicillin 500 mg PO TID x 10 days

D. Gabapentin 300 mg PO TID

E. Methotrexate 15 mg PO weekly

F. Carbamazepine 200 mg PO BID

---

210 What eye care advice should you give a Bell's Palsy patient? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Eye lubricant with ointment and drops

B. Tape the eyelid during sleep

C. Wear an eye patch 24/7

D. Use steroid eye drops daily

E. No eye care needed

F. Apply warm compresses to the eye

---

211 A diabetic patient on metformin, gliclazide, and empagliflozin. Obese with CVD. HbA1c is 12.6. What medication should you start?

A. Sitagliptin 100 mg PO daily

B. Pioglitazone 15 mg PO daily

C. Long-acting insulin (Glargine)

D. Acarbose 50 mg PO TID

E. Repaglinide 0.5 mg PO TID

F. Increase metformin dose

---

212 What are the most common side effects of insulin? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Hypoglycemia

B. Weight gain

C. Hepatotoxicity

D. Renal failure

E. Pulmonary fibrosis

F. Pancreatitis

---

213 When starting insulin in a Type 2 diabetic already on sulfonylurea (gliclazide), what should you do with the sulfonylurea?

A. Continue at the same dose
B. Double the dose
C. Taper and discontinue
D. Switch to metformin
E. Add another sulfonylurea

---

214 What is the first-line medication for Type 2 Diabetes?

A. Sulfonylurea (Glipizide)
B. Biguanide (Metformin)
C. DPP-4 inhibitor (Sitagliptin)
D. SGLT2 inhibitor (Empagliflozin)
E. GLP-1 agonist (Liraglutide)
F. Insulin Glargine

---

215 What are the side effects of metformin? (Select 3)

SELECT 3 CORRECT ANSWERS

A. GI disturbances (nausea, vomiting, diarrhea)
B. Vitamin B12 deficiency
C. Lactic acidosis (rare)
D. Hypoglycemia (common)
E. Weight gain
F. Pancreatitis

---

216 A child presents with barking cough, stridor, and respiratory distress (Croup). Weight 13.3 kg. What medication should you give to stabilize?

A. Albuterol nebulizer
B. Dexamethasone 0.6 mg/kg PO/IM single dose
C. Amoxicillin 90 mg/kg PO divided BID
D. Ipratropium nebulizer
E. Prednisone 2 mg/kg PO daily x 5 days
F. Azithromycin 10 mg/kg PO daily

---

217 If the croup patient is stressed with stridor, what additional treatment should be given?

- **A.** IV antibiotics
- **B.** Nebulized racemic epinephrine
- **C.** Oral prednisolone
- **D.** Inhaled budesonide
- **E.** IM ceftriaxone
- **F.** Nebulized albuterol

---

218 A pregnant woman at 34 weeks presents with severe unilateral leg pain. Ultrasound confirms DVT. What medication should you start?

- **A.** Warfarin 5 mg PO daily
- **B.** Rivaroxaban 15 mg PO BID
- **C.** Low molecular weight heparin (Enoxaparin 1 mg/kg SC q12h)
- **D.** Aspirin 81 mg PO daily
- **E.** Clopidogrel 75 mg PO daily
- **F.** Apixaban 5 mg PO BID

---

219 What anticoagulant should NOT be given in pregnancy and why?

- **A.** Heparin - causes osteoporosis
- **B.** Warfarin - crosses placenta causing teratogenicity
- **C.** Enoxaparin - causes bleeding
- **D.** Aspirin - causes Reye syndrome
- **E.** Fondaparinux - causes fetal hemorrhage

---

220 A woman with endometriosis on NSAIDs with no improvement. What is the best medication to give? (Select 1)

- **A.** Acetaminophen 1g PO QID
- **B.** Combined estrogen-progestin OCP (Yasmin)
- **C.** Morphine 15 mg PO q4h
- **D.** Gabapentin 300 mg PO TID
- **E.** Methotrexate 15 mg PO weekly

---

221 The endometriosis patient returns 3 months later with no improvement on OCP. What is the next step?

- **A.** Increase OCP dose
- **B.** GnRH agonist (Leuprolide 3.75 mg IM q4weeks x 6 months)
- **C.** Hysterectomy
- **D.** Add opioids
- **E.** Switch to acetaminophen
- **F.** Aromatase inhibitor immediately

---

222 A patient presents with a painful, fluctuant swelling on the back of the neck (carbuncle/abscess). What is the management?

- **A.** Oral antibiotics only
- **B.** Incision and drainage
- **C.** Warm compresses only
- **D.** Topical mupirocin
- **E.** IV antibiotics only
- **F.** Observation

---

223 What is the most common organism causing skin abscesses?

- **A.** Streptococcus pyogenes
- **B.** Staphylococcus aureus
- **C.** Pseudomonas aeruginosa
- **D.** E. coli
- **E.** Klebsiella pneumoniae

---

224 After incision and drainage of an abscess with 7 cm erythema, what antibiotic should you prescribe for suspected CA-MRSA?

- **A.** Amoxicillin 500 mg PO TID
- **B.** TMP-SMX DS (160/800 mg) PO BID x 7 days
- **C.** Cephalexin 500 mg PO QID
- **D.** Ciprofloxacin 500 mg PO BID
- **E.** Azithromycin 500 mg PO daily
- **F.** Penicillin V 500 mg PO QID

---

225 A patient presents with right flank pain radiating to the groin (renal colic). What medications should you give? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Ketorolac 30 mg IV
B. Morphine 5 mg IV
C. Acetaminophen 1g PO
D. Gabapentin 300 mg PO
E. Amitriptyline 25 mg PO
F. Prednisone 40 mg PO

---

226 How can you speed the passage of a kidney stone?

A. Tamsulosin 0.4 mg PO daily x 4 weeks
B. Furosemide 40 mg PO daily
C. Hydrochlorothiazide 25 mg PO daily
D. Spironolactone 25 mg PO daily
E. Atenolol 50 mg PO daily

---

227 What non-pharmacological advice prevents kidney stone recurrence? (Select 3)

SELECT 3 CORRECT ANSWERS

A. Sufficient fluid intake to produce >2L urine/day
B. Limit dietary sodium below 2300 mg/day
C. Increase fruit and vegetable intake
D. Decrease water intake
E. High-sodium diet
F. Avoid all dairy products

---

228 An old patient on lithium, ramipril, acetaminophen, and levothyroxine. No investigations done for a long time. What investigations should be ordered? (Select 5)

SELECT 5 CORRECT ANSWERS

A. Serum lithium levels
B. TSH
C. Renal function (BUN, creatinine)
D. CBC with differential
E. Serum electrolytes
F. Chest X-ray
G. Pulmonary function tests

---

229 The patient from Question 228 is brought to the ER confused, ataxic, with nausea. What single investigation should you order?

- **A.** CT head
- **B.** Serum lithium level
- **C.** Blood glucose
- **D.** Liver function tests
- **E.** Chest X-ray
- **F.** Lumbar puncture

---

230 What are the side effects of lithium? (Select 4)

SELECT 4 CORRECT ANSWERS

- **A.** Hypothyroidism
- **B.** Nephrogenic diabetes insipidus
- **C.** Tremors (fine)
- **D.** Cardiac arrhythmia
- **E.** Improved renal function
- **F.** Weight loss
- **G.** Hair growth

---

231 What monitoring is required for a patient on lithium? (Select 3)

SELECT 3 CORRECT ANSWERS

- **A.** Lithium concentration
- **B.** Thyroid and renal function every 6 months
- **C.** Serum calcium at baseline and every 12 months
- **D.** Weekly chest X-ray
- **E.** Daily blood glucose
- **F.** Monthly liver biopsy

---

232 A patient with osteoarthritis is taking acetaminophen. What other medications can be used? (Select 2)

SELECT 2 CORRECT ANSWERS

- **A.** Topical NSAIDs (Diclofenac 1% gel QID)
- **B.** Oral NSAIDs (Naproxen 500 mg PO BID)
- **C.** Opioids as first line
- **D.** Methotrexate
- **E.** Hydroxychloroquine
- **F.** Colchicine

---

233 What should be given to prevent NSAID-induced GI side effects?

- **A.** Omeprazole 20 mg PO daily
- **B.** Sucralfate 1g PO QID
- **C.** Misoprostol 200 mcg PO QID
- **D.** Famotidine 20 mg PO BID
- **E.** Bismuth subsalicylate

---

234 A 6-week postpartum mother cannot sleep after the baby sleeps and has depressive symptoms but does not want to harm the baby. What medication should you give?

- **A.** Fluoxetine 20 mg PO daily
- **B.** Sertraline 50 mg PO daily
- **C.** Amitriptyline 25 mg PO qhs
- **D.** Bupropion 150 mg PO daily
- **E.** Venlafaxine 37.5 mg PO daily
- **F.** Paroxetine 20 mg PO daily

---

235 Why is sertraline preferred during lactation?

- **A.** It has the longest half-life
- **B.** It has the lowest transfer into breast milk
- **C.** It is the cheapest option
- **D.** It causes the most weight loss
- **E.** It has no side effects
- **F.** It is the strongest antidepressant

---

236 A breastfeeding mother develops mastitis with systemic symptoms (fever, chills). What antibiotic should you prescribe?

- **A.** Dicloxacillin 500 mg PO QID x 10 days
- **B.** Metronidazole 500 mg PO TID x 7 days
- **C.** Ciprofloxacin 500 mg PO BID x 7 days
- **D.** Azithromycin 500 mg PO daily x 3 days
- **E.** Doxycycline 100 mg PO BID x 7 days
- **F.** TMP-SMX DS PO BID x 7 days

---

237 What non-pharmacological interventions increase breast milk production? (Select 4)

SELECT 4 CORRECT ANSWERS

A. Optimizing breastfeeding technique (position and latch-on)
B. Increasing breastfeeding frequency
C. Ensuring complete breast emptying
D. Reducing stress and getting adequate sleep
E. Supplementing with formula immediately
F. Restricting fluid intake
G. Avoiding breastfeeding at night

---

238 A 5-day-old neonate presents with jaundice. Bilirubin is 200 and below the treatment threshold for his age. He is healthy and feeding well. What non-pharmacological measures should you take? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Ensure adequate feeds and hydration
B. Continue breastfeeding
C. Start phototherapy immediately
D. Exchange transfusion
E. Stop breastfeeding
F. Start IV antibiotics

---

239 The neonate from Question 238 later develops an anal fissure after switching to formula. What topical treatment should you give?

A. Hydrocortisone 1% cream
B. Petroleum jelly (Vaseline)
C. Nitroglycerin 0.2% ointment
D. Mupirocin 2% ointment
E. Clotrimazole 1% cream

---

240 A child female 9 years old, weight 90 kg, height 135 cm. Parents concerned about weight for 4 years. What clinical tool should you use to tailor treatment?

A. Waist circumference
B. Body mass index (BMI) calculation
C. Skin fold thickness
D. Bioelectrical impedance
E. DEXA scan
F. Serum leptin level

---

241 What dietary modifications should you advise for the obese child? (Select 5)

SELECT 5 CORRECT ANSWERS

A. Eat moderate breakfast daily, avoid high-sugar foods
B. Decrease fast food to 1 or fewer meals per week
C. Avoid sugar-sweetened beverages
D. Eat 5 servings of fruits and vegetables daily
E. Decrease portion sizes
F. Skip meals to reduce calories
G. Drink diet soda freely
H. Eat only one meal per day

---

242 A child with febrile convulsion. Temperature 38.5°C, shaking lasted 2-3 minutes and stopped. Runny nose and cough. What medication should you give now?

A. Diazepam 0.2 mg/kg rectal
B. Phenytoin 20 mg/kg IV
C. Acetaminophen 15 mg/kg PO
D. Phenobarbital 20 mg/kg IV
E. Levetiracetam 20 mg/kg IV
F. Valproic acid 15 mg/kg IV

---

243 Is there a daily medication to prevent future febrile seizures?

A. Yes, phenobarbital daily
B. Yes, valproic acid daily
C. Yes, carbamazepine daily
D. No proven prophylaxis treatment exists
E. Yes, levetiracetam daily

---

244 If the febrile seizure child presented in status epilepticus, what medication would you give?

A. Acetaminophen 15 mg/kg PO
B. Diazepam 0.1-0.2 mg/kg IV or Lorazepam 0.05-0.1 mg/kg IV
C. Phenytoin 5 mg/kg IV
D. Carbamazepine 10 mg/kg PO
E. Gabapentin 10 mg/kg PO
F. Topiramate 5 mg/kg PO

---

245 A 26-year-old presents with recurrent boils in the axilla (Hidradenitis Suppurativa). What non-pharmacological advice should you give? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Smoking cessation
B. Weight reduction
C. Apply heat packs daily
D. Increase sugar intake
E. Wear tight-fitting clothes
F. Avoid all exercise

---

246 What topical medication should you give for hidradenitis suppurativa?

A. Mupirocin 2% ointment BID
B. Topical clindamycin 1% solution BID
C. Hydrocortisone 1% cream BID
D. Benzoyl peroxide 5% gel daily
E. Tretinoin 0.025% gel daily

---

247 The hidradenitis patient returns with 2 large boils. What oral medication should you give?

A. Amoxicillin 500 mg PO TID
B. Doxycycline 100 mg PO BID for several months
C. Ciprofloxacin 500 mg PO BID
D. Metronidazole 500 mg PO TID
E. Fluconazole 150 mg PO weekly

---

248 A 45-year-old with rosacea presents with papules, pustules, and telangiectasia on her face. What topical treatment should you give?

A. Tretinoin 0.025% gel daily
B. Metronidazole 0.75% gel BID
C. Hydrocortisone 2.5% cream BID
D. Benzoyl peroxide 10% gel daily
E. Clindamycin 1% lotion BID

---

249 If topical treatment fails for rosacea, what oral medication should you prescribe?

A. Isotretinoin 0.5 mg/kg/day
B. Doxycycline 100 mg PO BID x 4-12 weeks
C. Prednisone 40 mg PO daily
D. Methotrexate 15 mg PO weekly
E. Fluconazole 150 mg PO weekly

---

250 What non-pharmacological advice should you give a rosacea patient? (Select 4)

SELECT 4 CORRECT ANSWERS

A. Avoid triggers of flushing (keep a diary)
B. Use gentle skincare
C. Use adequate sun protection
D. Avoid spicy foods and alcohol
E. Scrub face vigorously daily
F. Use hot water for face washing
G. Apply heavy makeup to cover lesions

---

251 A patient with fibromyalgia and HTN. BP normal, exercise and diet good. On venlafaxine 37.5 mg. What non-pharmacological advice should you give? (Select 4)

SELECT 4 CORRECT ANSWERS

A. Good sleep hygiene
B. Initiate exercise program (aquatic aerobic)
C. Relaxation techniques and stress reduction
D. Maintain healthy weight
E. Complete bed rest
F. Avoid all physical activity
G. Increase caffeine intake

---

252 What are the first-line medications for fibromyalgia? (Select 3)

SELECT 3 CORRECT ANSWERS

A. TCA (Amitriptyline 10 mg PO qhs)
B. SNRI (Duloxetine 30 mg PO daily)
C. Anticonvulsant (Pregabalin 25 mg PO qhs)
D. Opioid (Morphine 15 mg PO q4h)
E. NSAID (Ibuprofen 400 mg PO TID)
F. Benzodiazepine (Diazepam 5 mg PO TID)

253 A patient with genital herpes presents with multiple small painful ulcers. What is the treatment?

- **A.** Valacyclovir 1g PO BID x 7 days
- **B.** Amoxicillin 500 mg PO TID x 10 days
- **C.** Fluconazole 150 mg PO single dose
- **D.** Metronidazole 500 mg PO BID x 7 days
- **E.** Doxycycline 100 mg PO BID x 7 days
- **F.** Ceftriaxone 500 mg IM single dose

---

254 What advice should you give a patient with genital herpes? (Select 3)

SELECT 3 CORRECT ANSWERS

- **A.** Use latex condoms with every sexual encounter
- **B.** Communicate with sexual partners before first encounter
- **C.** Avoid oral sex if ulcers around the mouth
- **D.** Herpes is curable with antibiotics
- **E.** No precautions needed between outbreaks
- **F.** Share towels to build partner immunity

---

255 A patient with psoriasis. What is the first-line topical treatment?

- **A.** Topical corticosteroids (Betamethasone 0.05% BID)
- **B.** Topical methotrexate
- **C.** Topical cyclosporine
- **D.** Topical tacrolimus only
- **E.** Topical antibiotics
- **F.** Topical retinoids only

---

256 What other first-line topical options exist for psoriasis? (Select 2)

SELECT 2 CORRECT ANSWERS

- **A.** Calcipotriene 0.005% ointment BID (Vitamin D analogue)
- **B.** Salicylic acid 3% cream (Keratolytic)
- **C.** Topical metformin
- **D.** Topical insulin
- **E.** Topical aspirin
- **F.** Topical lithium

---

257 For severe psoriasis with joint involvement, what systemic medication is best?

A. Methotrexate 2.5-22.5 mg PO/SC/IM weekly
B. Amoxicillin 500 mg PO TID
C. Prednisone 60 mg PO daily long-term
D. Gabapentin 300 mg PO TID
E. Fluoxetine 20 mg PO daily

---

258 A child with diaper dermatitis. What non-pharmacological measures should you advise? (Select 3)

SELECT 3 CORRECT ANSWERS

A. Frequent diaper changes
B. Air exposure
C. Gentle skin cleansing
D. Use alcohol-based wipes
E. Keep diaper on continuously
F. Apply talcum powder liberally

---

259 What pharmacological treatments are used for diaper dermatitis? (Select 3)

SELECT 3 CORRECT ANSWERS

A. Topical hydrocortisone 1% ointment BID x 3-7 days
B. Topical clotrimazole ointment BID x 2 weeks
C. Zinc oxide cream with every diaper change
D. Oral fluconazole 6 mg/kg
E. IV antibiotics
F. Topical metronidazole gel

---

260 An old man with TIA a few days ago. Heart rate irregularly irregular. BP 165/100 on ramipril 5 mg. What anticoagulant should you give?

A. Aspirin 81 mg PO daily
B. Clopidogrel 75 mg PO daily
C. Apixaban 5 mg PO BID
D. Heparin IV infusion
E. Warfarin 2 mg PO daily
F. Enoxaparin 1 mg/kg SC BID

---

261 What medication should you give to control the heart rate in atrial fibrillation? (Select for chronic control)

A. Oral beta-blocker (Metoprolol)

B. **

C. IV Adenosine

D. Oral flecainide

E. IV Amiodarone

F. Oral digoxin

---

262 A patient presents with signs of MI at a rural hospital. The closest cardiac center is 4 hours away. You started aspirin. What is the next step?

A. Wait for transfer to cardiac center

B. Fibrinolytic therapy (Tenecteplase)

C. Start heparin infusion only

D. Give morphine and observe

E. Perform PCI at rural hospital

F. Discharge with follow-up

---

263 What are absolute contraindications for fibrinolytic therapy in MI? (Select 4)

SELECT 5 CORRECT ANSWERS

A. History of any intracranial hemorrhage

B. Ischemic stroke within preceding 3 months

C. Aortic dissection

D. Active bleeding

E. Age over 70

F. Mild hypertension

G. Diabetes mellitus

H. Significant closed head trauma within 3 months

---

264 A 40-year-old male smoker (15 cigarettes/day), smokes cannabis 1g/day, drinks wine twice weekly. Presents with persistent vomiting for 2-3 days. What is the cause?

A. Peptic ulcer disease

B. Cannabinoid hyperemesis syndrome

C. Alcohol gastritis

D. Gastroenteritis

E. Pancreatitis

F. Small bowel obstruction

---

265 What IV fluid should you give the cannabinoid hyperemesis patient?

- **A.** D5W
- **B.** Normal saline or Lactated Ringer's
- **C.** Half normal saline
- **D.** D10W
- **E.** Albumin 5%
- **F.** Hypertonic saline

---

266 What antiemetics can you give for cannabinoid hyperemesis? (Select 2)

SELECT 2 CORRECT ANSWERS

- **A.** Ondansetron 4 mg IV q8h
- **B.** Metoclopramide 10 mg IV q8h
- **C.** Domperidone 10 mg PO TID
- **D.** Scopolamine patch
- **E.** Dexamethasone 8 mg IV
- **F.** Promethazine 25 mg PO

---

267 A 17-year-old boy with fibromyalgia smokes marijuana since age 14. He read that cannabis is good for fibromyalgia. What are the contraindications to prescribing medical cannabis? (Select 4)

SELECT 5 CORRECT ANSWERS

- **A.** Patient under 25 years of age
- **B.** Substance abuse disorder
- **C.** History of psychosis
- **D.** Cardiovascular disease
- **E.** Mild headache
- **F.** Family history of diabetes
- **G.** Pregnancy and breastfeeding

---

268 A 35-year-old homosexual presents asking for HIV pre-exposure prophylaxis (PrEP). What are the common side effects? (Select 4)

SELECT 4 CORRECT ANSWERS

A. Nausea

B. Diarrhea

C. Headache

D. Myalgia

E. Permanent liver failure

F. Blindness

G. Cardiac arrest

P. . What are the common side effects? (Select 4)

---

269 What advice should you give a patient on HIV PrEP? (Select 4)

SELECT 4 CORRECT ANSWERS

A. Encourage condom use until drug achieves intracellular concentrations

B. Strict adherence to routine monitoring (HIV testing, STIs, creatinine)

C. Encourage other risk reduction methods

D. Strict adherence to medication

E. Unprotected sex is completely safe on PrEP

F. No follow-up needed once started

G. Donate blood freely

---

270 A patient on amiodarone. What are the serious side effects? (Select 4)

SELECT 5 CORRECT ANSWERS

A. Bradycardia and hypotension

B. Hepatotoxicity

C. Pulmonary toxicity (interstitial pneumonitis/fibrosis)

D. Hypo or hyperthyroidism

E. Improved vision

F. Weight loss

G. QT prolongation

---

271 What monitoring is required for a patient on amiodarone? (Select 4)

SELECT 5 CORRECT ANSWERS

A. TSH (baseline, 3 months, then yearly)

B. Liver function tests (baseline, 6 months, then annually)

C. Chest X-ray and PFTs (baseline, then yearly)

D. ECG (baseline, during loading, then annually)

E. Eye examination yearly

F. Weekly blood glucose

G. Monthly bone density scan

---

272 A patient with 3rd degree heart block. P waves dissociated from QRS complexes. What is the immediate treatment?

A. Epinephrine 1 mg IV

B. Atropine 0.5 mg IV (may repeat q3-5min, max 3 mg)

C. Amiodarone 300 mg IV

D. Adenosine 6 mg IV rapid push

E. Lidocaine 1 mg/kg IV

F. Verapamil 5 mg IV

---

273 What is the definitive treatment for 3rd degree heart block?

A. Lifelong atropine

B. Permanent implantable pacemaker

C. Amiodarone maintenance

D. Beta-blocker therapy

E. Calcium channel blocker therapy

F. Digoxin therapy

---

274 A patient with active TB. What is the treatment regimen? (Select 4 medications)

SELECT 4 CORRECT ANSWERS

A. Isoniazid

B. Rifampin

C. Ethambutol

D. Pyrazinamide

E. Azithromycin

F. Ciprofloxacin

G. Metronidazole

---

275 What supplement should be given with isoniazid to prevent peripheral neuropathy?

- **A.** Vitamin B12
- **B.** Pyridoxine (Vitamin B6) 50-100 mg/day
- **C.** Folic acid
- **D.** Vitamin D
- **E.** Thiamine
- **F.** Vitamin C

---

276 What is the duration of TB treatment?

- **A.** 2 weeks
- **B.** 1 month
- **C.** 3 months
- **D.** At least 6 months
- **E.** 1 year
- **F.** Lifelong

---

277 H. pylori eradication therapy. What is the recommended regimen?

- **A.** PPI + Amoxicillin only x 7 days
- **B.** PPI + Amoxicillin + Metronidazole + Clarithromycin x 14 days
- **C.** PPI + Metronidazole only x 7 days
- **D.** H2 blocker + Amoxicillin x 10 days
- **E.** Sucralfate + Bismuth x 7 days

---

278 A patient with watery diarrhea after finishing levofloxacin (C. difficile). What is the first-line treatment?

- **A.** Metronidazole 500 mg PO TID x 10 days
- **B.** Fidaxomicin 200 mg PO BID x 10 days
- **C.** Ciprofloxacin 500 mg PO BID x 7 days
- **D.** Amoxicillin 500 mg PO TID x 10 days
- **E.** Loperamide 4 mg PO then 2 mg PRN
- **F.** Rifaximin 550 mg PO TID x 14 days

---

279 What advice should you give while the C. difficile patient waits for investigation results? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Contact precautions and hand hygiene

B. Avoid anti-diarrheal agents

C. Take loperamide freely

D. Continue levofloxacin

E. Restrict all oral intake

F. No precautions needed

---

280 A patient with sleep apnea. What is the definitive treatment?

A. Benzodiazepine at bedtime

B. Positive airway pressure (CPAP)

C. Oral melatonin

D. Antihistamine at bedtime

E. Oxygen therapy only

F. Stimulant medication

---

281 What behavioral modifications should be advised for sleep apnea? (Select 4)

SELECT 4 CORRECT ANSWERS

A. Weight loss and exercise

B. Avoid sleeping supine

C. Alcohol avoidance

D. Avoid sedating medications (BDZ, opiates, antihistamines)

E. Sleep in a cold room

F. Increase caffeine intake

G. Sleep more hours

---

282 A patient with polymyalgia rheumatica on steroids. What is the initial treatment?

A. Prednisone 15 mg PO daily

B. Methotrexate 15 mg PO weekly

C. Ibuprofen 800 mg PO TID

D. Hydroxychloroquine 400 mg PO daily

E. Adalimumab 40 mg SC biweekly

---

283 How should steroids be tapered in polymyalgia rheumatica?

A. Stop abruptly after 2 weeks

B. Taper within 4-8 weeks (>10mg: lower by 2.5mg q2-4 weeks; <10mg: lower by 1mg q2-4 weeks)

C. Double the dose then stop

D. Continue same dose indefinitely

E. Switch to NSAID immediately

---

284 A boy with lice nits on his hair. What medication should you use?

A. Ivermectin 200 mcg/kg PO

B. Permethrin 1% (leave 10 minutes, wash, repeat in 7 days)

C. Lindane 1% shampoo

D. Malathion 0.5% lotion

E. Dimeticone 4% lotion

F. Spinosad 0.9% suspension

---

285 What other interventions should be done for lice management? (Select 3)

SELECT 3 CORRECT ANSWERS

A. Examine and treat household members if infested

B. Wash clothing and linen in hot water and high-heat dryer

C. Vacuum furniture and carpeting

D. Shave the child's head completely

E. Fumigate the entire house

F. Isolate the child for 2 weeks

---

286 A child with umbilicated lesions (Molluscum Contagiosum). What treatments can be used? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Cryotherapy with liquid nitrogen

B. Curettage with topical anesthetics

C. Oral acyclovir

D. IV antibiotics

E. Systemic steroids

F. Radiation therapy

---

287 A 38-year-old female with toenail onychomycosis. Topical treatment failed for one year. What oral medications can you prescribe? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Terbinafine 250 mg PO daily x 12 weeks

B. Itraconazole 200 mg PO daily x 12 weeks

C. Fluconazole 50 mg PO daily x 4 weeks

D. Ketoconazole 200 mg PO daily x 6 months

E. Griseofulvin 500 mg PO daily x 4 weeks

---

288 What investigation should be done before starting oral antifungal therapy for onychomycosis?

A. CBC

B. Liver function tests

C. Renal function tests

D. Chest X-ray

E. ECG

F. Serum calcium

---

289 A 25-year-old man with hypopigmented patches on his trunk (Tinea Versicolor). What topical treatment should you give?

A. Ketoconazole 2% shampoo

B. Hydrocortisone 1% cream

C. Mupirocin 2% ointment

D. Acyclovir 5% cream

E. Calamine lotion

F. Benzoyl peroxide 5% gel

---

290 What oral medication can be used for tinea versicolor if topical fails?

A. Fluconazole 400 mg PO once

B. Amoxicillin 500 mg PO TID

C. Prednisone 40 mg PO daily

D. Metronidazole 500 mg PO TID

E. Doxycycline 100 mg PO BID

---

291 A 60-year-old with urgency, frequency, nocturia 8 times. Prostate and abdominal exam normal (Urinary Incontinence/Overactive Bladder). What medication should you give?

- **A.** Tamsulosin 0.4 mg PO daily
- **B.** Oxybutynin 5 mg PO daily
- **C.** Finasteride 5 mg PO daily
- **D.** Sildenafil 50 mg PO PRN
- **E.** Desmopressin 0.2 mg PO qhs
- **F.** Furosemide 20 mg PO daily

---

292 The urinary incontinence patient returns 2 weeks later with BP 90/60. What should you do?

- **A.** Increase oxybutynin dose
- **B.** Add another anticholinergic
- **C.** Correct hypotension and switch to Mirabegron 25 mg PO daily
- **D.** Start IV fluids and continue oxybutynin
- **E.** Add midodrine

---

293 A woman with hot flashes wants non-hormonal medications for menopause. What can you prescribe? (Select 4)

SELECT 4 CORRECT ANSWERS

- **A.** SSRI: Paroxetine 7.5 mg PO daily
- **B.** SNRI: Venlafaxine 37.5 mg PO daily
- **C.** Anticonvulsant: Gabapentin 300 mg PO TID
- **D.** Alpha-2 agonist: Clonidine 0.1 mg PO daily
- **E.** Opioid: Morphine 15 mg PO q4h
- **F.** Benzodiazepine: Diazepam 5 mg PO TID
- **G.** Stimulant: Methylphenidate 10 mg PO BID

---

294 If the menopausal woman wants hormonal therapy and has an intact uterus, what should you prescribe?

- **A.** Estrogen alone (CEE 0.625 mg)
- **B.** Estrogen + Progesterone (CEE 0.625 mg + MPA 2.5 mg)
- **C.** Progesterone alone
- **D.** Testosterone gel
- **E.** GnRH agonist
- **F.** Tamoxifen 20 mg daily

---

295 What are absolute contraindications to hormone replacement therapy? (Select 3)

SELECT 4 CORRECT ANSWERS

A. History of breast cancer

B. Coronary artery disease

C. Previous VTE or stroke

D. Mild hot flashes

E. Age over 50

F. Family history of osteoporosis

G. Active liver disease

---

296 A woman presents with bruises on her elbow and forearm. She is sad and says she was "playing with her partner" (Intimate Partner Violence). What should you do? (Select 3)

SELECT 3 CORRECT ANSWERS

A. Express empathy and acknowledge the situation

B. Assess safety of the patient

C. Ask if she wants referral for help and support

D. Confront the partner immediately

E. Ignore the bruises

F. Discharge without further action

---

297 What questions should you ask before sending the IPV patient home? (Select 3)

SELECT 4 CORRECT ANSWERS

A. Do you feel safe in your relationship?

B. Are your friends aware you have been hurt?

C. Do you have a safe place to go in an emergency?

D. What is your partner's phone number?

E. What did you do to provoke the violence?

F. Is there an accessible weapon?

---

298 A patient with grief after her daughter died one month ago. Does not meet criteria for depression. Sleep disturbance but does not affect work. What medication should you give for sleep?

- **A.** Zopiclone 3.75 mg PO qhs
- **B.** Lorazepam 1 mg PO qhs
- **C.** Trazodone 50 mg PO qhs
- **D.** None - not affecting daily activities
- **E.** Amitriptyline 25 mg PO qhs
- **F.** Quetiapine 25 mg PO qhs

---

299 The grief patient returns 7 months later with worse sleep, fatigue, guilt, and more symptoms. What is the most important thing to assess?

- **A.** Sleep quality
- **B.** Appetite changes
- **C.** Suicidal ideation
- **D.** Work performance
- **E.** Social activities
- **F.** Exercise habits

---

300 A patient with serotonin syndrome (tremor, myoclonus, hyperreflexia, fever, agitation). What is the treatment? (Select 3)

SELECT 4 CORRECT ANSWERS

- **A.** Discontinue serotonergic agents
- **B.** Lorazepam 1-2 mg IV
- **C.** Cyproheptadine 12 mg PO
- **D.** Increase SSRI dose
- **E.** Add another serotonergic agent
- **F.** Give acetaminophen for fever
- **G.** Active cooling if temperature >41.1°C

---

301 An 18-year-old teenager was hit on the face, broke his teeth, and got a 3 cm laceration. Last vaccination was at age 6. What vaccination should he receive today? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Tdap 0.5 mL IM
B. Human tetanus immunoglobulin 250 units IM
C. MMR vaccine
D. Hepatitis B vaccine
E. Pneumococcal vaccine
F. HPV vaccine

---

302 What is the initial wound management for a facial laceration? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Primary closure immediately
B. Wound saline irrigation and debridement
C. Leave wound open without cleaning
D. Apply topical antibiotics only
E. Closure by secondary intention
F. Apply superglue

---

303 A 26-year-old lady presents with a cat bite with fever and inflammation around the bite. What is the causative organism?

A. Staphylococcus aureus
B. Pasteurella multocida
C. Bartonella henselae
D. Streptococcus pyogenes
E. E. coli
F. Pseudomonas aeruginosa

---

304 What antibiotic should you prescribe for an infected cat bite?

A. Ciprofloxacin 500 mg PO BID x 7 days
B. Amoxicillin-clavulanate 875/125 mg PO BID x 5-14 days
C. Azithromycin 500 mg PO daily x 3 days
D. Metronidazole 500 mg PO TID x 7 days
E. Doxycycline 100 mg PO BID x 7 days
F. TMP-SMX DS PO BID x 7 days

---

305 A burn patient arrives at the ER. Weight 70 kg, 40% TBSA burn. Using the Parkland formula, what is the total crystalloid needed for the first 24 hours?

- **A.** 5,600 mL
- **B.** 8,400 mL
- **C.** 11,200 mL
- **D.** 14,000 mL
- **E.** 16,800 mL

---

306 How should the Parkland formula fluid be distributed?

- **A.** 100% over 24 hours evenly
- **B.** 50% over first 8 hours, 50% over next 16 hours
- **C.** 75% over first 8 hours, 25% over next 16 hours
- **D.** 25% over first 8 hours, 75% over next 16 hours
- **E.** All in the first 4 hours

---

307 What other measures are needed in burn management? (Select 4)

SELECT 4 CORRECT ANSWERS

- **A.** Tetanus prophylaxis
- **B.** Bladder catheterization to monitor urine output
- **C.** Nasogastric tube for ileus
- **D.** Wound care and dressing
- **E.** Prophylactic systemic antibiotics for all burns
- **F.** Immediate skin grafting
- **G.** Oral feeding immediately

---

308 A 1-year-old boy weighing 9 kg with diarrhea lost 5% of his weight. Calculate the fluid deficit.

- **A.** 225 mL
- **B.** 450 mL
- **C.** 675 mL
- **D.** 900 mL
- **E.** 1350 mL

309 Using the 4-2-1 rule, what is the hourly maintenance fluid for a 9 kg child?

- **A.** 18 mL/hr
- **B.** 27 mL/hr
- **C.** 36 mL/hr
- **D.** 45 mL/hr
- **E.** 54 mL/hr

---

310 What is the best fluid for oral rehydration in a mildly dehydrated child?

- **A.** Normal saline IV
- **B.** Oral rehydration therapy (ORT)
- **C.** D5W IV
- **D.** Apple juice
- **E.** Cow's milk
- **F.** Sports drinks

---

311 A 36-year-old with BMI 31, HbA1c 6.1 (prediabetes). What non-pharmacological measures should you advise? (Select 4)

SELECT 4 CORRECT ANSWERS

- **A.** Medical nutrition therapy (diet modification)
- **B.** Regular exercise (aerobic or resistance)
- **C.** Weight reduction
- **D.** Smoking cessation
- **E.** Start insulin immediately
- **F.** Bed rest
- **G.** Avoid all carbohydrates

---

312 What parameters should you follow up in a prediabetic patient? (Select 2)

SELECT 2 CORRECT ANSWERS

- **A.** HbA1c annually
- **B.** Fasting blood sugar annually
- **C.** Weekly finger-prick glucose
- **D.** Daily urine glucose
- **E.** Monthly insulin levels

313 Syphilis treatment: A patient with primary syphilis (painless ulcer). What is the treatment?

A. Penicillin G benzathine 2.4 million units IM once

B. Doxycycline 100 mg PO BID x 14 days

C. Azithromycin 1g PO single dose

D. Ceftriaxone 500 mg IM single dose

E. Metronidazole 500 mg PO BID x 7 days

F. Ciprofloxacin 500 mg PO BID x 7 days

---

314 What are the complications of untreated syphilis? (Select 3)

SELECT 3 CORRECT ANSWERS

A. Aortic aneurysm

B. CNS manifestations (dementia, seizures)

C. Argyll Robertson pupil

D. Improved immunity

E. Spontaneous resolution always

F. Diabetes mellitus

---

315 A young man with scrotal swelling and gonococcal infection (epididymitis). What is the treatment? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Ceftriaxone 250 mg IM single dose

B. Doxycycline 100 mg PO BID x 10-14 days

C. Metronidazole 500 mg PO BID x 7 days

D. Fluconazole 150 mg PO single dose

E. Ciprofloxacin 500 mg PO BID x 7 days

---

316 A child with Kawasaki disease. What is the treatment? (Select 2)

SELECT 2 CORRECT ANSWERS

A. IVIG (Intravenous Immunoglobulin)

B. Aspirin

C. Ibuprofen

D. Prednisone

E. Methotrexate

F. Azithromycin

---

317 What long-term follow-up is needed for Kawasaki disease? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Echocardiography
B. Stress ECG
C. Weekly chest X-ray
D. Monthly liver biopsy
E. Daily blood cultures

---

318 A pregnant woman with severe hypertension and proteinuria (preeclampsia). What medication should you give to prevent seizures?

A. Phenytoin 100 mg IV
B. Magnesium sulfate 4g IV bolus then 1g/hr infusion
C. Diazepam 10 mg IV
D. Levetiracetam 500 mg IV
E. Carbamazepine 200 mg PO
F. Valproic acid 500 mg IV

---

319 How do you monitor magnesium sulfate therapy? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Deep tendon reflexes
B. Respiratory pattern
C. Serum calcium levels
D. Liver enzymes hourly
E. Daily chest X-ray

---

320 What is the antidote for magnesium sulfate toxicity?

A. Naloxone
B. Flumazenil
C. Calcium gluconate
D. Protamine sulfate
E. Vitamin K
F. N-acetylcysteine

---

321 A 2-year-old child ingested many chewable multivitamin tablets. What is the most dangerous element?

- **A.** Vitamin C
- **B.** Vitamin B12
- **C.** Iron
- **D.** Zinc
- **E.** Folic acid
- **F.** Vitamin D

---

322 The child deteriorates after 3 hours. After stabilizing ABC, what specific treatment should you give?

- **A.** Activated charcoal
- **B.** IV Deferoxamine 15 mg/kg/hr
- **C.** N-acetylcysteine
- **D.** Whole bowel irrigation
- **E.** Naloxone
- **F.** Flumazenil

---

323 A patient on haloperidol develops rigidity, high fever, elevated CK, and altered mental status (Neuroleptic Malignant Syndrome). What is the management? (Select 4)

SELECT 4 CORRECT ANSWERS

- **A.** Stop the offending agent
- **B.** IV fluids
- **C.** Cooling measures
- **D.** Dantrolene
- **E.** Increase haloperidol dose
- **F.** Add another antipsychotic
- **G.** Give acetaminophen for fever

---

324 How do you prevent NMS recurrence when restarting antipsychotics? (Select 3)

SELECT 4 CORRECT ANSWERS

- **A.** Wait at least 2 weeks after recovery
- **B.** Use a low-potency agent
- **C.** Start at low dose and increase slowly
- **D.** Restart the same medication at the same dose
- **E.** Avoid concomitant lithium
- **F.** Use the highest dose immediately

---

325 An alcoholic patient presents confused with ataxia and ophthalmoplegia (Wernicke Encephalopathy). What is the treatment?

- **A.** Oral thiamine 100 mg daily
- **B.** IV Thiamine 500 mg TID x 72 hours then reassess
- **C.** IV glucose first then thiamine
- **D.** Oral multivitamin only
- **E.** IV folate 5 mg daily
- **F.** IM Vitamin B12

---

326 An 18-year-old with a corneal abrasion. What topical antibiotic should you prescribe?

- **A.** Erythromycin 0.5% ophthalmic ointment QID x 5 days
- **B.** Hydrocortisone 1% eye drops
- **C.** Pilocarpine 2% drops
- **D.** Timolol 0.5% drops
- **E.** Atropine 1% drops
- **F.** Oral amoxicillin

---

327 If the corneal abrasion is from a contact lens wearer, what antibiotic is preferred?

- **A.** Erythromycin ointment
- **B.** Ciprofloxacin 0.3% ophthalmic drops QID x 5 days
- **C.** Gentamicin drops
- **D.** Chloramphenicol drops
- **E.** Neomycin drops

---

328 A patient with allergic rhinitis and conjunctivitis. What medications should you prescribe? (Select 2)

SELECT 2 CORRECT ANSWERS

- **A.** Intranasal glucocorticoid (Fluticasone furoate nasal spray)
- **B.** Antihistamine eye drops (Azelastine 0.05%)
- **C.** Oral antibiotics
- **D.** Nasal decongestant long-term
- **E.** Oral steroids long-term
- **F.** IV antihistamines

---

329 What non-pharmacological advice for allergic conjunctivitis? (Select 3)

SELECT 3 CORRECT ANSWERS

A. Avoid contact lenses

B. Cool compresses

C. Avoid rubbing eyes

D. Use hot compresses

E. Wear contact lenses continuously

F. Apply makeup to cover redness

---

330 A case of phimosis in a child. What topical treatment should you prescribe?

A. Hydrocortisone 1% cream BID

B. Betamethasone 0.05% cream BID x 4-8 weeks

C. Mupirocin 2% ointment BID

D. Clotrimazole 1% cream BID

E. Tretinoin 0.025% cream daily

F. Tacrolimus 0.1% ointment BID

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331 A patient with erythema migrans after a tick bite (Lyme Disease). What is the treatment?

A. Amoxicillin 500 mg PO TID x 14 days

B. Doxycycline 100 mg PO BID x 10 days

C. Azithromycin 500 mg PO daily x 5 days

D. Ciprofloxacin 500 mg PO BID x 7 days

E. Metronidazole 500 mg PO TID x 10 days

F. Cephalexin 500 mg PO QID x 10 days

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332 A child with whooping cough (Pertussis). What antibiotic should you prescribe?

A. Amoxicillin 50 mg/kg PO divided TID x 10 days

B. Azithromycin 10 mg/kg PO day 1 then 5 mg/kg x 4 days

C. Cephalexin 25 mg/kg PO BID x 10 days

D. Doxycycline 2 mg/kg PO BID x 7 days

E. Ciprofloxacin 10 mg/kg PO BID x 7 days

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333 What is the rationale for treating pertussis with antibiotics?

A. To cure the cough immediately

B. To decrease transmission to others

C. To prevent pneumonia

D. To reduce fever

E. To improve appetite

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334 A 20-year-old post-operative patient (appendectomy) becomes agitated and removes IV lines (Delirium). What non-pharmacological measures improve outcome? (Select 3)

SELECT 3 CORRECT ANSWERS

A. Maintaining adequate hydration and nutrition

B. Enhancing mobility and range of motion

C. Cognitive stimulation with reassurance from familiar persons

D. Physical restraints as first line

E. Keeping the room dark and quiet at all times

F. Isolating the patient

---

335 What medications can be given for acute delirium? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Haloperidol 1 mg IM (max 5 mg/day)

B. Risperidone 0.5 mg PO BID

C. Diazepam 10 mg IV

D. Morphine 5 mg IV

E. Methylphenidate 10 mg PO

F. Lithium 300 mg PO

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336 A man with history of tibio-fibular fracture had a cast for 3 months. After removal, he has leg redness, shiny skin, and paresthesia (Complex Regional Pain Syndrome). What specialists should you refer to? (Select 3)

SELECT 3 CORRECT ANSWERS

A. Pain management specialist

B. Physical and occupational therapist

C. Psychologist

D. Dermatologist

E. Gastroenterologist

F. Endocrinologist

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337 What medications from different classes can treat Complex Regional Pain Syndrome? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Amitriptyline 10 mg PO qhs

B. Pregabalin 25-150 mg PO daily

C. Metformin 500 mg PO BID

D. Omeprazole 20 mg PO daily

E. Atenolol 50 mg PO daily

F. Furosemide 20 mg PO daily

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338 A 49-year-old man with chest pain exaggerated by activity and relieved by rest (Stable Angina). Now stable and not in pain. What medications should you give? (Select 4)

SELECT 4 CORRECT ANSWERS

A. Beta-blocker (Metoprolol 25 mg PO BID)

B. Antiplatelet (Aspirin 325 mg)

C. Statin (Atorvastatin 80 mg PO daily)

D. Calcium channel blocker

E. Opioid (Morphine 5 mg PO q4h)

F. Benzodiazepine (Diazepam 5 mg PO TID)

G. Sublingual nitroglycerin PRN

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339 What non-pharmacological advice should you give the angina patient? (Select 4)

SELECT 4 CORRECT ANSWERS

A. Maintain healthy weight

B. Safe exercise plan

C. Healthy diet limited in saturated fat and salt

D. Smoking cessation

E. Avoid all physical activity

F. Increase alcohol intake

G. Bed rest permanently

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340 A patient with skin induration on the back of the hand and wrist (Cellulitis). What is the first-line antibiotic?

A. Cephalexin 500 mg PO QID x 5-10 days

B. Ciprofloxacin 500 mg PO BID x 7 days

C. Metronidazole 500 mg PO TID x 7 days

D. Azithromycin 500 mg PO daily x 3 days

E. Doxycycline 100 mg PO BID x 7 days

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341 The cellulitis patient does not improve. What should you give for suspected MRSA?

A. Amoxicillin 500 mg PO TID
B. TMP-SMX 160/800 mg PO q12h
C. Cephalexin 500 mg PO QID (higher dose)
D. Penicillin V 500 mg PO QID
E. Erythromycin 500 mg PO QID

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342 A pregnant woman with nausea and vomiting (Hyperemesis Gravidarum). What medications can you give? (Select 3)

SELECT 3 CORRECT ANSWERS

A. Doxylamine 10 mg + Pyridoxine 10 mg PO QID (Diclectin)
B. Ondansetron 4-8 mg PO/IV q8h
C. Metoclopramide 5 mg PO 30 min before meals
D. Metronidazole 500 mg PO TID
E. Isotretinoin 20 mg PO daily
F. Misoprostol 200 mcg PO QID

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343 What non-pharmacological advice should you give for hyperemesis gravidarum? (Select 4)

SELECT 4 CORRECT ANSWERS

A. Eat small amounts every 1-2 hours to avoid empty stomach
B. Eliminate spicy, high-fat, and acidic foods
C. Drink fluids 30 minutes before or after solid food
D. Avoid lying down after eating
E. Eat large meals three times daily
F. Drink hot beverages with meals
G. Increase exposure to strong odors

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344 A woman with clear symptoms of IBS with diarrhea, bloating, undigested vegetables in stool, no weight loss. What medications can treat her? (Select 3)

SELECT 3 CORRECT ANSWERS

A. Loperamide 2 mg 45 minutes before meals
B. Rifaximin 550 mg PO TID x 2 weeks
C. Hyoscyamine 0.25 mg PO TID PRN
D. Omeprazole 20 mg PO daily
E. Metformin 500 mg PO BID
F. Prednisone 40 mg PO daily

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345 The IBS patient returns after cholecystectomy with worsening diarrhea. What medication should you give?

A. Loperamide 4 mg PO PRN
B. Cholestyramine 4g PO BID
C. Omeprazole 20 mg PO daily
D. Metoclopramide 10 mg PO TID
E. Ondansetron 4 mg PO TID
F. Prednisone 20 mg PO daily

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346 What non-pharmacological measures help IBS? (Select 4)

SELECT 5 CORRECT ANSWERS

A. Low FODMAP diet
B. Regular meals
C. Exclusion of gas-producing foods
D. Exercise and physical activity
E. High-fat diet
F. Increase dairy intake
G. Stress reduction and CBT

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347 A patient with scabies. What is the first-line treatment?

A. Permethrin 5% cream (massage all skin, leave 8-14 hours, wash off)
B. Permethrin 1% cream (leave 10 minutes)
C. Lindane 1% (first line for all patients)
D. Oral ivermectin only
E. Topical hydrocortisone 1%
F. Oral fluconazole

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348 What non-pharmacological measures are needed for scabies? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Wash clothing at 65°C and heat dry or seal in plastic bag for 2 weeks
B. Identify and treat potential human contacts
C. No treatment of contacts needed
D. Fumigate the entire house
E. Burn all clothing
F. Isolate patient for 1 month

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349 A patient with vaginal candidiasis after antibiotic use. What treatment should you give? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Fluconazole 150 mg PO single dose
B. Clotrimazole 500 mg intravaginal single dose
C. Metronidazole 500 mg PO BID x 7 days
D. Doxycycline 100 mg PO BID x 7 days
E. Ciprofloxacin 500 mg PO BID x 7 days

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350 A breastfeeding woman needs contraception. What should you prescribe and why?

A. Combined OCP - safe in lactation
B. Progestin-only pill (Norethindrone 0.35 mg) - neutral effect on breast milk
C. Depo-Provera 150 mg IM - increases milk production
D. Copper IUD - increases milk production
E. Estrogen patch - safe in lactation

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351 A female in a rural hospital with an open fracture of the left leg. What is the immediate management? (Select 5)

SELECT 5 CORRECT ANSWERS

A. Analgesia as needed
B. Reduction and immobilization
C. Immediate irrigation and debridement
D. Antibiotics (Cefazolin 2g IV q8h)
E. Tetanus prophylaxis
F. Discharge with oral antibiotics only
G. Wait for orthopedic consultation before any treatment
H. Apply a cast immediately

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352 What serious complications might occur with an open fracture? (Select 4)

SELECT 4 CORRECT ANSWERS

A. Arterial injury
B. Nerve injury
C. Compartment syndrome
D. Infection (osteomyelitis)
E. Improved healing
F. Spontaneous resolution
G. Weight gain

353 A patient with lateral epicondylitis (tennis elbow). What non-pharmacological measures should you advise? (Select 3)

SELECT 3 CORRECT ANSWERS

A. Activity modification
B. Ice application
C. Counter-force bracing
D. Increase repetitive movements
E. Apply heat continuously
F. Complete arm immobilization for 6 months

---

354 What pharmacological treatments can be used for epicondylitis? (Select 2)

SELECT 2 CORRECT ANSWERS

A. NSAIDs (Naproxen 500 mg PO BID)
B. Glucocorticoid injection (Triamcinolone - only once)
C. Opioids as first line
D. Methotrexate
E. Colchicine
F. Hydroxychloroquine

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355 A patient with Raynaud's phenomenon. What is the first-line pharmacological treatment?

A. Propranolol 40 mg PO BID
B. Amlodipine 5 mg PO daily x 4-6 weeks
C. Warfarin 5 mg PO daily
D. Aspirin 81 mg PO daily
E. Prednisone 10 mg PO daily
F. Methotrexate 15 mg PO weekly

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356 What non-pharmacological advice should you give for Raynaud's? (Select 3)

SELECT 3 CORRECT ANSWERS

A. Minimize cold exposure, dress warmly
B. Avoid vasoconstricting drugs and smoking
C. Avoid vibrating tools
D. Expose hands to cold water daily
E. Increase caffeine intake
F. Wear tight gloves

357 A bedridden patient develops a pressure ulcer. What risk factor modifications should be implemented? (Select 4)

SELECT 4 CORRECT ANSWERS

A. Pressure minimization (repositioning)
B. Prevent excessive moisture
C. Improve nutritional status
D. Early mobilization and physiotherapy
E. Keep patient in same position
F. Restrict all oral intake
G. Apply pressure to the wound

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358 A meningitis patient's close contact asks if they need treatment. What prophylaxis should you give?

A. Rifampin 600 mg PO q12h x 4 doses
B. Amoxicillin 500 mg PO TID x 7 days
C. Doxycycline 100 mg PO BID x 7 days
D. Metronidazole 500 mg PO TID x 7 days
E. Fluconazole 150 mg PO single dose

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359 What are alternative prophylaxis options for meningitis contacts? (Select 2)

SELECT 2 CORRECT ANSWERS

A. Single dose Ciprofloxacin 500 mg PO
B. Ceftriaxone 250 mg IM single dose
C. Azithromycin 1g PO single dose
D. Vancomycin 1g IV single dose
E. Metronidazole 2g PO single dose

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360 A woman with a positive pregnancy test is on an ACE inhibitor and TMP-SMX prophylaxis. What medications should be stopped? (Select 2)

SELECT 2 CORRECT ANSWERS

A. ACE inhibitor
B. TMP-SMX
C. Prenatal vitamins
D. Folic acid
E. Iron supplements

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361 What UTI prophylaxis can be given in pregnancy instead of TMP-SMX?

- **A.** Ciprofloxacin 250 mg PO qhs
- **B.** Cephalexin 250 mg PO qhs
- **C.** Nitrofurantoin 100 mg PO daily
- **D.** Doxycycline 100 mg PO daily
- **E.** Metronidazole 500 mg PO daily

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362 A woman took 20 tablets of lorazepam after a fight with her boyfriend. RR 8, SpO2 99%, BP 110/60. A nurse comes with flumazenil. What should you do?

- **A.** Give flumazenil immediately
- **B.** Avoid flumazenil (risk of withdrawal seizures in chronic BDZ users)
- **C.** Give naloxone instead
- **D.** Give activated charcoal immediately
- **E.** Intubate immediately

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363 An ectopic pregnancy is diagnosed. What is the medical treatment?

- **A.** Methotrexate 50 mg/m² IM
- **B.** Misoprostol 800 mcg vaginal
- **C.** Mifepristone 200 mg PO
- **D.** Oxytocin IV infusion
- **E.** Progesterone 200 mg PO daily

SECTION TWO

Answers and Clinical Reasoning

1 **Answer:** B

The patient presents with classic Parkinson's disease features: cogwheel rigidity, resting tremor, and unsteady gait that affect daily activities. Carbidopa/Levodopa is the most effective first-line treatment for Parkinson's disease when symptoms impair daily functioning. Levodopa is converted to dopamine in the brain, directly replacing the deficient neurotransmitter. Carbidopa prevents peripheral conversion of levodopa, reducing side effects like nausea. Starting with a half-tablet TID allows gradual titration. Dopamine agonists (C, E) are alternatives for younger patients or milder symptoms. Amantadine (A) has modest benefit. Benztropine (D) mainly helps tremor. Selegiline (F) is an adjunct, not first-line monotherapy.

2 **Answers:** A C E

Early side effects of Carbidopa/Levodopa occur at initiation and are related to peripheral dopaminergic stimulation. Nausea (A) is the most common early side effect due to dopamine stimulating the chemoreceptor trigger zone. Somnolence (C) occurs because dopamine affects the sleep-wake cycle. Dizziness (E) results from orthostatic hypotension caused by peripheral vasodilation. Dyskinesia (B), wearing-off (D), and impulse control disorders (F) are LATE complications that develop after prolonged use due to pulsatile dopaminergic stimulation and receptor sensitization.

3 **Answers:** B D F

Late side effects of Carbidopa/Levodopa develop after years of therapy. Wearing-off (B) occurs when the duration of benefit from each dose shortens as disease progresses and dopaminergic neurons continue to degenerate. Severe dyskinesia (D) — involuntary choreiform movements — results from chronic pulsatile stimulation of dopamine receptors causing receptor hypersensitivity. Impulse control disorders (F) — gambling, hypersexuality, compulsive shopping — result from chronic dopaminergic stimulation of the mesolimbic reward pathway. These are distinct from early side effects which are mainly GI and autonomic.

4 **Answers:** B C

Starting low and titrating slowly (B) allows the body to adapt to dopaminergic stimulation, minimizing nausea, dizziness, and orthostatic hypotension. Divided dosing (C) provides more stable plasma levels throughout the day, reducing peak-dose side effects and minimizing troughs. Starting high (A) increases side effect risk. Combining with opioids (D) is dangerous. Giving the full dose at bedtime (E) would cause excessive sedation and orthostatic hypotension. Grapefruit juice (F) has no beneficial role.

5 **Answer:** B

The patient is a 24-year-old male with non-scarring, non-scaly alopecia and a family history of baldness — classic androgenic alopecia (male pattern hair loss). Minoxidil 5% topical is the first-line treatment because it is safe, available OTC, and works by prolonging the anagen (growth) phase and increasing blood flow to hair follicles. Finasteride (A) is second-line after minoxidil failure. Dutasteride (C) is not first-line. Intralesional triamcinolone (D) is for alopecia areata. Spironolactone (E) is used in females. Baricitinib (F) is for alopecia areata.

6 **Answer:** D

Minoxidil requires a minimum of 6 months of consistent use before assessing therapeutic benefit. Hair growth is a slow process — the hair cycle takes months to shift from telogen (resting) to anagen (growth) phase. Patients may initially experience increased shedding (telogen effluvium) in the first 2-8 weeks as miniaturized hairs are replaced. Evaluating too early (A, B, C) would lead to premature discontinuation before the drug has had adequate time to work.

7 **Answer:** C

After 5 months of minoxidil without improvement (minimum 6 months recommended, but close enough to consider failure), the next step for a male wanting aggressive treatment is finasteride 1 mg daily. Finasteride is a 5-alpha reductase inhibitor that blocks conversion of testosterone to dihydrotestosterone (DHT), the hormone responsible for follicular miniaturization in androgenic alopecia. It is the most effective oral medication for male pattern hair loss. Baricitinib (A) and intralesional triamcinolone (B) are for alopecia areata. Dutasteride (D) is off-label. Spironolactone (E) is for females.

8 **Answer:** C

Alopecia areata is an autoimmune condition causing patchy hair loss, often triggered by stress. First-line treatment for limited/patchy alopecia areata is intralesional corticosteroid injection (triamcinolone acetonide). This delivers high-concentration anti-inflammatory medication directly to the affected area, suppressing the local immune attack on hair follicles while minimizing systemic side effects. Minoxidil (A) is adjunctive. Finasteride (B) is for androgenic alopecia. Oral prednisone (D) has too many systemic side effects for localized disease. Baricitinib (E) is for refractory cases.

9 **Answer:** C

After intralesional triamcinolone for alopecia areata, hair regrowth typically begins at 6-8 weeks. The corticosteroid suppresses the local autoimmune inflammatory process, allowing follicles to re-enter the anagen growth phase. The hair cycle requires time — follicles must transition from telogen to anagen and produce visible hair shafts. Expecting results at 1-2 weeks (A) or 3-4 weeks (B) is too early. Injections are typically repeated every 4-6 weeks if needed.

10 **Answer:** C

When intralesional corticosteroids fail after 6 months in alopecia areata, baricitinib — a Janus kinase (JAK) inhibitor — is the next medication to consider. JAK inhibitors block the JAK-STAT signaling pathway that mediates the autoimmune T-cell attack on hair follicles. Baricitinib is FDA-approved for severe alopecia areata. Finasteride (A) is for androgenic alopecia. Minoxidil (B) is adjunctive only. Methotrexate (D), cyclosporine (E), and azathioprine (F) are less targeted and have more side effects.

11 **Answers:** A B C E F G H

Multiple factors can increase INR in a warfarin patient:

- **Fever (A)**: increases warfarin metabolism variability and vitamin K turnover
- **Hyperthyroidism (B)**: increases catabolism of clotting factors
- **Heart failure (C)**: hepatic congestion impairs clotting factor synthesis
- **Liver disease (E)**: decreased production of clotting factors and impaired warfarin metabolism
- **Diarrhea (F)**: reduces vitamin K absorption from the gut
- **Wrong dose (G)**: accidental overdose directly increases INR
- **TMP-SMX (H)**: inhibits CYP2C9, the enzyme that metabolizes warfarin, increasing its effect

Hypothyroidism (D) would decrease INR. Vitamin K supplementation (I) would decrease INR.

12 **Answer:** B

An elderly patient with poor vision who cannot differentiate between warfarin tablets (which are color-coded by dose) needs a practical solution. A pill organizer or blister pack (compliance aid) pre-sorted by a pharmacist ensures the correct dose is taken each day without requiring the patient to identify individual tablets. This is the simplest, safest, and most cost-effective solution. Switching to a DOAC (A) may not be appropriate for all patients. Having family administer (C) creates dependency. Increasing font size (D) doesn't solve the tablet identification problem.

13 **Answer:** C

A 10-year-old with chest tightness triggered by exercise and relieved by rest has exercise-induced bronchoconstriction (EIB). The first-line treatment is a short-acting beta-2 agonist (SABA) — albuterol — used as a rescue inhaler before exercise. Albuterol relaxes bronchial smooth muscle by stimulating beta-2 adrenergic receptors, preventing or relieving bronchospasm. Montelukast (A) is second-line. Fluticasone (B) is for persistent asthma. Ipratropium (D) is not first-line for EIB. Prednisone (E) is for acute exacerbations. Theophylline (F) is rarely used in children.

14 **Answer:** B

Albuterol (SABA) should be administered 5-20 minutes before exercise for exercise-induced bronchoconstriction. This timing allows the medication to reach peak bronchodilatory effect before the exercise stimulus triggers bronchoconstriction. The onset of action of inhaled albuterol is approximately 5 minutes, with peak effect at 15-30 minutes. Administering immediately before (A) may not allow sufficient onset time. Waiting 30-60 minutes (C) or 1-2 hours (D) means the peak effect may have passed. After exercise (E) is reactive, not preventive.

15 **Answers:** B C E

For a pregnant patient on phenytoin:

- **Folic acid 4 mg daily (B)**: Phenytoin is a folate antagonist and increases the risk of neural tube defects. High-dose folic acid (4-5 mg, not the standard 0.4 mg) is essential throughout pregnancy.
- **Phenytoin level monitoring every 4 weeks (C)**: Pregnancy increases blood volume and alters protein binding, causing phenytoin levels to drop. Regular monitoring prevents subtherapeutic levels and breakthrough seizures.
- **Counseling (E)**: The patient must understand teratogenic risks (fetal hydantoin syndrome: craniofacial abnormalities, nail/digit hypoplasia, growth restriction) and the importance of seizure trigger avoidance.

Abrupt discontinuation (A) risks status epilepticus, which is more dangerous than the drug. Valproic acid (D) is MORE teratogenic.

16 **Answers:** A B D

Serious side effects of phenytoin (excluding teratogenicity):

- **Hepatotoxicity (A)**: Phenytoin can cause drug-induced liver injury, ranging from mild transaminase elevation to fulminant hepatic failure.
- **Agranulocytosis (B)**: A rare but life-threatening idiosyncratic reaction causing severe neutropenia and susceptibility to infections.
- **Stevens-Johnson Syndrome (D)**: A severe mucocutaneous hypersensitivity reaction that can be fatal, more common in patients with HLA-B*1502 allele.

Weight gain (C) is not a serious side effect. Hair growth/hirsutism (E) and acne (G) are cosmetic side effects, not serious. Cardiac arrhythmias (F) occur mainly with IV phenytoin infusion, not chronic oral use.

17 **Answer:** B

A patient on high-dose statin presenting with muscle aches, myalgia, and fatigue needs serum CK (creatine kinase) to assess for statin-induced myopathy. CK is released from damaged skeletal muscle cells and is the definitive biomarker for muscle injury. Elevated CK confirms myositis or rhabdomyolysis. ESR (A) is nonspecific for inflammation. LFTs (C) assess liver, not muscle. Serum lactate (D) is for sepsis/hypoperfusion. Troponin (E) is for cardiac muscle. Myoglobin (F) is less specific. ANA (G) is for autoimmune conditions.

18 **Answer:** C

CK at 20 times the upper limit of normal indicates rhabdomyolysis — the most severe form of statin-induced muscle injury. The spectrum is: myalgia (muscle pain, normal CK) → myositis (muscle pain, CK elevated <10x ULN) → rhabdomyolysis (CK >10x ULN with potential renal failure from myoglobin). Rhabdomyolysis can cause acute kidney injury from myoglobin precipitation in renal tubules. The statin must be immediately discontinued. Myalgia (A) has normal CK. Myositis (B) has moderately elevated CK. Polymyositis (D) and dermatomyositis (F) are autoimmune. Fibromyalgia (E) has normal CK.

19 **Answer:** B

An 8-month-old with frequent regurgitation on formula who is otherwise healthy and growing normally likely has physiologic reflux, but if treatment is warranted (persistent symptoms affecting feeding), PPIs are the pharmacological choice for infant GERD. PPIs (e.g., omeprazole, lansoprazole) suppress gastric acid production by irreversibly inhibiting the H+/K+ ATPase pump in parietal cells, reducing esophageal acid exposure. H2 blockers (A) are less effective. Prokinetics (C) have significant side effects in infants. Antacids (D) are not recommended in infants. Sucralfate (E) is not standard for infant GERD.

20 **Answer:** C

PPI therapy in infants with GERD should be trialed for 8 weeks before assessing benefit and determining whether to continue. This duration allows adequate time for esophageal mucosal healing and symptom improvement. If no improvement after 8 weeks, the diagnosis should be reconsidered and the PPI discontinued. Two weeks (A) and 4 weeks (B) are too short for full assessment. Twelve weeks (D) and 6 months (E) are unnecessarily long before initial reassessment.

21 **Answers:** A C E F H

Red flag findings in an infant with regurgitation that change management:

- **Bilious vomiting (A)**: Suggests intestinal obstruction (e.g., malrotation with volvulus) — a surgical emergency.
- **Weight loss/faltering growth (C)**: Indicates the reflux is pathological and affecting nutrition.
- **Projectile vomiting (E)**: Suggests pyloric stenosis, especially in infants 2-8 weeks old.
- **Hepatosplenomegaly (F)**: Suggests metabolic disease, infection, or malignancy.
- **Bulging fontanelle (H)**: Suggests increased intracranial pressure (meningitis, hydrocephalus) — vomiting may be from a CNS cause.

Occasional spit-up after feeds (B), normal stool pattern (D), and smiling after feeds (G) are normal findings that do not warrant a change in management.

22 **Answers:** A B F

An unconscious child with unknown ingestion, GCS 11, bradycardia (HR 60), respiratory depression (RR 8), and hypotension (BP 90/60) needs immediate empiric treatment:

- **Naloxone IV (A)**: Reverses opioid toxicity — the bradycardia, respiratory depression, and decreased consciousness are consistent with opioid ingestion. Given empirically when the substance is unknown.
- **Dextrose IV (B)**: Hypoglycemia can cause unconsciousness and must be ruled out/treated immediately in any altered mental status.
- **IV fluid (F)**: Addresses hypotension (BP 90/60) and supports circulation.
- **Thiamine IV (C)**: Typically given before glucose in adults to prevent Wernicke encephalopathy, though less relevant in children. Still part of the "coma cocktail."

Activated charcoal (D) is contraindicated in a patient with decreased consciousness (aspiration risk). Gastric lavage (E) is rarely indicated. Ipecac (G) is no longer recommended.

23 **Answers:** A B C E G

Preventing accidental pediatric poisoning:

- **Keep harmful products locked up and out of reach (A)**: The most important prevention measure — children are curious and explore by putting things in their mouths.
- **Do not take medicine in view of small children (B)**: Children imitate adults; seeing adults take pills encourages them to do the same.
- **Do not solely rely on child-resistant containers (C)**: These are "child-resistant," not "child-proof" — determined children can open them.
- **Properly dispose of unwanted medication (E)**: Unused medications left accessible are a poisoning risk.
- **Always re-secure childproof caps (G)**: Leaving caps loose defeats the purpose of child-resistant packaging.

Calling medicine "candy" (D) is DANGEROUS — it encourages children to seek out and ingest medications. Leaving medications on the counter (F) increases access.

24 **Answers:** B D

Six days after unprotected sex, the options are limited:

- **Copper IUD (B)**: Effective up to 5-7 days after unprotected intercourse. It is the MOST effective form of emergency contraception (>99%) and provides ongoing contraception. It works by creating a toxic environment for sperm and preventing implantation.
- **Ulipristal 30 mg (D)**: A selective progesterone receptor modulator effective up to 5 days (120 hours) after unprotected sex. At 6 days, it's at the edge of its window but still considered.

Levonorgestrel (C) is only effective up to 72 hours (3 days) — too late at 6 days. Combined OCPs (A) as emergency contraception (Yuzpe method) are only effective up to 72 hours. Mifepristone (E) is an abortifacient, not standard EC. Medroxyprogesterone (F) is not emergency contraception.

25 **Answer:** C

For a patient with BMI of 45 (morbidly obese), the copper IUD is the best emergency contraception option. Oral emergency contraceptives (levonorgestrel and ulipristal) have significantly reduced efficacy in obese women because the drug is distributed into a larger volume of distribution, resulting in lower serum concentrations. The copper IUD's efficacy is independent of body weight because it acts locally within the uterus. It remains >99% effective regardless of BMI.

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26 **Answers:** B D

In an elderly long-term care resident who hasn't been reviewed in 4 years:

- **Benzodiazepine (B)**: Should be stopped/tapered in elderly patients. Benzodiazepines increase fall risk, cause cognitive impairment, sedation, and dependence. They are on the Beers Criteria list of potentially inappropriate medications for older adults.
- **Rabeprazole (D)**: A PPI that should be reassessed. Long-term PPI use without clear indication should be stopped due to risks of osteoporosis, C. difficile infection, B12 deficiency, and hypomagnesemia.

Levothyroxine (A) is needed for hypothyroidism — should not be stopped. Allopurinol (C) is appropriate for gout prophylaxis. Acetaminophen (E) at 1g QID equals 4g/day — at the maximum but acceptable if needed for pain, though monitoring is warranted.

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27 **Answers:** B C

Rationale for stopping/tapering benzodiazepines in elderly:

- **Addictive (B)**: Benzodiazepines cause physical and psychological dependence, even at therapeutic doses. Tolerance develops, requiring dose escalation. Withdrawal can cause seizures and delirium.
- **Increased fall risk (C)**: Benzodiazepines cause sedation, impaired balance, muscle relaxation, and slowed reflexes — all contributing to falls. Falls in the elderly can cause hip fractures, head injuries, and death.

Hepatotoxicity (A) is not a major benzodiazepine concern. Renal failure (D) is not caused by benzodiazepines. Hypertension (E) is not a side effect — benzodiazepines actually lower blood pressure.

---

28 **Answers:** A B C F

Long-term risks of PPI use:

- **Osteoporosis and fracture risk (A)**: PPIs reduce calcium absorption by decreasing gastric acid, leading to decreased bone mineral density over years.
- **C. difficile infection (B)**: Gastric acid is a barrier against ingested pathogens. PPIs reduce this barrier, increasing susceptibility to C. difficile.
- **Decreased iron and Vitamin B12 absorption (C)**: Gastric acid is required for iron absorption and for cleaving B12 from food proteins. Chronic acid suppression impairs both.
- **Microscopic colitis (F)**: PPIs are associated with microscopic colitis, causing chronic watery diarrhea.

Increased HDL (D) is not a PPI effect. Hypertension (E) is not associated. Weight loss (G) is not a typical PPI effect.

29 **Answer:** B

A 23-year-old female with moderate acne that has failed topical treatments needs oral therapy. Isotretinoin is the most effective treatment for moderate-to-severe acne that has failed other treatments. It works by dramatically reducing sebum production, normalizing follicular keratinization, reducing Cutibacterium acnes colonization, and having anti-inflammatory effects. It is the only medication that can produce long-term remission. She is on OCP, which is important because isotretinoin is highly teratogenic and requires reliable contraception. Doxycycline (A) and minocycline (F) are oral antibiotics — reasonable but less effective than isotretinoin for refractory cases. Spironolactone (C) is an option for females but less effective.

30 **Answer:** C

The most important PREVENTABLE side effect of isotretinoin is teratogenicity. Isotretinoin is FDA Pregnancy Category X — it causes severe birth defects including craniofacial, cardiac, thymic, and CNS malformations. This is preventable through strict pregnancy prevention programs (iPLEDGE in the US): two forms of contraception, monthly pregnancy tests, and informed consent. Hepatotoxicity (A) and hyperlipidemia (B) are monitored but less catastrophic. Depression (D) is debated. Dry skin (E) is common but not serious. Photosensitivity (F) is manageable.

31 **Answers:** A B C D

Monitoring for isotretinoin:

- **CBC (A)**: Isotretinoin can cause leukopenia, thrombocytopenia, and anemia.
- **Liver function tests (B)**: Isotretinoin is hepatotoxic and can elevate transaminases.
- **Lipid profile (C)**: Isotretinoin commonly causes hypertriglyceridemia and hypercholesterolemia, which can lead to pancreatitis if severe.
- **Pregnancy test (D)**: Monthly pregnancy tests are mandatory due to severe teratogenicity. Must be negative before each prescription refill.

Serum calcium (E), thyroid function (F), and pulmonary function (G) are not routinely monitored with isotretinoin.

32 **Answer:** C

HIV post-exposure prophylaxis (PEP) requires a three-drug antiretroviral regimen. The standard regimen is typically two NRTIs (e.g., tenofovir + emtricitabine) plus an integrase inhibitor (e.g., raltegravir or dolutegravir). Three drugs are used because a single or dual regimen may not adequately suppress viral replication if HIV transmission has occurred, and resistance could develop. The three-drug approach mirrors the principle of combination antiretroviral therapy (cART) used in HIV treatment, providing maximum viral suppression.

33 **Answer:** C

HIV PEP should be given for 28 days (4 weeks). This duration is based on animal studies and clinical experience showing that a 4-week course provides adequate time to prevent viral establishment in the body. The virus takes time to integrate into host cell DNA and establish reservoirs — the 28-day course aims to suppress replication during this critical window. Shorter courses (A, B) may be insufficient. Longer courses (D, E, F) are unnecessary and increase side effect burden and non-adherence.

34 **Answers:** A B C

Advice after needle stick injury to prevent transmission:

- **Practice safe sex and use condoms (A)**: Until seroconversion is ruled out (at least 6 months of follow-up), the healthcare worker could potentially transmit HIV, HBV, or HCV to sexual partners.
- **Do not donate blood or organs (B)**: During the window period, the worker may be infectious but test negative. Donating could transmit infection to recipients.
- **Do not share personal objects like toothbrushes and razors (C)**: These items can have blood contamination and transmit bloodborne pathogens.

Avoiding all physical contact (D) is excessive — casual contact doesn't transmit these infections. Quarantine (E) is unnecessary. Stopping work (F) is not required unless performing exposure-prone procedures.

35 **Answer:** A

An elderly patient with X-ray changes consistent with RA and a rheumatology appointment in 3 weeks needs bridging therapy. Low-dose oral prednisone (≤10 mg daily) is appropriate as a bridge to control inflammation and symptoms until the rheumatologist can initiate disease-modifying therapy (DMARDs). It provides rapid anti-inflammatory relief. Dexamethasone IV (B) and hydrocortisone IV (D) are unnecessarily aggressive for outpatient management. Methylprednisolone IM (C) gives less predictable dosing. Intra-articular triamcinolone (F) is for single-joint involvement, not polyarticular RA.

36 **Answers:** A B C E G H

Systemic side effects of glucocorticoids:

- **Hyperglycemia (A)**: Glucocorticoids increase hepatic gluconeogenesis and cause insulin resistance.
- **Immunosuppression (B)**: They suppress T-cell function, neutrophil migration, and cytokine production, increasing infection risk.
- **Fluid retention and increased BP (C)**: Mineralocorticoid effects cause sodium and water retention.
- **Gastritis/peptic ulcer (E)**: Glucocorticoids reduce prostaglandin synthesis, decreasing gastric mucosal protection.
- **Insomnia and mood changes (G)**: CNS effects include euphoria, insomnia, psychosis, and depression.
- **Decreased bone density (H)**: Glucocorticoids inhibit osteoblast activity and calcium absorption, causing osteoporosis.

Hypoglycemia (D) is incorrect — they cause hyperglycemia. Weight loss (F) is incorrect — they cause weight gain (Cushingoid features).

37 **Answer:** B

Methotrexate is the first-line DMARD (disease-modifying antirheumatic drug) for rheumatoid arthritis. It inhibits dihydrofolate reductase, reducing purine and pyrimidine synthesis, which suppresses the proliferation of rapidly dividing immune cells responsible for joint inflammation and destruction. It is the anchor drug in RA treatment — effective, well-studied, and relatively affordable. TNF inhibitors (A) and JAK inhibitors (C) are used when methotrexate fails. Hydroxychloroquine (E) is less potent. Cyclosporine (F) has more toxicity.

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38 **Answers:** A B C

Non-pharmacological measures for hypertension:

- **Weight reduction (A)**: Each kg of weight loss reduces BP by approximately 1 mmHg. Obesity increases cardiac output and peripheral resistance.
- **Regular exercise (B)**: 150 minutes/week of moderate aerobic exercise reduces systolic BP by 5-8 mmHg through improved endothelial function and reduced peripheral resistance.
- **DASH diet (C)**: Dietary Approaches to Stop Hypertension — rich in fruits, vegetables, whole grains, and low-fat dairy; low in saturated fat and sodium. Reduces systolic BP by 8-14 mmHg.

Increasing caffeine (D) can raise BP. Bed rest (E) is counterproductive. High sodium (F) worsens hypertension. Stress management (G) is helpful but the three listed are the most evidence-based.

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39 **Answers:** A B E

When starting ramipril (ACE inhibitor), these investigations are essential:

- **Serum potassium (A)**: ACE inhibitors reduce aldosterone secretion, decreasing potassium excretion. This can cause hyperkalemia, especially in patients with renal impairment or on potassium-sparing diuretics.
- **Serum creatinine (B)**: ACE inhibitors dilate the efferent arteriole of the glomerulus, reducing glomerular filtration pressure. A rise in creatinine >30% from baseline indicates renal artery stenosis and requires discontinuation.
- **eGFR (E)**: Estimated glomerular filtration rate assesses overall kidney function and helps determine if the ACE inhibitor is causing renal impairment.

TSH (C), LFTs (D), calcium (F), and uric acid (G) are not specifically required for ACE inhibitor monitoring.

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40 **Answer:** B

Renal function and potassium should be checked 1-2 weeks after starting an ACE inhibitor. This timing allows detection of:

- Significant creatinine rise (>30% suggests bilateral renal artery stenosis)
- Hyperkalemia from reduced aldosterone

Checking immediately (A) wouldn't show drug effects. Waiting 1 month (C) or longer (D, E) risks missing early nephrotoxicity or dangerous hyperkalemia. The 1-2 week window is the standard of care recommended by all major guidelines for ACE inhibitor initiation. Repeat monitoring should also occur after any dose change.

41 **Answers:** A B C

Findings requiring urgent urology referral in BPH:

- **Abnormal rectal exam with irregular/nodular prostate (A)**: An irregular, hard, or nodular prostate on DRE raises suspicion for prostate cancer, which requires urgent biopsy and evaluation.
- **Palpable urinary bladder (B)**: Indicates urinary retention — the bladder is distended and not emptying. This can cause hydronephrosis and renal failure if untreated, requiring catheterization and surgical evaluation.
- **Significant weight loss (C)**: Unexplained weight loss in a patient with urinary symptoms suggests possible malignancy (prostate, bladder, or renal cancer).

Mild urinary frequency (D) is a common BPH symptom manageable in primary care. Normal DRE (E) is reassuring. Bilateral pedal edema (F) has many causes and is not specific to urological emergency.

42 **Answer:** B

Tamsulosin is the first-line treatment for BPH. It selectively blocks alpha-1A adrenergic receptors in the prostate and bladder neck smooth muscle, causing relaxation and improving urine flow. It provides rapid symptom relief within days to weeks. Tamsulosin is preferred over non-selective alpha blockers because it has less effect on vascular alpha-1B receptors, causing less orthostatic hypotension. Finasteride (A) is a 5-alpha reductase inhibitor used for large prostates (>40g) but takes 6 months for effect. Tadalafil (C) is an alternative but not first-line. Anticholinergics (D) can worsen retention.

43 **Answer:** B

The patient has bulimia nervosa — characterized by binge eating, compensatory behaviors (purging causing dental erosion/enamel wearing), and food restriction. CBT is the first-line treatment for bulimia nervosa. It addresses the dysfunctional thoughts and behaviors around food, body image, and self-worth. CBT for bulimia has the strongest evidence base, with remission rates of 30-50%. It helps patients identify triggers for binge-purge cycles and develop healthier coping strategies. Fluoxetine (A) is second-line after CBT failure. Olanzapine (C) is for anorexia. Bupropion (F) is CONTRAINDICATED in eating disorders due to seizure risk.

44 **Answers:** A B C D E

Factors causing CBT failure in bulimia nervosa:

- **Non-adherence (A)**: CBT requires active participation, homework, and food diaries. Non-compliance undermines treatment.
- **Lack of motivation (B)**: Patients ambivalent about change or in the pre-contemplation stage will not engage meaningfully in therapy.
- **Cognitive instability (C)**: Severe malnutrition affects cognitive function, making it difficult to engage in the cognitive restructuring required by CBT.
- **Substance use disorder (D)**: Active substance abuse impairs judgment, motivation, and the ability to implement behavioral changes.
- **Comorbid anxiety or depression (E)**: Untreated psychiatric comorbidities interfere with CBT engagement and perpetuate the binge-purge cycle.

High intelligence (F) would actually help with CBT. Young age (G) is not inherently a barrier.

45 **Answer:** B

After 6 months of failed CBT for bulimia nervosa, fluoxetine is the best pharmacological treatment. Fluoxetine is the ONLY FDA-approved medication for bulimia nervosa. It reduces binge-purge frequency by modulating serotonin pathways involved in appetite regulation, mood, and impulse control. The therapeutic dose for bulimia is typically 60 mg/day (higher than for depression), but starting at 20 mg and titrating up is standard. Olanzapine (A) is for anorexia. Bupropion (C) is CONTRAINDICATED in eating disorders (seizure risk). Topiramate (D), lithium (E), and valproic acid (F) are not first-line for bulimia.

46 **Answers:** A B C D E G

Important history questions for an infertile couple:

- **Fertility history in previous relationships (A)**: Determines if either partner has previously conceived, helping identify which partner may have the issue.
- **Sexual history and frequency (B)**: Infrequent intercourse or poor timing relative to ovulation is a common cause of infertility.
- **Menstrual history (C)**: Irregular or absent periods suggest anovulation — the most common female cause of infertility.
- **Smoking, marijuana, alcohol use (D)**: All impair fertility — smoking damages oocytes and sperm, marijuana disrupts the hypothalamic-pituitary axis, alcohol is gonadotoxic.
- **Medical and surgical history (E)**: Conditions like PCOS, endometriosis, pelvic surgery, undescended testes, or chemotherapy affect fertility.
- **History of chemotherapy or radiation (G)**: Both are gonadotoxic and can cause permanent infertility.

Favorite hobbies (F) and blood type (H) are not relevant to fertility assessment.

47 **Answer:** B or F

For anovulatory infertility:

- **Clomiphene citrate (B)**: A selective estrogen receptor modulator (SERM) that blocks estrogen feedback at the hypothalamus, increasing GnRH, FSH, and LH release, which stimulates ovulation. It is the classic first-line ovulation induction agent.
- **Letrozole (F)**: An aromatase inhibitor that reduces estrogen production, similarly increasing FSH release and stimulating ovulation. Recent evidence shows letrozole may have higher live birth rates than clomiphene, especially in PCOS.

Metformin (A) is adjunctive in PCOS but not a primary ovulation inducer. Progesterone (C) supports pregnancy but doesn't induce ovulation. Estradiol (D) would suppress ovulation. Goserelin (E) suppresses the HPG axis.

48 **Answer:** B

A man who goes to the gym daily and takes "some pills" with oligospermia is most likely using anabolic-androgenic steroids (AAS). AAS suppress the hypothalamic-pituitary-gonadal axis through negative feedback — exogenous testosterone suppresses GnRH, LH, and FSH production. Without FSH, spermatogenesis ceases, causing oligospermia or azoospermia. Without LH, endogenous testosterone production drops, causing testicular atrophy. This is a very common cause of male infertility in gym-goers. Creatine (A), whey protein (C), multivitamins (D), caffeine (E), and fish oil (F) do not significantly affect sperm production.

49 **Answers:** A B D

Side effects of anabolic-androgenic steroids:

- **Gynecomastia (A)**: Excess testosterone is aromatized to estrogen, causing breast tissue growth in males.
- **Erythrocytosis (B)**: Androgens stimulate erythropoietin production and directly stimulate bone marrow, increasing red blood cell mass. This increases thrombotic risk (stroke, PE, MI).
- **Hepatotoxicity (D)**: Especially with oral 17-alpha-alkylated steroids (e.g., oxandrolone, stanozolol), which can cause cholestasis, peliosis hepatis, and hepatocellular carcinoma.

Improved fertility (C) is incorrect — AAS cause infertility. Decreased aggression (E) is incorrect — AAS cause increased aggression ("roid rage"). Testicular hypertrophy (F) is incorrect — AAS cause testicular ATROPHY due to HPG axis suppression.

50 **Answer:** B

A patient assigned female at birth seeking masculinization (deeper voice, masculine features) is a transgender male seeking gender-affirming hormone therapy. Testosterone is the first-line treatment. Testosterone enanthate IM is the most commonly used formulation, starting at a low dose (50 mg weekly) and titrating based on serum testosterone levels and clinical response. Effects include voice deepening, facial hair growth, fat redistribution, muscle mass increase, and cessation of menses. Estradiol (A) would feminize. Spironolactone (C) is an anti-androgen used in transfeminine care. Finasteride (D) blocks DHT. GnRH agonist (E) suppresses sex hormones. Progesterone (F) is not masculinizing.

51 **Answers:** A B C D

Monitoring for testosterone therapy:

- **Serum testosterone (A)**: Must be monitored to ensure levels are in the male physiologic range (300-1000 ng/dL) and to guide dose adjustments.
- **Hematocrit (B)**: Testosterone stimulates erythropoiesis. Polycythemia (hematocrit >54%) increases thrombotic risk and requires dose reduction or phlebotomy.
- **Lipid profile (C)**: Testosterone can decrease HDL and increase LDL, increasing cardiovascular risk.
- **Liver function (D)**: Testosterone, especially oral formulations, can cause hepatotoxicity.

Serum calcium (E), thyroid function (F), and pulmonary function (G) are not specifically affected by testosterone therapy.

52 **Answer:** B

The presentation — chest pain worse lying down, improved sitting/leaning forward, recent respiratory infection, diffuse ST elevation on ECG — is classic acute pericarditis. First-line treatment is the combination of:

- **NSAID (Ibuprofen)**: Reduces pericardial inflammation and pain. Ibuprofen is preferred over other NSAIDs because it has the fewest coronary side effects.
- **Colchicine**: Reduces pericardial inflammation by inhibiting microtubule assembly and neutrophil migration. It significantly reduces recurrence rates (from ~30% to ~15%).

Aspirin + clopidogrel (A) is for ACS. Prednisone (C) is reserved for refractory cases. Nitroglycerin (D) is for angina. Metoprolol (E) is not indicated. Heparin (F) could worsen pericardial effusion/tamponade.

53 **Answer:** D

Colchicine should be continued for 3 months in acute pericarditis. This duration is based on the COPE and CORE trials, which demonstrated that 3 months of colchicine significantly reduces the recurrence rate of pericarditis. The typical dose is 0.5 mg BID (or 0.5 mg daily if <70 kg). Stopping too early (A, B) increases recurrence risk. Six months (E) is the duration for recurrent pericarditis, not the first episode. Twelve months (F) is excessive for a first episode.

54 **Answer:** B

A patient with a history of peptic ulcer who is now on NSAID therapy needs gastroprotection. A PPI (pantoprazole) is the best choice for preventing NSAID-induced peptic ulcer recurrence. PPIs suppress gastric acid production by >90%, allowing mucosal healing and preventing NSAID-related gastric and duodenal ulcers. Sucralfate (A) provides a physical barrier but is less effective than PPIs for NSAID gastroprotection. Misoprostol (C) is effective but causes diarrhea and cramping, limiting tolerability. Bismuth (D) is not standard prophylaxis. Famotidine (E) is less effective than PPIs.

55 **Answer:** B

When pericarditis does not respond to NSAIDs + colchicine after one week (persistent pain and ECG changes), the next step is systemic corticosteroids. Prednisone at 0.5 mg/kg/day is used for 2 weeks, then gradually tapered over weeks to months to prevent rebound pericarditis. Corticosteroids are reserved for refractory cases because they increase recurrence risk if used as first-line therapy. Morphine (A) treats pain but not inflammation. Azathioprine (C) and methotrexate (D) are for steroid-dependent recurrent pericarditis. Anakinra (E) is for colchicine/steroid-resistant cases.

56 **Answers:** A B C D G

Conditions associated with panic attacks and self-harm:

- **Borderline personality disorder (A)**: Characterized by emotional instability, self-harm, impulsivity, and fear of abandonment. Panic attacks and self-harm are core features.
- **Substance use disorders (B)**: Substances can trigger panic attacks, and self-harm often co-occurs with substance abuse.
- **Bipolar disorder (C)**: Manic and depressive episodes can include panic attacks, and self-harm occurs during depressive or mixed episodes.
- **Major depression (D)**: Panic attacks commonly co-occur with depression, and self-harm is a feature of severe depression.
- **Schizophrenia (G)**: Command hallucinations can lead to self-harm, and panic attacks can occur.

Diabetes (E) and hypothyroidism (F) are medical conditions not typically associated with this presentation.

57 **Answers:** B C

For panic attacks with self-harm:

- **SSRI antidepressants (B)**: First-line long-term treatment for panic disorder. SSRIs (e.g., sertraline, paroxetine) reduce the frequency and severity of panic attacks by modulating serotonin neurotransmission. They also treat underlying depression and reduce self-harm behavior.
- **Benzodiazepines (C)**: Provide rapid short-term relief of acute panic symptoms while waiting for SSRIs to take effect (2-4 weeks). They act on GABA receptors to produce anxiolytic, sedative effects. Used as a bridge therapy, then tapered.

Antipsychotics (A) are not first-line for panic disorder. Mood stabilizers (D) are for bipolar disorder. Opioids (E) are not indicated. Stimulants (F) would worsen anxiety and panic.

58 **Answer:** B

A 66-year-old with SOB, orthopnea, PND, on digoxin and furosemide, with hypotension, tachycardia, and hypoxia presents with acute decompensated heart failure. BNP (or NT-proBNP) is the key blood investigation. BNP is released from ventricular myocytes in response to volume overload and wall stress. Elevated BNP (>400 pg/mL) confirms heart failure as the cause of dyspnea, distinguishing it from pulmonary causes. It also helps guide treatment response.

Troponin (A) is for myocardial infarction. D-dimer (C) is for pulmonary embolism. Serum lactate (D) is for sepsis/tissue hypoperfusion. Procalcitonin (E) is for bacterial infection. CRP (F) is a nonspecific inflammatory marker.

59 **Answers:** A B C D E

Initial therapy for acute decompensated heart failure with hypotension:

- **Position at 45 degrees (A)**: Reduces venous return (preload), decreasing pulmonary congestion and improving breathing.
- **High-flow O2 via non-rebreather mask (B)**: SpO2 is 84% — severe hypoxia requires immediate high-flow oxygen to prevent organ damage.
- **IV Furosemide 160 mg bolus (C)**: Double the patient's daily oral dose (80 mg BID = 160 mg/day) given IV for rapid diuresis, reducing fluid overload and pulmonary congestion. IV route ensures bioavailability in the setting of gut edema.
- **Morphine (D)**: Reduces anxiety, dyspnea perception, and preload through venodilation. Controversial in recent guidelines but still used in severe cases.
- **Dobutamine (E)**: An inotrope needed because BP is 90/60 (cardiogenic shock). Dobutamine increases cardiac contractility and output via beta-1 stimulation.

Oral metoprolol (F) is contraindicated in acute decompensation with hypotension — it would worsen cardiac output. IV normal saline bolus (G) would worsen fluid overload. Sublingual nitroglycerin (H) is contraindicated with systolic BP <90.

60 **Answers:** A B C D

Discharge medications for heart failure (guideline-directed medical therapy):

- **ACE inhibitor — Ramipril (A)**: Reduces afterload, prevents cardiac remodeling, and improves survival by blocking the RAAS system.
- **Beta-blocker — Metoprolol (B)**: Reduces heart rate, myocardial oxygen demand, and prevents adverse remodeling. Proven mortality benefit (MERIT-HF trial). Started at low dose when euvolemic and stable.
- **Potassium-sparing diuretic — Spironolactone (C)**: An aldosterone antagonist that reduces fibrosis, prevents hypokalemia, and improves survival (RALES trial).
- **Statin — Atorvastatin (D)**: For cardiovascular risk reduction, especially if ischemic cardiomyopathy is the underlying cause.

Amlodipine (E) is not standard HF therapy. Digoxin loading dose (F) is inappropriate — digoxin is used at maintenance doses for symptom control, not loading. Aspirin 325 mg (G) is too high a dose and not specifically part of HF therapy unless indicated for other reasons.

61 **Answers:** A B C

Acute COPD exacerbation with fever, increased/purulent sputum, and dyspnea (Anthonisen Type 1 — all three cardinal symptoms):

- **Short-acting beta agonist — Albuterol (A)**: Rapid bronchodilation by relaxing airway smooth muscle via beta-2 receptor stimulation. First-line rescue therapy.
- **Short-acting muscarinic antagonist — Ipratropium (B)**: Blocks acetylcholine at muscarinic receptors in airway smooth muscle, providing additional bronchodilation. Synergistic with albuterol.
- **Systemic glucocorticoids — Prednisone (C)**: Reduces airway inflammation, shortens recovery time, improves FEV1, and reduces treatment failure. Typically prednisone 40 mg daily for 5 days.

An antibiotic (e.g., amoxicillin-clavulanate, azithromycin, or doxycycline) should also be given since all three Anthonisen criteria are present with purulent sputum. LABA (D) is for maintenance, not acute exacerbation. Montelukast (E) is for asthma. Theophylline (F) has a narrow therapeutic index and is not first-line.

62 **Answers:** A B

After a COPD exacerbation, maintenance therapy should include:

- **ICS — Budesonide (A)**: Inhaled corticosteroids reduce airway inflammation and decrease exacerbation frequency, especially in patients with frequent exacerbations or eosinophilic phenotype.
- **LABA — Formoterol (B)**: Long-acting beta-2 agonist provides sustained bronchodilation (12+ hours), improving airflow, exercise tolerance, and quality of life.

The combination ICS/LABA (e.g., budesonide/formoterol) is standard maintenance therapy for COPD patients with exacerbation history. SABA PRN only (C) is insufficient after an exacerbation. Oral prednisone daily (D) has too many long-term side effects. Daily azithromycin (E) is only for select patients with frequent exacerbations. Theophylline (F) is third-line.

63 **Answers:** A B C D

Non-pharmacological advice for COPD:

- **Smoking cessation (A)**: The single most important intervention. It is the only measure proven to slow the decline in FEV1 and reduce mortality in COPD.
- **Influenza and pneumococcal vaccination (B)**: COPD patients are at high risk for respiratory infections that trigger exacerbations. Annual influenza and pneumococcal vaccines reduce hospitalizations and mortality.
- **Pulmonary rehabilitation (C)**: A structured program of exercise training, education, and behavioral change that improves exercise capacity, reduces dyspnea, and decreases hospitalizations.
- **Education about proper inhaler technique (D)**: Up to 90% of patients use inhalers incorrectly, leading to suboptimal drug delivery. Proper technique is essential for medication efficacy.

Avoiding all physical activity (E) worsens deconditioning. High-sodium diet (F) promotes fluid retention. Bed rest (G) accelerates muscle wasting and deconditioning.

64 **Answer:** B

The patient has anterior uveitis (iritis) — perilimbic congestion (ciliary flush), severe eye pain, slit lamp showing +3 cells in the anterior chamber, normal IOP, and chronic back/hip problems (suggesting ankylosing spondylitis, a common association). Cyclopentolate is a cycloplegic/mydriatic agent that relieves symptoms by:

1. Relaxing the ciliary muscle spasm (cycloplegia) — reducing pain

2. Dilating the pupil (mydriasis) — preventing posterior synechiae (iris adhesion to the lens)

This is the immediate symptom-relieving medication a non-ophthalmologist can give. Prednisolone drops (A) are the definitive treatment but are typically initiated by the ophthalmologist. Timolol (C) is for glaucoma. Pilocarpine (D) constricts the pupil — contraindicated. Erythromycin (E) and ciprofloxacin (F) are for bacterial infections.

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65 **Answer:** B

The ophthalmologist will prescribe topical corticosteroid drops — prednisolone acetate 1% — as the definitive treatment for anterior uveitis. Prednisolone suppresses the autoimmune inflammatory process in the anterior chamber by inhibiting prostaglandin and leukotriene synthesis, reducing leukocyte migration, and stabilizing cell membranes. The frequency (BID to QID) depends on severity. This is combined with the cyclopentolate already started. Timolol (C) and latanoprost (D) are for glaucoma. Ofloxacin (E) is an antibiotic for bacterial infections.

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66 **Answers:** A B

In amphetamine overdose with severe hypertension (BP 240/120), benzodiazepines are first-line for agitation:

- **Diazepam (A)** and **Lorazepam (B)**: Benzodiazepines enhance GABA activity, producing sedation and reducing sympathetic hyperactivity. In stimulant toxicity, agitation drives the hypertension and tachycardia — treating agitation with benzodiazepines often reduces BP significantly. They also prevent seizures, which are a risk with amphetamine overdose.

Haloperidol (C) should be avoided — it lowers the seizure threshold and can worsen hyperthermia. Propranolol (D) is contraindicated in stimulant overdose — unopposed alpha stimulation can worsen hypertension. Phenytoin (E) is not first-line for stimulant-related seizures. Ketamine (F) is not standard first-line.

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67 **Answers:** A B

For severe hypertension (240/120) in amphetamine overdose not controlled by benzodiazepines:

- **Nitroprusside (A)**: A direct vasodilator (arterial and venous) that rapidly reduces BP. It is titratable and has an immediate onset, making it ideal for hypertensive emergencies.
- **Phentolamine (B)**: A non-selective alpha-adrenergic blocker that directly counteracts the alpha-mediated vasoconstriction caused by amphetamines.

Propranolol (C) and atenolol (F) are CONTRAINDICATED — beta-blockers in stimulant overdose cause unopposed alpha-receptor stimulation, paradoxically worsening hypertension and coronary vasoconstriction. Enalaprilat (D) has too slow an onset for this emergency. Clonidine (E) is not first-line for hypertensive emergency.

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68 **Answer:** F (or B)

A 57-year-old with community-acquired pneumonia and a penicillin allergy (to cloxacillin) needs an alternative antibiotic. The allergy is to a penicillin-class drug, so:

- **Azithromycin (F)** or **Clarithromycin (B)**: Macrolide antibiotics are first-line alternatives for CAP in patients with penicillin allergy. They cover typical organisms (Streptococcus pneumoniae) and atypical organisms (Mycoplasma, Chlamydophila, Legionella).

Amoxicillin (A) is a penicillin — contraindicated with penicillin allergy. Cephalexin (C) has cross-reactivity risk with penicillin allergy (though low, ~1-2%). Ciprofloxacin (D) is a fluoroquinolone but not first-line for typical CAP. Doxycycline (E) is an alternative but macrolides are preferred.

69 **Answer:** A

The patient developed a white lesion in his mouth after using his wife's "puffer" — an inhaled corticosteroid (ICS). Oral candidiasis (thrush) is the most common local side effect of ICS. The corticosteroid deposits in the oropharynx, suppressing local immune defenses and promoting Candida albicans overgrowth. This is why patients are instructed to rinse their mouth with water and spit after each ICS use. Treatment is with oral nystatin suspension or fluconazole. Herpes (B), aphthous ulcers (C), leukoplakia (D), lichen planus (E), and SCC (F) have different presentations and are not related to ICS use.

70 **Answer:** B

Nifedipine is a dihydropyridine calcium channel blocker that causes peripheral edema as a common side effect. The mechanism is precapillary arteriolar vasodilation without corresponding venous dilation, creating a pressure gradient that forces fluid into the interstitial space. This is NOT fluid overload — it is a local hemodynamic effect. It is dose-dependent and occurs in up to 30% of patients on higher doses. Heart failure (A) would have other signs (JVD, crackles, orthopnea). DVT (C) is usually unilateral. Nephrotic syndrome (D) would have proteinuria. Liver cirrhosis (E) would have other stigmata.

71 **Answer:** C

The patient was inappropriately given furosemide at a walk-in clinic for nifedipine-induced edema. Furosemide caused hyponatremia (Na 125) and hypokalemia — both dangerous electrolyte disturbances. The edema was NOT from fluid overload, so diuretics were not indicated. The correct action is to stop furosemide immediately. The hyponatremia is diuretic-induced and will correct once the offending agent is removed. Increasing furosemide (A) would worsen electrolyte abnormalities. Adding spironolactone (B) adds another diuretic. Hypertonic saline (D) is for severe symptomatic hyponatremia. The underlying issue — nifedipine edema — should be addressed differently.

72 **Answers:** B C

Managing nifedipine-induced peripheral edema:

- **Combine with ACEI or ARB (B)**: ACE inhibitors and ARBs cause postcapillary venodilation, which counteracts the precapillary arteriolar dilation caused by nifedipine. This equalizes the pressure gradient and reduces edema by up to 60%.
- **Switch to a non-dihydropyridine CCB — diltiazem or verapamil (C)**: These cause less peripheral edema because they have less peripheral vasodilatory effect compared to dihydropyridines (nifedipine, amlodipine).

Adding a loop diuretic (A) is inappropriate — the edema is not from volume overload. Doubling the dose (D) would worsen edema. Compression stockings alone (E) are insufficient. Discontinuing all antihypertensives (F) is dangerous.

73 **Answer:** B

A patient with menorrhagia (8 days of bleeding, heavy flow), enlarged uterus (14-week size suggesting fibroids), dizziness, and fatigue has iron deficiency anemia from chronic blood loss. Oral iron supplementation with ferrous sulfate is the standard treatment. Each 325 mg tablet contains 65 mg of elemental iron. Iron replaces depleted stores and supports hemoglobin synthesis. It should be taken on an empty stomach with vitamin C to enhance absorption. Vitamin B12 (A) is for B12 deficiency. Folic acid (C) is for folate deficiency. Erythropoietin (D) is for renal anemia. IV iron (E) is for oral iron intolerance or malabsorption. Blood transfusion (F) is for severe symptomatic anemia.

74 **Answers:** A B C D G

Factors interfering with oral iron absorption:

- **Celiac disease (A)**: Damages duodenal mucosa where iron is primarily absorbed, causing malabsorption.
- **H. pylori infection (B)**: Reduces gastric acid production and causes chronic gastritis,

impairing iron absorption. H. pylori also competes for iron for its own metabolism.

- **PPI use (C)**: PPIs reduce gastric acid, which is needed to convert ferric iron (Fe^{3+}) to the absorbable ferrous form (Fe^{2+}). Chronic PPI use significantly reduces non-heme iron absorption.
- **Calcium supplementation (D)**: Calcium directly competes with iron for absorption at the duodenal enterocyte. They should be taken at different times of day.
- **Atrophic gastritis (G)**: Causes achlorhydria (absent gastric acid), severely impairing iron solubilization and absorption.

Vitamin C supplementation (E) actually ENHANCES iron absorption by reducing ferric to ferrous iron and forming a soluble chelate. High-protein diet (F) does not significantly impair iron absorption.

75 **Answer:** C

A 60-year-old male with erectile dysfunction and comorbidities (CVD, HTN, DM) should receive a PDE5 inhibitor as first-line treatment. Sildenafil inhibits phosphodiesterase-5, preventing breakdown of cGMP in the corpus cavernosum. This enhances nitric oxide-mediated smooth muscle relaxation and increases blood flow to the penis during sexual stimulation. It is effective, well-tolerated, and has cardiovascular safety data. Testosterone (A) is only indicated if testosterone levels are low. Alprostadil (B) is second-line (intracavernosal injection). Tadalafil (D) and vardenafil (E) are also PDE5 inhibitors but sildenafil is the classic first choice. Yohimbine (F) has limited evidence.

76 **Answer:** C

Nitrates are ABSOLUTELY contraindicated with PDE5 inhibitors. Both medications cause vasodilation through the nitric oxide-cGMP pathway. PDE5 inhibitors prevent cGMP breakdown while nitrates increase cGMP production — the combination causes synergistic, profound, and potentially fatal hypotension. Patients must not use nitrates within 24 hours of sildenafil/vardenafil or 48 hours of tadalafil. Beta-blockers (A), ACE inhibitors (B), CCBs (D), thiazides (E), and statins (F) can be safely used with PDE5 inhibitors, though mild additive hypotension may occur.

77 **Answer:** B

This patient has benzodiazepine withdrawal syndrome — he stopped lorazepam abruptly 2 days ago and now presents with hypertension (170/110), hyperreflexia, and likely anxiety/agitation. Benzodiazepine withdrawal can be life-threatening, causing seizures, delirium, and autonomic instability. Diazepam is the treatment of choice because:

1. It has a long half-life (20-100 hours) providing smooth, self-tapering coverage
2. Active metabolites provide extended duration of action
3. It can be given PO or IV depending on severity

Lorazepam (A) has a shorter half-life and is less ideal for withdrawal management. Alprazolam (C) is short-acting and highly addictive. Phenobarbital (D) is second-line. Carbamazepine (E) and gabapentin (F) are adjuncts, not first-line.

78 **Answers:** B C

Benzodiazepine weaning plan:

- **Avoid sudden discontinuation (B)**: Abrupt cessation causes withdrawal syndrome — seizures, delirium tremens-like symptoms, psychosis, and potentially death. The risk is proportional to dose and duration of use.
- **Switch to long-acting BDZ and slowly taper (C)**: Converting to diazepam (long half-life) provides stable plasma levels, preventing interdose withdrawal. Then gradually reduce the dose by 10-25% every 1-2 weeks over weeks to months, depending on duration of prior use.

Stopping abruptly (A) is dangerous. Switching to opioids (D) introduces a new addiction. Doubling then stopping (E) is irrational. Stimulants (F) would worsen anxiety and are not appropriate.

79 **Answer:** C

The maximum daily dose of acetaminophen for a healthy adult is 4 grams (4000 mg) per day. This equals 1000 mg (two 500 mg tablets) four times daily. Exceeding this dose risks hepatotoxicity because acetaminophen is metabolized by CYP2E1 to the toxic metabolite NAPQI, which is normally detoxified by glutathione. When glutathione stores are depleted (overdose, chronic alcohol use, malnutrition), NAPQI accumulates and causes hepatocellular necrosis. For patients with liver disease, chronic alcohol use, or malnutrition, the maximum is reduced to 2-3 grams/day.

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80 **Answers:** A B C

Monitoring for chronic high-dose acetaminophen:

- **Liver function tests (A)**: Acetaminophen is the most common cause of acute liver failure. Regular LFTs (AST, ALT, bilirubin) detect early hepatotoxicity before clinical symptoms appear.
- **CBC (B)**: Chronic acetaminophen use can rarely cause thrombocytopenia and other blood dyscrasias. CBC monitors for these effects.
- **Kidney function tests (C)**: Chronic acetaminophen use is associated with analgesic nephropathy and can contribute to chronic kidney disease, especially with long-term use.

Thyroid function (D) and lipid profile (E) are not affected by acetaminophen. Serum acetaminophen level (F) is for acute overdose assessment, not routine chronic monitoring.

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81 **Answers:** A B

ACE inhibitor monitoring:

- **Serum potassium (A)**: ACE inhibitors reduce aldosterone, which normally promotes potassium excretion. Reduced aldosterone leads to potassium retention and risk of hyperkalemia, especially in patients with renal impairment, diabetes, or on potassium-sparing diuretics.
- **Serum creatinine (B)**: ACE inhibitors dilate the efferent arteriole, reducing intraglomerular pressure. A mild creatinine rise (<30%) is expected and acceptable. A rise >30% suggests renal artery stenosis and requires drug discontinuation.

Serum calcium (C), TSH (D), uric acid (E), and magnesium (F) are not specifically affected by ACE inhibitors.

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82 **Answer:** B

The concern about vaccines causing autism originated from a fraudulent 1998 study by Andrew Wakefield, which was retracted and the author lost his medical license. Multiple large-scale epidemiological studies involving millions of children have conclusively shown NO association between vaccines (including MMR) and autism. The physician should:

1. Acknowledge the mother's concern with empathy
2. Provide evidence-based information
3. Explain the benefits of vaccination and risks of non-vaccination
4. Allow the mother to make an informed decision

Agreeing to delay (A) puts the child at risk. Being confrontational (C) damages the therapeutic relationship. Referral (D) is unnecessary. Reporting to CPS (E) is inappropriate. Alternative medicine (F) is not evidence-based.

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83 **Answers:** A B C G

Common side effects of DTaP vaccine:

- **Erythema and pain at injection site (A)**: Local inflammatory reaction at the injection site is the most common side effect, occurring in up to 50% of recipients. It is self-limiting.
- **Excessive crying and irritability (B)**: Common in infants, lasting 24-48 hours. Caused by local pain and the immune response.
- **Fever (C)**: Low-grade fever occurs in 10-30% of recipients as part of the normal immune response to the vaccine antigens.
- **Drowsiness (G)**: Common in infants following vaccination, lasting 24-48 hours.

Permanent neurological damage (D) is extremely rare, not common. Autism (E) has no association with vaccines. Anaphylaxis (F) is rare (~1 per million doses), not common.

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84 **Answers:** D (All three: Ramipril Isotretinoin Minocycline)

All three medications are contraindicated in pregnancy:

- **Ramipril**: ACE inhibitor — causes renal tubular dysgenesis, oligohydramnios, fetal hypotension, and skull ossification defects, especially in 2nd/3rd trimesters.
- **Isotretinoin**: Retinoid — the most potent human teratogen in clinical use. Causes craniofacial, cardiac, thymic, and CNS malformations. Absolutely contraindicated at any gestational age.
- **Minocycline**: Tetracycline antibiotic — causes permanent teeth discoloration (yellow-brown), enamel hypoplasia, and inhibits fetal bone growth when used after 16 weeks gestation.

85 **Answer:** B

ACE inhibitors in pregnancy (especially 2nd and 3rd trimesters) cause:

- **Renal tubular dysgenesis**: Fetal kidneys depend on the renin-angiotensin system for development. ACE inhibitors block angiotensin II, impairing renal tubular development and function.
- **Oligohydramnios**: Fetal renal impairment reduces urine output, which is the primary source of amniotic fluid in later pregnancy. Reduced amniotic fluid causes limb contractures, pulmonary hypoplasia, and fetal death.
- Additional effects include fetal hypotension, skull ossification defects, and growth restriction.

Neural tube defects (A) are associated with folate deficiency/valproic acid. Cardiac defects (C) are associated with lithium. Gestational diabetes (D) and preterm labor (E) are not ACE inhibitor effects.

86 **Answers:** A B C

Safe antihypertensives in pregnancy:

- **Labetalol (A)**: Combined alpha/beta blocker. First-line for hypertension in pregnancy. Safe profile with extensive clinical experience.
- **Nifedipine (B)**: Dihydropyridine CCB. Second-line option. Extended-release formulation preferred. Effective for both chronic and acute hypertension in pregnancy.
- **Hydralazine (C)**: Direct arteriolar vasodilator. Used IV for acute severe hypertension/preeclampsia. Long track record of safety in pregnancy.

Ramipril (D), candesartan (E), and lisinopril (F) are ACE inhibitors/ARBs — CONTRAINDICATED in pregnancy (fetal renal toxicity). Atenolol (G) is associated with fetal growth restriction and is generally avoided.

87 **Answer:** B

A 7-year-old with a small pimple that became yellow (honey-crusted) on the chest has impetigo — a superficial bacterial skin infection, most commonly caused by Staphylococcus aureus or Streptococcus pyogenes. For localized, non-bullous impetigo, topical mupirocin is first-line treatment. Mupirocin inhibits bacterial isoleucyl-tRNA synthetase, blocking protein synthesis. It achieves high local concentrations with minimal systemic absorption. Oral antibiotics (A, F) are reserved for extensive or systemic disease. Doxycycline (C) is contraindicated in children <8. IV antibiotics (D) are for severe/systemic infection. Hydrocortisone (E) is a steroid that would worsen infection.

88 **Answer:** B

Doxycycline (and all tetracyclines) are traditionally contraindicated in children under 8 years old because they chelate calcium and deposit in developing teeth and bones. In teeth, this causes permanent yellow-gray-brown discoloration and enamel hypoplasia. In bones, it can temporarily inhibit growth. The critical period is during tooth development (in utero through age 8). Note: Recent evidence suggests doxycycline may cause less staining than other tetracyclines, and short courses may be acceptable for serious infections (e.g., Rocky Mountain spotted fever), but it remains generally avoided in young children.

89 **Answer:** C

This patient has anaphylaxis — difficulty breathing, wheezing, urticaria, vomiting, and hypotension after a restaurant visit (likely food allergy). Epinephrine IM is the FIRST and MOST IMPORTANT medication in anaphylaxis. It works through:

- **Alpha-1**: Vasoconstriction → reverses hypotension and reduces mucosal edema
- **Beta-1**: Increases heart rate and contractility → improves cardiac output
- **Beta-2**: Bronchodilation → relieves wheezing; stabilizes mast cells → reduces further mediator release

Given IM in the anterolateral thigh for fastest absorption. The patient is on a beta-blocker, which may blunt epinephrine's effect — glucagon may be needed. Diphenhydramine (A) is adjunctive, not first-line. Methylprednisolone (B) takes hours to work. Albuterol (D) only treats bronchospasm. NS bolus (E) is supportive but not first. Atropine (F) treats bradycardia but doesn't address the underlying anaphylaxis.

90 **Answers:** A B C

Initial treatments for anaphylaxis in a patient on beta-blocker:

- **Oxygen (A)**: The patient is hypoxic from bronchospasm and distributive shock. High-flow oxygen is essential.
- **IV normal saline bolus (B)**: BP is 80/50 — distributive shock from massive vasodilation. Aggressive fluid resuscitation (1-2L bolus) restores intravascular volume.
- **Glucagon (C)**: CRITICAL in patients on beta-blockers. Beta-blockers blunt the effect of epinephrine by blocking beta receptors. Glucagon bypasses beta receptors entirely, directly activating adenylyl cyclase to increase cAMP, producing inotropic, chronotropic, and vasopressor effects. Dose: 1-5 mg IV over 5 minutes.

Oral antihistamine (D) is too slow in anaphylaxis. Inhaled corticosteroid (E) and oral prednisone (F) are too slow and not appropriate for acute anaphylaxis.

91 **Answer:** B

Every patient who has experienced anaphylaxis must be discharged with an epinephrine auto-injector (EpiPen). Anaphylaxis can recur (biphasic reaction in up to 20% of cases) and future exposures to the allergen can cause equally severe or worse reactions. The EpiPen allows immediate self-administration of epinephrine before emergency services arrive, which is life-saving. Patients should carry two auto-injectors at all times and be trained on proper use. Oral prednisone taper (A) may be given but is not the most critical discharge item. Daily antihistamine (C) is adjunctive. Inhaled corticosteroid (D) and antibiotics (E) are not indicated

for anaphylaxis discharge.

92 **Answers:** A B C D

Discharge instructions after anaphylaxis follow the AFEW mnemonic:

- **Anaphylaxis emergency action plan (A)**: A written plan detailing how to recognize anaphylaxis symptoms and steps to take, including when to use the EpiPen and when to call emergency services.
- **Further evaluation by an allergist (B)**: Referral for allergy testing to identify the specific trigger, education on avoidance strategies, and consideration of desensitization therapy.
- **Epinephrine auto-injector prescription (C)**: Two EpiPens prescribed with training on proper technique (inject into anterolateral thigh through clothing if needed).
- **Written information about anaphylaxis (D)**: Educational materials about triggers, recognition of early symptoms, and importance of immediate treatment.

Avoiding all restaurants permanently (E) is impractical. Daily steroids (F) are not indicated. Weekly allergy shots (G) may be considered later by the allergist but are not a discharge instruction.

93 **Answers:** B C

A traveler returning from Mexico with 6 episodes of watery diarrhea, no blood, and no fever has uncomplicated traveler's diarrhea:

- **Loperamide (B)**: An opioid receptor agonist that slows intestinal motility, increases water absorption, and reduces stool frequency. Safe and effective for non-bloody, non-febrile diarrhea. Provides symptomatic relief.
- **Oral rehydration solution (C)**: Essential to replace fluid and electrolyte losses from diarrhea. Prevents dehydration, which is the main danger of acute diarrhea.

Ciprofloxacin (A) is reserved for moderate-severe traveler's diarrhea (fever, bloody stools, >4 episodes/day with significant symptoms). This patient has mild disease without red flags. IV antibiotics (D) are unnecessary. Metronidazole (E) is for parasitic infections (Giardia, amoebiasis). Bismuth subsalicylate (F) is an alternative but less effective than loperamide.

94 **Answers:** A B C

Prevention advice for traveler's diarrhea:

- **Drink bottled beverages (A)**: Tap water in endemic areas may be contaminated with enteric pathogens. Bottled, sealed beverages are safe. Avoid ice cubes made from tap water.
- **Eat fruits that you peel just prior to eating (B)**: The peel protects the inner fruit from contamination. Peeling immediately before eating ensures the interior is uncontaminated. Pre-peeled or pre-cut fruits may be contaminated.
- **Eat thoroughly cooked food served hot (C)**: Cooking kills most enteric pathogens. Food should be steaming hot when served. Buffet food that has been sitting at room temperature is risky.

Drinking tap water freely (D) is the primary risk factor. Eating street food without concern (E) is risky due to poor hygiene. Prophylactic antibiotics daily (F) are not routinely recommended due to resistance concerns and side effects.

95 **Answers:** A | B (or F)

Intractable hiccups in palliative care:

- **Baclofen (A)**: A GABA-B receptor agonist that suppresses the hiccup reflex arc by inhibiting the phrenic nerve and diaphragmatic spasm. It is one of the most effective medications for persistent hiccups.
- **Gabapentin (B)**: Modulates calcium channels and GABA pathways, reducing neuronal excitability in the hiccup reflex arc. Effective for chronic hiccups, especially those of central origin.
- **Metoclopramide (F)**: A dopamine antagonist and prokinetic that can treat hiccups caused by gastric distension or vagal irritation by promoting gastric emptying.

Morphine (C) is for pain/dyspnea but not specifically for hiccups. Diazepam (D) has limited evidence. Ondansetron (E) is an antiemetic but not effective for hiccups.

96 **Answers:** A | B

Dyspnea management in palliative care:

- **Opioids (A)**: First-line for dyspnea in palliative care. Low-dose morphine reduces the sensation of breathlessness by acting on opioid receptors in the brainstem respiratory center, reducing the ventilatory response to hypercapnia and hypoxia. It also reduces anxiety associated with dyspnea. Does NOT hasten death at appropriate doses.
- **Benzodiazepines (B)**: Address the anxiety component of dyspnea, which often worsens the sensation of breathlessness. Lorazepam or midazolam reduce the panic and distress associated with air hunger, especially when opioids alone are insufficient.

NSAIDs (C) do not treat dyspnea. Antibiotics (D) are only relevant if infection is the cause. Antihistamines (E) and stimulants (F) are not indicated for palliative dyspnea.

97 **Answers:** A | C | D | E

A patient with frequent tension-type headaches (band-like, no photophobia/vomiting, 3x/week), asthma, GERD, and weak insurance needs affordable medications:

- **Acetaminophen (A)**: First-line acute treatment for tension-type headache. Cheap, effective, and safe. Avoids NSAIDs which would worsen GERD.
- **Mirtazapine (C)**: An antidepressant with evidence for chronic headache prophylaxis. Affordable generic available.
- **Amitriptyline (D)**: A tricyclic antidepressant that is the gold standard prophylactic for chronic tension-type headache. Very inexpensive, well-studied, and effective at low doses (10-25 mg qhs).
- **Gabapentin (E)**: An anticonvulsant with evidence for headache prophylaxis. Affordable generic available.

Sumatriptan (B) is for migraines, not tension-type headaches. Topiramate (F) is more expensive and has more side effects. Botox (G) is very expensive. Ergotamine (H) is for migraines.

98 **Answer:** B

An 8-year-old with unilateral throbbing headache, nausea, worsened by noise, relieved by sleep, occurring weekly has pediatric migraine. First-line acute treatment in children is acetaminophen at 15 mg/kg. It is safe, effective, well-tolerated, and available in child-friendly formulations (liquid, chewable). It should be given early at headache onset for maximum efficacy. Ibuprofen (A) at 10 mg/kg is also first-line but acetaminophen is listed as the answer here. Sumatriptan (C) is second-line in children and the SC route is rarely used first. Propranolol (D), amitriptyline (E), and topiramate (F) are prophylactic medications, not acute treatments.

99 **Answers:** A B

The child has tablet phobia (fear of swallowing pills) and needs alternative formulations:

- **Ibuprofen syrup (A)**: Available as a liquid suspension (100 mg/5 mL), easy to administer to children who cannot swallow tablets. Effective for acute migraine.
- **Sumatriptan nasal spray (B)**: Available as a 5 mg or 20 mg nasal spray, bypassing the need for oral administration entirely. FDA-approved for adolescent migraine. Also has faster onset than oral formulations due to nasal mucosal absorption.

Acetaminophen suppository (C) is available but less practical and less accepted by older children. Morphine elixir (D) is inappropriate for pediatric migraine. Diazepam rectal gel (E) is for seizures. Prednisone liquid (F) is not for acute migraine.

100 **Answers:** A B C

A 75-year-old with multiple medication issues:

- **Stop metformin (A)**: GFR is 29 (<30 mL/min), which is a contraindication to metformin. Metformin is renally cleared, and accumulation in renal failure causes lactic acidosis — a rare but potentially fatal complication. Metformin should be stopped when GFR <30.
- **Hold ramipril (B)**: Potassium is 5.7 (hyperkalemia). Ramipril (ACE inhibitor) reduces aldosterone, causing potassium retention. With impaired renal function (GFR 29), potassium excretion is already compromised. Continuing ramipril risks life-threatening hyperkalemia (cardiac arrhythmias).
- **Hold metoprolol (C)**: HR is 40 (severe bradycardia). Metoprolol is a beta-blocker that slows heart rate. Bradycardia causes dizziness and falls — this patient already fell. The bradycardia is likely contributing to her symptoms.

Increasing metformin (D) is dangerous. Adding furosemide (E) and digoxin (F) are not appropriate without addressing the current medication problems first.

101 **Answer:** B

A child with otalgia (ear pain) and discharge has acute otitis media (AOM) with perforation. High-dose amoxicillin (90 mg/kg/day divided BID) is first-line because:

1. It covers the most common pathogens: Streptococcus pneumoniae, Haemophilus influenzae, and Moraxella catarrhalis

2. The high dose (90 mg/kg vs standard 45 mg/kg) overcomes intermediate penicillin resistance in S. pneumoniae

3. Amoxicillin has excellent middle ear penetration

4. It is safe, well-tolerated, and available in liquid formulation for children

Standard dose (45 mg/kg) (A) is insufficient given increasing pneumococcal resistance. Azithromycin (C) is second-line for penicillin-allergic patients. Cephalexin (D) has poor middle ear penetration. Ciprofloxacin otic drops (E) are for otitis externa, not media.

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102 **Answer:** B

A child with AOM who returns after 3 days with high-grade fever, worsening otalgia, and increased discharge despite antibiotics raises concern for acute mastoiditis. Mastoiditis is the most common serious complication of AOM — infection spreads from the middle ear to the mastoid air cells. Signs include:

- Persistent/worsening symptoms despite antibiotics
- Postauricular swelling, erythema, and tenderness
- Protrusion of the auricle
- High fever

This is an emergency because it can progress to intracranial complications (meningitis, brain abscess, sigmoid sinus thrombosis). Meningitis (A) and brain abscess (C) are complications of mastoiditis itself. Cholesteatoma (D) is a chronic complication. Labyrinthitis (E) and facial nerve palsy (F) are less common.

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103 **Answers:** B C

Management of acute mastoiditis:

- **Piperacillin-tazobactam IV (B)**: Broad-spectrum IV antibiotic covering gram-positive, gram-negative, and anaerobic organisms. Mastoiditis requires IV antibiotics because oral therapy has failed and the infection involves bone (osteitis). Alternative IV options include ceftriaxone or ampicillin-sulbactam.
- **Urgent referral to otolaryngologist (C)**: Mastoiditis may require surgical intervention — myringotomy with tube placement, mastoidectomy, or drainage of subperiosteal abscess. ENT evaluation is essential.

Oral amoxicillin-clavulanate (A) is insufficient — oral antibiotics have already failed. Watchful waiting (D) is dangerous in mastoiditis. Topical drops only (E) cannot treat bone infection. Oral prednisone (F) is not the primary treatment.

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104 **Answer:** C

An elderly woman with osteoporosis (T-score -2.5 at vertebra) and family history — the most important non-pharmacological advice is fall prevention. The reason: osteoporotic fractures occur when weakened bones are subjected to trauma, and the most common trauma in the elderly is FALLS. Preventing falls prevents fractures, which cause significant morbidity and mortality (hip fractures have 20-30% one-year mortality in the elderly). Fall prevention includes:

- Home safety assessment (remove rugs, improve lighting, install grab bars)
- Balance and strength training
- Vision correction
- Medication review (stop sedatives)
- Appropriate footwear

Calcium (A) and vitamin D (B) supplementation are important but secondary to fall prevention. Weight-bearing exercise (D) helps bone density. Smoking cessation (E) and alcohol avoidance (F) are beneficial but less impactful than fall prevention.

105 **Answer:** B

Alendronate is the first-line pharmacological treatment for osteoporosis. It is a bisphosphonate that:

1. Binds to hydroxyapatite in bone
2. Is taken up by osteoclasts during bone resorption
3. Inhibits farnesyl pyrophosphate synthase (mevalonate pathway)
4. Induces osteoclast apoptosis
5. Reduces bone resorption, increasing bone mineral density

Weekly dosing (70 mg) improves adherence compared to daily dosing. Must be taken on an empty stomach with a full glass of water, remaining upright for 30 minutes to prevent esophageal irritation. Raloxifene (A) is second-line. Teriparatide (C) is for severe osteoporosis or bisphosphonate failure. Denosumab (D) is second-line. Calcitonin (E) is rarely used. Zoledronic acid (F) is an alternative bisphosphonate.

106 **Answers:** A B C

Benefits of breastfeeding to the baby:

- **Meets nutritional needs until 6 months (A)**: Breast milk provides the perfect balance of proteins, fats, carbohydrates, vitamins, and minerals for infant growth. Exclusive breastfeeding is recommended for the first 6 months.
- **Enhances GI tract growth, motility and maturity (B)**: Growth factors in breast milk (EGF, IGF) promote intestinal epithelial development, tight junction formation, and mucosal immune maturation.
- **Decreases risk of necrotizing enterocolitis and respiratory infections (C)**: IgA antibodies, lactoferrin, lysozyme, and oligosaccharides in breast milk provide passive immunity and promote beneficial gut microbiome, reducing NEC (especially in preterm infants) and respiratory infections.

Guaranteeing higher IQ (D) is overstated — modest cognitive benefits exist but are not guaranteed. Preventing all childhood illnesses (E) is false. Eliminating need for supplementation (F) is false — vitamin D supplementation is still needed.

107 **Answers:** A B C

Maternal benefits of breastfeeding:

Money savings (A): Formula feeding costs approximately $1,200-$1,500 per year. Breastfeeding is essentially free, saving significant money for the family. No bottles, sterilization equipment, or formula purchases needed.

- **Enhanced weight loss after pregnancy (B)**: Breastfeeding burns approximately 500 extra calories per day for milk production. This helps mothers return to pre-pregnancy weight faster. Lactation mobilizes fat stores deposited during pregnancy.
- **Decreased risk of breast, ovarian, and endometrial cancers (C)**: Breastfeeding reduces lifetime estrogen exposure through lactational amenorrhea. Each year of breastfeeding reduces breast cancer risk by ~4.3%. It also promotes terminal differentiation of breast epithelial cells, making them more resistant to malignant transformation.

Complete prevention of postpartum depression (D) is false — breastfeeding may reduce risk but does not prevent it. Guaranteed contraception (E) is false — lactational amenorrhea method is only ~98% effective under strict conditions. Preventing all future cancers (F) is false.

108 **Answer:** B

The most important step for successful breastfeeding in the delivery room is immediate and uninterrupted skin-to-skin contact between mother and newborn. This:

1. Triggers the newborn's innate feeding reflexes (rooting, sucking)
2. Colonizes the baby with maternal skin flora
3. Regulates the baby's temperature, heart rate, and breathing
4. Stimulates maternal oxytocin release, promoting milk let-down and uterine contraction
5. Facilitates the first breastfeed within the first hour of life ("golden hour")
6. Promotes maternal-infant bonding

Formula supplementation (A) undermines breastfeeding establishment. Separation for assessment (C) disrupts bonding and feeding reflexes. Bottle feeding first (D) causes nipple confusion. Delayed feeding (E) misses the critical window when the baby is most alert.

109 **Answer:** B

All exclusively breastfed infants should receive vitamin D 400 IU daily supplementation starting from birth. Breast milk contains insufficient vitamin D (~25 IU/L) to meet the infant's needs (400 IU/day). Vitamin D is essential for calcium absorption and bone mineralization. Deficiency causes rickets — softening and weakening of bones. Risk factors for deficiency include limited sun exposure, dark skin pigmentation, and northern latitudes. Iron drops (A) are recommended starting at 4 months in exclusively breastfed infants, not from birth. Vitamin C (C), calcium (D), zinc (E), and multivitamins (F) are not routinely supplemented in breastfed infants.

110 **Answer:** B

A patient with hepatic encephalopathy (HE) and deteriorating consciousness with chronic liver disease should receive lactulose as first-line treatment. Lactulose works by:

1. Being metabolized by colonic bacteria to lactic acid and acetic acid, lowering colonic pH

2. Acidic environment converts ammonia (NH_3) to ammonium (NH_4^+), which cannot be absorbed

3. Osmotic laxative effect increases fecal nitrogen excretion

4. Reduces ammonia-producing bacterial populations

Lactulose is the cornerstone of HE treatment. Rifaximin (A) is second-line, added when lactulose alone fails. Neomycin (C) has ototoxicity and nephrotoxicity risks. Metronidazole (D) has neurotoxicity with prolonged use. Flumazenil (E) is for benzodiazepine overdose. Naloxone (F) is for opioid overdose.

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111 **Answer:** B

Lactulose dosing in hepatic encephalopathy is titrated to achieve 2-3 soft stools per day. This is the optimal target because:

1. It ensures adequate fecal nitrogen/ammonia excretion

2. Too few stools (A) means insufficient ammonia clearance

3. Too many stools (C) causes dehydration, electrolyte imbalances (hyponatremia, hypokalemia), and worsening encephalopathy

4. The dose is individualized — typically 30-45 mL every 1-2 hours initially until bowel movement, then adjusted to maintenance

Maximum dose regardless of output (D) risks severe diarrhea and dehydration. Titrating based on ammonia levels only (E) is impractical — ammonia levels correlate poorly with clinical severity.

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112 **Answer:** B

When lactulose alone fails to improve hepatic encephalopathy, rifaximin is added. Rifaximin is a non-absorbable antibiotic that:

1. Acts locally in the gut lumen

2. Reduces ammonia-producing bacteria (urease-producing organisms)

3. Has minimal systemic absorption (<0.4%), reducing side effect risk

4. The RFHE trial showed rifaximin + lactulose reduced HE recurrence by 58% compared to lactulose alone

Rifaximin is added TO lactulose, not as a replacement. Neomycin (A) is effective but limited by nephrotoxicity and ototoxicity with long-term use. Metronidazole (C) causes peripheral neuropathy with prolonged use. Vancomycin (D) is not standard for HE. Lactulose enema (E) may be used in patients who cannot take oral medications but is not the next step here.

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113 **Answers:** A B C

Causes of recurrent urethritis:

- **Nonadherence to treatment (A)**: Incomplete antibiotic courses allow surviving organisms to repopulate, causing recurrence. Patients may stop antibiotics when symptoms improve.
- **Atypical pathogen or resistant organism (B)**: Initial treatment may not cover all causative organisms. Chlamydia, Mycoplasma genitalium, and Trichomonas vaginalis may not respond to standard gonorrhea treatment. Antimicrobial resistance is increasing.
- **Infected untreated sexual partners (C)**: If sexual partners are not simultaneously treated, they reinfect the patient upon resuming sexual activity — the "ping-pong" effect.

Excessive water intake (D), tight underwear (E), and frequent urination (F) do not cause recurrent urethritis.

114 **Answer:** C

Patients with urethritis should abstain from sexual activity for 7 days after treatment initiation. This allows:

1. Sufficient time for the antibiotic to eradicate the infection

2. Resolution of inflammation and infectivity

3. Prevention of transmission to sexual partners

Both the patient AND their partners should complete treatment and wait 7 days before resuming sexual activity. Immediately (A) risks transmission. Three days (B) may be insufficient for complete eradication. Fourteen days (D) and 30 days (E) are unnecessarily long. Waiting for test of cure (F) is ideal but not always practical.

115 **Answer:** C

A patient presenting with acute ischemic stroke symptoms (right hemiparesis, 3 hours onset) has BP 170/90. In acute ischemic stroke, blood pressure should NOT be lowered unless it exceeds 185/110 mmHg (if thrombolysis is planned) or 220/120 mmHg (if thrombolysis is not planned). The rationale:

1. Elevated BP in acute stroke is a compensatory mechanism (Cushing response) to maintain cerebral perfusion to the ischemic penumbra

2. Lowering BP aggressively reduces perfusion to already compromised brain tissue

3. This can extend the infarct size and worsen outcomes

At 170/90, the BP is below the treatment threshold. Labetalol (A) and nicardipine (B) would be used if BP were ≥185/110. Amlodipine (D) is too slow-acting. Nitroprusside (F) causes unpredictable drops.

116 **Answer:** C

The stroke patient with symptom onset 3 hours ago (within the 4.5-hour window) should receive fibrinolytic therapy. Tenecteplase (or alteplase) is a tissue plasminogen activator (tPA) that:

1. Converts plasminogen to plasmin

2. Plasmin dissolves the fibrin clot occluding the cerebral artery

3. Restores blood flow to the ischemic penumbra

4. Must be given within 4.5 hours of symptom onset for maximum benefit

Tenecteplase is increasingly preferred over alteplase due to single-bolus dosing, higher fibrin specificity, and non-inferior outcomes. Aspirin (A) is given after 24 hours post-thrombolysis. Heparin (B) is not first-line for acute ischemic stroke. Clopidogrel (D) is for secondary prevention. Warfarin (E) is for long-term anticoagulation. Nimodipine (F) is for subarachnoid hemorrhage vasospasm.

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117 **Answers:** A B C

Absolute contraindications for fibrinolytic therapy in stroke:

- **Intracranial hemorrhage (A)**: Fibrinolytics dissolve clots — giving them to a patient with intracranial bleeding would cause catastrophic hemorrhage expansion and death. CT must rule out hemorrhage before administration.
- **Previous stroke in the last 3 months (B)**: Recent stroke creates a vulnerable area of damaged brain tissue and blood-brain barrier disruption. Fibrinolytics increase the risk of hemorrhagic transformation of the recent infarct.
- **Severe closed head/facial trauma in last 3 months (C)**: Recent trauma may have caused occult intracranial vascular injury. Fibrinolytics could cause intracranial hemorrhage from these damaged vessels.

Mild headache (D) is not a contraindication. Age over 60 (E) is not an absolute contraindication. Hypertension 150/90 (F) is below the threshold. Diabetes (G) is not a contraindication.

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118 **Answers:** A B

The patient on fluphenazine (typical antipsychotic) has akathisia — a subjective feeling of inner restlessness with an inability to sit still, manifesting as pacing and constant movement. Treatment:

- **Propranolol (A)**: A beta-blocker that is first-line for akathisia. The mechanism is not fully understood but likely involves blocking peripheral and central beta-adrenergic receptors that mediate the restlessness. Very effective with rapid onset.
- **Benztropine (B)**: An anticholinergic that helps akathisia by restoring the dopamine-acetylcholine balance disrupted by antipsychotic dopamine blockade.

Haloperidol (C) is another typical antipsychotic — would worsen akathisia. Chlorpromazine (D) is also a typical antipsychotic. Olanzapine (E) could be considered as a switch but not added. Lithium (F) is not for akathisia.

119 **Answer:** B

The patient now has drug-induced parkinsonism (pseudo-parkinsonism) — cogwheel rigidity and unsteady gait caused by dopamine receptor blockade from the typical antipsychotic. Benztropine is the treatment of choice because:

1. It is an anticholinergic that restores the dopamine-acetylcholine balance in the basal ganglia

2. When dopamine is blocked by antipsychotics, relative acetylcholine excess causes parkinsonian symptoms

3. Benztropine blocks muscarinic receptors, reducing this acetylcholine excess

4. Effective for rigidity, tremor, and bradykinesia

Propranolol (A) is better for akathisia than parkinsonism. Levodopa/Carbidopa (C) is for true Parkinson's disease and would counteract the antipsychotic. Haloperidol (D) would worsen symptoms. Diazepam (E) is not specific treatment. Amantadine (F) is an alternative but benztropine is preferred.

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120 **Answer:** C

After multiple extrapyramidal side effects (akathisia, then pseudo-parkinsonism), the best approach is switching to an atypical (second-generation) antipsychotic. Atypical antipsychotics (e.g., risperidone, olanzapine, quetiapine, aripiprazole) have:

1. Lower affinity for D2 receptors in the nigrostriatal pathway

2. Serotonin 5-HT2A antagonism that modulates dopamine release

3. Significantly lower rates of EPS compared to typical antipsychotics

4. Better tolerability and adherence

Increasing the typical antipsychotic dose (A) would worsen EPS. Adding another typical (B) compounds the problem. Adding lithium (D) doesn't address EPS. Discontinuing all medications (E) risks psychotic relapse. Permanent benzodiazepine (F) is not appropriate.

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121 **Answers:** B D

Antipsychotics causing the MOST weight gain:

- **Olanzapine (B)**: Causes the most weight gain among commonly used atypical antipsychotics — average 4-5 kg in the first 6-8 weeks. Mechanism involves histamine H1 receptor blockade (appetite stimulation), serotonin 5-HT2C blockade, and metabolic effects increasing insulin resistance.
- **Clozapine (D)**: Causes comparable or greater weight gain than olanzapine. Also causes metabolic syndrome, dyslipidemia, and diabetes. Reserved for treatment-resistant schizophrenia due to agranulocytosis risk.

Aripiprazole (A) is weight-neutral. Ziprasidone (C) is weight-neutral or causes minimal gain. Haloperidol (E) is a typical antipsychotic with minimal weight gain. Lurasidone (F) is weight-neutral.

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122 **Answer:** B (or A)

A patient on warfarin for AF who is frustrated with regular INR monitoring can switch to a DOAC (direct oral anticoagulant). Apixaban is preferred because:

1. No routine coagulation monitoring required (unlike warfarin's INR checks)

2. Fixed dosing — predictable pharmacokinetics

3. Fewer drug-food interactions than warfarin

4. Lower bleeding risk than warfarin (ARISTOTLE trial showed apixaban superior to warfarin)

5. Rapid onset — no bridging with heparin needed

Dabigatran (A) is also acceptable but has higher GI bleeding risk and requires dose adjustment for renal function. Clopidogrel (C) and aspirin (D) are antiplatelet agents, not anticoagulants — insufficient for AF stroke prevention. Heparin (E) requires injection and monitoring. Fondaparinux (F) is injectable.

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123 **Answers:** A B C G

Contraindications to DOACs:

- **Acute bleeding (A)**: DOACs are anticoagulants — giving them during active bleeding would worsen hemorrhage. Any active significant bleeding is an absolute contraindication.
- **Moderate to severe hepatic impairment (B)**: DOACs are partially hepatically metabolized. Liver dysfunction impairs drug clearance and also impairs production of clotting factors, creating dual bleeding risk.
- **Pregnancy (G)**: DOACs cross the placenta and have not been studied in pregnancy. Animal studies show reproductive toxicity. LMWH is the anticoagulant of choice in pregnancy.

Mild hypertension (D) is not a contraindication. Age over 65 (E) is not a contraindication — DOACs are commonly used in elderly patients. Diabetes (F) is not a contraindication.

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124 **Answer:** B

A GFR of 10 mL/min represents severe renal impairment (Stage 5 CKD). DOACs are significantly renally cleared:

- Dabigatran: 80% renal excretion — contraindicated if CrCl <30
- Rivaroxaban: 36% renal excretion — contraindicated if CrCl <15
- Apixaban: 27% renal excretion — use with caution if CrCl <25

With GFR 10, drug accumulation occurs, leading to supratherapeutic levels and high bleeding risk. Warfarin is preferred in severe renal impairment because it is hepatically metabolized. Hepatotoxicity (A) is not the primary concern. Hyperkalemia (C) is associated with heparin, not DOACs. Weekly monitoring (D) defeats the purpose of DOACs. DOACs are oral, not IV only (E).

125 **Answers:** A B C

Non-pharmacological management of Type 2 Diabetes:

- **Weight reduction through diet and exercise (A)**: Even 5-7% weight loss significantly improves insulin sensitivity, reduces HbA1c by 0.5-1%, and can delay or prevent progression from prediabetes to diabetes. Caloric restriction and balanced macronutrient intake are essential.
- **Regular exercise (B)**: 150 minutes/week of moderate aerobic exercise improves glucose uptake by skeletal muscle independent of insulin, enhances insulin sensitivity for 24-72 hours post-exercise, and reduces cardiovascular risk.
- **Psychological intervention (C)**: Diabetes distress, depression, and anxiety are common in diabetic patients and impair self-management. CBT and motivational interviewing improve adherence to diet, exercise, and medication.

Unlimited carbohydrate intake (D) worsens glycemic control. Bed rest (E) and avoiding physical activity (F) worsen insulin resistance and cardiovascular risk.

126 **Answers:** A B C D E

Chronic follow-up investigations for diabetic patients:

- **HbA1c every 3-6 months (A)**: Reflects average blood glucose over 2-3 months. Target <7% for most adults. Guides medication adjustments.
- **Serum creatinine annually (B)**: Monitors for diabetic nephropathy — diabetes is the leading cause of CKD. Detects declining renal function early.
- **Urinary albumin-to-creatinine ratio annually (C)**: Detects microalbuminuria — the earliest sign of diabetic nephropathy. Early detection allows intervention with ACE inhibitors/ARBs to slow progression.
- **Dilated eye examination annually (D)**: Screens for diabetic retinopathy — the leading cause of blindness in working-age adults. Early detection allows laser photocoagulation to prevent vision loss.
- **Comprehensive foot examination annually (E)**: Screens for diabetic neuropathy, peripheral vascular disease, and foot deformities. Prevents diabetic foot ulcers and amputations.

Weekly blood glucose only (F) is insufficient for comprehensive monitoring. Annual chest X-ray (G) is not specifically indicated for diabetes.

127 **Answer:** C

A patient with hyperthyroidism symptoms (tachycardia, palpitations, heat intolerance, weight loss, fine tremors) needs a beta-blocker for rapid symptom relief. Atenolol (or propranolol) works by:

1. Blocking beta-1 receptors — reducing heart rate and palpitations
2. Reducing peripheral conversion of T4 to T3 (propranolol specifically)
3. Alleviating tremor, anxiety, and heat intolerance
4. Providing rapid symptomatic relief within hours while waiting for antithyroid drugs to take effect (which take weeks)

Methimazole (A) and PTU (B) are definitive treatments but take 4-6 weeks to normalize thyroid function. Radioactive iodine (D) is definitive but takes months. Levothyroxine (E) is for hypothyroidism — would worsen hyperthyroidism. Prednisone (F) is for thyroid storm or thyroiditis, not routine hyperthyroidism.

128 **Answer:** B

Methimazole is the long-term antithyroid medication for hyperthyroidism. It works by:

1. Inhibiting thyroid peroxidase (TPO) enzyme

2. Blocking iodine organification and coupling of iodotyrosines

3. Preventing synthesis of T3 and T4

4. Gradually normalizing thyroid hormone levels over 4-8 weeks

Methimazole is preferred over PTU because of once-daily dosing, fewer side effects, and lower risk of hepatotoxicity. PTU is reserved for first trimester pregnancy and thyroid storm. Propranolol (A) only treats symptoms, not the underlying disease. Levothyroxine (C) is for hypothyroidism. Hydrocortisone (D) is not standard. Lithium (E) has antithyroid effects but is not first-line.

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129 **Answers:** A B C

Monitoring for methimazole:

- **TFT — TSH, free T4 (A)**: Essential to monitor treatment response. Initially checked every 4-6 weeks until euthyroid, then every 3-6 months. TSH may remain suppressed for months even after T4 normalizes, so free T4 is the primary monitoring parameter initially.
- **CBC with differential (B)**: Methimazole can cause agranulocytosis (0.1-0.3% incidence) — a life-threatening drop in neutrophils. Patients must be educated to report sore throat, fever, or mouth ulcers immediately. Baseline CBC and periodic monitoring are recommended.
- **Liver function tests (C)**: Methimazole can cause cholestatic hepatitis. LFTs should be checked at baseline and if symptoms of hepatotoxicity develop (jaundice, dark urine, abdominal pain).

Serum calcium (D) is not specifically affected. Chest X-ray (E) and weekly ECG (F) are not required for methimazole monitoring.

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130 **Answer:** B

A patient with hypothyroidism (elevated TSH of 10, weight gain, constipation) needs thyroid hormone replacement. Levothyroxine (synthetic T4) is the treatment of choice because:

1. It is identical to endogenous T4

2. Long half-life (7 days) allows once-daily dosing

3. Peripheral conversion to T3 provides physiologic T3 levels

4. Dose is weight-based: 1.6 mcg/kg/day for full replacement

5. Must be taken on an empty stomach, 30-60 minutes before breakfast, with water only

Methimazole (A) and PTU (D) are for hyperthyroidism — would worsen hypothyroidism. Liothyronine (C) is synthetic T3 — short half-life requires multiple daily doses and causes T3 level fluctuations. Prednisone (E) is not treatment for hypothyroidism.

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131 **Answer:** C

TSH should be checked 4-6 weeks after starting levothyroxine because:

1. Levothyroxine has a long half-life (7 days) — steady-state is reached in 5-6 half-lives (~5-6 weeks)

2. TSH response to thyroid hormone changes is slow — the pituitary takes weeks to adjust TSH secretion

3. Checking too early gives inaccurate results and may lead to inappropriate dose changes

After each dose adjustment, wait another 4-6 weeks before rechecking. Once stable, TSH can be monitored every 6-12 months. One week (A) and 2 weeks (B) are too early. Three months (D), 6 months (E), and 1 year (F) are too late for initial assessment.

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132 **Answers:** A B

A 35-year-old woman with depression (weeping, insomnia, low energy, no self-harm) needs antidepressant therapy. Two medications from different classes:

- **SSRI — Paroxetine 20 mg/day (A)**: Selective serotonin reuptake inhibitors are first-line for depression. They increase serotonin availability in the synaptic cleft by blocking the serotonin transporter (SERT). Well-tolerated with a favorable side effect profile.
- **SNRI — Venlafaxine 37.5 mg/day (B)**: Serotonin-norepinephrine reuptake inhibitors block both SERT and NET, increasing both serotonin and norepinephrine. Effective for depression with prominent fatigue and pain.

Note: In practice, you would choose ONE antidepressant, not combine two. The question asks for options from different classes. TCAs (C) have more side effects. MAOIs (D) have dietary restrictions and drug interactions. Benzodiazepines (E) are not antidepressants. Stimulants (F) are not first-line for depression.

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133 **Answer:** C

Antidepressants typically take 2-4 weeks to show clinical improvement. The reason for this delay:

1. While serotonin/norepinephrine reuptake is blocked immediately, downstream neuroplastic changes take weeks

2. Serotonin autoreceptor desensitization must occur before net serotonin transmission increases

3. Neurogenesis in the hippocampus (BDNF-mediated) takes weeks

4. Receptor density changes and second messenger system adaptations require time

Patients must be counseled about this delay to prevent premature discontinuation. Side effects often appear before therapeutic benefits, which can be discouraging. Full response may take 6-8 weeks. One to three days (A) and 1 week (B) are too early. Two to three months (D) and 6 months (E) are longer than typical onset.

---

134 **Answer:** C

After remission of a first depressive episode, antidepressant treatment should continue for at least 6 months (some guidelines say 6-12 months). This is the continuation phase, which:

1. Prevents relapse — the risk of relapse is highest in the first 6 months after remission

2. Allows consolidation of neuroplastic changes

3. Premature discontinuation leads to relapse in ~50% of patients

For recurrent depression (≥3 episodes), maintenance therapy for 2+ years or lifelong may be recommended. Two weeks (A) and 1 month (B) are far too short. One year (D) may be appropriate for some patients. Lifelong (E) is for recurrent depression, not first episodes.

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135 **Answers:** A B C

First-line treatments for smoking cessation:

- **Nicotine replacement therapy (A)**: Patches, gum, lozenges, inhalers, and nasal spray deliver nicotine without the harmful combustion products. They reduce withdrawal symptoms and cravings by partially stimulating nicotinic receptors. Doubles quit rates compared to placebo.
- **Varenicline (B)**: A partial agonist at alpha-4-beta-2 nicotinic acetylcholine receptors. It reduces cravings (partial agonism) and blocks the rewarding effects of smoking (competitive antagonism). Most effective single agent — triples quit rates.
- **Bupropion (C)**: An atypical antidepressant that inhibits norepinephrine and dopamine reuptake. Reduces withdrawal symptoms and cravings. Also blocks nicotinic receptors. Doubles quit rates.

Clonidine (D) and nortriptyline (E) are second-line agents. Fluoxetine (F) is not specifically indicated for smoking cessation.

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136 **Answer:** C

Bupropion is contraindicated in patients with a history of seizures. This patient has a history of head trauma and seizures, making bupropion dangerous. Bupropion lowers the seizure threshold in a dose-dependent manner because it inhibits GABA-mediated inhibition and increases excitatory neurotransmission (dopamine and norepinephrine). The seizure risk is approximately 0.4% at therapeutic doses but increases significantly with higher doses, predisposing conditions, or concurrent medications that lower seizure threshold. Nicotine patch (A), varenicline (B), nicotine gum (D), and nicotine lozenges (E) are all safe in patients with seizure history.

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137 **Answer:** B

Bupropion is contraindicated in seizure patients because:

1. It inhibits GABAergic inhibitory neurotransmission

2. It increases dopaminergic and noradrenergic excitatory neurotransmission

3. This combination lowers the seizure threshold

4. The risk is dose-dependent — higher doses carry greater risk

5. Additional risk factors include eating disorders, alcohol withdrawal, CNS tumors, head trauma, and concurrent medications that lower seizure threshold

Hepatotoxicity (A) is not the primary concern. Cardiac arrhythmias (C) are not a major bupropion side effect. Renal failure (D) is not caused by bupropion. Drug interactions with antiepileptics (E) exist but are not the primary reason for contraindication.

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138 **Answers:** A B C

A Crohn's disease patient with vomiting, diarrhea, hypoglycemia (glucose 2.1), and hemodynamic instability (BP 80/50, pulse 125) needs immediate resuscitation:

- **Isotonic IV fluid (A)**: NS or RL to restore intravascular volume and treat hypotension/tachycardia. The patient is in hypovolemic shock from fluid losses (vomiting, diarrhea).
- **Correct hypoglycemia with D50W IV (B)**: Glucose of 2.1 mmol/L is critically low and can cause seizures, brain damage, and death. Immediate IV dextrose (25-50 mL of D50W) is life-saving.
- **Correct electrolyte disturbances (C)**: Vomiting and diarrhea cause significant electrolyte losses — hypokalemia, hyponatremia, hypomagnesemia, metabolic acidosis. These must be identified and corrected to prevent cardiac arrhythmias.

Starting oral feeding immediately (D) is inappropriate in an acutely ill, vomiting patient. IV antibiotics (E) may be needed but are not the immediate priority. Urgent colonoscopy (F) is contraindicated in an unstable patient.

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139 **Answers:** A B C

Acute gout treatment (3 medication options):

- **Prednisone 30 mg PO daily (A)**: Systemic corticosteroid that suppresses the inflammatory response to monosodium urate crystals. Used when NSAIDs and colchicine are contraindicated or insufficient. Effective for patients with renal impairment or GI issues.
- **Naproxen 500 mg PO BID (B)**: NSAIDs are first-line for acute gout. They inhibit cyclooxygenase (COX-1 and COX-2), reducing prostaglandin-mediated inflammation. Naproxen is preferred over indomethacin due to better tolerability. Should be started at full dose and continued until the flare resolves.
- **Colchicine 0.6 mg PO daily (C)**: Inhibits microtubule polymerization, preventing neutrophil migration and phagocytosis of urate crystals. Most effective when started within 12-24 hours of flare onset. Low-dose regimen (1.2 mg then 0.6 mg one hour later) is as effective as high-dose with fewer GI side effects.

Allopurinol (D) and febuxostat (E) are urate-lowering therapies for chronic gout prevention — they should NOT be started during an acute flare as they can worsen the attack. Probenecid (F) is a uricosuric agent for chronic management.

140 **Answers:** A B

Preventing gout recurrence:

- **Maintain healthy weight and regular exercise (A)**: Obesity increases uric acid production and decreases renal excretion. Weight loss reduces serum urate levels. Exercise improves metabolic health and reduces insulin resistance, which impairs renal urate excretion.
- **Dietary moderation in purine intake (B)**: High-purine foods (red meat, organ meats, shellfish, beer) are metabolized to uric acid. Reducing intake lowers serum urate. Also limit fructose-sweetened beverages (increase urate production) and alcohol (especially beer — contains purines and impairs renal urate excretion).

Increasing red meat (C) increases purine load. Increasing alcohol (D) impairs urate excretion. Daily aspirin (E) at low doses actually increases urate levels by competing for renal tubular secretion. Avoiding all exercise (F) worsens metabolic syndrome.

141 **Answer:** B

An 8-year-old with ADHD who has failed non-pharmacological management needs first-line medication. Methylphenidate is the first-line stimulant for ADHD because:

1. It blocks dopamine and norepinephrine reuptake transporters (DAT and NET)
2. Increases dopamine and norepinephrine in the prefrontal cortex
3. Improves attention, concentration, impulse control, and executive function
4. Rapid onset of action (30-60 minutes)
5. Extensive safety and efficacy data in children
6. Starting dose of 5 mg BID allows careful titration

Atomoxetine (A) is a non-stimulant — second-line when stimulants fail or are contraindicated. Dextroamphetamine (C) is an alternative stimulant but methylphenidate is tried first. Clonidine (D) and guanfacine (E) are alpha-2 agonists — adjuncts or alternatives. Bupropion (F) is not first-line for pediatric ADHD.

142 **Answers:** A B C D

Monitoring for a child on methylphenidate:

- **Weight (A)**: Methylphenidate suppresses appetite, causing weight loss or poor weight gain. Weight should be plotted on growth charts at each visit.
- **Height (B)**: Stimulants may cause growth deceleration (0.5-1 cm/year reduction). Height velocity should be monitored on growth charts. Drug holidays during summer may allow catch-up growth.
- **Heart rate (C)**: Methylphenidate increases sympathetic activity, causing tachycardia. Baseline and periodic heart rate monitoring detects cardiovascular effects.
- **Blood pressure (D)**: Stimulants can increase BP by 2-4 mmHg on average. Significant hypertension requires dose reduction or discontinuation.

Serum drug levels (E) are not routinely monitored — dose is titrated clinically. Monthly liver enzymes (F) are not required. Chest X-ray (G) is not indicated.

143 **Answers:** A B D

Side effects of methylphenidate:

- **Cardiovascular events and increased blood pressure (A)**: Stimulants increase sympathetic tone, raising heart rate and blood pressure. Rare but serious events include sudden cardiac death in patients with underlying structural heart disease. Pre-treatment cardiac screening is recommended.
- **Growth retardation (B)**: Appetite suppression leads to decreased caloric intake and reduced growth velocity. Average height reduction is 1-2 cm over 2-3 years. May be partially reversible with drug holidays.
- **New onset psychosis or mania (D)**: Stimulants increase dopamine in the mesolimbic pathway, which can trigger psychotic symptoms (hallucinations, paranoia) or manic episodes, especially in patients with predisposition to bipolar disorder or psychosis.

Weight gain (C) is incorrect — stimulants cause weight LOSS. Improved appetite (E) is incorrect — they suppress appetite. Excessive sleepiness (F) is incorrect — stimulants cause insomnia.

144 **Answer:** B

After 2 months of minimal response to methylphenidate at maximum dose, the next step is switching to an alternative stimulant — amphetamine or dextroamphetamine. The rationale:

1. Approximately 30% of ADHD patients respond better to amphetamines than methylphenidate and vice versa

2. Amphetamines have a different mechanism — they actively release dopamine and norepinephrine from presynaptic vesicles (in addition to reuptake blockade)

3. This provides a more potent dopaminergic effect

4. Trying both stimulant classes before moving to non-stimulants is standard practice

Doubling the dose (A) beyond maximum is dangerous. Adding an SSRI (C) is not standard. Benzodiazepines (D) are not for ADHD. Discontinuing all medications (E) abandons treatment. Lithium (F) is for bipolar disorder.

145 **Answers:** A B

A 16-year-old with schizophrenia (paranoid delusions, flat affect) needs antipsychotic medication:

- **Risperidone (A)**: An atypical antipsychotic that blocks D2 and 5-HT2A receptors. FDA-approved for adolescent schizophrenia (age 13+). Effective for positive symptoms (delusions, hallucinations) and has some benefit for negative symptoms (flat affect). Starting at low dose (0.25-0.5 mg) and titrating minimizes side effects.
- **Olanzapine (B)**: Another atypical antipsychotic with D2 and 5-HT2A blockade. FDA-approved for adolescent schizophrenia (age 13+). Effective but carries significant metabolic side effects (weight gain, diabetes, dyslipidemia).

Fluoxetine (C) is an antidepressant — not for schizophrenia. Lithium (D) is for bipolar disorder. Methylphenidate (E) is for ADHD and could worsen psychosis. Diazepam (F) is not antipsychotic.

146 **Answers:** A B

The patient has acute dystonia — sustained involuntary muscle contractions causing neck spasm (torticollis) and involuntary leg movements. This is an extrapyramidal side effect of antipsychotics occurring within hours to days of starting treatment. Treatment:

- **Benztropine (A)**: Anticholinergic that rapidly reverses dystonia by restoring dopamine-acetylcholine balance. Can be given IV for rapid effect or PO for maintenance.
- **Diphenhydramine (B)**: An antihistamine with anticholinergic properties that also effectively treats acute dystonia. Given IM for rapid onset. Often used when benztropine is unavailable.

Both work within 15-30 minutes when given parenterally. Haloperidol (C) would worsen dystonia. Increasing antipsychotic dose (D) would worsen it. Diazepam (E) is less effective than anticholinergics. Phenytoin (F) is not indicated.

147 **Answer:** B

In hyperkalemia (K=7.2), the FIRST medication is calcium gluconate because:

1. It does NOT lower potassium levels

2. It stabilizes the cardiac membrane by antagonizing the effect of potassium on cardiac myocytes

3. It raises the threshold potential, reducing the risk of fatal arrhythmias (peaked T waves, widened QRS, sine wave, VF, asystole)

4. Onset is within 1-3 minutes, lasting 30-60 minutes

5. This buys time while other treatments lower potassium

Insulin with glucose (A) shifts potassium intracellularly but takes 15-30 minutes — not fast enough to prevent immediate cardiac death. Albuterol (C) also shifts potassium but takes 15-30 minutes. Furosemide (D) removes potassium but takes hours. Sodium bicarbonate (E) is less reliable. Kayexalate (F) takes hours to work.

148 **Answers:** A B

Medications that shift potassium into cells:

- **Insulin with glucose (A)**: Insulin activates the Na+/K+ ATPase pump on cell membranes, driving potassium into cells. Glucose is co-administered to prevent hypoglycemia. Typical dose: 10 units regular insulin + 25g dextrose (D50W). Onset 15-30 minutes, lowers K by 0.5-1.2 mEq/L.
- **Beta-2 agonist — Albuterol (B)**: Stimulates beta-2 receptors on skeletal muscle cells, activating the Na+/K+ ATPase pump and driving potassium intracellularly. Given as high-dose nebulization (10-20 mg). Onset 15-30 minutes, lowers K by 0.5-1.0 mEq/L.

Calcium gluconate (C) stabilizes the heart but does NOT shift potassium. Furosemide (D) removes potassium through renal excretion, not intracellular shift. Patiromer (E) and sodium polystyrene sulfonate (F) bind potassium in the GI tract — removal, not shifting.

149 **Answer:** C

A woman with restless leg syndrome (RLS) and LOW serum ferritin should receive iron supplementation FIRST. Iron deficiency is a well-established cause/exacerbator of RLS because:

1. Iron is a cofactor for tyrosine hydroxylase — the rate-limiting enzyme in dopamine synthesis

2. Low brain iron impairs dopaminergic function in the basal ganglia

3. Ferritin <75 mcg/L is associated with worsening RLS symptoms

4. Iron repletion alone may resolve or significantly improve RLS symptoms

Treating the underlying cause (iron deficiency) before adding symptomatic medications is the correct approach. Gabapentin (A) and pramipexole (B) are used when iron stores are adequate but symptoms persist. Ropinirole (D) is a dopamine agonist — second-line. Clonazepam (E) is not first-line. Pregabalin (F) is an alternative to gabapentin.

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150 **Answers:** A B

Other medications for RLS (after iron repletion):

- **Gabapentin (A)**: An alpha-2-delta calcium channel ligand that is now considered first-line pharmacotherapy for RLS. It reduces sensory symptoms (paresthesias, urge to move) and improves sleep quality. Does not cause augmentation (worsening of symptoms with long-term use), unlike dopamine agonists.
- **Pramipexole (B)**: A dopamine agonist (D2/D3 receptor) that directly stimulates dopaminergic pathways. Effective for motor symptoms of RLS. However, long-term use carries risk of augmentation (earlier onset, increased intensity, spread to upper limbs) and impulse control disorders.

Methylphenidate (C) is a stimulant — would worsen RLS. Fluoxetine (D) can worsen RLS (serotonergic medications exacerbate symptoms). Haloperidol (E) blocks dopamine — would worsen RLS. Propranolol (F) is not indicated for RLS.

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151 **Answer:** D

Plantar fasciitis is a clinical diagnosis — no imaging investigation is needed. The diagnosis is made based on:

1. Classic history: pain with first steps in the morning, pain after prolonged standing/activity

2. Tenderness at the medial calcaneal tubercle on examination

3. Pain reproduced by dorsiflexion of the toes (windlass test)

Imaging is not recommended because:

- X-ray (A) may show a heel spur, but this is an incidental finding present in 50% of asymptomatic people and does not correlate with symptoms
- MRI (B) is reserved for refractory cases or suspected alternative diagnoses
- Ultrasound (C) may show fascial thickening but is not needed for diagnosis
- CT (E) and bone scan (F) are not indicated

152 **Answers:** A B C D

Non-pharmacological treatment for plantar fasciitis:

- **Stretching exercises for plantar fascia and calf muscles (A)**: The cornerstone of treatment. Stretching the gastrocnemius-soleus complex and plantar fascia reduces tension on the calcaneal insertion. Wall stretches and towel stretches performed multiple times daily are most effective.
- **Avoiding flat shoes and barefoot walking (B)**: Flat shoes and bare feet provide no arch support, increasing strain on the plantar fascia. Supportive shoes with cushioned soles and arch support reduce mechanical stress.
- **Using silicone heel shoe inserts (C)**: Heel cups and orthotic inserts redistribute pressure, cushion the heel, and support the arch, reducing plantar fascia strain during weight-bearing.
- **Decreasing aggravating physical activities (D)**: Relative rest from high-impact activities (running, jumping) allows inflammation to resolve. Activity modification, not complete cessation, is recommended.

Increasing running distance (E) worsens the condition. High heels (F) alter biomechanics. Complete bed rest (G) is unnecessary and causes deconditioning.

153 **Answer:** B

An elderly patient with headache, visual disturbance, and temporal tenderness has giant cell arteritis (GCA). When visual symptoms are present, this is an EMERGENCY because:

1. GCA causes inflammation of the ophthalmic/posterior ciliary arteries
2. This can cause irreversible blindness from anterior ischemic optic neuropathy
3. Once vision is lost, it rarely recovers

High-dose IV methylprednisolone (1000 mg/day for 3 days) is given immediately when visual symptoms are present to:

- Rapidly suppress vascular inflammation
- Prevent progression to complete blindness
- Protect the contralateral eye

Oral prednisone 1 mg/kg (A) is appropriate for GCA WITHOUT visual symptoms. With visual involvement, IV pulse therapy is mandatory for maximum anti-inflammatory effect. Ibuprofen (C) and acetaminophen (D) are insufficient for GCA. Methotrexate (E) is a steroid-sparing agent for long-term management. Tocilizumab (F) is adjunctive for refractory/relapsing GCA but not the immediate treatment.

154 **Answers:** A B C D G

Long-term steroid side effects to counsel the GCA patient about:

- **Blood glucose impairment/DM (A)**: Steroids increase hepatic gluconeogenesis and cause insulin resistance, leading to steroid-induced diabetes or worsening of existing diabetes.
- **Immunosuppression/infection (B)**: Steroids suppress T-cell function, neutrophil migration, and cytokine production, increasing susceptibility to bacterial, viral, and fungal infections including opportunistic infections.
- **Osteoporosis (C)**: Steroids inhibit osteoblast function, increase osteoclast activity, and reduce calcium absorption. GCA patients require long-term steroids, making osteoporosis prevention (bisphosphonates, calcium, vitamin D) essential.
- **Blood pressure elevation (D)**: Mineralocorticoid effects cause sodium and water retention, expanding intravascular volume and raising blood pressure.
- **Glaucoma/cataracts (G)**: Steroids increase intraocular pressure (open-angle glaucoma) and cause posterior subcapsular cataracts with prolonged use. Regular ophthalmologic monitoring is needed.

Weight loss (E) is incorrect — steroids cause weight GAIN. Improved wound healing (F) is incorrect — steroids IMPAIR wound healing.

155 **Answers:** A B C D E H

Non-pharmacological sleep hygiene advice:

- **Avoid caffeinated drinks especially before sleep (A)**: Caffeine blocks adenosine receptors, preventing sleep onset. Half-life is 5-6 hours — drinking 5 coffees daily significantly disrupts sleep architecture. Avoid caffeine at least 6 hours before bedtime.
- **Avoid daytime sleep (B)**: Napping reduces sleep pressure (homeostatic sleep drive), making it harder to fall asleep at night. If napping is necessary, limit to 20-30 minutes before 3 PM.
- **Schedule fixed sleep and wake times (C)**: Consistent sleep-wake times synchronize the circadian rhythm, improving sleep quality and reducing sleep onset latency. Maintain the schedule even on weekends.
- **Avoid eating or exercise 3 hours before bedtime (D)**: Late meals cause gastric discomfort and GERD. Exercise raises core body temperature and stimulates cortisol/adrenaline — both inhibit sleep onset.
- **Keep bedroom for sleep and sex only (E)**: This strengthens the psychological association between the bedroom and sleep (stimulus control therapy). Working, watching TV, or eating in bed weakens this association.
- **Avoid using screens before bedtime (H)**: Blue light from phones/laptops suppresses melatonin secretion by stimulating melanopsin receptors in the retina, delaying sleep onset.

Using phone/laptop in bed (F) worsens insomnia. Alcohol before bed (G) disrupts sleep architecture — causes early awakening and reduces REM sleep.

156 **Answer:** C

A 55-year-old woman wanting a non-addictive sleep medication should receive ramelteon. Ramelteon is a melatonin receptor agonist (MT1 and MT2) that:

1. Promotes sleep onset by mimicking endogenous melatonin's effect on the suprachiasmatic nucleus

2. Has NO abuse potential — it is not a controlled substance

3. Does not cause physical dependence or withdrawal

4. Does not impair next-day cognitive function

5. Safe for long-term use

Lorazepam (A), diazepam (D), and alprazolam (E) are benzodiazepines — highly addictive with tolerance, dependence, and withdrawal. Zopiclone (B) and zolpidem (F) are "Z-drugs" — while marketed as less addictive than benzodiazepines, they still carry significant dependence and abuse potential and are controlled substances.

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157 **Answers:** A B C D G H

Non-pharmacological options for nocturnal enuresis:

- **Void 4-7 times per day including before bed (A)**: Regular voiding prevents bladder overdistension and ensures the bladder is empty at bedtime, reducing nighttime accidents.
- **Avoid high-sugar and caffeine drinks in the evening (B)**: Caffeine is a diuretic and bladder irritant. Sugar-sweetened beverages increase urine production. Avoiding these in the evening reduces nighttime urine volume.
- **Provide smaller fluid intake after 5 PM (C)**: Reducing evening fluid intake decreases nighttime urine production without causing dehydration. Adequate fluids should be consumed earlier in the day.
- **Use a bed-wetting alarm (D)**: The most effective long-term treatment for nocturnal enuresis. The alarm sounds when moisture is detected, conditioning the child to wake when the bladder is full. Success rates of 60-80% with sustained improvement.
- **Educate parents this is not uncommon and not the child's fault (G)**: Nocturnal enuresis affects 15% of 5-year-olds and 5% of 10-year-olds. It is involuntary and not behavioral. Reassurance reduces parental frustration and child's shame/anxiety.
- **Wake child at night to use toilet (H)**: Scheduled waking (lifting) helps empty the bladder during the night, reducing wet nights while other interventions take effect.

Punishing the child (E) is harmful — increases anxiety and worsens enuresis. Restricting all fluids after noon (F) causes dehydration.

158 **Answer:** B

When non-pharmacological measures fail for nocturnal enuresis, desmopressin is first-line pharmacotherapy. Desmopressin is a synthetic analogue of antidiuretic hormone (ADH/vasopressin) that:

1. Acts on V2 receptors in the renal collecting ducts

2. Increases water reabsorption, concentrating urine

3. Reduces nighttime urine production

4. Taken 1 hour before bedtime

5. Effective in 60-70% of children

6. Must restrict fluid intake in the evening to prevent hyponatremia

Oxybutynin (A) is an anticholinergic for overactive bladder — not first-line for primary nocturnal enuresis. Imipramine (C) is a TCA that was historically used but has cardiac toxicity risk in children — not first-line. Furosemide (D) is a diuretic — would worsen enuresis. Tolterodine (E) is for overactive bladder.

---

159 **Answer:** B

The concerning drug interaction is between clarithromycin and cyclosporine. Clarithromycin is a potent inhibitor of CYP3A4 and P-glycoprotein — the primary enzymes responsible for cyclosporine metabolism. When clarithromycin is given:

1. CYP3A4 inhibition reduces cyclosporine hepatic metabolism

2. P-glycoprotein inhibition reduces cyclosporine intestinal and renal excretion

3. Cyclosporine blood levels increase dramatically (2-5 fold)

4. This leads to cyclosporine toxicity

This is clinically significant in transplant patients because cyclosporine has a narrow therapeutic index — small increases in levels cause significant toxicity. Interactions with acetaminophen (A), ibuprofen (C), vitamin D (D), and calcium (E) are not clinically significant.

---

160 **Answer:** B

The most common symptom of cyclosporine toxicity from this interaction is tremor. Cyclosporine toxicity manifests as:

1. **Tremor** — the earliest and most common neurotoxic sign, caused by cyclosporine's effect on the nervous system

2. Nephrotoxicity — elevated creatinine, decreased urine output

3. Hypertension — from renal vasoconstriction

4. Headache

5. Gingival hyperplasia

6. Hirsutism

Tremor is often the first clinical sign that alerts the physician to supratherapeutic cyclosporine levels. Headache (A) can occur but is less specific. Rash (C), diarrhea (D), cough (E), and blurred vision (F) are not the most common presentations of cyclosporine toxicity.

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161 **Answer:** B

To avoid cyclosporine toxicity when co-administering clarithromycin:

1. Reduce the cyclosporine dose by approximately 50% when starting clarithromycin

2. Monitor cyclosporine trough levels closely (every 2-3 days initially)

3. Adjust dose based on levels

4. Return to original dose when clarithromycin is completed

Alternatively, consider using azithromycin instead of clarithromycin — azithromycin does NOT significantly inhibit CYP3A4 and has minimal interaction with cyclosporine. Increasing the dose (A) would worsen toxicity. Adding another antibiotic (C) doesn't address the interaction. Giving both at the same time (D) doesn't prevent the interaction. Switching to IV cyclosporine (E) doesn't avoid the metabolic interaction.

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162 **Answers:** B C

A patient who received high-dose midazolam (a benzodiazepine) and develops respiratory depression (RR 6):

- **Flumazenil (B)**: A competitive benzodiazepine receptor antagonist that reverses the sedative and respiratory depressant effects of midazolam. It binds to the GABA-A receptor benzodiazepine site without activating it, displacing midazolam. Onset is 1-2 minutes IV. Note: In this acute procedural sedation setting (not chronic BDZ use), flumazenil is safe to use.
- **Non-invasive ventilation (C)**: Provides respiratory support (BiPAP/CPAP) while waiting for flumazenil to take effect and as a safety measure. Maintains oxygenation and ventilation without intubation.

Naloxone (A) reverses opioids, not benzodiazepines — ketamine is not an opioid. Epinephrine (D) is for anaphylaxis/cardiac arrest. Atropine (E) is for bradycardia. Dantrolene (F) is for malignant hyperthermia and NMS.

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163 **Answers:** A B

Preventing respiratory depression from sedation:

- **Don't combine sedative medications (A)**: Combining benzodiazepines with other sedatives (ketamine, opioids, propofol) produces synergistic respiratory depression. Each agent alone may be safe, but combinations dramatically increase risk. If combination is necessary, reduce doses of each agent.
- **Start at lower doses with slow titration (B)**: Titrating to effect allows the minimum effective dose to be used, reducing the risk of oversedation. Wait for peak effect of each dose before giving more.

Using higher doses for faster effect (C) increases overdose risk. Avoiding monitoring (D) is dangerous — continuous pulse oximetry, capnography, and vital signs are essential during procedural sedation. General anesthesia (E) is not always necessary and carries its own risks.

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164 **Answer:** B

St. John's Wort (Hypericum perforatum) is a potent inducer of:

1. CYP3A4, CYP2C9, CYP1A2 enzymes

2. P-glycoprotein transporter

This accelerates the metabolism of warfarin (primarily metabolized by CYP2C9), reducing its plasma concentration and anticoagulant effect. The result is subtherapeutic INR and increased risk of thromboembolic events (stroke, DVT, PE). This is treatment FAILURE, not increased effect. St. John's Wort does NOT increase warfarin effect (A). It definitely has significant interactions (C). It does not cause allergy (D) or toxic metabolite formation (E).

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165 **Answers:** A B C D

Medications with treatment failure when combined with St. John's Wort:

- **Warfarin (A)**: CYP2C9 induction increases warfarin metabolism → subtherapeutic INR → thromboembolism risk.
- **Hormonal contraceptives (B)**: CYP3A4 induction increases estrogen/progestin metabolism → contraceptive failure → unintended pregnancy. This is a critical interaction.
- **Cyclosporine (C)**: CYP3A4 and P-glycoprotein induction dramatically reduces cyclosporine levels → organ transplant rejection. Multiple case reports of acute rejection.
- **Digoxin (D)**: P-glycoprotein induction increases digoxin intestinal and renal excretion → subtherapeutic levels → loss of rate control in AF or worsening heart failure.

Acetaminophen (E), vitamin D (F), and calcium supplements (G) are not significantly affected by St. John's Wort enzyme induction.

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166 **Answer:** C

A patient with somatic symptom disorder (multiple somatic complaints, many normal investigations, doctor shopping) needs a structured therapeutic relationship. The most important step after reassurance and counseling is scheduling regular visits NOT contingent on symptoms because:

1. It provides consistent support and reduces the need to develop new symptoms to access care

2. It prevents doctor shopping and unnecessary investigations

3. It establishes a single physician as the primary point of contact

4. It allows gradual building of trust and therapeutic alliance

5. It shifts the focus from symptom-driven visits to wellness-focused care

Ordering more investigations (A) reinforces illness behavior. Immediate specialist referral (B) perpetuates doctor shopping. Prescribing opioids (D) is inappropriate and addictive. Discharge from care (E) is abandonment. Hospital admission (F) is unnecessary.

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167 **Answer:** C

First-line pharmacological treatment for somatic symptom disorder is an SSRI antidepressant because:

1. SSRIs modulate serotonin pathways involved in pain perception and mood regulation

2. They treat comorbid depression and anxiety, which are present in 50-70% of patients

3. They reduce the intensity and frequency of somatic symptoms

4. They have a favorable side effect profile

5. Evidence supports fluoxetine, sertraline, and citalopram for somatic symptom disorders

Benzodiazepines (A) are addictive and not effective long-term. Opioids (B) are contraindicated — risk of dependence and reinforcing illness behavior. Antipsychotics (D) are not first-line. Mood stabilizers (E) and stimulants (F) are not indicated.

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168 **Answers:** A B C F G

History questions to predict cardiovascular risk in an obese patient:

- **Smoking (A)**: The single most important modifiable cardiovascular risk factor. Smoking damages endothelium, promotes atherosclerosis, increases thrombosis, and causes coronary vasospasm.
- **Family history of premature CHD (B)**: First-degree relative with CHD before age 55 (male) or 65 (female) is a major non-modifiable risk factor indicating genetic predisposition to atherosclerosis and dyslipidemia.
- **Sedentary lifestyle (C)**: Physical inactivity is an independent cardiovascular risk factor. It worsens insulin resistance, dyslipidemia, hypertension, and obesity.
- **Dyslipidemia history (F)**: Elevated LDL, low HDL, and elevated triglycerides accelerate atherosclerosis and increase MI/stroke risk.
- **Obstructive sleep apnea (G)**: OSA causes intermittent hypoxia, sympathetic activation, systemic inflammation, and endothelial dysfunction — all increasing cardiovascular risk. Highly prevalent in obese patients.

Favorite color (D) and shoe size (E) have no relevance to cardiovascular risk assessment.

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169 **Answer:** B

Central (visceral/abdominal) obesity is the single best clinical assessment for predicting cardiac risk because:

1. Visceral fat is metabolically active — it secretes pro-inflammatory cytokines (TNF-alpha, IL-6), adipokines, and free fatty acids

2. It directly causes insulin resistance, dyslipidemia, and systemic inflammation

3. Waist circumference >102 cm (men) or >88 cm (women) indicates high cardiovascular risk

4. Waist-to-hip ratio is a stronger predictor of MI than BMI alone

5. Two patients with identical BMI can have vastly different cardiovascular risk based on fat distribution

BMI (A) does not distinguish between visceral and subcutaneous fat or muscle mass. Skin fold thickness (C) measures subcutaneous fat only. Body fat percentage (D) and grip strength (E) are less predictive of cardiovascular events.

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170 **Answer:** A

Bacterial vaginosis (BV) presents with fishy odor, thin grayish-green discharge, no fever, and positive whiff test. It is caused by overgrowth of anaerobic bacteria (Gardnerella vaginalis, Prevotella, Mobiluncus) replacing normal Lactobacillus flora. Metronidazole is first-line because:

1. It is highly effective against anaerobic bacteria

2. It achieves high vaginal tissue concentrations

3. 7-day oral course has cure rates of 80-90%

4. Alternative: metronidazole 0.75% vaginal gel for 5 days

Fluconazole (B) is for candidiasis. Azithromycin (C) is for chlamydia. Doxycycline (D) is for chlamydia/PID. Clindamycin (E) is an alternative for BV but not first-line. Ciprofloxacin (F) is not indicated for BV.

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171 **Answers:** A B

Precautions for metronidazole:

- **Avoid alcohol during treatment and 24 hours after (A)**: Metronidazole inhibits aldehyde dehydrogenase, causing a disulfiram-like reaction when combined with alcohol. Symptoms include severe nausea, vomiting, flushing, headache, abdominal cramps, and tachycardia. Alcohol must be avoided during the entire course and for at least 24-48 hours after completion.
- **Complete the full course of antibiotics (B)**: Incomplete courses lead to treatment failure, recurrence, and potential antibiotic resistance. Even if symptoms resolve early, the full 7-day course must be completed to eradicate the infection.

Taking with grapefruit juice (C) is not specifically required. Avoiding sunlight completely (D) is not necessary — metronidazole does not cause significant photosensitivity. Taking on an empty stomach only (E) is incorrect — metronidazole can be taken with food to reduce GI side effects.

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172 **Answers:** A B C D H

Absolute contraindications to combined OCP:

- **Age ≥35 and smoking ≥15 cigarettes/day (A)**: The combination of estrogen, age, and smoking dramatically increases the risk of arterial thromboembolism (MI, stroke). Estrogen is prothrombotic, smoking damages endothelium, and age increases baseline cardiovascular risk.
- **History of VTE (B)**: Estrogen in COCPs increases hepatic production of clotting factors (II, VII, IX, X) and reduces antithrombin III. Prior VTE indicates thrombotic predisposition — estrogen would cause recurrence.
- **Known ischemic heart disease (C)**: Estrogen promotes thrombosis and vasoconstriction in already diseased coronary arteries, risking MI.
- **Migraine with aura (D)**: Aura indicates cortical spreading depression with transient cerebral ischemia. Estrogen increases stroke risk 2-4 fold in these patients. Migraine WITHOUT aura is not an absolute contraindication.
- **Current breast cancer (H)**: Breast cancer is often hormone-receptor positive. Estrogen and progesterone in COCPs can stimulate tumor growth and progression.

Mild acne (E), family history of diabetes (F), and BMI 26 (G) are not absolute contraindications.

173 **Answer:** B

For ONE missed COCP pill:

1. Take the missed pill as soon as remembered, even if it means taking two pills on the same day

2. Continue the rest of the pack as normal

3. No backup contraception needed (if only 1 pill missed)

4. No emergency contraception needed

The rationale: Missing one pill causes a small dip in hormone levels, but it is insufficient to trigger ovulation. The remaining pills maintain suppression of the hypothalamic-pituitary-ovarian axis. Since she hasn't had sex in the last month, there is no pregnancy risk regardless. Stopping the pack (A) is unnecessary. Emergency contraception (C) is not needed for one missed pill. Taking 3 pills (D) is excessive. Discarding the pack (E) is wasteful and unnecessary.

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174 **Answers:** A B

For TWO missed pills in the FIRST week:

- **Take remaining pills at usual time (A)**: Resume the pack normally. The most recently missed pill should be taken as soon as remembered (even if two pills are taken on the same day). Earlier missed pills are discarded.
- **Use backup contraception (B)**: Two missed pills in the first week means hormone levels dropped enough to potentially allow follicular development and ovulation. Backup contraception (condoms) is needed for 7 consecutive days of active pills to re-establish ovarian suppression.

Additionally, if unprotected sex occurred in the previous 5 days, emergency contraception should be considered. Stopping the pack entirely (C) increases pregnancy risk. Taking all missed pills at once (D) causes unnecessary side effects. No action needed (E) is incorrect — two missed pills require intervention.

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175 **Answer:** B or F

Essential tremor (familial tremor that improves with alcohol) has two first-line medications:

- **Propranolol (B)**: A non-selective beta-blocker that reduces tremor amplitude by blocking peripheral beta-2 adrenergic receptors in skeletal muscle. The extended-release formulation provides consistent tremor control. Reduces tremor by 50-60% in most patients.
- **Primidone (F)**: An anticonvulsant metabolized to phenobarbital and PEMA. The mechanism for tremor reduction is unclear but may involve GABAergic enhancement. Starting at 25 mg qhs minimizes initial side effects (sedation, dizziness). Equally effective as propranolol.

Levodopa/Carbidopa (A) is for Parkinson's disease. Gabapentin (C) is second-line. Topiramate (D) is second-line. Clonazepam (E) is not first-line for essential tremor.

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176 **Answers:** A B C D

Factors that aggravate essential tremor:

- **Caffeine (A)**: A CNS stimulant that increases sympathetic activity and enhances physiologic tremor. Caffeine blocks adenosine receptors, increasing norepinephrine release, which worsens tremor amplitude.
- **Stress (B)**: Psychological stress activates the sympathetic nervous system, increasing catecholamine release. This amplifies the tremor through beta-adrenergic stimulation of skeletal muscle.
- **Fatigue (C)**: Physical and mental fatigue impairs motor control and increases physiologic tremor. Sleep deprivation worsens tremor through similar mechanisms.
- **Cold (D)**: Cold exposure causes peripheral vasoconstriction and increases muscle tension/shivering, which amplifies tremor. Cold also increases sympathetic tone.

Sleep (E) actually improves tremor — essential tremor disappears during sleep. Relaxation (F) reduces tremor. Warm environment (G) does not worsen tremor.

177 **Answers:** A B

Gonorrhea treatment requires dual therapy:

- **Ceftriaxone 500 mg IM (A)**: A third-generation cephalosporin that covers Neisseria gonorrhoeae, including most resistant strains. Single IM dose ensures compliance and achieves bactericidal levels. This is the backbone of gonorrhea treatment.
- **Doxycycline 100 mg PO BID x 7 days (B)**: Added to cover presumptive co-infection with Chlamydia trachomatis, which co-exists in 20-40% of gonorrhea cases. Doxycycline also provides additional coverage against gonorrhea.

Metronidazole (C) is for anaerobes/BV/trichomoniasis. Fluconazole (D) is for candidiasis. Ciprofloxacin (E) is no longer recommended due to widespread gonococcal resistance. Penicillin G (F) is for syphilis, not gonorrhea.

178 **Answer:** C

Gonorrhea patients should abstain from sexual activity for 7 days after treatment initiation because:

1. It takes time for the antibiotic to fully eradicate the organism
2. The patient remains potentially infectious during this period
3. Sexual contact before eradication risks transmitting the infection to partners
4. Both the patient AND all sexual partners must complete treatment and wait 7 days

Immediately (A) risks transmission. Three days (B) may be insufficient. Fourteen days (D) and 30 days (F) are unnecessarily long. Waiting for test of cure (E) is ideal but the standard recommendation is 7 days.

179 **Answers:** A B C D

Advice for gonorrhea patients:

- **All sexual partners should be evaluated and treated (A)**: Partners from the preceding 60 days should be notified, tested, and treated presumptively to prevent reinfection and ongoing transmission. Expedited partner therapy (providing prescriptions for partners) is an option.
- **Consistent condom use (B)**: Latex condoms significantly reduce transmission of gonorrhea and other STIs when used correctly and consistently with every sexual encounter.
- **HIV counseling and testing (C)**: Gonorrhea increases HIV transmission risk 3-5 fold by disrupting mucosal barriers and recruiting HIV target cells. All STI patients should be offered HIV testing.
- **Return at 3 months for retesting (D)**: Reinfection rates are high (10-20% within 3 months). Retesting at 3 months detects reinfection from untreated partners or new exposures.

No follow-up needed (E) is incorrect. Avoiding all sexual contact permanently (F) is impractical. Not notifying partners (G) is unethical and perpetuates transmission.

180 **Answer:** B

The safe DEET concentration in insect repellent for children >2 months is 10-30%. DEET (N,N-diethyl-meta-toluamide) is the most effective insect repellent:

1. 10% DEET provides ~2 hours of protection
2. 30% DEET provides ~5 hours of protection
3. Concentrations above 30% do not significantly increase duration and increase toxicity risk
4. For children >2 months, 10-30% is safe when applied correctly
5. Apply to exposed skin and clothing, avoid hands/eyes/mouth
6. Do NOT use on infants <2 months

5% (A) provides inadequate protection. 50% (C), 75% (D), and 100% (E) are unnecessarily high concentrations with increased toxicity risk. DEET can be used in children (F) — it is safe at appropriate concentrations.

181 **Answers:** A B C

Non-pharmacological mosquito bite prevention:

- **Wear light-colored, long-sleeved shirts and pants (A)**: Light colors are less attractive to mosquitoes (they are attracted to dark colors). Long sleeves and pants reduce exposed skin area available for biting. Loose-fitting clothing is preferred as mosquitoes can bite through tight fabric.
- **Avoid outdoor exposure between dusk and dawn (B)**: Many mosquito species (especially Anopheles — malaria vector, and Culex — West Nile vector) are most active during twilight and nighttime hours. Limiting outdoor exposure during peak biting times reduces risk.
- **Sleep within insecticide-treated bed nets (C)**: Bed nets treated with permethrin or deltamethrin provide a physical and chemical barrier against mosquitoes during sleep. They are the cornerstone of malaria prevention in endemic areas, reducing malaria transmission by 50%.

Dark-colored clothing (D) attracts mosquitoes. Perfumed products (E) attract mosquitoes — floral scents mimic nectar. Open windows at night (F) allows mosquito entry.

182 **Answers:** A B C

Side effects of gentamicin (aminoglycoside):

- **Nephrotoxicity (A)**: Gentamicin accumulates in proximal renal tubular cells, causing acute tubular necrosis. Risk increases with prolonged therapy, high trough levels, dehydration, and concurrent nephrotoxic drugs. Usually reversible if detected early.
- **Ototoxicity — hearing loss (B)**: Gentamicin damages cochlear hair cells (sensorineural hearing loss) and vestibular hair cells (vertigo, ataxia). This is often IRREVERSIBLE. Risk increases with duration, high peak levels, and concurrent ototoxic drugs.
- **Neurotoxicity (C)**: Gentamicin can cause neuromuscular blockade by inhibiting presynaptic acetylcholine release and blocking postsynaptic receptors. This can cause respiratory paralysis, especially with concurrent neuromuscular blockers or in myasthenia gravis patients.

Hepatotoxicity (D) is not a significant aminoglycoside effect. Cardiotoxicity (E) is not typical. Pulmonary fibrosis (F) is not associated with aminoglycosides.

183 **Answers:** A B C

Monitoring for gentamicin:

- **Renal function — BUN/creatinine (A)**: Monitored every 2-3 days during therapy. Rising creatinine indicates nephrotoxicity and requires dose adjustment or discontinuation. Urine output should also be monitored.
- **Serum drug concentration (B)**: Peak levels (drawn 30 minutes after IV infusion) ensure therapeutic efficacy. Trough levels (drawn just before next dose) ensure levels are low enough to prevent toxicity. Target trough <1 mcg/mL for conventional dosing.
- **Hearing test (C)**: Baseline and periodic audiometry detects early ototoxicity before irreversible damage occurs. Patients should also be asked about tinnitus, hearing changes, and dizziness/vertigo, which may indicate vestibular toxicity. Early detection allows drug discontinuation before permanent damage occurs.

Liver function tests (D) are not specifically required for aminoglycosides. Chest X-ray (E) is not indicated. Weekly ECG (F) is not necessary for gentamicin monitoring.

184 **Answer:** B

During a measles outbreak, an unimmunized person should receive two doses of MMR vaccine separated by at least 28 days because:

1. Two doses provide 97% immunity against measles (vs 93% with one dose)

2. The second dose is not a booster but catches the 5-7% who fail to respond to the first dose (primary vaccine failure)

3. During an outbreak, maximum protection is critical

4. The 28-day minimum interval allows the immune response from the first dose to mature before the second dose stimulates a stronger anamnestic response

A single dose (A) provides inadequate protection during an outbreak. Three doses (C) are not required. Measles-only vaccine (D) is not standard — MMR provides broader protection. Immunoglobulin only (E) provides temporary passive immunity but not lasting protection. Adults can receive MMR (F is incorrect).

185 **Answers:** A B C

Contraindications to MMR vaccine:

- **Severe allergic reaction/anaphylaxis to previous dose or component (A)**: History of anaphylaxis to a previous MMR dose or to vaccine components (gelatin, neomycin) is an absolute contraindication. Re-administration could cause fatal anaphylaxis.
- **Pregnancy or attempting to become pregnant (B)**: MMR contains live attenuated viruses that could theoretically infect the fetus. Women should avoid pregnancy for at least 4 weeks after vaccination. Although no cases of congenital rubella syndrome from vaccine virus have been documented, the theoretical risk warrants avoidance.
- **Immunodeficiency (C)**: Severely immunocompromised patients (HIV with CD4 <200, chemotherapy, high-dose steroids, congenital immunodeficiency) cannot mount an adequate immune response and the live vaccine could cause disseminated infection.

Mild URI (D) is NOT a contraindication — vaccination can proceed. History of febrile seizure (E) is not a contraindication. Age over 50 (F) is not a contraindication. Breastfeeding (G) is not a contraindication — MMR is safe during lactation.

186 **Answer:** B

The Dix-Hallpike maneuver is the diagnostic test for BPPV:

1. Patient sits upright on the exam table

2. Head is turned 45 degrees to one side

3. Patient is rapidly laid back with head hanging 20 degrees below the table

4. Observe for nystagmus and reproduction of vertigo

5. Positive test: rotatory/torsional nystagmus with latency (2-5 seconds), duration <1 minute, and fatigability (decreases with repetition)

6. The affected side is the side that reproduces symptoms

The nystagmus is caused by displaced otoconia (calcium carbonate crystals) in the posterior semicircular canal moving with gravity. Epley maneuver (A) is the TREATMENT, not diagnostic test. Semont maneuver (C) is also treatment. Brandt-Daroff exercises (D) are rehabilitation exercises. Romberg test (E) tests proprioception. Rinne test (F) tests hearing.

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187 **Answers:** A B

Non-pharmacological treatment for BPPV:

- **Epley maneuver (A)**: Also called canalith repositioning procedure. A series of head position changes that use gravity to guide displaced otoconia out of the posterior semicircular canal back into the utricle where they are reabsorbed. Success rate of 80-90% after 1-3 treatments. The most widely used and studied treatment for BPPV.
- **Semont maneuver (B)**: Also called the liberatory maneuver. An alternative repositioning technique where the patient is rapidly moved from lying on the affected side to lying on the opposite side. Uses inertia to dislodge otoconia. Success rate of 70-90%.

Bed rest for 2 weeks (C) is counterproductive — prolonged immobility worsens vestibular compensation. Cervical collar (D) and traction (E) are not indicated. Immediate surgery (F) is reserved for rare refractory cases.

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188 **Answers:** A B

Medications for BPPV symptoms:

- **Betahistine 8-16 mg PO daily (A)**: A histamine H1 agonist and H3 antagonist that improves microcirculation in the inner ear and modulates vestibular neuron firing. Reduces vertigo intensity and frequency. Widely used in Europe and Canada for vestibular disorders.
- **Dimenhydrinate 50-100 mg PO q4-6h (B)**: An antihistamine with anticholinergic properties that suppresses vestibular input and reduces nausea/vomiting associated with vertigo. Provides symptomatic relief during acute episodes. Should be used short-term only as chronic use impairs vestibular compensation.

Propranolol (C) is for essential tremor/migraine. Amitriptyline (D) is for chronic pain/depression. Carbamazepine (E) is for seizures/trigeminal neuralgia. Phenytoin (F) is for seizures. Note: Medications are adjunctive — repositioning maneuvers are the primary treatment.

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189 **Answers:** A B

In an immunocompromised child (on steroids for kidney disease), LIVE vaccines are contraindicated:

- **MMR (A)**: Contains live attenuated measles, mumps, and rubella viruses. In immunocompromised patients, the attenuated virus can replicate uncontrollably, causing disseminated infection that can be fatal.
- **Varicella (B)**: Contains live attenuated varicella-zoster virus. In immunosuppressed patients, it can cause disseminated varicella (widespread vesicular rash, pneumonia, hepatitis, encephalitis).

Inactivated influenza (C), pneumococcal conjugate (D), hepatitis B (E), and DTaP (F) are all inactivated/subunit vaccines that are SAFE in immunocompromised patients. They cannot cause infection because they contain no live organisms. They may have reduced immunogenicity but are still recommended.

190 **Answer:** D

The immunocompromised child should receive the inactivated influenza vaccine because:

1. It contains killed virus — cannot cause infection regardless of immune status
2. Influenza is particularly dangerous in immunocompromised patients
3. Even with reduced immune response, partial protection is beneficial
4. Annual vaccination is recommended for all immunocompromised patients

MMR (A) and varicella (B) are live vaccines — contraindicated. Live attenuated influenza (nasal spray) (C) is contraindicated — it contains live virus. BCG (E) is a live vaccine — contraindicated. Oral polio vaccine (F) is a live vaccine — contraindicated. Only inactivated/killed vaccines are safe.

191 **Answers:** A B

Meniere's disease treatment with two different medication classes:

- **Betahistine (A)**: A histamine analogue (H1 agonist, H3 antagonist) that improves inner ear microcirculation and reduces endolymphatic pressure. It decreases the frequency and severity of vertigo attacks. First-line medical therapy for Meniere's disease in many countries.
- **Hydrochlorothiazide (B)**: A thiazide diuretic that reduces endolymphatic fluid volume (endolymphatic hydrops is the pathological hallmark of Meniere's disease). By reducing fluid accumulation in the endolymphatic sac, it decreases the frequency of vertigo episodes, hearing fluctuation, and aural fullness.

Propranolol (C) is for essential tremor/migraine. Amitriptyline (D) is for depression/chronic pain. Carbamazepine (E) is for seizures. Gabapentin (F) is for neuropathic pain.

192 **Answers:** A B C

Non-pharmacological management for Meniere's disease:

- **Vestibular rehabilitation therapy (A)**: A physiotherapy program that promotes central vestibular compensation through exercises that challenge balance and gaze stability. Improves functional balance, reduces fall risk, and helps the brain adapt to asymmetric vestibular input between episodes.
- **Salt restriction (B)**: Limiting sodium to <1500-2000 mg/day reduces fluid retention throughout the body, including the endolymphatic system. High sodium intake increases endolymphatic pressure, triggering Meniere's attacks.
- **Limiting caffeine and alcohol (C)**: Caffeine is a vasoconstrictor that may reduce inner ear blood flow and is a diuretic that can cause fluid shifts. Alcohol directly affects vestibular function and can trigger vertigo episodes. Both should be minimized.

High-sodium diet (D) worsens endolymphatic hydrops. Increased caffeine (E) worsens symptoms. Bed rest for 6 months (F) prevents vestibular compensation and causes deconditioning.

193 **Answer:** B

An elderly patient with dementia and MMSE 20/30 (mild-moderate dementia) whose wife wants help with memory should receive a cholinesterase inhibitor. Donepezil works by:

1. Inhibiting acetylcholinesterase enzyme in the synaptic cleft
2. Preventing breakdown of acetylcholine
3. Increasing acetylcholine availability in the cerebral cortex and hippocampus
4. Acetylcholine is the primary neurotransmitter involved in memory and learning
5. In Alzheimer's disease, cholinergic neurons degenerate — donepezil compensates for this loss

Donepezil provides modest but meaningful improvement in cognition, function, and behavior. It does not cure or halt disease progression. SSRI (A) is for depression. Benzodiazepines (C) worsen cognition in dementia. Typical antipsychotics (D) increase mortality in dementia patients. Mood stabilizers (E) and stimulants (F) are not indicated.

194 **Answer:** B

A dementia patient with aggression and behavioral disturbance needs a low-dose atypical antipsychotic. Olanzapine at a low dose (2.5 mg) is used because:

1. It has sedating properties that reduce agitation and aggression
2. D2 and 5-HT2A receptor blockade reduces psychotic symptoms (hallucinations, delusions) that may drive aggression
3. Low doses minimize side effects in elderly patients
4. Should be used for the shortest duration possible

IMPORTANT: All antipsychotics carry a BLACK BOX WARNING for increased mortality in elderly dementia patients (1.6-1.7x increased risk of death, mainly from cardiovascular events and infections). They should only be used when non-pharmacological measures fail and the behavior poses a safety risk.

Haloperidol 5 mg (A) is too high a dose and typical antipsychotics have more EPS. Diazepam (C) worsens confusion and increases fall risk. Methylphenidate (D) could worsen agitation. Lithium (E) and carbamazepine (F) are not first-line for dementia-related aggression.

195 **Answers:** A B C D H

Non-pharmacological strategies for dementia agitation:

- **Distraction and redirection (A)**: When agitation begins, redirecting the patient's attention to a pleasant activity (music, looking at photos, simple tasks) interrupts the escalation cycle without confrontation.
- **Structured routines (B)**: Consistent daily schedules reduce confusion and anxiety. Predictable mealtimes, activities, and bedtimes provide a sense of security and reduce behavioral disturbances.
- **Providing calm, reassuring responses (C)**: Agitated dementia patients respond to emotional tone more than words. A calm, gentle, non-confrontational approach de-escalates agitation. Arguing or correcting the patient worsens behavior.
- **Establishing consistent sleep and wake times (D)**: Sleep disturbances are common in dementia and contribute to "sundowning" (evening agitation). Regular sleep-wake cycles improve circadian rhythm and reduce nighttime confusion.
- **Exposure to bright light during early part of day (H)**: Bright light therapy (2500+ lux) in the morning helps reset the circadian rhythm, improving sleep quality and reducing sundowning behavior.

Frequent environmental changes (E) increase confusion. Physical restraints (F) increase agitation, injury, and mortality. Loud music (G) can overstimulate and worsen agitation.

196 **Answers:** A B C D

Alcohol withdrawal conditions the patient is at risk for:

- **Alcohol tremulousness (A) — 6-36 hours**: The earliest manifestation. Tremor, anxiety, insomnia, tachycardia, and diaphoresis from sympathetic hyperactivity as GABA inhibition is lost and glutamate excitation is unopposed.
- **Withdrawal seizures (B) — 6-48 hours**: Generalized tonic-clonic seizures from CNS hyperexcitability. Occur in 5-10% of withdrawal patients. Can be the first sign of withdrawal.
- **Alcoholic hallucinosis (C) — 12-48 hours**: Visual, auditory, or tactile hallucinations (classically "bugs crawling on skin") with intact sensorium. Patient knows hallucinations are not real (unlike delirium tremens).
- **Delirium tremens (D) — 48-96 hours**: The most severe and life-threatening form. Characterized by confusion, agitation, hallucinations, autonomic instability (fever, tachycardia, hypertension, diaphoresis), and seizures. Mortality 5-15% without treatment.

Alcohol-induced hepatitis (E) is not an immediate withdrawal complication. Wernicke encephalopathy (F) is from thiamine deficiency, not directly from withdrawal timing.

197 **Answer:** B

Benzodiazepines are the cornerstone of alcohol withdrawal management because:

1. They act on GABA-A receptors, replacing the GABAergic effect of alcohol

2. They reduce CNS hyperexcitability that causes withdrawal symptoms

3. They prevent and treat withdrawal seizures

4. They prevent progression to delirium tremens

5. They reduce autonomic hyperactivity (tachycardia, hypertension, diaphoresis)

6. Long-acting benzodiazepines (diazepam, chlordiazepoxide) provide smooth, self-tapering coverage

Symptom-triggered dosing (using CIWA-Ar scale) is preferred over fixed-dose regimens. Antipsychotics (A) lower seizure threshold — dangerous in withdrawal. SSRIs (C) are not effective for acute withdrawal. Anticonvulsants (D) are adjunctive but not first-line. Opioids (E) and stimulants (F) are not indicated.

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198 **Answers:** A B C G

Additional medications for alcohol withdrawal:

- **Isotonic IV fluid (A)**: Alcohol withdrawal patients are often dehydrated from poor oral intake, vomiting, diaphoresis, and the diuretic effect of alcohol. IV fluids restore intravascular volume and correct electrolyte abnormalities.
- **Thiamine 100 mg IM/IV (B)**: MUST be given BEFORE glucose to prevent precipitating Wernicke encephalopathy. Chronic alcoholics are thiamine-deficient because alcohol impairs thiamine absorption, reduces hepatic storage, and increases urinary excretion. Thiamine is a cofactor for pyruvate dehydrogenase — without it, glucose metabolism produces toxic metabolites that damage the mammillary bodies and thalamus.
- **Glucose (D50W) IV (C)**: Alcoholics are prone to hypoglycemia from depleted glycogen stores, impaired gluconeogenesis, and poor nutritional intake. Hypoglycemia can cause seizures and brain damage. Always give thiamine FIRST.
- **Multivitamins with folate (G)**: Chronic alcoholics have multiple nutritional deficiencies — folate (megaloblastic anemia), B12, magnesium, zinc, and other micronutrients. Multivitamin supplementation addresses these deficiencies.

Naloxone (D) is for opioid overdose, not alcohol withdrawal. Flumazenil (E) reverses benzodiazepines — would worsen withdrawal. Phenytoin (F) is not effective for alcohol withdrawal seizures (benzodiazepines are preferred).

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199 **Answers:** A | B (or A | C)

Hyperkalemia (K=6) in digoxin overdose — IV medications with different mechanisms:

- **Insulin with glucose (A)**: SHIFTS potassium intracellularly by activating Na+/K+ ATPase. This is a redistribution mechanism — moves potassium from extracellular to intracellular space without removing it from the body.
- **Loop diuretics with normal saline (B)**: REMOVES potassium from the body through renal excretion. Furosemide blocks the Na-K-2Cl cotransporter in the loop of Henle, increasing potassium excretion in urine. This is an elimination mechanism.
- **Calcium gluconate IV (C)**: STABILIZES the cardiac membrane — does not lower potassium but protects against arrhythmias. However, CAUTION in digoxin toxicity — calcium can worsen digoxin-induced cardiac toxicity. Some sources recommend avoiding calcium in digoxin toxicity or using it very cautiously.

NOTE: In digoxin toxicity specifically, calcium gluconate is controversial because hypercalcemia potentiates digoxin's toxic effects on the heart ("stone heart"). The definitive treatment is digoxin-specific antibody fragments (Fab).

200 **Answer:** D

The antidote for digoxin toxicity is digoxin-specific antibody fragments (Digibind/DigiFab). Mechanism:

1. Fab fragments bind free digoxin in the bloodstream with high affinity
2. This creates large antibody-digoxin complexes that cannot enter cells
3. Digoxin is pulled out of tissue binding sites down the concentration gradient
4. The complexes are renally excreted
5. Onset of action is 20-30 minutes
6. Indications: life-threatening arrhythmias, hyperkalemia >5.5, hemodynamic instability, serum digoxin >10 ng/mL

Naloxone (A) is for opioid overdose. Flumazenil (B) is for benzodiazepine overdose. N-acetylcysteine (C) is for acetaminophen overdose. Protamine sulfate (E) reverses heparin. Vitamin K (F) reverses warfarin. Activated charcoal (G) may help if ingestion was recent (<1 hour). Deferoxamine (H) is for iron overdose.

201 **Answer:** B

A patient with anorexia nervosa (low BMI, distorted body image) who has failed nutritional therapy and CBT should receive olanzapine. Olanzapine is the best-studied medication for anorexia nervosa because:

1. It promotes weight gain through H1 and 5-HT2C receptor blockade (appetite stimulation)
2. It reduces obsessive thoughts about food, weight, and body image
3. It decreases anxiety and agitation around meals
4. It has the most evidence for weight restoration in anorexia
5. Low starting dose (2.5 mg) minimizes metabolic side effects

Fluoxetine (A) is for bulimia, not anorexia — SSRIs are ineffective in underweight anorexic patients because serotonin synthesis requires adequate tryptophan from nutrition. Amitriptyline (C) has cardiac risks in malnourished patients. Bupropion (D) is CONTRAINDICATED in eating disorders. Risperidone (E) has less evidence than olanzapine. Mirtazapine (F) is an alternative but less studied.

202 **Answers:** A B

Contraindicated medications in anorexia nervosa:

- **Bupropion (A)**: Contraindicated because it lowers the seizure threshold. Anorexic patients are at increased seizure risk due to electrolyte imbalances (hypokalemia, hyponatremia, hypomagnesemia), hypoglycemia, and malnutrition. The combination significantly increases seizure risk. This is an FDA black box warning.
- **TCAs (B)**: Contraindicated due to cardiotoxicity risk. Malnourished anorexic patients often have prolonged QTc, electrolyte abnormalities, and bradycardia. TCAs further prolong QTc and can cause fatal cardiac arrhythmias (torsades de pointes) in this vulnerable population.

SSRIs causing weight gain (C) is incorrect — SSRIs are generally weight-neutral and are not contraindicated for that reason. Olanzapine causing hypoglycemia (D) is not the primary concern. Benzodiazepines causing respiratory depression (E) is not specific to anorexia. Metformin (F) would worsen weight loss and is not typically used.

203 **Answers:** A B

Herpes zoster (shingles) treatment:

- **Valacyclovir (A)**: An antiviral prodrug converted to acyclovir, which inhibits viral DNA polymerase. It must be started within 72 hours of rash onset for maximum benefit. It reduces viral replication, shortens the duration of acute pain, accelerates rash healing, and reduces the risk of post-herpetic neuralgia. Valacyclovir has better bioavailability than acyclovir, allowing less frequent dosing.
- **Ibuprofen (B)**: An NSAID for acute pain management. Herpes zoster causes significant neuropathic and inflammatory pain. NSAIDs address the inflammatory component. Stronger analgesics (acetaminophen, opioids) may be needed for severe pain.

Acyclovir 200 mg 5 times daily (C) is the dose for herpes simplex, not zoster — zoster requires 800 mg 5 times daily. Amoxicillin (D) is an antibiotic — not effective against viruses. Prednisone (E) is controversial and not routinely recommended. Metronidazole (F) is for anaerobic bacteria.

204 **Answers:** A B C F

Preventing herpes zoster spread:

- **Keep the rash covered (A)**: The vesicular fluid contains varicella-zoster virus (VZV). Covering the rash with a non-adherent dressing prevents direct contact transmission to susceptible individuals.
- **Wash hands often (B)**: Hand hygiene prevents fomite transmission. VZV can survive on surfaces briefly, and touching the rash then touching others can transmit the virus.
- **Avoid contact with pregnant women who never had chickenpox (C)**: VZV from shingles can cause primary varicella (chickenpox) in non-immune individuals. Primary varicella in pregnancy causes congenital varicella syndrome (limb hypoplasia, skin scarring, CNS abnormalities, eye defects).
- **Avoid contact with immunocompromised individuals (F)**: Immunocompromised people who contract VZV can develop disseminated varicella with pneumonia, hepatitis, and encephalitis, which can be fatal.

Sharing towels (D) would spread the virus. Exposing the rash to sunlight (E) does not promote healing and increases transmission risk.

205 **Answers:** A B

Post-herpetic neuralgia (PHN) — neuropathic pain persisting >90 days after herpes zoster rash:

- **Pregabalin (A)**: An alpha-2-delta calcium channel ligand that reduces excitatory neurotransmitter release (glutamate, substance P) from hyperexcitable dorsal horn neurons. FDA-approved for PHN. Reduces burning pain, allodynia, and sleep disturbance.
- **Amitriptyline (B)**: A tricyclic antidepressant that inhibits serotonin and norepinephrine reuptake in descending pain inhibitory pathways. Also blocks sodium channels in peripheral nerves, reducing ectopic firing. Effective for neuropathic pain at doses lower than those used for depression.

Ibuprofen (C) is ineffective for neuropathic pain — NSAIDs target inflammatory pain pathways, not neuropathic mechanisms. Metformin (D) is for diabetes. Omeprazole (E) is for GERD. Atenolol (F) is for hypertension.

206 **Answer:** B

The recombinant zoster vaccine (Shingrix) prevents herpes zoster:

1. Recommended for all adults ≥50 years old
2. Two-dose series (0 and 2-6 months apart)
3. Contains recombinant VZV glycoprotein E with AS01B adjuvant
4. >90% efficacy in preventing shingles across all age groups
5. Also reduces post-herpetic neuralgia risk by >85%
6. It is NOT a live vaccine — safe in immunocompromised patients
7. Preferred over the older live attenuated zoster vaccine (Zostavax), which is no longer available

Varicella vaccine (A) is for preventing chickenpox in children, not shingles in elderly. Pneumococcal (C), influenza (D), MMR (E), and hepatitis B (F) vaccines do not prevent herpes zoster.

207 **Answers:** A B C D G

Advice for chronic hepatitis B carrier to prevent spread:

- **Have sexual and household contacts vaccinated (A)**: Hepatitis B vaccine is 95% effective at preventing infection. All close contacts should be tested and vaccinated if non-immune. This is the most important prevention measure.
- **Use barrier protection during sexual intercourse (B)**: HBV is transmitted through sexual contact via infected body fluids. Condoms significantly reduce transmission risk.
- **Do not share toothbrushes or razors (C)**: These items can have microscopic blood contamination. HBV is highly infectious — as few as 10 viral particles can cause infection.
- **Do not donate blood, organs, or sperm (D)**: HBV is transmitted through blood and body fluids. Donation would transmit the virus to recipients.
- **Clean blood spills with detergent or bleach (G)**: HBV survives on environmental surfaces for up to 7 days. Proper decontamination with 1:10 bleach solution kills the virus and prevents environmental transmission.

Avoiding all social contact (E) is unnecessary — HBV is not transmitted through casual contact. Home isolation (F) is not required.

208 **Answer:** B

A chronic hepatitis B carrier should receive the hepatitis A vaccine because:

1. Co-infection with hepatitis A in a patient with chronic hepatitis B causes significantly worse outcomes

2. Superimposed hepatitis A on chronic HBV can cause fulminant hepatic failure with high mortality

3. The already-compromised liver cannot handle the additional inflammatory insult

4. Hepatitis A vaccine is safe, effective (>95% protection), and well-tolerated

5. Two-dose series provides lifelong immunity

Hepatitis B vaccine booster (A) is not useful — the patient is already chronically infected and a carrier. The vaccine cannot clear established chronic infection. Pneumococcal (C) and influenza (D) vaccines are recommended for general health but are not specifically indicated because of HBV status. HPV vaccine (E) is for HPV prevention, not related to HBV.

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209 **Answers:** A B

Bell's Palsy (idiopathic facial nerve paralysis) treatment:

- **Prednisolone (A)**: The cornerstone of treatment. High-dose corticosteroids reduce inflammation and edema of the facial nerve within the narrow fallopian canal. This decompresses the nerve and improves recovery. Must be started within 72 hours of onset. Improves complete recovery rates from ~65% to ~85%.
- **Valacyclovir (B)**: Added because herpes simplex virus (HSV-1) reactivation in the geniculate ganglion is believed to be a common cause of Bell's palsy. Antiviral therapy combined with steroids may provide additional benefit, especially in severe cases (complete paralysis). The evidence for antivirals alone is weak, but combination therapy is standard practice.

Amoxicillin (C) is an antibiotic — Bell's palsy is not bacterial. Gabapentin (D) is for neuropathic pain. Methotrexate (E) is for autoimmune diseases. Carbamazepine (F) is for seizures/trigeminal neuralgia.

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210 **Answers:** A B

Eye care in Bell's palsy is critical because:

- **Eye lubricant with ointment and drops (A)**: The paralyzed orbicularis oculi muscle cannot close the eyelid, causing corneal exposure. Artificial tears (drops during the day, ointment at night) keep the cornea moist and prevent exposure keratopathy, corneal ulceration, and potential blindness.
- **Tape the eyelid during sleep (B)**: During sleep, the eye remains open (lagophthalmos). Taping the eyelid shut with hypoallergenic tape prevents overnight corneal drying and damage. This is essential until facial nerve function recovers.

Wearing an eye patch 24/7 (C) is excessive and impairs vision in the functioning eye. Steroid eye drops (D) are not indicated unless there is corneal inflammation. No eye care (E) risks corneal damage. Warm compresses (F) are not the primary intervention.

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211 **Answer:** C

A diabetic patient already on metformin, gliclazide (sulfonylurea), and empagliflozin (SGLT2 inhibitor) with HbA1c of 12.6% (severely uncontrolled) needs insulin. The rationale:

1. HbA1c >10% with symptoms indicates significant beta-cell failure — oral agents alone are insufficient

2. Triple oral therapy has failed to achieve glycemic control

3. Basal insulin (glargine) provides 24-hour background insulin coverage

4. It suppresses hepatic glucose production overnight, reducing fasting glucose

5. Once-daily injection improves adherence

6. Start at 10 units or 0.1-0.2 units/kg at bedtime and titrate based on fasting glucose

Sitagliptin (A) is a DPP-4 inhibitor — adding a fourth oral agent when HbA1c is 12.6% is insufficient. Pioglitazone (B) causes fluid retention and weight gain — not ideal in an obese patient with CVD. Acarbose (D) has modest HbA1c reduction (~0.5%). Repaglinide (E) is similar to sulfonylureas — redundant with gliclazide. Increasing metformin (F) alone won't achieve adequate control at this HbA1c level.

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212 **Answers:** A B

Most common side effects of insulin:

- **Hypoglycemia (A)**: The most dangerous and common side effect. Insulin lowers blood glucose — if the dose exceeds the body's glucose needs (missed meals, excessive exercise, dose error), blood glucose drops below 4 mmol/L. Symptoms include tremor, sweating, confusion, seizures, and coma. Severe hypoglycemia can be fatal.
- **Weight gain (B)**: Insulin is an anabolic hormone that promotes glucose uptake into cells, lipogenesis, and protein synthesis. It also reduces glycosuria (glucose lost in urine), meaning calories that were previously excreted are now retained. Average weight gain is 2-4 kg in the first year.

Hepatotoxicity (C) is not a significant insulin side effect. Renal failure (D) is not caused by insulin. Pulmonary fibrosis (E) is not associated. Pancreatitis (F) is associated with GLP-1 agonists, not insulin.

---

213 **Answer:** C

When starting basal insulin in a Type 2 diabetic already on a sulfonylurea (gliclazide), the sulfonylurea should be tapered and discontinued because:

1. Both insulin and sulfonylureas increase circulating insulin levels

2. Sulfonylureas stimulate pancreatic beta-cells to secrete insulin

3. Combining both significantly increases hypoglycemia risk

4. The exogenous insulin replaces the need for sulfonylurea-stimulated endogenous insulin

5. Gradual tapering prevents rebound hyperglycemia

Continuing at the same dose (A) increases hypoglycemia risk. Doubling the dose (B) is dangerous. Switching to metformin (D) — metformin should already be continued alongside insulin as it works by a different mechanism (reducing hepatic glucose production). Adding another sulfonylurea (E) is irrational.

214 **Answer:** B

Metformin is the first-line medication for Type 2 Diabetes because:

1. It reduces hepatic glucose production (primary mechanism)

2. It improves insulin sensitivity in peripheral tissues

3. It does NOT cause hypoglycemia (does not stimulate insulin secretion)

4. It is weight-neutral or causes modest weight loss

5. It has cardiovascular benefits (UKPDS trial showed reduced MI risk)

6. It is inexpensive and well-tolerated

7. It has decades of safety data

8. It can be combined with all other diabetes medications

Sulfonylureas (A) cause hypoglycemia and weight gain. DPP-4 inhibitors (C) are less effective. SGLT2 inhibitors (D) are second-line. GLP-1 agonists (E) are injectable and expensive. Insulin (F) is reserved for advanced disease.

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215 **Answers:** A B C

Side effects of metformin:

- **GI disturbances — nausea, vomiting, diarrhea (A)**: The most common side effects, occurring in 20-30% of patients. Caused by metformin's effect on intestinal serotonin and bile acid metabolism. Minimized by starting at low dose, titrating slowly, and taking with food. Extended-release formulation reduces GI side effects.
- **Vitamin B12 deficiency (B)**: Occurs in 10-30% of long-term users. Metformin interferes with B12 absorption in the terminal ileum by altering calcium-dependent membrane action. Can cause megaloblastic anemia and peripheral neuropathy. B12 levels should be monitored periodically.
- **Lactic acidosis (C)**: Rare but potentially fatal (mortality ~50%). Metformin inhibits mitochondrial complex I, and in conditions of tissue hypoxia (renal failure, sepsis, heart failure), lactate accumulates. Contraindicated when GFR <30. Risk is very low (~3-10 per 100,000 patient-years) with appropriate patient selection.

Hypoglycemia (D) is NOT common — metformin does not stimulate insulin secretion. Weight gain (E) is incorrect — metformin is weight-neutral or causes modest weight loss. Pancreatitis (F) is not associated with metformin.

216 **Answer:** B

A child with croup (barking cough, stridor, respiratory distress) should receive dexamethasone because:

1. Croup is caused by parainfluenza virus causing subglottic airway inflammation and edema

2. Dexamethasone reduces subglottic inflammation and edema

3. Single dose of 0.6 mg/kg (max 10 mg) is effective — for this 13.3 kg child: 0.6 × 13.3 = ~8 mg

4. Onset within 2-3 hours, lasting 24-72 hours due to long half-life

5. Oral route is preferred (less distressing than IM)

6. Reduces need for hospitalization, intubation, and return visits

7. Single dose is as effective as multiple doses

Albuterol (A) is for asthma/bronchospasm — not effective in croup (upper airway obstruction). Amoxicillin (C) is for bacterial infections — croup is viral. Ipratropium (D) is for COPD/asthma. Prednisone 5-day course (E) is unnecessary — single-dose dexamethasone is sufficient. Azithromycin (F) is an antibiotic.

---

217 **Answer:** B

A croup patient with stridor at rest (moderate-severe croup) needs nebulized racemic epinephrine in addition to dexamethasone:

1. Racemic epinephrine causes mucosal vasoconstriction via alpha-1 adrenergic stimulation

2. This rapidly reduces subglottic edema and improves airway patency

3. Onset within 10-30 minutes

4. Duration is short (1-2 hours) — "rebound" worsening can occur

5. Patient must be observed for 2-4 hours after administration for rebound stridor

6. Does not replace dexamethasone — used as adjunctive rescue therapy

IV antibiotics (A) are not indicated — croup is viral. Oral prednisolone (C) is an alternative steroid but dexamethasone is already given. Inhaled budesonide (D) is less effective than systemic dexamethasone. IM ceftriaxone (E) is for bacterial infections. Nebulized albuterol (F) targets lower airway bronchospasm, not upper airway edema.

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218 **Answer:** C

A pregnant woman at 34 weeks with confirmed DVT should receive LMWH because:

1. LMWH (enoxaparin) does NOT cross the placenta — safe for the fetus

2. Predictable pharmacokinetics allow weight-based dosing without routine monitoring

3. Lower risk of heparin-induced thrombocytopenia (HIT) compared to unfractionated heparin

4. Lower risk of osteoporosis with long-term use compared to UFH

5. Can be self-administered subcutaneously

6. Treatment dose: 1 mg/kg every 12 hours

Warfarin (A) crosses the placenta — CONTRAINDICATED (see Q219). Rivaroxaban (B) and apixaban (F) are DOACs — contraindicated in pregnancy (cross placenta, insufficient safety data). Aspirin (D) is antiplatelet, not anticoagulant — insufficient for DVT. Clopidogrel (E) is antiplatelet — not for DVT treatment.

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219 **Answer:** B

Warfarin is contraindicated in pregnancy because:

1. It freely crosses the placenta (small molecule, highly protein-bound but free fraction crosses)

2. **First trimester**: Causes warfarin embryopathy — nasal hypoplasia, stippled epiphyses (chondrodysplasia punctata), limb hypoplasia, CNS abnormalities. Risk is highest during weeks 6-12.

3. **Second/third trimester**: Causes fetal CNS hemorrhage, microcephaly, optic atrophy, and mental retardation

4. **Any trimester**: Risk of fetal and placental hemorrhage

5. The teratogenic mechanism involves inhibition of vitamin K-dependent proteins essential for bone and cartilage development (osteocalcin, matrix Gla protein)

Heparin causing osteoporosis (A) is a side effect but not the reason it's contraindicated — heparin is actually the PREFERRED anticoagulant in pregnancy. Enoxaparin causing bleeding (C) is a risk but it's still the treatment of choice. Aspirin causing Reye syndrome (D) is in children, not fetuses. Fondaparinux (E) has limited pregnancy data but does not cross the placenta significantly.

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220 **Answer:** B

A woman with endometriosis whose NSAIDs have failed should receive combined OCP as the next step:

1. Estrogen-progestin OCPs suppress ovulation and reduce estrogen-driven endometrial tissue growth

2. They create a pseudo-pregnancy state with decidualization and atrophy of endometrial implants

3. Continuous (non-cyclic) use is preferred — skipping the placebo week eliminates withdrawal bleeding and associated pain

4. They reduce dysmenorrhea, pelvic pain, and dyspareunia

5. Well-tolerated, affordable, and provide contraception simultaneously

6. Yasmin contains drospirenone (anti-androgenic progestin) which may have additional benefits

Acetaminophen (A) is a weaker analgesic than NSAIDs — unlikely to help if NSAIDs failed. Morphine (C) is inappropriate for chronic endometriosis pain. Gabapentin (D) is for neuropathic pain. Methotrexate (E) is not used for endometriosis.

---

221 **Answer:** B

When OCPs fail for endometriosis after 3 months, the next step is a GnRH agonist:

1. Leuprolide initially stimulates then downregulates GnRH receptors in the pituitary

2. This suppresses FSH and LH, creating a hypoestrogenic state ("medical menopause")

3. Without estrogen, endometrial implants atrophy and become inactive

4. Significantly reduces pain in 80-90% of patients

5. Limited to 6 months due to bone density loss from estrogen deprivation

6. "Add-back therapy" with low-dose norethindrone can be given to reduce bone loss and vasomotor symptoms

Increasing OCP dose (A) is unlikely to help if standard dose failed. Hysterectomy (C) is definitive but reserved for severe refractory cases. Opioids (D) are not appropriate for chronic management. Acetaminophen (E) already failed. Aromatase inhibitors (F) are third-line.

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222 **Answer:** B

A painful, fluctuant swelling (carbuncle/abscess) requires incision and drainage (I&D) as the primary treatment:

1. Abscesses are walled-off collections of pus — antibiotics alone cannot penetrate the abscess cavity

2. I&D evacuates the purulent material, removes the nidus of infection, and allows healing

3. The cavity is irrigated and packed with gauze to prevent premature closure

4. I&D alone is curative for most uncomplicated abscesses <5 cm

5. Antibiotics are added only if there is surrounding cellulitis >2 cm, systemic signs, immunocompromise, or failed I&D

Oral antibiotics only (A) are insufficient — the abscess wall prevents antibiotic penetration. Warm compresses only (C) may help small abscesses drain spontaneously but are insufficient for established abscesses. Topical mupirocin (D) cannot penetrate deep infections. IV antibiotics only (E) still cannot penetrate the abscess. Observation (F) allows progression.

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223 **Answer:** B

Staphylococcus aureus is the most common organism causing skin abscesses:

1. S. aureus colonizes the skin and nares of 30% of the population

2. It produces virulence factors: coagulase (forms fibrin wall around abscess), protein A (evades immune system), Panton-Valentine leukocidin (PVL — destroys neutrophils)

3. Community-acquired MRSA (CA-MRSA) is increasingly common, especially the USA300 strain

4. CA-MRSA produces PVL, which is associated with recurrent skin abscesses

5. S. aureus accounts for >75% of all skin and soft tissue abscesses

Streptococcus pyogenes (A) more commonly causes cellulitis and erysipelas than abscesses. Pseudomonas (C) is associated with hot tub folliculitis and immunocompromised patients. E. coli (D) causes urinary and intra-abdominal infections. Klebsiella (E) is a hospital-acquired pathogen.

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224 **Answer:** B

After I&D of an abscess with 7 cm surrounding erythema (cellulitis >2 cm), antibiotics are indicated. For suspected CA-MRSA:

1. TMP-SMX (trimethoprim-sulfamethoxazole) has excellent activity against CA-MRSA

2. It inhibits sequential steps in folate synthesis — trimethoprim inhibits dihydrofolate reductase, sulfamethoxazole inhibits dihydropteroate synthase

3. Oral formulation allows outpatient treatment

4. Well-tolerated and inexpensive

5. Alternative: doxycycline 100 mg BID or clindamycin 300 mg TID

Amoxicillin (A) does not cover MRSA. Cephalexin (C) does not cover MRSA. Ciprofloxacin (D) has variable MRSA coverage. Azithromycin (E) has poor MRSA activity. Penicillin V (F) does not cover MRSA.

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225 **Answers:** A B

Renal colic (kidney stone) pain management:

- **Ketorolac (A)**: An IV NSAID that is first-line for renal colic. It inhibits prostaglandin synthesis, reducing ureteral smooth muscle spasm, inflammation, and pain. It also reduces renal blood flow and ureteral edema. Studies show NSAIDs are as effective as opioids for renal colic with fewer side effects.
- **Morphine (B)**: An opioid analgesic for severe pain not controlled by NSAIDs alone. Renal colic is one of the most painful conditions — opioids provide rapid relief for breakthrough pain. IV morphine has onset within 5-10 minutes. Used as adjunctive therapy when NSAIDs alone are insufficient.

Acetaminophen (C) is less effective than NSAIDs for renal colic. Gabapentin (D) is for chronic neuropathic pain, not acute colic. Amitriptyline (E) is for chronic pain. Prednisone (F) is not indicated for renal colic.

226 **Answer:** A

Tamsulosin is used as medical expulsive therapy (MET) to facilitate kidney stone passage:

1. It is an alpha-1A adrenergic receptor antagonist
2. Alpha-1 receptors are concentrated in the distal ureter
3. Blocking these receptors relaxes ureteral smooth muscle
4. This reduces ureteral spasm and increases the lumen diameter
5. Facilitates passage of stones 5-10 mm in size
6. Reduces time to stone passage and need for analgesics
7. Most effective for distal ureteral stones

Furosemide (B) increases urine output but does not relax the ureter. Hydrochlorothiazide (C) is used for prevention of calcium stones (reduces urinary calcium), not acute passage. Spironolactone (D) is not indicated. Atenolol (E) has no role in stone passage.

227 **Answers:** A B C

Non-pharmacological advice to prevent kidney stone recurrence:

- **Sufficient fluid intake to produce >2L urine/day (A)**: The single most important prevention measure. Dilute urine reduces the concentration of stone-forming salts (calcium, oxalate, uric acid, phosphate) below their saturation point, preventing crystal formation. Aim for 2.5-3L of fluid intake daily.
- **Limit dietary sodium below 2300 mg/day (B)**: High sodium intake increases urinary calcium excretion (natriuresis causes calciuresis). Reducing sodium to <2300 mg/day significantly reduces calcium stone recurrence.
- **Increase fruit and vegetable intake (C)**: Fruits and vegetables provide citrate (a natural stone inhibitor that binds calcium), potassium (reduces urinary calcium), and phytate (inhibits crystal formation). They also alkalinize urine, reducing uric acid stone risk.

Decreasing water intake (D) concentrates urine and promotes stone formation. High-sodium diet (E) increases calcium excretion. Avoiding all dairy (F) is incorrect — moderate calcium intake actually REDUCES stone risk by binding oxalate in the gut.

228 **Answers:** A B C D E

Investigations for a patient on lithium, ramipril, acetaminophen, and levothyroxine with no recent monitoring:

- **Serum lithium levels (A)**: Lithium has a narrow therapeutic index (0.6-1.2 mEq/L). Levels must be monitored regularly — toxicity occurs at >1.5 mEq/L and can be fatal.
- **TSH (B)**: Both lithium and levothyroxine affect thyroid function. Lithium causes hypothyroidism in 20-30% of patients. TSH monitors both lithium's thyroid effects and levothyroxine dosing adequacy.
- **Renal function — BUN, creatinine (C)**: Lithium causes nephrogenic diabetes insipidus and chronic tubulointerstitial nephropathy. Ramipril also affects renal function. Both require renal monitoring.
- **CBC with differential (D)**: Baseline monitoring for all chronic medications. Lithium can cause leukocytosis.
- **Serum electrolytes (E)**: Lithium affects sodium and potassium handling. Ramipril causes hyperkalemia. Electrolyte imbalances affect lithium clearance — hyponatremia increases lithium levels.

Chest X-ray (F) and pulmonary function tests (G) are not specifically indicated for these medications.

229 **Answer:** B

The patient presents with confusion, ataxia, and nausea — classic signs of lithium toxicity. The single most important investigation is serum lithium level because:

1. These symptoms (confusion, ataxia, nausea, tremor) are hallmarks of lithium toxicity

2. Lithium has a narrow therapeutic index — small changes in levels cause toxicity

3. The level determines severity and guides management:

- Mild toxicity (1.5-2.5 mEq/L): GI symptoms, tremor
- Moderate toxicity (2.5-3.5 mEq/L): confusion, ataxia, dysarthria
- Severe toxicity (>3.5 mEq/L): seizures, coma, death

4. Treatment decisions (IV fluids vs hemodialysis) depend on the level

CT head (A) may be considered but lithium toxicity is more likely given the medication history. Blood glucose (C) should be checked but is less specific. LFTs (D), chest X-ray (E), and lumbar puncture (F) are not the priority.

230 **Answers:** A B C D

Side effects of lithium:

- **Hypothyroidism (A)**: Lithium inhibits thyroid hormone synthesis and release by blocking iodine uptake, iodotyrosine coupling, and thyroid hormone secretion. Occurs in 20-30% of patients. Requires regular TSH monitoring and levothyroxine replacement if needed.
- **Nephrogenic diabetes insipidus (B)**: Lithium accumulates in collecting duct cells and inhibits aquaporin-2 expression, making the kidney resistant to ADH. Causes polyuria (up to 3-6L/day) and polydipsia. Occurs in 40% of long-term users.
- **Tremors — fine (C)**: A fine postural tremor of the hands is the most common neurological side effect. Caused by lithium's effect on neuromuscular transmission. Worsens with higher levels. Can be treated with propranolol.
- **Cardiac arrhythmia (D)**: Lithium can cause T-wave flattening/inversion, sinus node dysfunction, and rarely first-degree AV block. ECG monitoring is recommended.

Improved renal function (E) is incorrect — lithium damages kidneys. Weight loss (F) is incorrect — lithium causes weight GAIN. Hair growth (G) is incorrect — lithium can cause hair loss.

231 **Answers:** A B C

Monitoring required for lithium:

- **Lithium concentration (A)**: Checked 5-7 days after initiation or dose change (steady state), then every 3-6 months when stable. Therapeutic range 0.6-1.2 mEq/L. Drawn 12 hours post-dose (trough level).
- **Thyroid and renal function every 6 months (B)**: TSH monitors for lithium-induced hypothyroidism. Creatinine and eGFR monitor for lithium nephrotoxicity. Both are checked at baseline, then every 6 months.
- **Serum calcium at baseline and every 12 months (C)**: Lithium causes hyperparathyroidism in 10-25% of patients by altering the calcium-sensing receptor set point. This leads to hypercalcemia, which can cause kidney stones, bone loss, and neuropsychiatric symptoms.

Weekly chest X-ray (D) is unnecessary. Daily blood glucose (E) is not required. Monthly liver biopsy (F) is not indicated — lithium is not hepatotoxic.

232 **Answers:** A B

Additional medications for osteoarthritis beyond acetaminophen:

- **Topical NSAIDs — Diclofenac gel (A)**: First-line adjunct to acetaminophen. Applied directly over the affected joint, achieving local anti-inflammatory concentrations with minimal systemic absorption. Reduces GI and cardiovascular risks compared to oral NSAIDs. Particularly effective for hand and knee OA.
- **Oral NSAIDs — Naproxen (B)**: Used when topical NSAIDs and acetaminophen are insufficient. Naproxen inhibits COX-1 and COX-2, reducing prostaglandin-mediated inflammation and pain. It has the most favorable cardiovascular safety profile among NSAIDs. Should be used at the lowest effective dose for the shortest duration.

Opioids as first line (C) are not appropriate — they are reserved for severe refractory pain due to addiction risk. Methotrexate (D) is for inflammatory arthritis (RA), not OA. Hydroxychloroquine (E) is for RA/lupus. Colchicine (F) is for gout.

233 **Answer:** A

A PPI (omeprazole) should be co-prescribed with oral NSAIDs to prevent GI side effects:

1. NSAIDs inhibit COX-1, reducing prostaglandin E2 and prostacyclin production in the gastric mucosa

2. These prostaglandins normally stimulate mucus and bicarbonate secretion and maintain mucosal blood flow

3. Without this protection, gastric acid damages the mucosa, causing gastritis, ulcers, and GI bleeding

4. PPIs suppress acid production by >90%, compensating for the lost mucosal protection

5. Indicated especially in patients >65, history of GI bleeding, concurrent anticoagulants/steroids, or high-dose/prolonged NSAID use

Sucralfate (B) provides a physical barrier but is less effective than PPIs. Misoprostol (C) is a prostaglandin analogue that is effective but causes diarrhea and cramping — poorly tolerated. Famotidine (D) is less effective than PPIs. Bismuth (E) is not standard gastroprotection.

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234 **Answer:** B

A 6-week postpartum mother with depressive symptoms (insomnia, depressive mood) but no harm ideation has postpartum depression. Sertraline is the preferred antidepressant because:

1. It is an SSRI with strong evidence for postpartum depression

2. It has the lowest transfer into breast milk among SSRIs (see Q235)

3. It is safe for the breastfeeding infant

4. Effective for depression and comorbid anxiety

5. Well-tolerated with a favorable side effect profile

6. Starting dose 50 mg with titration to 100-200 mg as needed

Fluoxetine (A) has a long half-life and active metabolites — higher breast milk transfer. Amitriptyline (C) is a TCA with anticholinergic side effects and cardiac risk. Bupropion (D) has limited lactation data. Venlafaxine (E) has higher breast milk transfer. Paroxetine (F) has low breast milk transfer but is associated with withdrawal symptoms and has more drug interactions.

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235 **Answer:** B

Sertraline is preferred during lactation because:

1. It has the lowest relative infant dose (RID) among SSRIs — approximately 0.5-2% of the maternal weight-adjusted dose

2. Infant serum levels are usually undetectable

3. Its primary metabolite (desmethylsertraline) is less active and also has low breast milk transfer

4. Multiple studies show no adverse effects on breastfed infants

5. No reported developmental or behavioral effects in exposed infants

6. The American Academy of Pediatrics considers it compatible with breastfeeding

Longest half-life (A) is incorrect — fluoxetine has the longest half-life. Cheapest option (C) is not the primary reason. Most weight loss (D) is not the reason. No side effects (E) is incorrect — all SSRIs have side effects. Strongest antidepressant (F) is not accurate — efficacy is similar among SSRIs.

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236 **Answer:** A

A breastfeeding mother with mastitis (fever, chills — systemic symptoms) needs antibiotics. Dicloxacillin is the first-line choice because:

1. Mastitis is most commonly caused by Staphylococcus aureus from the infant's oral flora entering through cracked nipples

2. Dicloxacillin is a penicillinase-resistant penicillin with excellent anti-staphylococcal activity

3. It is safe during breastfeeding — minimal transfer into breast milk

4. Breastfeeding should CONTINUE during treatment — it helps drain the infected breast and does not harm the infant

5. Alternative: cephalexin 500 mg QID if dicloxacillin unavailable

Metronidazole (B) covers anaerobes, not staphylococci. Ciprofloxacin (C) is avoided in breastfeeding — accumulates in breast milk and affects infant cartilage. Azithromycin (D) is not first-line for mastitis. Doxycycline (E) is avoided in breastfeeding — teeth discoloration risk. TMP-SMX (F) is avoided in breastfeeding if infant is <2 months (kernicterus risk).

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237 **Answers:** A B C D

Non-pharmacological interventions to increase breast milk production:

- **Optimizing breastfeeding technique — position and latch-on (A)**: Poor latch causes inadequate breast stimulation and incomplete milk removal, both of which reduce milk production. Proper positioning (cradle, cross-cradle, football hold) and deep latch ensure effective milk transfer and stimulate prolactin release.
- **Increasing breastfeeding frequency (B)**: Breast milk production works on supply-and-demand. More frequent feeding (8-12 times/day) increases prolactin release and stimulates more milk production. "The more you feed, the more you make."
- **Ensuring complete breast emptying (C)**: Residual milk in the breast contains feedback inhibitor of lactation (FIL), a whey protein that locally suppresses milk production. Complete emptying removes FIL and signals the breast to produce more milk.
- **Reducing stress and getting adequate sleep (D)**: Stress and fatigue increase cortisol and catecholamines, which inhibit oxytocin release (the let-down reflex). Adequate rest and relaxation promote oxytocin release and milk flow.

Supplementing with formula immediately (E) reduces breast stimulation and milk production. Restricting fluid intake (F) can reduce milk supply. Avoiding nighttime breastfeeding (G) reduces prolactin stimulation — prolactin levels are highest at night.

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238 **Answers:** A B

A 5-day-old neonate with jaundice but bilirubin below the treatment threshold:

- **Ensure adequate feeds and hydration (A)**: Adequate feeding promotes intestinal motility and stool passage, which eliminates bilirubin through the GI tract. Dehydration concentrates bilirubin and reduces hepatic blood flow, worsening jaundice. Feeding every 2-3 hours is recommended.
- **Continue breastfeeding (B)**: Breastfeeding jaundice (not breast milk jaundice) occurs from inadequate intake in the first week. The solution is to increase breastfeeding frequency, not stop it. Breast milk provides nutrition, hydration, and promotes stooling. Stopping breastfeeding is counterproductive and undermines breastfeeding establishment.

Phototherapy (C) is not indicated — bilirubin is below the treatment threshold. Exchange transfusion (D) is for severe hyperbilirubinemia. Stopping breastfeeding (E) is incorrect. IV antibiotics (F) are not indicated without signs of infection.

239 **Answer:** B

Petroleum jelly (Vaseline) (B): For an infant with an anal fissure after switching to formula, petroleum jelly is the appropriate topical treatment because:

1. It provides a protective barrier over the fissure, reducing friction during bowel movements

2. It lubricates the anal canal, making stool passage less painful

3. It is safe, non-toxic, and gentle for neonatal skin

4. It promotes moist wound healing

5. No prescription needed

Hydrocortisone (A) is a steroid — not first-line for infant anal fissures and can cause skin thinning. Nitroglycerin 0.2% ointment (C) is used for chronic adult anal fissures — not appropriate for neonates. Mupirocin (D) is an antibiotic for bacterial infections. Clotrimazole (E) is an antifungal.

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240 **Answer:** B

For a 9-year-old female (weight 90 kg, height 135 cm) with parental concern about weight, BMI is the clinical tool used to tailor treatment because:

1. In children, BMI is calculated the same way as adults (weight in kg / height in m^2)

2. However, it is plotted on age- and sex-specific growth charts (BMI-for-age percentiles)

3. Overweight: 85th-94th percentile; Obese: ≥95th percentile

4. This child's BMI = $90/(1.35)^2 = 49.4$ kg/m^2 — severely obese

5. BMI percentile guides the intensity of intervention (lifestyle vs pharmacological vs surgical)

6. Serial BMI tracking monitors treatment response

Waist circumference (A) is useful in adults but not the primary tool in pediatric obesity assessment. Skin fold thickness (C) requires specialized calipers and training. Bioelectrical impedance (D) is not standard clinical practice. DEXA scan (E) is for bone density. Serum leptin (F) is a research tool, not clinical practice.

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241 **Answers:** A B C D E

Dietary modifications for the obese child:

- **Eat moderate breakfast daily, avoid high-sugar foods (A)**: Breakfast prevents compensatory overeating later in the day. Avoiding high-sugar foods reduces insulin spikes and empty calorie consumption.
- **Decrease fast food to 1 or fewer meals per week (B)**: Fast food is calorie-dense, nutrient-poor, high in saturated fat, sodium, and sugar. Reducing frequency significantly decreases caloric intake.
- **Avoid sugar-sweetened beverages (C)**: Sodas, juices, and sports drinks are the single largest source of added sugar in children's diets. Liquid calories do not trigger satiety signals, leading to overconsumption. Replacing with water can reduce intake by 200-400 calories/day.
- **Eat 5 servings of fruits and vegetables daily (D)**: Fruits and vegetables are nutrient-dense, high in fiber, and low in calories. Fiber promotes satiety and reduces overall caloric intake.
- **Decrease portion sizes (E)**: Children are served increasingly large portions. Using smaller plates, measuring portions, and avoiding second helpings reduces caloric intake without requiring specific food restrictions.

Skipping meals (F) causes compensatory overeating and metabolic slowing. Diet soda freely (G) may perpetuate sweet taste preferences. Eating only one meal per day (H) is dangerous for a growing child.

242 **Answer:** C

A child with a febrile convulsion that has already stopped (lasted 2-3 minutes, self-resolved) with temperature 38.5°C and URI symptoms:

1. The seizure has already stopped — no anticonvulsant is needed
2. The priority is treating the fever and the underlying cause (viral URI)
3. Acetaminophen 15 mg/kg reduces fever and provides comfort
4. Simple febrile seizures (<15 minutes, generalized, single episode in 24 hours) do not require anticonvulsant treatment
5. The child should be observed and parents reassured

Diazepam (A) is for active seizures — the seizure has stopped. Phenytoin (B), phenobarbital (D), levetiracetam (E), and valproic acid (F) are anticonvulsants for status epilepticus or epilepsy — not indicated for a simple febrile seizure that has resolved.

243 **Answer:** D

There is no recommended daily medication to prevent future febrile seizures because:

1. Simple febrile seizures are benign — they do not cause brain damage, epilepsy, or developmental delay

2. The risk of recurrence is 30-35%, but recurrences are also benign

3. Daily anticonvulsants (phenobarbital, valproic acid) have significant side effects that outweigh the minimal risk of febrile seizures

4. Phenobarbital causes cognitive impairment and behavioral problems in children

5. Valproic acid causes hepatotoxicity and is teratogenic

6. The risk-benefit ratio does not favor prophylactic treatment

7. Parent education and reassurance are the mainstay of management

Phenobarbital (A), valproic acid (B), carbamazepine (C), and levetiracetam (E) are all inappropriate for febrile seizure prophylaxis.

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244 **Answer:** B

If the febrile seizure child presented in status epilepticus (seizure lasting >5 minutes or recurrent seizures without recovery):

1. Benzodiazepines are first-line for status epilepticus

2. **Diazepam 0.1-0.2 mg/kg IV** — rapid onset (1-3 minutes), lipophilic so crosses BBB quickly

3. **Lorazepam 0.05-0.1 mg/kg IV** — longer duration of action (12-24 hours vs 15-20 minutes for diazepam), preferred by many guidelines

4. If no IV access: rectal diazepam 0.5 mg/kg, buccal/intranasal midazolam 0.2 mg/kg

5. Benzodiazepines enhance GABA-A receptor activity, rapidly terminating seizure activity

Acetaminophen (A) treats fever but does not stop seizures. Phenytoin 5 mg/kg (C) is too low a dose and is second-line. Carbamazepine (D), gabapentin (E), and topiramate (F) are oral maintenance medications — not for acute status epilepticus.

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245 **Answers:** A B

Non-pharmacological advice for hidradenitis suppurativa (HS):

- **Smoking cessation (A)**: Smoking is the strongest modifiable risk factor for HS. Nicotine promotes follicular plugging, alters immune function, and worsens inflammation. Smoking cessation significantly reduces disease severity and flare frequency.
- **Weight reduction (B)**: Obesity worsens HS through mechanical friction in skin folds, hormonal changes (hyperandrogenism), and chronic systemic inflammation. Weight loss reduces friction, improves hormonal balance, and decreases inflammatory markers.

Additional non-pharmacological measures include wearing loose-fitting clothing, using gentle cleansers, and avoiding shaving affected areas. Heat packs (C) may provide temporary comfort but are not a primary recommendation. Increasing sugar (D) worsens inflammation. Tight-fitting clothes (E) increase friction and worsen HS. Avoiding all exercise (F) is counterproductive — exercise aids weight loss.

246 **Answer:** B

For mild hidradenitis suppurativa, topical clindamycin is first-line because:

1. It provides local antibiotic and anti-inflammatory effects

2. Clindamycin inhibits bacterial protein synthesis at the 50S ribosomal subunit

3. It reduces the bacterial biofilm in hair follicles

4. It has anti-inflammatory properties independent of its antibiotic effect

5. Applied twice daily to affected areas

6. Minimal systemic absorption reduces side effect risk

Mupirocin (A) is for impetigo/MRSA decolonization — not standard for HS. Hydrocortisone (C) is a weak steroid that does not address the underlying follicular occlusion and infection. Benzoyl peroxide (D) may be adjunctive but is not first-line. Tretinoin (E) is for acne vulgaris.

---

247 **Answer:** B

For moderate HS with multiple lesions, oral antibiotics are needed. Doxycycline is preferred because:

1. It has both antibiotic and anti-inflammatory properties

2. It inhibits matrix metalloproteinases (MMPs) that contribute to tissue destruction

3. It reduces neutrophil chemotaxis and pro-inflammatory cytokines

4. Broad-spectrum coverage against HS-associated bacteria

5. Prolonged courses (10-12 weeks or longer) are needed because HS is a chronic inflammatory condition, not a simple infection

6. Well-tolerated with once or twice daily dosing

Amoxicillin (A) lacks the anti-inflammatory properties of tetracyclines. Ciprofloxacin (C) is not first-line for HS. Metronidazole (D) is for anaerobic infections. Fluconazole (E) is an antifungal — HS is not fungal.

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248 **Answer:** B

For papulopustular rosacea, topical metronidazole is first-line because:

1. It has anti-inflammatory and immunomodulatory properties

2. It reduces reactive oxygen species (free radicals) that drive rosacea inflammation

3. It inhibits neutrophil-generated oxidative tissue damage

4. Applied twice daily for 8-12 weeks initially

5. Well-tolerated with minimal side effects

6. Alternative topical options: azelaic acid 15% gel, ivermectin 1% cream

Tretinoin (A) is for acne vulgaris and can worsen rosacea by causing irritation. Hydrocortisone (C) is CONTRAINDICATED in rosacea — steroids cause steroid rosacea and rebound flares. Benzoyl peroxide 10% (D) is too irritating for rosacea-prone skin. Clindamycin (E) is for acne, not first-line for rosacea.

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249 **Answer:** B

When topical treatment fails for rosacea, oral doxycycline is the next step:

1. Sub-antimicrobial dose doxycycline (40 mg modified-release daily) is preferred — anti-inflammatory without antibiotic resistance concerns

2. Standard dose (100 mg BID) is also effective

3. Doxycycline inhibits MMPs, reduces inflammatory cytokines, and suppresses neutrophil function

4. Duration is 4-12 weeks, then reassess

5. Maintenance with topical therapy after oral course

Isotretinoin (A) is reserved for severe, refractory rosacea — not second-line. Prednisone (C) is contraindicated — causes rebound flares. Methotrexate (D) is not indicated for rosacea. Fluconazole (E) is an antifungal.

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250 **Answers:** A B C D

Non-pharmacological advice for rosacea:

- **Avoid triggers of flushing — keep a diary (A)**: Common triggers include sun exposure, heat, spicy foods, alcohol, hot beverages, stress, and certain skincare products. A diary helps identify individual triggers for avoidance.
- **Use gentle skincare (B)**: Rosacea skin has impaired barrier function. Gentle, fragrance-free, non-comedogenic cleansers and moisturizers protect the barrier. Avoid harsh scrubs, astringents, and alcohol-based products.
- **Use adequate sun protection (C)**: UV radiation is the most common rosacea trigger. Daily broad-spectrum SPF 30+ sunscreen (mineral/physical preferred — zinc oxide, titanium dioxide) is essential. Wear hats and seek shade.
- **Avoid spicy foods and alcohol (D)**: Capsaicin in spicy foods and alcohol cause vasodilation and flushing, directly triggering rosacea flares. Red wine is the most commonly reported alcoholic trigger.

Scrubbing face vigorously (E) damages the impaired skin barrier and worsens inflammation. Hot water (F) triggers flushing — lukewarm water is recommended. Heavy makeup (G) can clog pores and irritate sensitive rosacea skin.

251 **Answers:** A B C D

Non-pharmacological advice for fibromyalgia:

- **Good sleep hygiene (A)**: Sleep disturbance is a core feature of fibromyalgia. Non-restorative sleep worsens pain sensitivity and fatigue. Sleep hygiene measures (consistent schedule, dark/quiet room, avoiding screens) improve sleep quality and reduce symptom severity.
- **Initiate exercise program — aquatic aerobic (B)**: Exercise is the most effective non-pharmacological treatment for fibromyalgia. Aquatic (pool-based) exercise is particularly beneficial because warm water reduces joint stress, provides resistance, and promotes muscle relaxation. Start low and progress gradually.
- **Relaxation techniques and stress reduction (C)**: Stress amplifies central sensitization in fibromyalgia. Techniques include mindfulness meditation, progressive muscle relaxation, deep breathing, and yoga. These reduce cortisol, sympathetic tone, and pain perception.
- **Maintain healthy weight (D)**: Obesity worsens fibromyalgia symptoms through increased mechanical stress, systemic inflammation, and sleep apnea. Weight management improves pain, function, and quality of life.

Complete bed rest (E) worsens deconditioning and pain. Avoiding all physical activity (F) is counterproductive. Increasing caffeine (G) disrupts sleep.

252 **Answers:** A B C

First-line medications for fibromyalgia:

- **TCA — Amitriptyline 10 mg PO qhs (A)**: Inhibits serotonin and norepinephrine reuptake in descending pain inhibitory pathways. Also blocks sodium channels and NMDA receptors. Improves pain, sleep, and fatigue. Low doses (10-50 mg) are used — lower than antidepressant doses.
- **SNRI — Duloxetine 30 mg PO daily (B)**: FDA-approved for fibromyalgia. Increases serotonin and norepinephrine in descending pain inhibitory pathways. Effective for pain, fatigue, and comorbid depression/anxiety. Starting dose 30 mg, target 60 mg.
- **Anticonvulsant — Pregabalin 25 mg PO qhs (C)**: FDA-approved for fibromyalgia. Binds alpha-2-delta subunit of voltage-gated calcium channels, reducing excitatory neurotransmitter release (glutamate, substance P). Reduces pain, improves sleep, and reduces anxiety.

Opioids (D) are NOT recommended — they worsen central sensitization and have addiction risk. NSAIDs (E) are ineffective because fibromyalgia pain is central (not peripheral/inflammatory). Benzodiazepines (F) impair sleep architecture and cause dependence.

253 **Answer:** A

Genital herpes (multiple small painful ulcers) treatment:

1. Caused by herpes simplex virus (HSV-2 primarily, HSV-1 increasingly)

Valacyclovir is the preferred antiviral — better bioavailability than acyclovir (55% vs 15-20%), allowing less frequent dosing. It is converted to acyclovir, which is then phosphorylated by viral thymidine kinase and incorporated into viral DNA, causing chain termination and halting viral replication.

For first episode: Valacyclovir 1g BID x 7-10 days

For recurrent episodes: Valacyclovir 500 mg BID x 3 days

For suppressive therapy (≥6 outbreaks/year): Valacyclovir 500 mg-1g daily

Amoxicillin (B) is an antibiotic — ineffective against viruses. Fluconazole (C) is an antifungal. Metronidazole (D) is for anaerobic bacteria/BV. Doxycycline (E) is for chlamydia. Ceftriaxone (F) is for gonorrhea.

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254 **Answers:** A B C

Advice for genital herpes patients:

- **Use latex condoms with every sexual encounter (A)**: Condoms reduce HSV transmission by 30-50%. While not 100% effective (HSV can shed from areas not covered by condoms), consistent use significantly reduces partner transmission risk.
- **Communicate with sexual partners before first encounter (B)**: Ethical obligation to disclose HSV status. Partners can make informed decisions about risk acceptance. Discussion should include transmission risk, condom use, and suppressive therapy options.
- **Avoid oral sex if ulcers around the mouth (C)**: HSV-1 oral lesions can be transmitted to the partner's genitals during oral sex, causing genital herpes. Avoid oral-genital contact during active oral outbreaks.

Herpes is curable with antibiotics (D) is FALSE — herpes is a lifelong viral infection with no cure. Antivirals reduce symptoms and transmission but do not eradicate the virus from dorsal root ganglia. No precautions between outbreaks (E) is incorrect — asymptomatic viral shedding occurs and can transmit infection. Sharing towels (F) is not a significant transmission route but is not recommended.

255 **Answer:** A

Topical corticosteroids are first-line for psoriasis because:

1. They suppress the immune-mediated inflammatory process driving keratinocyte hyperproliferation

2. They reduce erythema, scaling, and plaque thickness

3. Potency is matched to body site — high potency (betamethasone 0.05%) for thick plaques on trunk/extremities, low potency (hydrocortisone 1%) for face/flexures

4. Applied twice daily for 2-4 weeks, then tapered to prevent rebound

5. Most effective when used under occlusion for thick plaques

6. Rapid symptom relief — improvement within 1-2 weeks

Topical methotrexate (B) does not exist as a standard formulation. Topical cyclosporine (C) is used in ophthalmology, not psoriasis. Topical tacrolimus (D) is used for facial/flexural psoriasis but is not first-line for body plaques. Topical antibiotics (E) are not indicated unless secondary infection. Topical retinoids alone (F) are adjunctive, not first-line monotherapy.

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256 **Answers:** A B

Other first-line topical options for psoriasis:

- **Calcipotriene — Vitamin D analogue (A)**: Binds to vitamin D receptors on keratinocytes, inhibiting proliferation and promoting differentiation. Also modulates T-cell function, reducing inflammation. Often combined with betamethasone (Enstilar, Taclonex) for synergistic effect. Applied BID, with improvement in 4-8 weeks.
- **Salicylic acid — Keratolytic (B)**: Breaks down the thick, adherent scale characteristic of psoriatic plaques by dissolving intercellular cement (desmosomes) between corneocytes. This descaling improves penetration of other topical agents (corticosteroids, vitamin D analogues). Used as adjunctive therapy.

Topical metformin (C), topical insulin (D), topical aspirin (E), and topical lithium (F) are not used in psoriasis treatment.

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257 **Answer:** A

For severe psoriasis with joint involvement (psoriatic arthritis), methotrexate is the best systemic medication because:

1. It treats BOTH skin and joint disease simultaneously

2. It inhibits dihydrofolate reductase, reducing purine/pyrimidine synthesis in rapidly dividing immune cells

3. It also has anti-inflammatory effects through adenosine release

4. Weekly dosing improves adherence

5. Folic acid 1 mg daily (except on methotrexate day) reduces side effects

6. Monitoring: CBC, LFTs, renal function at baseline and regularly

7. First-line systemic DMARD for psoriatic arthritis per GRAPPA and EULAR guidelines

Amoxicillin (B) is an antibiotic — psoriasis is not infectious. Long-term prednisone (C) is avoided in psoriasis — withdrawal causes severe rebound flares (pustular psoriasis). Gabapentin (D) is for neuropathic pain. Fluoxetine (E) is an antidepressant.

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258 **Answers:** A B C

Non-pharmacological measures for diaper dermatitis:

- **Frequent diaper changes (A)**: The primary cause of diaper dermatitis is prolonged skin contact with urine and feces. Urine raises skin pH, activating fecal lipases and proteases that damage the skin barrier. Changing diapers every 1-2 hours (or immediately after soiling) minimizes exposure.
- **Air exposure (B)**: Allowing the diaper area to air-dry between changes reduces moisture accumulation. "Diaper-free time" several times daily promotes skin healing by reducing occlusion and maceration.
- **Gentle skin cleansing (C)**: Use warm water and soft cloths or fragrance-free, alcohol-free wipes. Avoid vigorous rubbing — pat dry gently. Harsh soaps and wipes with alcohol/fragrance further damage the compromised skin barrier.

Alcohol-based wipes (D) cause stinging and further skin barrier damage. Keeping diaper on continuously (E) worsens the condition. Talcum powder (F) is NOT recommended — inhalation risk causes respiratory problems, and talc can accumulate in skin folds causing granulomas.

259 **Answers:** A B C

Pharmacological treatments for diaper dermatitis:

- **Topical hydrocortisone 1% ointment BID x 3-7 days (A)**: Low-potency steroid that reduces inflammation and erythema. Used for moderate-severe dermatitis. Limited to 3-7 days to avoid skin atrophy in the thin diaper area. Ointment base preferred over cream for better barrier protection.
- **Topical clotrimazole ointment BID x 2 weeks (B)**: An antifungal that treats Candida albicans superinfection, which occurs in >50% of diaper dermatitis lasting >3 days. Candidal involvement is suggested by satellite lesions, beefy red erythema, and involvement of skin folds.
- **Zinc oxide cream with every diaper change (C)**: A barrier cream that protects the skin from further irritant exposure. Zinc oxide creates a physical barrier between the skin and urine/feces, allows healing underneath, and has mild antiseptic and astringent properties.

Oral fluconazole (D) is reserved for severe/refractory candidal diaper dermatitis. IV antibiotics (E) are not indicated. Topical metronidazole (F) is not used for diaper dermatitis.

260 **Answer:** C

An elderly man with TIA, atrial fibrillation (irregularly irregular heart rate), and hypertension needs anticoagulation:

1. AF with TIA indicates high stroke risk — CHA_2DS_2-VASc score is elevated (age, TIA, hypertension)

2. Anticoagulation is mandatory to prevent cardioembolic stroke

3. Apixaban is preferred over warfarin because:

- No routine INR monitoring needed
- Lower bleeding risk (ARISTOTLE trial)
- Fixed dosing
- Fewer drug-food interactions
- Better quality of life

Aspirin (A) and clopidogrel (B) are antiplatelet agents — insufficient for AF stroke prevention. Heparin IV (D) is for acute treatment, not long-term prevention. Warfarin (E) requires monitoring the patient wants to avoid. Enoxaparin (F) requires daily injections.

261 **Answer:** A (or B)

For chronic heart rate control in atrial fibrillation:

- **Metoprolol (A)**: A beta-1 selective blocker that slows AV nodal conduction, reducing ventricular rate. First-line for rate control in AF. Target resting heart rate <110 bpm (lenient) or <80 bpm (strict). Well-tolerated, inexpensive, and available in long-acting formulations.
- **Diltiazem (B)**: A non-dihydropyridine CCB that slows AV nodal conduction through calcium channel blockade. Alternative first-line when beta-blockers are contraindicated (asthma, severe COPD). Also effective for rate control.

IV adenosine (C) is for SVT termination, not AF rate control. Oral flecainide (D) is for rhythm control, not rate control, and is contraindicated in structural heart disease. IV amiodarone (E) is for acute rate/rhythm control, not chronic management. Oral digoxin (F) is second-line — less effective during exercise and has a narrow therapeutic index.

262 **Answer:** B

A patient with MI at a rural hospital 4 hours from the nearest cardiac center:

1. Primary PCI (percutaneous coronary intervention) is the gold standard for STEMI
2. However, PCI must be performed within 120 minutes of first medical contact
3. If PCI cannot be performed within 120 minutes, fibrinolytic therapy is indicated
4. 4 hours to the cardiac center exceeds the 120-minute window
5. Tenecteplase is preferred — single IV bolus, weight-based dosing, easy to administer
6. It dissolves the coronary thrombus by converting plasminogen to plasmin
7. Must be given within 12 hours of symptom onset (ideally within 30 minutes of arrival)
8. After fibrinolysis, transfer to PCI-capable center for angiography within 3-24 hours

Waiting for transfer (A) wastes critical time — "time is muscle." Heparin alone (C) is insufficient. Morphine and observation (D) does not address the occlusion. PCI at rural hospital (E) is not available. Discharge (F) is dangerous.

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263 **Answers:** A B C D H

Absolute contraindications for fibrinolytic therapy in MI:

- **History of any intracranial hemorrhage (A)**: Fibrinolytics dissolve clots systemically — prior intracranial hemorrhage indicates vascular vulnerability, and fibrinolytics could cause fatal recurrent hemorrhage.
- **Ischemic stroke within preceding 3 months (B)**: Recent ischemic stroke has a damaged blood-brain barrier and fragile vasculature. Fibrinolytics can cause hemorrhagic transformation of the recent infarct.
- **Aortic dissection (C)**: Fibrinolytics would dissolve the thrombus sealing the dissection flap, causing fatal hemorrhage into the false lumen or pericardium.
- **Active bleeding (D)**: Fibrinolytics would worsen any active hemorrhage by dissolving hemostatic clots throughout the body.
- **Significant closed head trauma within 3 months (H)**: Recent head trauma may have caused occult intracranial vascular injury. Fibrinolytics could cause intracranial hemorrhage.

Age over 70 (E) is not an absolute contraindication. Mild hypertension (F) is not a contraindication — only BP >180/110 is a relative contraindication. Diabetes (G) is not a contraindication.

264 **Answer:** B

A 40-year-old who smokes cannabis 1g/day presenting with persistent vomiting for 2-3 days has cannabinoid hyperemesis syndrome (CHS):

1. CHS occurs in chronic, heavy cannabis users (typically daily use for >1 year)

2. Paradoxically, cannabis is antiemetic at low doses but causes hyperemesis with chronic heavy use

3. Mechanism: chronic CB1 receptor stimulation in the gut causes delayed gastric emptying and visceral hypersensitivity

4. Classic triad: cyclic vomiting, abdominal pain, compulsive hot bathing/showering (pathognomonic — hot water provides temporary relief)

5. Diagnosis is clinical — requires chronic cannabis use and exclusion of other causes

6. The ONLY definitive treatment is cannabis cessation

Peptic ulcer (A) would have epigastric pain and relationship to meals. Alcohol gastritis (C) is possible but cannabis use is more prominent. Gastroenteritis (D) would have diarrhea. Pancreatitis (E) would have severe epigastric pain radiating to back. Small bowel obstruction (F) would have distension and obstipation.

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265 **Answer:** B

The CHS patient with persistent vomiting for 2-3 days needs isotonic crystalloid fluid replacement:

1. Prolonged vomiting causes significant dehydration and electrolyte losses

2. Normal saline (0.9% NaCl) or Lactated Ringer's replaces intravascular volume and electrolytes

3. These are isotonic solutions that stay in the intravascular space effectively

4. Correct dehydration, hypochloremic metabolic alkalosis, and hypokalemia from vomiting

5. Monitor urine output to guide fluid resuscitation

D5W (A) is hypotonic and provides minimal volume expansion — inappropriate for dehydration. Half normal saline (C) is hypotonic. D10W (D) is for hypoglycemia. Albumin (E) is for specific indications like cirrhosis. Hypertonic saline (F) is for severe hyponatremia.

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266 **Answers:** A B

Antiemetics for cannabinoid hyperemesis:

- **Ondansetron (A)**: A 5-HT3 receptor antagonist that blocks serotonin-mediated stimulation of the chemoreceptor trigger zone and vagal afferents. Effective for acute nausea/vomiting in CHS.
- **Metoclopramide (B)**: A dopamine D2 antagonist and prokinetic that blocks the chemoreceptor trigger zone and promotes gastric emptying. Addresses both the nausea and the delayed gastric emptying seen in CHS.

Note: Traditional antiemetics may have limited efficacy in CHS. Capsaicin cream (0.075%) applied to the abdomen is an emerging treatment — it stimulates TRPV1 receptors similar to hot water bathing. Hot showers provide temporary relief but are not a definitive treatment. The only definitive treatment is complete cannabis cessation.

Domperidone (C) is an alternative prokinetic but less commonly used acutely. Scopolamine (D) is for motion sickness. Dexamethasone (E) is for chemotherapy-induced nausea. Promethazine (F) is an alternative but causes more sedation.

267 **Answers:** A B C D G

Contraindications to prescribing medical cannabis:

- **Patient under 25 years of age (A)**: The brain continues developing until age 25, particularly the prefrontal cortex (executive function, decision-making). Cannabis exposure during brain development causes permanent structural and functional changes, impaired cognition, and increased psychosis risk.
- **Substance abuse disorder (B)**: Cannabis has addictive potential. Patients with existing substance use disorders have higher risk of cannabis dependence and cross-addiction.
- **History of psychosis (C)**: Cannabis (particularly high-THC strains) increases dopamine in the mesolimbic pathway, triggering or worsening psychotic episodes. Risk is dose-dependent and highest in those with personal or family history of psychosis.
- **Cardiovascular disease (D)**: Cannabis causes tachycardia, orthostatic hypotension, and increased myocardial oxygen demand. It increases MI risk 4.8-fold in the first hour after use.
- **Pregnancy and breastfeeding (G)**: THC crosses the placenta and is excreted in breast milk. It affects fetal neurodevelopment, causes low birth weight, and impairs infant neurobehavioral development.

Mild headache (E) and family history of diabetes (F) are not contraindications.

268 **Answers:** A B C D

Common side effects of HIV PrEP (tenofovir/emtricitabine):

- **Nausea (A)**: The most common side effect, occurring in 10-20% of patients. Usually mild and self-limiting within the first month as the body adjusts. Taking with food helps.
- **Diarrhea (B)**: GI disturbance from the medication's effect on intestinal epithelium. Usually resolves within 2-4 weeks.
- **Headache (C)**: Common in the first few weeks of therapy. Typically mild and self-resolving.
- **Myalgia (D)**: Muscle aches can occur, usually mild and transient.

These are collectively called the "start-up syndrome" — they occur in the first 1-4 weeks and typically resolve spontaneously. Permanent liver failure (E), blindness (F), and cardiac arrest (G) are not common side effects. Tenofovir can cause renal tubular dysfunction and decreased bone mineral density with long-term use, which is why monitoring is important.

269 **Answers:** A B C D

Advice for patients on HIV PrEP:

- **Encourage condom use until drug achieves intracellular concentrations (A)**: PrEP takes 7 days of daily use to achieve protective levels in rectal tissue and 21 days in vaginal tissue. Condoms provide protection during this loading period and also protect against other STIs.
- **Strict adherence to routine monitoring (B)**: HIV testing every 3 months (to detect breakthrough infection), STI screening, renal function (creatinine/eGFR), and hepatitis B status must be monitored regularly.
- **Encourage other risk reduction methods (C)**: PrEP is part of a comprehensive prevention strategy — condom use, reducing number of partners, avoiding shared needles, and treating STIs all reduce HIV risk.
- **Strict adherence to medication (D)**: PrEP efficacy is directly proportional to adherence. Daily dosing achieves >99% protection with perfect adherence but drops to ~75% with 4 doses/week. Missing doses significantly reduces protection.

Unprotected sex being completely safe (E) is FALSE — PrEP is not 100% effective and does not protect against other STIs. No follow-up needed (F) is dangerous. Donating blood freely (G) is inappropriate during PrEP use.

270 **Answers:** A B C D G

Serious side effects of amiodarone:

- **Bradycardia and hypotension (A)**: Amiodarone blocks sodium, potassium, and calcium channels and has anti-adrenergic effects. This slows conduction and heart rate, potentially causing symptomatic bradycardia and hypotension.
- **Hepatotoxicity (B)**: Amiodarone and its metabolite (desethylamiodarone) accumulate in hepatocytes, causing phospholipidosis and hepatocellular damage. Ranges from mild transaminase elevation (15-30%) to fatal hepatitis (<3%).
- **Pulmonary toxicity (C)**: The most feared side effect. Amiodarone accumulates in type II pneumocytes, causing interstitial pneumonitis and fibrosis. Incidence 2-7%. Can be fatal. Presents with cough, dyspnea, and diffuse infiltrates.
- **Hypo or hyperthyroidism (D)**: Amiodarone contains 37% iodine by weight. It causes hypothyroidism (by Wolff-Chaikoff effect — excess iodine inhibits thyroid hormone synthesis) or hyperthyroidism (by Jod-Basedow effect — excess iodine stimulates hormone production in autonomous thyroid tissue).
- **QT prolongation (G)**: Amiodarone blocks potassium channels (IKr), prolonging repolarization and QT interval. This paradoxically increases risk of torsades de pointes, though the risk is lower than with other class III antiarrhythmics.

Improved vision (E) is incorrect — amiodarone causes corneal microdeposits (>90%) and optic neuropathy. Weight loss (F) is incorrect — amiodarone can cause weight changes.

271 **Answers:** A B C D E

Monitoring for amiodarone:

- **TSH — baseline, 3 months, then yearly (A)**: Detects amiodarone-induced thyroid dysfunction. Both hypo- and hyperthyroidism can occur at any time during therapy.
- **Liver function tests — baseline, 6 months, then annually (B)**: Detects hepatotoxicity. Transaminases >3x ULN require dose reduction or discontinuation.
- **Chest X-ray and PFTs — baseline, then yearly (C)**: Detects pulmonary toxicity early. Decreased DLCO (diffusion capacity) is the earliest sign. New infiltrates on CXR suggest amiodarone pneumonitis.
- **ECG — baseline, during loading, then annually (D)**: Monitors QT interval, heart rate, and conduction. QTc >500 ms requires dose reduction or discontinuation.
- **Eye examination yearly (E)**: Detects corneal microdeposits (nearly universal but usually asymptomatic) and rare but serious optic neuropathy that can cause permanent vision loss.

Weekly blood glucose (F) is not specifically required. Monthly bone density scan (G) is not indicated for amiodarone.

272 **Answer:** B

Third-degree (complete) heart block with P waves dissociated from QRS complexes requires immediate treatment:

1. Atropine is the first-line pharmacological treatment
2. It blocks muscarinic (M2) receptors in the SA and AV nodes
3. This increases sinus node firing rate and AV conduction
4. Dose: 0.5 mg IV, may repeat every 3-5 minutes to maximum 3 mg
5. Atropine is a temporizing measure while preparing for definitive treatment (pacing)
6. It may be less effective in infranodal block (wide QRS) but should still be tried

Epinephrine (A) is for cardiac arrest, not first-line for heart block. Amiodarone (C) would worsen bradycardia. Adenosine (D) slows AV conduction — CONTRAINDICATED in heart block. Lidocaine (E) is for ventricular arrhythmias. Verapamil (F) slows AV conduction — CONTRAINDICATED.

273 **Answer:** B

The definitive treatment for third-degree heart block is a permanent pacemaker because:

1. Complete heart block means NO electrical communication between atria and ventricles

2. The ventricles are driven by an escape rhythm (junctional or ventricular) which is unreliable

3. This escape rhythm can fail at any time, causing asystole and sudden death

4. A permanent pacemaker provides reliable, consistent ventricular pacing

5. Dual-chamber pacemakers (DDD) restore AV synchrony

6. Pacemaker implantation is a Class I indication for third-degree heart block

Lifelong atropine (A) is impractical and unreliable. Amiodarone (C) would worsen conduction. Beta-blockers (D) and calcium channel blockers (E) are CONTRAINDICATED — they worsen heart block. Digoxin (F) slows AV conduction — CONTRAINDICATED.

---

274 **Answers:** A B C D

Active TB treatment uses four drugs (RIPE regimen):

- **Isoniazid (A)**: Inhibits mycolic acid synthesis in the mycobacterial cell wall. The most potent bactericidal drug against actively growing M. tuberculosis. Used throughout the entire 6-month course.
- **Rifampin (B)**: Inhibits bacterial RNA polymerase, blocking RNA transcription. Bactericidal against both actively growing and semi-dormant bacilli. Essential for sterilizing activity. Used throughout the entire 6-month course.
- **Ethambutol (C)**: Inhibits arabinosyl transferase, blocking arabinogalactan synthesis in the cell wall. Bacteriostatic. Used in the initial 2-month intensive phase to prevent resistance emergence.
- **Pyrazinamide (D)**: Converted to pyrazinoic acid, which disrupts membrane function in acidic environments (inside macrophages). Bactericidal against intracellular, semi-dormant bacilli. Used in the initial 2-month intensive phase.

Four drugs are used initially to prevent resistance emergence. Azithromycin (E), ciprofloxacin (F), and metronidazole (G) are not part of standard TB treatment.

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275 **Answer:** B

Pyridoxine (vitamin B6) is given with isoniazid to prevent peripheral neuropathy because:

1. Isoniazid inhibits pyridoxine phosphokinase, the enzyme that converts pyridoxine to its active form (pyridoxal-5-phosphate)

2. Pyridoxal-5-phosphate is a cofactor for many enzymes in amino acid metabolism and neurotransmitter synthesis

3. Deficiency causes peripheral neuropathy — numbness, tingling, and burning in hands and feet

4. Risk factors: malnutrition, diabetes, HIV, alcoholism, pregnancy, renal failure

5. Supplementation with 25-50 mg/day (up to 100 mg in high-risk patients) prevents this complication

6. Should be given to ALL patients on isoniazid

Vitamin B12 (A) does not prevent isoniazid neuropathy. Folic acid (C) is for folate deficiency. Vitamin D (D) is for bone health. Thiamine (E) is for Wernicke encephalopathy. Vitamin C (F) does not prevent isoniazid neuropathy.

276 **Answer:** D

TB treatment duration is at least 6 months:

1. **Intensive phase (months 1-2)**: All four drugs (RIPE) — rapidly kills actively growing bacilli and reduces bacterial load by 99%

2. **Continuation phase (months 3-6)**: Isoniazid + Rifampin only — eliminates semi-dormant (persister) bacilli that survive the intensive phase

3. The 6-month duration is necessary because:

- M. tuberculosis has a slow doubling time (15-20 hours vs 20 minutes for E. coli)
- Semi-dormant bacilli are metabolically inactive and resistant to killing
- Shorter courses have unacceptable relapse rates

4. Some forms require longer treatment: TB meningitis (12 months), bone/joint TB (9-12 months)

Two weeks (A), 1 month (B), and 3 months (C) are insufficient. One year (E) is for complicated TB. Lifelong (F) is not necessary for drug-susceptible TB.

277 **Answer:** B

H. pylori eradication uses quadruple therapy (bismuth-based or concomitant):

1. **PPI (e.g., omeprazole 20 mg BID)**: Raises gastric pH, enhancing antibiotic efficacy and promoting mucosal healing

2. **Amoxicillin 1g BID**: Disrupts cell wall synthesis

3. **Metronidazole 500 mg BID**: Damages bacterial DNA through reactive metabolites

4. **Clarithromycin 500 mg BID**: Inhibits protein synthesis at 50S ribosome

5. **14-day duration**: Longer courses (14 days vs 7-10 days) achieve higher eradication rates (>90% vs 70-80%)

6. Quadruple therapy is now preferred over triple therapy due to increasing clarithromycin resistance

PPI + amoxicillin only (A) is insufficient. PPI + metronidazole only (C) is insufficient. H2 blocker + amoxicillin (D) is less effective than PPI-based regimens. Sucralfate + bismuth (E) lacks antibiotics.

278 **Answer:** B

C. difficile infection after antibiotic use (levofloxacin) — fidaxomicin is now first-line because:

1. It is a narrow-spectrum macrocyclic antibiotic that selectively targets C. difficile

2. It spares normal gut flora, reducing recurrence rates

3. The EXTEND trial showed fidaxomicin had significantly lower recurrence rates than vancomycin (13% vs 27%)

4. It inhibits bacterial RNA polymerase

5. Achieves high fecal concentrations with minimal systemic absorption

6. Current IDSA/SHEA 2021 guidelines recommend fidaxomicin over oral vancomycin for initial and recurrent episodes

Metronidazole (A) is no longer first-line — inferior efficacy and higher recurrence rates. Ciprofloxacin (C) would worsen C. difficile. Amoxicillin (D) is not effective against C. difficile. Loperamide (E) is CONTRAINDICATED — slowing motility retains toxin in the colon. Rifaximin (F) is used as a "chaser" regimen for recurrent CDI, not first-line.

279 **Answers:** A B

Advice while awaiting C. difficile investigation results:

- **Contact precautions and hand hygiene (A)**: C. difficile spores are resistant to alcohol-based hand sanitizers. SOAP AND WATER handwashing is required. Contact precautions (gown and gloves) prevent transmission to other patients. The patient should be in a private room if possible.
- **Avoid anti-diarrheal agents (B)**: Loperamide and other anti-motility agents are CONTRAINDICATED because they slow intestinal motility, retaining C. difficile toxins in the colon. This worsens mucosal damage, increases risk of toxic megacolon, and delays toxin clearance. Diarrhea is actually a protective mechanism that helps eliminate the toxin.

Continuing levofloxacin (D) would worsen C. difficile by further disrupting normal flora. Restricting all oral intake (E) is unnecessary. No precautions needed (F) is incorrect — C. difficile is highly transmissible through spores.

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280 **Answer:** B

CPAP is the definitive treatment for obstructive sleep apnea because:

1. It delivers continuous positive pressure through a nasal/facial mask
2. This pneumatically splints the upper airway open, preventing collapse during sleep
3. It eliminates apneas, hypopneas, and oxygen desaturations
4. Improves sleep quality, daytime alertness, and cognitive function
5. Reduces cardiovascular risk (hypertension, arrhythmias, stroke, MI)
6. Effective from the first night of use
7. Titrated during a sleep study to determine optimal pressure

Benzodiazepines (A) are CONTRAINDICATED — they relax upper airway muscles and worsen apnea. Oral melatonin (C) helps sleep onset but does not treat airway obstruction. Antihistamines (D) cause sedation and worsen apnea. Oxygen alone (E) does not prevent airway collapse. Stimulants (F) do not address the underlying obstruction.

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281 **Answers:** A B C D

Behavioral modifications for sleep apnea:

- **Weight loss and exercise (A)**: Obesity is the strongest modifiable risk factor. Excess pharyngeal fat narrows the airway. A 10% weight loss can reduce AHI (apnea-hypopnea index) by 26-50%. Exercise improves upper airway muscle tone even without weight loss.
- **Avoid sleeping supine (B)**: In the supine position, gravity causes the tongue and soft palate to fall posteriorly, worsening airway obstruction. Lateral sleeping reduces AHI by 50% in position-dependent OSA. Tennis ball technique or positional therapy devices help maintain lateral position.
- **Alcohol avoidance (C)**: Alcohol relaxes upper airway dilator muscles (genioglossus), increases airway collapsibility, and prolongs apnea duration. It also suppresses the arousal response to hypoxia. Avoid alcohol within 3-4 hours of bedtime.
- **Avoid sedating medications (D)**: Benzodiazepines, opioids, and sedating antihistamines all relax upper airway muscles and suppress respiratory drive, worsening OSA severity and oxygen desaturation.

Sleeping in a cold room (E) does not treat OSA. Increasing caffeine (F) disrupts sleep quality. Sleeping more hours (G) does not address the underlying obstruction.

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282 **Answer:** A

Polymyalgia rheumatica (PMR) initial treatment:

1. Low-dose prednisone (12.5-25 mg/day, typically 15 mg) is the standard initial treatment
2. PMR is an inflammatory condition causing bilateral shoulder and hip girdle pain/stiffness
3. Dramatic response to low-dose steroids is characteristic — improvement within 24-72 hours
4. This rapid response is actually a diagnostic criterion — if no improvement within 1 week, reconsider the diagnosis
5. ESR and CRP normalize within 2-4 weeks
6. The initial dose is maintained for 2-4 weeks before tapering begins

Methotrexate (B) is a steroid-sparing agent added if relapse occurs during tapering — not initial treatment. Ibuprofen (C) is insufficient for PMR. Hydroxychloroquine (D) is for RA/lupus. Adalimumab (E) is a biologic — not first-line for PMR.

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283 **Answer:** B

Steroid tapering in PMR:

1. After initial response (2-4 weeks at 15 mg), begin gradual taper

2. **Above 10 mg**: Reduce by 2.5 mg every 2-4 weeks

3. **Below 10 mg**: Reduce by 1 mg every 2-4 weeks (slower taper because adrenal suppression recovery is slower at lower doses)

4. Total treatment duration is typically 1-2 years

5. Monitor symptoms and inflammatory markers (ESR/CRP) at each dose reduction

6. If relapse occurs during taper, increase to the last effective dose and taper more slowly

7. Approximately 50% of patients relapse during tapering

Stopping abruptly (A) causes adrenal crisis and disease flare. Doubling then stopping (C) is irrational. Continuing indefinitely (D) causes cumulative steroid toxicity. Switching to NSAID immediately (E) is insufficient for PMR.

---

284 **Answer:** B

Head lice treatment with permethrin 1%:

1. Permethrin is a synthetic pyrethroid insecticide

2. It disrupts sodium channel function in lice nerve cells, causing paralysis and death

3. Applied to clean, towel-dried hair

4. Left on for 10 minutes, then rinsed

5. Repeat in 7-10 days to kill newly hatched nymphs (permethrin does not reliably kill all eggs/nits)

6. Available over-the-counter

7. Safe for children >2 months of age

8. Most common first-line treatment worldwide

Ivermectin oral (A) is second-line for resistant cases. Lindane (C) is no longer recommended due to neurotoxicity risk. Malathion (D) is second-line. Dimeticone (E) is an alternative physical treatment. Spinosad (F) is for resistant cases.

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285 **Answers:** A B C

Additional interventions for lice management:

- **Examine and treat household members if infested (A)**: Lice spread through direct head-to-head contact. All household members should be examined, and those with live lice or nits within 1 cm of the scalp should be treated simultaneously to prevent re-infestation.
- **Wash clothing and linen in hot water and high-heat dryer (B)**: Lice and nits are killed by temperatures >60°C (130°F). All bedding, towels, clothing, and hats worn in the previous 48 hours should be machine washed in hot water and dried on high heat for 20+ minutes.
- **Vacuum furniture and carpeting (C)**: Lice that fall off the head can survive for 1-2 days on furniture. Vacuuming removes stray lice and reduces environmental contamination. Focus on areas where the infested person sat or lay.

Shaving the head (D) is unnecessary and psychologically traumatic for children. Fumigating the house (E) is excessive — lice cannot survive long off the human host. Isolating the child for 2 weeks (F) is unnecessary — the child can return to school after first treatment.

286 **Answers:** A B

Molluscum contagiosum treatment in children:

- **Cryotherapy with liquid nitrogen (A)**: Freezing destroys the infected epidermal cells containing the molluscum contagiosum virus (poxvirus). Applied for 5-10 seconds per lesion. May require multiple sessions. Effective but can be painful — topical anesthetic (EMLA cream) applied beforehand helps.
- **Curettage with topical anesthetics (B)**: Physical removal of the molluscum body (central core) using a small curette after applying topical anesthetic (EMLA cream). Quick, effective, and provides immediate clearance. The most definitive single-session treatment.

Note: Molluscum contagiosum is self-limiting and resolves spontaneously in 6-12 months in immunocompetent children. Treatment is considered for cosmetic reasons, to prevent spread, or if lesions are symptomatic.

Oral acyclovir (C) is for herpes viruses — molluscum is a poxvirus. IV antibiotics (D) are not indicated. Systemic steroids (E) would worsen viral infection. Radiation (F) is never appropriate.

287 **Answers:** A B

Oral antifungals for toenail onychomycosis after topical failure:

- **Terbinafine (A)**: First-line oral antifungal for dermatophyte onychomycosis. It inhibits squalene epoxidase, blocking ergosterol synthesis in the fungal cell membrane. Accumulates in nail keratin and persists for months after treatment. Cure rates 70-80% for toenails. 12-week course for toenails (6 weeks for fingernails).
- **Itraconazole (B)**: A triazole antifungal that inhibits lanosterol 14-alpha-demethylase (CYP51), blocking ergosterol synthesis. Can be given as continuous therapy (200 mg daily x 12 weeks) or pulse therapy (200 mg BID x 1 week per month for 3-4 months). Cure rates 60-70%.

Fluconazole 50 mg x 4 weeks (C) is insufficient dose and duration. Ketoconazole (D) is no longer recommended orally due to hepatotoxicity risk. Griseofulvin (E) requires very prolonged courses (12-18 months) and has lower cure rates.

288 **Answer:** B

Before starting oral antifungal therapy for onychomycosis, LFTs must be checked because:

1. Both terbinafine and itraconazole are hepatotoxic

2. Terbinafine causes hepatocellular injury in ~1% of patients (rarely fulminant hepatic failure)

3. Itraconazole causes cholestatic hepatitis

4. Baseline LFTs identify pre-existing liver disease that would contraindicate treatment

5. LFTs should be rechecked at 4-6 weeks during treatment

6. Treatment should be stopped if transaminases rise >3x ULN or if symptoms of hepatitis develop (jaundice, nausea, dark urine)

CBC (A) is not specifically required. Renal function (C) is less critical as these drugs are hepatically metabolized. Chest X-ray (D), ECG (E), and serum calcium (F) are not indicated.

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289 **Answer:** A

Tinea versicolor (pityriasis versicolor) — caused by Malassezia furfur (a lipophilic yeast) — first-line topical treatment:

1. Ketoconazole 2% shampoo is applied to affected areas

2. Left on for 5-10 minutes, then rinsed

3. Used daily for 1-2 weeks

4. Ketoconazole inhibits ergosterol synthesis in the fungal cell membrane

5. Shampoo formulation covers large body surface areas efficiently

6. Alternative topicals: selenium sulfide 2.5% lotion, zinc pyrithione shampoo

Note: Hypopigmentation may persist for months after successful treatment because the yeast produces azelaic acid, which inhibits melanocyte tyrosinase. Repigmentation occurs gradually with sun exposure.

Hydrocortisone (B) is a steroid — would worsen fungal infection. Mupirocin (C) is antibacterial. Acyclovir (D) is antiviral. Calamine (E) is for itch relief. Benzoyl peroxide (F) is for acne.

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290 **Answer:** A

When topical treatment fails for tinea versicolor, oral fluconazole is used:

1. Single dose of 400 mg (or 300 mg weekly x 2 weeks as alternative)

2. Fluconazole inhibits fungal CYP51 (lanosterol 14-alpha-demethylase), blocking ergosterol synthesis

3. It achieves high concentrations in skin and sebaceous secretions

4. Single-dose regimen improves adherence

5. Alternative: itraconazole 200 mg daily x 5-7 days

6. Exercise after taking the medication (to promote sweating and drug delivery to skin) may improve efficacy

Amoxicillin (B) is antibacterial. Prednisone (C) would worsen fungal infection. Metronidazole (D) is for anaerobic bacteria. Doxycycline (E) is antibacterial.

291 **Answer:** B

A 60-year-old with urgency, frequency, and nocturia (8 times) with normal prostate and abdominal exam has overactive bladder (OAB). Oxybutynin is first-line because:

1. It is an antimuscarinic/anticholinergic that blocks M3 muscarinic receptors on the detrusor muscle

2. This reduces involuntary detrusor contractions (detrusor overactivity)

3. Increases bladder capacity and reduces urgency, frequency, and urge incontinence

4. Available in immediate-release, extended-release, and transdermal formulations

5. Extended-release and transdermal have fewer anticholinergic side effects (dry mouth, constipation, blurred vision)

Tamsulosin (A) is for BPH — prostate is normal here. Finasteride (C) is for enlarged prostate. Sildenafil (D) is for erectile dysfunction. Desmopressin (E) is for nocturnal enuresis. Furosemide (F) would worsen frequency.

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292 **Answer:** C

The patient returns with hypotension (BP 90/60) — likely caused by oxybutynin's anticholinergic effects (vasodilation, reduced cardiac output). Management:

1. Correct the hypotension with IV fluids and/or dose adjustment

2. Discontinue oxybutynin

3. Switch to mirabegron — a beta-3 adrenergic agonist that:

- Relaxes the detrusor muscle through a DIFFERENT mechanism (beta-3 stimulation, not anticholinergic)
- Does NOT cause anticholinergic side effects (dry mouth, constipation, cognitive impairment, hypotension)
- Safe in elderly patients
- Does not worsen glaucoma or urinary retention
- Starting dose 25 mg, can increase to 50 mg

Increasing oxybutynin (A) would worsen hypotension. Adding another anticholinergic (B) compounds side effects. Continuing oxybutynin with IV fluids (D) does not address the cause. Adding midodrine (E) treats the symptom but not the cause.

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293 **Answers:** A B C D

Non-hormonal medications for menopausal hot flashes:

- **SSRI — Paroxetine 7.5 mg PO daily (A)**: The only FDA-approved non-hormonal medication for hot flashes. SSRIs modulate thermoregulatory centers in the hypothalamus by affecting serotonin pathways. Low-dose paroxetine reduces hot flash frequency by 50-65%.
- **SNRI — Venlafaxine 37.5 mg PO daily (B)**: Inhibits serotonin and norepinephrine reuptake, modulating hypothalamic thermoregulatory centers. Reduces hot flash frequency by 40-60%. Can be titrated to 75-150 mg for better effect. Also helps mood and sleep.
- **Anticonvulsant — Gabapentin 300 mg PO TID (C)**: Modulates calcium channels and GABA pathways in the thermoregulatory center. Reduces hot flashes by 45-50%. Also improves sleep quality. Particularly useful for nighttime hot flashes.
- **Alpha-2 agonist — Clonidine 0.1 mg PO daily (D)**: Stimulates alpha-2 adrenergic receptors in the hypothalamus, reducing sympathetic outflow and stabilizing the thermoregulatory set point. Reduces hot flashes by 30-40%. Less effective than SSRIs/SNRIs but an option when others are contraindicated.

Opioids (E) are not indicated for hot flashes. Benzodiazepines (F) are not effective and cause dependence. Stimulants (G) would worsen hot flashes by increasing sympathetic activity.

294 **Answer:** B

A menopausal woman with an intact uterus who wants hormonal therapy must receive combined estrogen PLUS progesterone because:

1. Estrogen alone (unopposed estrogen) stimulates endometrial proliferation

2. Without progesterone opposition, this leads to endometrial hyperplasia and increased endometrial cancer risk (5-10 fold increase)

3. Progesterone (medroxyprogesterone acetate/MPA) counteracts estrogen's proliferative effect on the endometrium

4. Continuous combined therapy (daily estrogen + daily progesterone) is preferred in postmenopausal women — avoids withdrawal bleeding

5. CEE 0.625 mg + MPA 2.5 mg is the standard regimen

Estrogen alone (A) is ONLY for women who have had a hysterectomy (no endometrium to protect). Progesterone alone (C) is less effective for hot flashes. Testosterone (D) is not standard HRT. GnRH agonist (E) would worsen menopausal symptoms. Tamoxifen (F) is for breast cancer prevention, not HRT.

295 **Answers:** A B C G

Absolute contraindications to HRT:

- **History of breast cancer (A)**: Estrogen and progesterone stimulate breast cancer cell proliferation through hormone receptor activation. HRT increases recurrence risk in breast cancer survivors.
- **Coronary artery disease (B)**: The WHI trial showed HRT increases cardiovascular events in women with existing CAD. Estrogen promotes thrombosis and inflammation in diseased coronary arteries.
- **Previous VTE or stroke (C)**: Estrogen increases hepatic production of clotting factors (II, VII, IX, X, fibrinogen) and reduces antithrombin III and protein S. This prothrombotic state increases VTE recurrence and stroke risk.
- **Active liver disease (G)**: Estrogen is hepatically metabolized. Active liver disease impairs metabolism, causing drug accumulation and worsening hepatic function. Also increases risk of cholestasis and hepatic vein thrombosis.

Mild hot flashes (D) are actually an indication for HRT. Age over 50 (E) is not a contraindication — HRT is most beneficial when started within 10 years of menopause. Family history of osteoporosis (F) is not a contraindication.

296 **Answers:** A B C

Managing suspected intimate partner violence (IPV):

- **Express empathy and acknowledge the situation (A)**: Validating the patient's experience without judgment creates a safe space. Statements like "I'm concerned about your safety" and "You don't deserve this" build trust and encourage disclosure. Many IPV victims feel shame and blame themselves.
- **Assess safety of the patient (B)**: Determine the immediate danger level — Is the abuser escalating? Are weapons involved? Are children at risk? Safety assessment guides the urgency of intervention and safety planning.
- **Ask if she wants referral for help and support (C)**: Offer resources — domestic violence hotlines, shelters, legal advocacy, counseling. Respect her autonomy — she may not be ready to leave. Provide information and let her decide. Document injuries carefully.

Confronting the partner (D) is dangerous — it can escalate violence and endanger the patient. Ignoring the bruises (E) is a missed opportunity to help. Discharging without action (F) is negligent.

297 **Answers:** A B C F

Questions before sending an IPV patient home:

- **Do you feel safe in your relationship? (A)**: Direct screening question that opens the conversation. Many patients will not disclose unless directly asked in a private, non-judgmental setting.
- **Are your friends aware you have been hurt? (B)**: Assesses the patient's support network. Isolated victims are at higher risk. Knowing who is aware helps identify potential safety resources.
- **Do you have a safe place to go in an emergency? (C)**: Assesses whether the patient has an escape plan. If not, provide information about shelters and safe houses. Help develop a safety plan including packed bag, important documents, and emergency contacts.
- **Is there an accessible weapon? (F)**: The presence of a firearm in the home increases homicide risk 5-fold in IPV situations. This is a critical lethality assessment question.

Asking for the partner's phone number (D) is inappropriate and could endanger the patient. Asking what she did to provoke violence (E) is victim-blaming and harmful.

---

298 **Answer:** D

A patient grieving her daughter's death one month ago with sleep disturbance that does not affect work does NOT need medication because:

1. Normal grief (bereavement) is a natural response to loss — not a psychiatric disorder
2. Sleep disturbance is expected during acute grief
3. Symptoms are not affecting daily functioning (she is still working)
4. It has only been one month — grief typically improves over 6-12 months
5. Prescribing sleep medication may interfere with the natural grieving process
6. Non-pharmacological support is appropriate: empathetic listening, validation, encouraging social support, and follow-up

Zopiclone (A), lorazepam (B), and trazodone (C) are unnecessary when symptoms are not impairing function. Amitriptyline (E) and quetiapine (F) are also not indicated. The key principle: do not medicalize normal grief.

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299 **Answer:** C

The grief patient returns 7 months later with worsening symptoms (sleep, fatigue, guilt) — this now meets criteria for major depressive disorder (complicated grief transitioning to depression). The most important thing to assess is suicidal ideation because:

1. Depression is the strongest risk factor for suicide
2. Bereavement increases suicide risk, especially in the first year
3. Guilt is a particularly concerning symptom — it can manifest as "I should have died instead"
4. Worsening trajectory (getting worse, not better) indicates pathological grief
5. Direct questioning about suicidal thoughts does NOT increase suicide risk — it opens the door for help
6. Ask: "Have you had thoughts of harming yourself or ending your life?"
7. Assessment guides urgency of intervention — passive ideation vs active plan vs intent

Sleep quality (A), appetite (B), work performance (D), social activities (E), and exercise (F) are all important but none are as immediately life-threatening as suicidal ideation.

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300 **Answers:** A B C G

Serotonin syndrome treatment:

- **Discontinue serotonergic agents (A)**: The most critical step. Serotonin syndrome is caused by excess serotonergic activity — removing the offending agent(s) allows serotonin levels to normalize. Common culprits: SSRIs, SNRIs, MAOIs, tramadol, triptans, linezolid, St. John's Wort.
- **Lorazepam 1-2 mg IV (B)**: Benzodiazepines treat agitation, muscle rigidity, and seizures associated with serotonin syndrome. They enhance GABA inhibition, counteracting the excitatory serotonergic excess. Also reduce autonomic instability.
- **Cyproheptadine 12 mg PO (C)**: A serotonin antagonist (5-HT2A blocker) that directly counteracts excess serotonin activity. It is the specific antidote for serotonin syndrome. Initial dose 12 mg, then 2 mg every 2 hours as needed. Only available orally — can be crushed and given via NG tube.
- **Active cooling if temperature >41.1°C (G)**: Severe hyperthermia (>41.1°C/106°F) is life-threatening and requires aggressive cooling — ice packs, cooling blankets, cold IV fluids. Hyperthermia in serotonin syndrome is caused by excessive muscle activity, not infection — antipyretics (acetaminophen) are ineffective.

Increasing SSRI dose (D) would worsen the syndrome. Adding another serotonergic agent (E) is dangerous. Acetaminophen for fever (F) is ineffective — the hyperthermia is from muscle hyperactivity, not prostaglandin-mediated fever.

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301 **Answers:** A B

An 18-year-old with a contaminated wound (facial laceration from trauma, broken teeth) whose last vaccination was at age 6 (12 years ago):

- **Tdap 0.5 mL IM (A)**: Tetanus-diphtheria-acellular pertussis booster. His last tetanus vaccination was 12 years ago — boosters are recommended every 10 years, and for contaminated wounds if >5 years since last dose. Tdap is preferred over Td in adolescents/adults who haven't received Tdap before.
- **Human tetanus immunoglobulin (TIG) 250 units IM (B)**: Provides immediate passive immunity with preformed anti-tetanus antibodies. Given for contaminated/tetanus-prone wounds when vaccination history is incomplete or >5 years since last dose. TIG provides protection while the vaccine stimulates active immunity (takes 2 weeks).

Both are given simultaneously but at different injection sites. MMR (C), hepatitis B (D), pneumococcal (E), and HPV (F) vaccines are not specifically indicated for wound management.

302 **Answers:** B E

Initial wound management for a facial laceration from a fight (contaminated wound):

- **Wound saline irrigation and debridement (B)**: The most important step in wound management. Copious irrigation with normal saline (minimum 250 mL under pressure) removes bacteria, debris, and foreign bodies, reducing infection risk by 80%. Debridement removes devitalized tissue that serves as a bacterial growth medium.
- **Closure by secondary intention (E)**: Contaminated wounds, bite wounds, and wounds presenting >6-12 hours after injury should NOT be primarily closed. They are left open to heal by secondary intention (granulation) to reduce infection risk. Delayed primary closure can be performed at 3-5 days if no infection develops.

Primary closure immediately (A) is contraindicated in contaminated wounds — traps bacteria and increases abscess/infection risk. Leaving wound open without cleaning (C) allows infection. Topical antibiotics only (D) are insufficient without irrigation. Superglue (F) is for clean, superficial lacerations only.

303 **Answer:** B

An infected cat bite with fever and inflammation is most commonly caused by Pasteurella multocida because:

1. Pasteurella multocida colonizes the oral flora of 70-90% of cats
2. Cat teeth are thin and sharp — they create deep puncture wounds that inoculate bacteria into deep tissues
3. Pasteurella causes rapidly progressive cellulitis — typically within 12-24 hours of the bite
4. Presents with intense pain, swelling, erythema, and purulent drainage
5. Can cause septic arthritis, osteomyelitis, and bacteremia if untreated

Staphylococcus aureus (A) is common in dog bites and human skin infections but not the primary pathogen in cat bites. Bartonella henselae (C) causes cat scratch disease (lymphadenopathy), not acute wound infection. Streptococcus pyogenes (D), E. coli (E), and Pseudomonas (F) are less common in cat bites.

304 **Answer:** B

Amoxicillin-clavulanate is the first-line antibiotic for infected cat bites because:

1. It covers Pasteurella multocida (the primary pathogen)

2. It covers Staphylococcus and Streptococcus species (secondary pathogens)

3. It covers anaerobes (Fusobacterium, Bacteroides) present in cat oral flora

4. Clavulanate inhibits beta-lactamase, extending coverage to resistant organisms

5. Good oral bioavailability and tissue penetration

6. Duration 5-14 days depending on severity

Ciprofloxacin (A) has poor gram-positive coverage. Azithromycin (C) has limited Pasteurella activity. Metronidazole (D) covers only anaerobes. Doxycycline (E) is an alternative for penicillin-allergic patients but not first-line. TMP-SMX (F) has poor Pasteurella coverage.

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305 **Answer:** C

Parkland formula calculation:

- **Formula**: 4 mL × body weight (kg) × %TBSA burned
- **Calculation**: 4 × 70 × 40 = 11,200 mL
- This is the total crystalloid (Lactated Ringer's) needed for the first 24 hours
- The formula estimates fluid requirements to replace massive fluid losses from burn-induced capillary leak, evaporative losses, and third-spacing
- Lactated Ringer's is preferred because it is isotonic and contains physiologic electrolytes

5,600 mL (A) uses 2 mL instead of 4 mL. 8,400 mL (B) uses 3 mL. 14,000 mL (D) uses 5 mL. 16,800 mL (E) uses 6 mL.

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306 **Answer:** B

Parkland formula fluid distribution:

1. **First 8 hours**: Give 50% of total (5,600 mL) — rate = 700 mL/hr

2. **Next 16 hours**: Give remaining 50% (5,600 mL) — rate = 350 mL/hr

3. The 8-hour clock starts from the TIME OF BURN, not time of hospital arrival

4. Front-loading fluid is necessary because capillary leak is maximal in the first 8-12 hours

5. Fluid rate is adjusted based on urine output:

- Adults: target 0.5-1 mL/kg/hr
- Children: target 1-1.5 mL/kg/hr

6. Urine output is the best indicator of adequate resuscitation

100% evenly (A) does not account for early capillary leak. 75%/25% (C) is too front-loaded. 25%/75% (D) is too back-loaded. All in first 4 hours (E) would cause fluid overload and pulmonary edema in the early period while leaving the patient under-resuscitated later.

307 **Answers:** A B C D

Additional burn management measures:

- **Tetanus prophylaxis (A)**: Burns are tetanus-prone wounds. Damaged tissue creates an anaerobic environment ideal for Clostridium tetani. Tetanus toxoid booster (if >5 years since last dose) and TIG (if incomplete vaccination) should be given.
- **Bladder catheterization to monitor urine output (B)**: Urine output is the most reliable indicator of adequate fluid resuscitation. Target 0.5-1 mL/kg/hr in adults. Foley catheter allows continuous hourly monitoring and guides fluid rate adjustments.
- **Nasogastric tube for ileus (C)**: Burns >20% TBSA cause paralytic ileus from splanchnic vasoconstriction and sympathetic activation. NG tube decompresses the stomach, prevents aspiration, and allows early enteral feeding when ileus resolves.
- **Wound care and dressing (D)**: Gentle debridement of loose skin, application of topical antimicrobials (silver sulfadiazine), and sterile non-adherent dressings protect the wound, reduce bacterial colonization, and promote healing.

Prophylactic systemic antibiotics for all burns (E) are NOT recommended — they promote resistant organisms without proven benefit. Antibiotics are reserved for documented infection. Immediate skin grafting (F) is premature — grafting is performed after wound stabilization. Oral feeding immediately (G) is inappropriate with ileus.

308 **Answer:** B

Fluid deficit calculation for a dehydrated child:

- **Formula**: % dehydration × body weight (kg) × 10 = deficit in mL
- **Calculation**: 5% × 9 kg × 10 = 450 mL
- Alternatively: 50 mL/kg × 9 kg = 450 mL
- This represents the volume of fluid lost that needs to be replaced
- The deficit is typically replaced over 24 hours (half in first 8 hours, half in next 16 hours) in addition to maintenance fluids
- Clinical signs of 5% dehydration: slightly dry mucous membranes, mildly decreased skin turgor, slightly decreased urine output

225 mL (A) represents 2.5% dehydration. 675 mL (C) represents 7.5%. 900 mL (D) represents 10%. 1350 mL (E) represents 15%.

309 **Answer:** C

Maintenance fluid calculation using the 4-2-1 rule (Holliday-Segar method):

- **First 10 kg**: 4 mL/kg/hr
- **Next 10 kg (11-20 kg)**: 2 mL/kg/hr
- **Each kg above 20 kg**: 1 mL/kg/hr

For a 9 kg child:

- 9 kg × 4 mL/kg/hr = **36 mL/hr**
- Daily maintenance = 36 × 24 = 864 mL/day

This formula estimates the fluid needed to replace insensible losses (respiration, skin evaporation) and urinary output under normal conditions. The total fluid plan = maintenance + deficit replacement + ongoing losses.

18 mL/hr (A) uses 2 mL/kg/hr. 27 mL/hr (B) uses 3 mL/kg/hr. 45 mL/hr (D) uses 5 mL/kg/hr. 54 mL/hr (E) uses 6 mL/kg/hr.

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310 **Answer:** B

For a mildly dehydrated child, oral rehydration therapy is the best fluid:

1. WHO-ORS contains: sodium 75 mEq/L, glucose 75 mmol/L, potassium 20 mEq/L, citrate 10 mmol/L

2. Glucose-sodium co-transport mechanism: glucose enhances sodium (and thus water) absorption in the small intestine even during diarrhea

3. ORT is as effective as IV fluids for mild-moderate dehydration

4. It is safer, cheaper, less invasive, and can be administered at home

5. Give 50-100 mL/kg over 4 hours for mild dehydration

6. Continue breastfeeding alongside ORT

Normal saline IV (A) is reserved for severe dehydration or inability to tolerate oral fluids. D5W IV (C) is hypotonic and does not replace electrolytes. Apple juice (D) has high osmolality and low sodium — can worsen diarrhea. Cow's milk (E) is inappropriate during acute diarrhea. Sports drinks (F) have too much sugar and too little sodium.

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311 **Answers:** A B C D

Non-pharmacological measures for prediabetes (HbA1c 6.1, BMI 31):

- **Medical nutrition therapy — diet modification (A)**: Reduce total caloric intake, limit refined carbohydrates and added sugars, increase fiber intake (25-30g/day), emphasize whole grains, lean proteins, and healthy fats. Mediterranean and DASH diets have the strongest evidence for diabetes prevention.
- **Regular exercise — aerobic or resistance (B)**: 150 minutes/week of moderate aerobic exercise (brisk walking, cycling) or 75 minutes of vigorous exercise. Resistance training 2-3 times/week. Exercise improves insulin sensitivity for 24-72 hours post-session and reduces diabetes risk by 58% (DPP trial).
- **Weight reduction (C)**: 5-7% weight loss (approximately 4-6 kg for this patient) reduces diabetes risk by 58%. Weight loss improves insulin sensitivity, reduces visceral fat, and normalizes glucose metabolism.
- **Smoking cessation (D)**: Smoking increases insulin resistance, visceral fat accumulation, and inflammation. Smokers have 30-40% higher risk of developing Type 2 diabetes. Cessation improves insulin sensitivity within weeks.

Starting insulin immediately (E) is inappropriate for prediabetes. Bed rest (F) worsens insulin resistance. Avoiding all carbohydrates (G) is unnecessary and unsustainable.

312 **Answers:** A B

Follow-up parameters for prediabetes:

- **HbA1c annually (A)**: Monitors glycemic status over the preceding 2-3 months. Prediabetes is HbA1c 5.7-6.4%. Annual testing detects progression to diabetes (≥6.5%) or improvement with lifestyle changes. Guides the decision to initiate pharmacotherapy (metformin) if lifestyle measures fail.
- **Fasting blood sugar annually (B)**: Complements HbA1c monitoring. Prediabetes is fasting glucose 100-125 mg/dL. Diabetes is ≥126 mg/dL. Some patients have discordant HbA1c and fasting glucose — monitoring both provides a more complete picture.

Weekly finger-prick glucose (C) is unnecessary for prediabetes — this level of monitoring is for diabetic patients on insulin. Daily urine glucose (D) is insensitive and outdated. Monthly insulin levels (E) are not standard clinical practice for prediabetes monitoring.

313 **Answer:** A

Primary syphilis (painless chancre) treatment:

1. Penicillin G benzathine is the ONLY proven effective treatment for syphilis

2. Single IM injection of 2.4 million units cures primary and secondary syphilis

3. Benzathine formulation provides sustained therapeutic levels for 2-3 weeks

4. Treponema pallidum has NEVER developed penicillin resistance

5. The Jarisch-Herxheimer reaction (fever, myalgia, headache) may occur within 24 hours due to rapid treponemal killing — self-limiting

6. For penicillin-allergic patients: doxycycline 100 mg BID x 14 days is an alternative

Doxycycline (B) is the alternative for penicillin-allergic patients but not first-line. Azithromycin (C) is no longer recommended due to increasing resistance. Ceftriaxone (D) has some activity but is not standard. Metronidazole (E) and ciprofloxacin (F) are not effective against T. pallidum.

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314 **Answers:** A B C

Complications of untreated syphilis (tertiary syphilis):

- **Aortic aneurysm (A)**: Cardiovascular syphilis — T. pallidum infects the vasa vasorum of the aorta, causing endarteritis obliterans. This weakens the aortic wall, leading to ascending aortic aneurysm, aortic regurgitation, and coronary ostial stenosis. Occurs 15-30 years after primary infection.
- **CNS manifestations — dementia, seizures (B)**: Neurosyphilis — T. pallidum invades the CNS, causing meningovascular syphilis (stroke), general paresis (progressive dementia, personality changes), and tabes dorsalis (posterior column degeneration causing ataxia, lightning pains, bladder dysfunction).
- **Argyll Robertson pupil (C)**: Pathognomonic for neurosyphilis — pupils that accommodate (constrict for near vision) but do NOT react to light. Caused by damage to the pretectal nuclei in the midbrain. Called "prostitute's pupil" — accommodates but doesn't react.

Improved immunity (D) is incorrect — syphilis does not confer protective immunity. Spontaneous resolution always (E) is incorrect — untreated syphilis progresses through stages. Diabetes (F) is not caused by syphilis.

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315 **Answers:** A B

Gonococcal epididymitis treatment:

- **Ceftriaxone 250 mg IM single dose (A)**: Covers Neisseria gonorrhoeae — the identified causative organism. Third-generation cephalosporin with excellent gonococcal activity. Single dose ensures compliance.
- **Doxycycline 100 mg PO BID x 10-14 days (B)**: Covers presumptive Chlamydia trachomatis co-infection (present in 20-40% of gonorrhea cases) and provides additional coverage for other epididymitis pathogens. Longer duration (10-14 days) than for uncomplicated urethritis because epididymitis involves deeper tissue infection.

Additionally: scrotal elevation, ice packs, and NSAIDs for pain relief. Metronidazole (C) is for anaerobes. Fluconazole (D) is antifungal. Ciprofloxacin (E) is no longer recommended for gonorrhea due to widespread resistance.

316 **Answers:** A B

Kawasaki disease treatment:

- **IVIG — Intravenous Immunoglobulin (A)**: The cornerstone of treatment. Given as a single high dose (2 g/kg over 10-12 hours) within the first 10 days of illness. IVIG reduces coronary artery aneurysm risk from 25% to <5%. Mechanism: broadly immunomodulatory — neutralizes pathogenic antibodies, modulates cytokine production, and suppresses T-cell activation.
- **Aspirin (B)**: Given in two phases:

1. High-dose (80-100 mg/kg/day divided QID) during acute phase — anti-inflammatory effect

2. Low-dose (3-5 mg/kg/day) for 6-8 weeks after fever resolves — antiplatelet effect to prevent coronary thrombosis

Note: This is the ONE exception where aspirin is used in children (normally avoided due to Reye syndrome risk)

Ibuprofen (C) is avoided — it antagonizes aspirin's antiplatelet effect. Prednisone (D) is reserved for IVIG-resistant cases. Methotrexate (E) is not standard. Azithromycin (F) is not indicated — Kawasaki is not bacterial.

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317 **Answers:** A B

Long-term follow-up for Kawasaki disease:

- **Echocardiography (A)**: The most important follow-up investigation. Monitors for coronary artery aneurysms — the most serious complication. Schedule: baseline, 2 weeks, 6-8 weeks after onset. If aneurysms are detected, more frequent monitoring with possible coronary angiography. Aneurysms may regress, persist, or progress to stenosis/thrombosis.
- **Stress ECG (B)**: For patients with coronary artery abnormalities, stress testing (exercise ECG or stress echocardiography) detects myocardial ischemia from coronary stenosis. Performed annually or as indicated based on aneurysm severity.

Weekly chest X-ray (C) is unnecessary. Monthly liver biopsy (D) is not indicated. Daily blood cultures (E) are not needed — Kawasaki is not an infectious endocarditis.

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318 **Answer:** B

Preeclampsia with severe features requires magnesium sulfate for seizure prophylaxis:

1. Magnesium sulfate is the ONLY proven medication to prevent eclamptic seizures

2. Loading dose: 4-6g IV over 15-20 minutes

3. Maintenance: 1-2 g/hr continuous infusion

4. Mechanism: blocks NMDA receptors (reducing neuronal excitability), causes cerebral vasodilation (reducing ischemia), and has anticonvulsant properties

5. The MAGPIE trial showed magnesium sulfate reduces eclampsia risk by 58%

6. Continue for 24-48 hours postpartum (eclampsia risk persists)

Phenytoin (A) is less effective than magnesium for eclampsia prevention. Diazepam (C) is less effective and causes neonatal depression. Levetiracetam (D) has insufficient evidence. Carbamazepine (E) is not used for eclampsia. Valproic acid (F) is teratogenic and not indicated.

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319 **Answers:** A B

Monitoring magnesium sulfate therapy:

- **Deep tendon reflexes (A)**: Loss of patellar reflexes is the earliest clinical sign of magnesium toxicity. Reflexes should be checked every 1-2 hours. Absent reflexes indicate serum magnesium >7 mEq/L — the infusion should be stopped immediately. This is the most practical bedside monitoring tool.
- **Respiratory pattern (B)**: Magnesium causes respiratory depression by blocking neuromuscular transmission at the diaphragm. Respiratory rate should be monitored hourly — rate <12/min indicates toxicity. Respiratory arrest occurs at serum magnesium >12 mEq/L.

Additional monitoring: urine output (>25 mL/hr ensures renal magnesium excretion), oxygen saturation, and serum magnesium levels (therapeutic range 4-7 mEq/L).

Serum calcium (C) is not the primary monitoring parameter. Hourly liver enzymes (D) are unnecessary. Daily chest X-ray (E) is not indicated.

320 **Answer:** C

The antidote for magnesium sulfate toxicity is calcium gluconate because:

1. Calcium directly antagonizes magnesium at the neuromuscular junction
2. It restores neuromuscular transmission blocked by excess magnesium
3. Dose: 1g (10 mL of 10% solution) IV over 2-3 minutes
4. Onset within 1-3 minutes

Should be kept at the bedside of every patient receiving magnesium sulfate infusion for immediate availability. The sequence of magnesium toxicity is:

- Loss of deep tendon reflexes (7-10 mEq/L)
- Respiratory depression (10-13 mEq/L)
- Cardiac arrest (>15 mEq/L)

Naloxone (A) reverses opioids. Flumazenil (B) reverses benzodiazepines. Protamine sulfate (D) reverses heparin. Vitamin K (E) reverses warfarin. N-acetylcysteine (F) is for acetaminophen overdose.

321 **Answer:** C

The most dangerous element in chewable multivitamin overdose in children is iron because:

1. Iron is the leading cause of fatal poisoning in children under 6

2. Toxic dose: >20 mg/kg elemental iron causes GI toxicity; >60 mg/kg is potentially lethal

3. Iron toxicity occurs in stages:

- Stage 1 (0-6 hrs): GI — vomiting, diarrhea, abdominal pain, GI hemorrhage
- Stage 2 (6-24 hrs): Apparent improvement (latent period)
- Stage 3 (12-48 hrs): Systemic toxicity — metabolic acidosis, hepatic failure, shock, coagulopathy
- Stage 4 (2-6 weeks): GI scarring and strictures

4. Free iron generates hydroxyl radicals through Fenton reaction, causing cellular destruction

Vitamin C (A) causes diarrhea but is not lethal. Vitamin B12 (B) is water-soluble with minimal toxicity. Zinc (D) can cause GI upset but is rarely fatal. Folic acid (E) is non-toxic. Vitamin D (F) can cause hypercalcemia but is less acutely dangerous than iron.

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322 **Answer:** B

After stabilizing ABC in iron poisoning, the specific treatment is IV deferoxamine:

1. Deferoxamine is an iron chelator that binds free iron in the bloodstream

2. Forms ferrioxamine complex — water-soluble and renally excreted

3. Dose: 15 mg/kg/hr continuous IV infusion (max 6-8 g/day)

4. Indications: serum iron >500 mcg/dL, metabolic acidosis, shock, altered mental status, or clinical deterioration

5. Characteristic "vin rosé" (pinkish-red) urine indicates iron chelation is occurring

6. Continue until clinical improvement and urine color normalizes

Activated charcoal (A) does NOT bind iron — it is ineffective for iron poisoning. N-acetylcysteine (C) is for acetaminophen overdose. Whole bowel irrigation (D) may be considered as adjunct for iron tablets visible on X-ray but is not the specific treatment. Naloxone (E) is for opioids. Flumazenil (F) is for benzodiazepines.

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323 **Answers:** A B C D

Neuroleptic Malignant Syndrome (NMS) management:

- **Stop the offending agent (A)**: The most critical step. NMS is caused by sudden dopamine receptor blockade (from antipsychotics) or dopamine agonist withdrawal. Removing the causative agent allows dopaminergic function to recover.
- **IV fluids (B)**: Aggressive hydration is essential because NMS causes severe dehydration from hyperthermia, diaphoresis, and rhabdomyolysis. IV fluids also prevent acute kidney injury from myoglobin precipitation in renal tubules.
- **Cooling measures (C)**: Hyperthermia (often >40°C) is life-threatening. Active cooling with ice packs, cooling blankets, and cold IV fluids reduces core temperature. Antipyretics are ineffective because the hyperthermia is from muscle rigidity, not prostaglandin-mediated fever.
- **Dantrolene (D)**: A direct-acting skeletal muscle relaxant that blocks ryanodine receptors on the sarcoplasmic reticulum, preventing calcium release and reducing muscle rigidity and heat generation. Dose: 1-2.5 mg/kg IV, repeat every 5-10 minutes up to 10 mg/kg/day. Alternative: bromocriptine (dopamine agonist).

Increasing haloperidol (E) would worsen NMS. Adding another antipsychotic (F) is dangerous. Acetaminophen (G) is ineffective for NMS hyperthermia.

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324 **Answers:** A B C E

Preventing NMS recurrence when restarting antipsychotics:

- **Wait at least 2 weeks after recovery (A)**: Allows complete resolution of NMS and recovery of dopaminergic function. Restarting too early risks recurrence because the dopamine system remains sensitized.
- **Use a low-potency agent (B)**: Low-potency antipsychotics (e.g., quetiapine, chlorpromazine) have weaker D2 blockade than high-potency agents (haloperidol, fluphenazine), reducing NMS risk. Atypical antipsychotics are generally preferred.
- **Start at low dose and increase slowly (C)**: Gradual titration allows the dopamine system to adapt to receptor blockade, reducing the risk of acute dopamine depletion that triggers NMS.
- **Avoid concomitant lithium (E)**: Lithium combined with antipsychotics increases NMS risk through unclear mechanisms — possibly by further impairing dopaminergic transmission and thermoregulation.

Restarting the same medication at the same dose (D) carries the highest recurrence risk. Using the highest dose immediately (F) is dangerous.

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325 **Answer:** B

Wernicke encephalopathy (classic triad: confusion, ataxia, ophthalmoplegia) in an alcoholic patient requires immediate high-dose IV thiamine:

1. Thiamine (vitamin B1) is the cofactor for pyruvate dehydrogenase, alpha-ketoglutarate dehydrogenase, and transketolase

2. Deficiency causes energy failure in metabolically active brain regions (mammillary bodies, thalamus, periaqueductal gray)

3. High-dose IV thiamine (500 mg TID) is given for at least 72 hours

4. IV route is essential — oral absorption is unreliable in alcoholics (impaired GI absorption)

5. Thiamine MUST be given BEFORE glucose — glucose metabolism without thiamine generates toxic metabolites that worsen brain damage

6. If untreated, Wernicke progresses to Korsakoff syndrome (irreversible anterograde amnesia, confabulation)

Oral thiamine (A) has inadequate bioavailability. IV glucose first (C) is DANGEROUS — precipitates Wernicke. Oral multivitamin only (D) is insufficient. IV folate (E) and IM B12 (F) do not treat Wernicke.

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326 **Answer:** A

Corneal abrasion treatment:

1. Erythromycin ophthalmic ointment is first-line prophylactic antibiotic for corneal abrasions

2. It prevents secondary bacterial infection of the exposed corneal stroma

3. Ointment formulation provides lubrication and a protective film over the abrasion

4. Erythromycin covers common ocular pathogens: Staphylococcus, Streptococcus, and Haemophilus

5. Applied QID (every 6 hours) for 5 days until re-epithelialization occurs

6. Safe, well-tolerated, and does not cause corneal toxicity

Hydrocortisone eye drops (B) are CONTRAINDICATED — steroids impair corneal healing and increase infection risk. Pilocarpine (C) is for glaucoma. Timolol (D) is for glaucoma. Atropine (E) may be used for pain (cycloplegia) but is not the antibiotic. Oral amoxicillin (F) does not achieve adequate corneal concentrations.

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327 **Answer:** B

Contact lens-related corneal abrasions require fluoroquinolone coverage because:

1. Contact lens wearers are at high risk for Pseudomonas aeruginosa keratitis

2. Pseudomonas is a gram-negative organism that thrives in the moist environment under contact lenses

3. Pseudomonas keratitis is aggressive — can cause corneal perforation within 24-48 hours

4. Ciprofloxacin 0.3% drops provide broad-spectrum coverage including Pseudomonas

5. Fluoroquinolones also cover Staphylococcus, Streptococcus, and other gram-negatives

6. Drops are preferred over ointment for contact lens wearers — ointment can trap bacteria

Erythromycin ointment (A) does NOT cover Pseudomonas — inadequate for contact lens wearers. Gentamicin (C) covers Pseudomonas but is more toxic to corneal epithelium. Chloramphenicol (D) has limited Pseudomonas coverage. Neomycin (E) causes allergic reactions and has limited Pseudomonas activity.

328 **Answers:** A B

Allergic rhinitis and conjunctivitis treatment:

- **Intranasal glucocorticoid — Fluticasone furoate (A)**: First-line for allergic rhinitis. Suppresses nasal mucosal inflammation by inhibiting multiple inflammatory mediators. Reduces sneezing, rhinorrhea, nasal congestion, and itching. Also reduces ocular symptoms through the naso-ocular reflex. Most effective single agent for allergic rhinitis.
- **Antihistamine eye drops — Azelastine 0.05% (B)**: First-line for allergic conjunctivitis. Azelastine is a dual-action agent — H1 antihistamine (blocks histamine-mediated itching, redness, tearing) and mast cell stabilizer (prevents further histamine release). Provides rapid relief within minutes.

Oral antibiotics (C) are not indicated — this is allergic, not infectious. Long-term nasal decongestants (D) cause rhinitis medicamentosa (rebound congestion) after 3-5 days. Long-term oral steroids (E) have excessive side effects. IV antihistamines (F) are unnecessary for outpatient allergic disease.

329 **Answers:** A B C

Non-pharmacological advice for allergic conjunctivitis:

- **Avoid contact lenses (A)**: Contact lenses trap allergens against the conjunctival surface, worsening inflammation. They also accumulate protein deposits that act as allergen reservoirs. Switch to glasses during allergy season or until symptoms resolve.
- **Cool compresses (B)**: Cold compresses constrict conjunctival blood vessels, reducing redness and swelling. They also soothe itching and provide immediate comfort. Apply for 10-15 minutes several times daily.
- **Avoid rubbing eyes (C)**: Rubbing mechanically degranulates mast cells, releasing more histamine and worsening itching, redness, and swelling. It creates a vicious itch-rub cycle. Rubbing can also cause corneal abrasion and introduce bacteria.

Hot compresses (D) would worsen vasodilation and swelling. Wearing contact lenses continuously (E) worsens allergen exposure. Applying makeup (F) introduces potential allergens and irritants.

330 **Answer:** B

Phimosis (inability to retract the foreskin) in a child — topical corticosteroid treatment:

1. Betamethasone 0.05% is a potent topical steroid applied to the tight foreskin band
2. Applied BID for 4-8 weeks with gentle retraction exercises
3. Mechanism: steroids thin the skin by reducing collagen synthesis, promoting epithelial atrophy, and reducing inflammation
4. This loosens the phimotic ring, allowing gradual retraction
5. Success rate: 65-95% with proper technique
6. Avoids the need for circumcision in most cases
7. Parents should be taught gentle retraction technique — never forceful

Hydrocortisone 1% (A) is too weak — insufficient potency to thin the phimotic band. Mupirocin (C) is an antibiotic — not indicated. Clotrimazole (D) is antifungal. Tretinoin (E) is for acne. Tacrolimus (F) is a calcineurin inhibitor — second-line if steroids fail.

331 **Answer:** B

Lyme disease (erythema migrans after tick bite) treatment:

1. Caused by Borrelia burgdorferi transmitted by Ixodes tick

2. Doxycycline is first-line for early Lyme disease in adults and children ≥8 years

3. It is bacteriostatic — inhibits protein synthesis at the 30S ribosomal subunit

4. 10-day course (some guidelines say 10-21 days) for early localized disease

5. Also provides prophylaxis against other tick-borne co-infections (Anaplasma, Ehrlichia)

6. For children <8 years or pregnant women: amoxicillin 500 mg TID x 14-21 days

Amoxicillin (A) is the alternative for children <8 and pregnant women but not first-line in adults. Azithromycin (C) is second-line with lower efficacy. Ciprofloxacin (D) is not effective against Borrelia. Metronidazole (E) is for anaerobes. Cephalexin (F) has inadequate Borrelia coverage.

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332 **Answer:** B

Pertussis (whooping cough) treatment:

1. Caused by Bordetella pertussis — gram-negative coccobacillus

2. Azithromycin is first-line in children because:

- Excellent activity against B. pertussis
- Once-daily dosing improves adherence
- Well-tolerated in children
- 5-day course (shorter than alternatives)
- Available in liquid formulation

3. Alternative: clarithromycin or erythromycin

4. Treatment is most effective when started in the catarrhal phase (first 1-2 weeks)

5. In the paroxysmal phase, antibiotics reduce transmission but may not shorten illness duration

Amoxicillin (A) is NOT effective against B. pertussis. Cephalexin (C) has no pertussis activity. Doxycycline (D) is avoided in children <8. Ciprofloxacin (E) is not standard for pertussis in children.

333 **Answer:** B

The primary rationale for treating pertussis with antibiotics is to reduce transmission:

1. B. pertussis is highly contagious — attack rate is 80-90% in susceptible household contacts

2. Transmission occurs through respiratory droplets during coughing

3. Antibiotics eradicate B. pertussis from the nasopharynx within 5 days

4. Without treatment, patients remain infectious for up to 3 weeks after cough onset

5. Reducing transmission protects vulnerable populations — unvaccinated infants (<6 months) who have the highest mortality from pertussis (90% of pertussis deaths occur in infants <4 months). Prophylactic antibiotics should also be given to close contacts, especially households with infants.

Curing the cough immediately (A) is incorrect — antibiotics do not significantly shorten the paroxysmal cough phase once established. Preventing pneumonia (C) is a secondary benefit but not the primary rationale. Reducing fever (D) is incorrect — pertussis typically causes minimal fever. Improving appetite (E) is not the primary goal.

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334 **Answers:** A B C

Non-pharmacological measures for post-operative delirium:

- **Maintaining adequate hydration and nutrition (A)**: Dehydration and malnutrition worsen delirium by causing electrolyte imbalances, hypoglycemia, and reduced cerebral perfusion. Ensuring adequate oral or IV fluids and early nutritional support promotes recovery.
- **Enhancing mobility and range of motion (B)**: Early mobilization prevents deconditioning, reduces ICU-acquired weakness, and promotes normal sleep-wake cycles. Even passive range of motion exercises help. Immobility is a major risk factor for prolonged delirium.
- **Cognitive stimulation with reassurance from familiar persons (C)**: Orientation cues (clocks, calendars, familiar objects), family presence, and gentle reorientation reduce confusion and agitation. Familiar voices and faces provide comfort and reduce fear in the delirious patient.

Physical restraints as first line (D) are CONTRAINDICATED — they increase agitation, injury risk, and prolong delirium. Restraints should be an absolute last resort. Keeping the room dark and quiet at all times (E) disrupts circadian rhythm — daytime light exposure is important. Isolating the patient (F) worsens confusion and agitation.

335 **Answers:** A B

Medications for acute delirium:

- **Haloperidol (A)**: The most studied antipsychotic for delirium. Low-dose haloperidol (0.5-1 mg IM/IV) reduces agitation, hallucinations, and disorganized thinking by blocking D2 receptors. Maximum 5 mg/day in elderly. Monitor QTc — haloperidol can prolong QT interval.
- **Risperidone (B)**: An atypical antipsychotic alternative with fewer extrapyramidal side effects than haloperidol. Low dose (0.25-0.5 mg BID) is effective for delirium. Available in oral dissolving tablet for patients who cannot swallow.

Important: Antipsychotics treat the SYMPTOMS of delirium — the underlying cause (infection, metabolic derangement, medication, pain) must be identified and treated.

Diazepam (C) worsens delirium (except in alcohol/benzodiazepine withdrawal delirium). Morphine (D) can worsen confusion. Methylphenidate (E) is a stimulant — inappropriate. Lithium (F) is for bipolar disorder.

336 **Answers:** A B C

Referrals for Complex Regional Pain Syndrome (CRPS):

- **Pain management specialist (A)**: CRPS requires multimodal pain management including nerve blocks (sympathetic ganglion blocks), spinal cord stimulation, intrathecal drug delivery, and medication optimization. Pain specialists coordinate these advanced interventions.
- **Physical and occupational therapist (B)**: The cornerstone of CRPS treatment. Graded motor imagery, mirror therapy, desensitization techniques, and progressive functional exercises restore limb function and reduce pain. Early aggressive rehabilitation improves outcomes significantly.
- **Psychologist (C)**: CRPS causes significant psychological distress — depression, anxiety, fear-avoidance behavior, and catastrophizing. CBT addresses maladaptive pain beliefs, improves coping strategies, and reduces disability. Psychological support is essential for comprehensive management.

Dermatologist (D) is not specifically needed. Gastroenterologist (E) is not relevant. Endocrinologist (F) is not indicated for CRPS.

337 **Answers:** A B

Medications from different classes for CRPS:

- **Amitriptyline (A)**: A tricyclic antidepressant that treats neuropathic pain by inhibiting serotonin and norepinephrine reuptake in descending pain inhibitory pathways and blocking sodium channels in peripheral nerves. Low doses (10-75 mg) are effective for neuropathic pain. Also improves sleep.
- **Pregabalin (B)**: An alpha-2-delta calcium channel ligand that reduces excitatory neurotransmitter release from hyperexcitable neurons. Effective for neuropathic pain, allodynia, and hyperalgesia in CRPS. Also reduces anxiety and improves sleep.

These represent two different mechanisms — monoamine reuptake inhibition and calcium channel modulation. Metformin (C) is for diabetes. Omeprazole (D) is for GERD. Atenolol (E) is for hypertension. Furosemide (F) is a diuretic.

338 **Answers:** A B C G

Stable angina management (patient currently stable and pain-free):

- **Beta-blocker — Metoprolol 25 mg PO BID (A)**: First-line anti-anginal. Reduces heart rate, contractility, and myocardial oxygen demand. Prevents angina episodes and improves exercise tolerance. Also provides mortality benefit post-MI.
- **Antiplatelet — Aspirin 325 mg (B)**: Irreversibly inhibits COX-1 in platelets, preventing thromboxane A2 production and platelet aggregation. Reduces risk of MI and cardiovascular death by 25-30%. Loading dose 325 mg, then maintenance 81 mg daily.
- **Statin — Atorvastatin 80 mg PO daily (C)**: High-intensity statin therapy stabilizes atherosclerotic plaques, reduces LDL, and has anti-inflammatory effects. Reduces cardiovascular events by 30-40% regardless of baseline cholesterol.
- **Sublingual nitroglycerin PRN (G)**: For acute angina episodes. Nitroglycerin causes venodilation (reducing preload) and coronary vasodilation, rapidly relieving chest pain. Patient should carry it at all times and use at onset of symptoms.

Calcium channel blocker (D) is added if beta-blocker alone is insufficient. Opioids (E) are not appropriate for chronic angina. Benzodiazepines (F) are not indicated.

339 **Answers:** A B C D

Non-pharmacological advice for stable angina:

- **Maintain healthy weight (A)**: Obesity increases cardiac workload, worsens dyslipidemia, and promotes insulin resistance. Weight loss reduces angina frequency and cardiovascular risk.
- **Safe exercise plan (B)**: Cardiac rehabilitation with supervised, graded exercise improves exercise tolerance, reduces angina threshold, and improves endothelial function. Patients should be taught to recognize warning symptoms and use nitroglycerin prophylactically before known triggers.
- **Healthy diet limited in saturated fat and salt (C)**: Mediterranean diet reduces cardiovascular events by 30%. Limiting saturated fat reduces LDL. Limiting sodium reduces blood pressure. Increasing fruits, vegetables, whole grains, and omega-3 fatty acids is beneficial.
- **Smoking cessation (D)**: Smoking causes endothelial dysfunction, coronary vasospasm, platelet activation, and accelerated atherosclerosis. Cessation reduces cardiovascular mortality by 36% — the single most impactful lifestyle modification.

Avoiding all physical activity (E) worsens deconditioning. Increasing alcohol (F) is harmful. Permanent bed rest (G) is counterproductive.

340 **Answer:** A

Cellulitis (skin induration, erythema, warmth) first-line treatment:

1. Cephalexin is a first-generation cephalosporin with excellent coverage of the two most common cellulitis pathogens: Staphylococcus aureus and Streptococcus pyogenes (Group A Strep)

2. It inhibits bacterial cell wall synthesis by binding penicillin-binding proteins

3. Good oral bioavailability and skin/soft tissue penetration

4. 500 mg QID for 5-10 days depending on severity and response

5. Well-tolerated with minimal side effects

6. Mark the borders of erythema with a pen to monitor progression/regression

Ciprofloxacin (B) has poor gram-positive coverage. Metronidazole (C) covers anaerobes only. Azithromycin (D) has limited skin infection evidence. Doxycycline (E) is used for MRSA cellulitis but is not first-line for uncomplicated cellulitis.

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341 **Answer:** B

When cellulitis does not improve on cephalexin, suspect MRSA:

1. MRSA produces PBP2a (altered penicillin-binding protein) that resists all beta-lactams including cephalexin

2. TMP-SMX has excellent MRSA activity

3. It inhibits sequential steps in folate synthesis — synergistic bactericidal effect

4. Good oral bioavailability and skin penetration

5. Well-tolerated and inexpensive

6. Alternative MRSA-active oral agents: doxycycline 100 mg BID, clindamycin 300 mg TID

Amoxicillin (A) does not cover MRSA. Higher dose cephalexin (C) still cannot overcome MRSA resistance — the mechanism is PBP alteration, not dose-dependent. Penicillin V (D) does not cover MRSA. Erythromycin (E) has high MRSA resistance rates.

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342 **Answers:** A B C

Medications for hyperemesis gravidarum:

- **Doxylamine 10 mg + Pyridoxine 10 mg PO QID — Diclectin (A)**: First-line treatment. Doxylamine is an H1 antihistamine that blocks the vomiting center. Pyridoxine (vitamin B6) reduces nausea through unclear mechanisms. The combination is the only FDA-approved medication for nausea/vomiting of pregnancy. Safe with extensive safety data.
- **Ondansetron 4-8 mg PO/IV q8h (B)**: A 5-HT3 antagonist used when first-line therapy fails. Highly effective antiemetic. Generally considered safe in pregnancy, though some studies suggest a small increased risk of cleft palate with first-trimester use. Used for moderate-severe symptoms.
- **Metoclopramide 5 mg PO 30 min before meals (C)**: A dopamine antagonist and prokinetic that reduces nausea and promotes gastric emptying. Safe in pregnancy. Used when doxylamine/pyridoxine is insufficient. Side effects include drowsiness and rarely extrapyramidal symptoms.

Metronidazole (D) is an antibiotic — not for nausea. Isotretinoin (E) is absolutely CONTRAINDICATED in pregnancy. Misoprostol (F) causes uterine contractions and abortion — CONTRAINDICATED.

343 **Answers:** A B C D

Non-pharmacological advice for hyperemesis gravidarum:

- **Eat small amounts every 1-2 hours to avoid empty stomach (A)**: An empty stomach increases gastric acid irritation and worsens nausea. Small, frequent meals maintain stable blood glucose and reduce gastric distension that triggers vomiting.
- **Eliminate spicy, high-fat, and acidic foods (B)**: These foods delay gastric emptying, increase acid production, and irritate the gastric mucosa. Bland, dry foods (crackers, toast, rice) are better tolerated.
- **Drink fluids 30 minutes before or after solid food (C)**: Separating liquids from solids reduces gastric distension and the sensation of fullness that triggers vomiting. Sipping small amounts of clear fluids throughout the day prevents dehydration.
- **Avoid lying down after eating (D)**: Remaining upright for 30-60 minutes after meals reduces gastroesophageal reflux and allows gravity to assist gastric emptying.

Large meals three times daily (E) causes gastric distension and vomiting. Hot beverages with meals (F) can worsen nausea. Strong odors (G) are a common nausea trigger — should be avoided.

344 **Answers:** A B C

IBS with diarrhea treatment:

- **Loperamide 2 mg 45 minutes before meals (A)**: An opioid receptor agonist that slows intestinal motility, increases water absorption, and increases anal sphincter tone. Taking it before meals prevents postprandial diarrhea (the gastrocolic reflex). Does not cross the blood-brain barrier — no CNS effects.
- **Rifaximin 550 mg PO TID x 2 weeks (B)**: A non-absorbable antibiotic that modulates gut microbiota. FDA-approved for IBS-D. Reduces bloating, abdominal pain, and loose stools. Works by reducing bacterial overgrowth and altering bacterial metabolism of bile acids. Can be repeated if symptoms recur.
- **Hyoscyamine 0.25 mg PO TID PRN (C)**: An anticholinergic/antispasmodic that relaxes intestinal smooth muscle, reducing cramping and abdominal pain. Taken as needed before meals or during symptom flares.

Omeprazole (D) is for GERD, not IBS. Metformin (E) actually causes diarrhea. Prednisone (F) is not indicated for IBS.

345 **Answer:** B

Post-cholecystectomy diarrhea worsening IBS-D is caused by bile acid malabsorption:

1. The gallbladder normally stores and concentrates bile acids, releasing them in response to meals
2. After cholecystectomy, bile continuously drips into the duodenum
3. Excess bile acids reach the colon, where they stimulate water and electrolyte secretion (secretory diarrhea)
4. Cholestyramine is a bile acid sequestrant that binds bile acids in the intestinal lumen
5. This prevents bile acids from reaching the colon and causing diarrhea
6. Dose: 4g mixed in water, taken before meals
7. Side effects: bloating, constipation (actually beneficial in this case)

Loperamide (A) treats symptoms but not the underlying bile acid mechanism. Omeprazole (C) does not address bile acid malabsorption. Metoclopramide (D) is a prokinetic — would worsen diarrhea. Ondansetron (E) is an antiemetic. Prednisone (F) is not indicated.

346 **Answers:** A B C D G

Non-pharmacological measures for IBS:

- **Low FODMAP diet (A)**: Fermentable Oligosaccharides, Disaccharides, Monosaccharides, and Polyols are poorly absorbed short-chain carbohydrates that are fermented by colonic bacteria, producing gas, bloating, and diarrhea. Eliminating high-FODMAP foods (onions, garlic, wheat, dairy, apples, beans) for 6-8 weeks, then systematic reintroduction, reduces symptoms in 70% of IBS patients.
- **Regular meals (B)**: Irregular eating patterns disrupt the migrating motor complex and gastrocolic reflex. Regular, moderate-sized meals at consistent times normalize intestinal motility.
- **Exclusion of gas-producing foods (C)**: Foods like beans, lentils, cruciferous vegetables (broccoli, cabbage, cauliflower), and carbonated beverages increase intestinal gas production through bacterial fermentation. Eliminating these reduces bloating, distension, and flatulence.
- **Exercise and physical activity (D)**: Regular moderate exercise (30 minutes, 3-5 times/week) improves intestinal transit time, reduces bloating, and decreases stress-related IBS symptoms. Exercise also improves mood and sleep quality, both of which affect IBS severity.
- **Stress reduction and CBT (G)**: The gut-brain axis plays a central role in IBS. Psychological stress increases visceral hypersensitivity and alters motility. CBT, mindfulness meditation, and gut-directed hypnotherapy reduce symptom severity by 50-70% and have lasting benefits.

High-fat diet (E) worsens IBS symptoms by stimulating the gastrocolic reflex. Increasing dairy (F) may worsen symptoms in lactose-intolerant patients.

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347 **Answer:** A

Scabies treatment with permethrin 5%:

1. Permethrin 5% is a synthetic pyrethroid that disrupts sodium channels in the Sarcoptes scabiei mite's nervous system, causing paralysis and death

2. Applied to ALL skin from neck to toes (including under nails, between fingers, genitalia, and skin folds) — not just symptomatic areas

3. Left on for 8-14 hours (typically overnight), then washed off

4. Repeat application in 7 days to kill newly hatched mites from surviving eggs

5. Safe for adults, children >2 months, and pregnant/breastfeeding women

6. Cure rate >95% with proper application

Permethrin 1% (B) is for head lice — insufficient concentration for scabies. Lindane (C) is second-line due to neurotoxicity risk — never first-line. Oral ivermectin only (D) is an alternative but not first-line topical treatment. Hydrocortisone (E) treats itch but not the mite. Fluconazole (F) is antifungal.

348 **Answers:** A B

Non-pharmacological measures for scabies:

- **Wash clothing at 65°C and heat dry or seal in plastic bag for 2 weeks (A)**: Scabies mites survive off the human body for 48-72 hours. Washing at ≥60°C kills mites and eggs. Items that cannot be washed should be sealed in plastic bags for 2 weeks — mites die without a human host within this period. This prevents re-infestation from contaminated fomites.
- **Identify and treat potential human contacts (B)**: Scabies is transmitted through prolonged skin-to-skin contact. ALL household members and sexual contacts should be treated SIMULTANEOUSLY, even if asymptomatic — they may be in the incubation period (4-6 weeks before symptoms appear). Failure to treat contacts is the most common cause of treatment failure.

No treatment of contacts (C) leads to re-infestation. Fumigating the house (D) is unnecessary — environmental spraying is not recommended. Burning clothing (E) is excessive. Isolating for 1 month (F) is unnecessary — patients are non-infectious 24 hours after treatment.

349 **Answers:** A B

Vaginal candidiasis treatment:

- **Fluconazole 150 mg PO single dose (A)**: A triazole antifungal that inhibits fungal CYP51 (lanosterol 14-alpha-demethylase), blocking ergosterol synthesis. Single oral dose is convenient, effective (cure rate >90%), and well-tolerated. Preferred by many patients over intravaginal therapy.
- **Clotrimazole 500 mg intravaginal single dose (B)**: An imidazole antifungal applied directly to the site of infection. Achieves high local concentrations with minimal systemic absorption. Single high-dose intravaginal treatment is as effective as oral fluconazole. Alternative: clotrimazole 200 mg intravaginal x 3 days or 100 mg x 7 days.

Metronidazole (C) is for bacterial vaginosis — not effective against Candida. Doxycycline (D) is antibacterial. Ciprofloxacin (E) is antibacterial. Note: Antibiotic use is the precipitating factor — antibiotics kill protective Lactobacillus, allowing Candida overgrowth.

350 **Answer:** B

A breastfeeding woman needs progestin-only contraception because:

1. Combined OCPs contain estrogen, which REDUCES breast milk production by suppressing prolactin
2. Progestin-only pills (POP/minipill) have NO effect on milk quantity or quality
3. Norethindrone 0.35 mg is taken daily without a hormone-free interval
4. Minimal transfer into breast milk — no adverse effects on infant growth or development
5. Can be started immediately postpartum (no waiting period)
6. Must be taken at the same time daily (3-hour window) for maximum efficacy

Combined OCP (A) is NOT safe in lactation — estrogen reduces milk supply. Depo-Provera (C) is progestin-only and safe but does not increase milk production — it is neutral. Copper IUD (D) is non-hormonal and safe but does not increase milk production. Estrogen patch (E) would reduce milk supply.

351 **Answers:** A B C D E

Immediate management of open fracture in a rural hospital:

- **Analgesia as needed (A)**: Open fractures are extremely painful. Adequate pain control with IV opioids (morphine) and/or IV NSAIDs (ketorolac) is essential for patient comfort and to facilitate examination and treatment.
- **Reduction and immobilization (B)**: Realigning the fracture and applying a splint reduces pain, prevents further soft tissue damage, reduces bleeding, and protects neurovascular structures. Traction splints for femoral fractures.
- **Immediate irrigation and debridement (C)**: Copious saline irrigation (minimum 6-9 liters for severe open fractures) removes bacteria, debris, and devitalized tissue. This is the single most important step in preventing osteomyelitis. Debridement removes non-viable tissue that serves as a bacterial growth medium.
- **Antibiotics — Cefazolin 2g IV q8h (D)**: IV antibiotics should be given within 1 hour of presentation. Cefazolin covers the most common organisms (Staphylococcus, Streptococcus). For severe (Gustilo III) fractures, add gentamicin for gram-negative coverage. Antibiotics reduce infection rates from 30% to <5%.
- **Tetanus prophylaxis (E)**: Open fractures are contaminated, tetanus-prone wounds. Tetanus toxoid and TIG should be given based on vaccination history.

Discharge with oral antibiotics only (F) is inadequate for open fractures. Waiting for orthopedic consultation (G) delays critical time-sensitive interventions. Applying a cast immediately (H) is inappropriate — casts are not used for open fractures as they prevent wound monitoring.

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352 **Answers:** A B C D

Serious complications of open fractures:

- **Arterial injury (A)**: Bone fragments can lacerate or compress adjacent arteries, causing hemorrhage or distal ischemia. Palpate distal pulses and assess capillary refill. Absent pulses require emergent vascular surgery.
- **Nerve injury (B)**: Bone fragments or displacement can stretch, compress, or transect peripheral nerves. Assess motor and sensory function distal to the fracture. Common examples: radial nerve in humeral shaft fractures, peroneal nerve in fibular neck fractures.
- **Compartment syndrome (C)**: Swelling within a closed fascial compartment increases pressure, compressing blood vessels and nerves. Causes ischemic necrosis of muscle within 6 hours. Classic sign: pain out of proportion to injury, pain with passive stretch. Requires emergent fasciotomy.
- **Infection — osteomyelitis (D)**: Open fractures expose bone to environmental bacteria. Without proper irrigation, debridement, and antibiotics, infection rates reach 30%. Osteomyelitis is difficult to treat and may require prolonged IV antibiotics and surgical debridement.

Improved healing (E) is incorrect — open fractures heal slower. Spontaneous resolution (F) does not occur. Weight gain (G) is not a complication.

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353 **Answers:** A B C

Non-pharmacological measures for lateral epicondylitis (tennis elbow):

- **Activity modification (A)**: Identify and reduce the repetitive wrist extension/gripping activities causing the condition. Modify technique (e.g., tennis stroke mechanics), use ergonomic tools, and take frequent breaks. Avoid aggravating movements during the acute phase.
- **Ice application (B)**: Apply ice for 15-20 minutes several times daily to reduce inflammation and pain. Ice causes vasoconstriction, reducing edema and inflammatory mediator release at the common extensor tendon origin.
- **Counter-force bracing (C)**: A forearm strap/brace applied 2-3 cm distal to the lateral epicondyle redistributes force away from the damaged tendon origin. It reduces strain on the extensor carpi radialis brevis during gripping and wrist extension activities.

Increasing repetitive movements (D) worsens the condition. Continuous heat (E) may increase inflammation in the acute phase. Complete arm immobilization for 6 months (F) causes severe stiffness, muscle atrophy, and is unnecessary.

354 **Answers:** A B

Pharmacological treatments for epicondylitis:

- **NSAIDs — Naproxen 500 mg PO BID (A)**: Reduces inflammation and pain at the tendon origin. Oral NSAIDs provide systemic anti-inflammatory effect. Topical NSAIDs (diclofenac gel) are an alternative with fewer systemic side effects. Used for 2-4 weeks during the acute phase.
- **Glucocorticoid injection — Triamcinolone — only once (B)**: Intralesional corticosteroid injection provides rapid pain relief (within 1 week) by suppressing local inflammation. However, limited to ONE injection because repeated injections cause tendon weakening, atrophy, and increased rupture risk. Provides short-term benefit (4-6 weeks) but may not improve long-term outcomes.

Opioids (C) are not appropriate for this condition. Methotrexate (D) is for autoimmune diseases. Colchicine (E) is for gout. Hydroxychloroquine (F) is for RA/lupus.

355 **Answer:** B

Raynaud's phenomenon first-line pharmacological treatment:

1. Amlodipine is a dihydropyridine calcium channel blocker
2. It relaxes vascular smooth muscle by blocking L-type calcium channels
3. This causes vasodilation of digital arteries, counteracting the vasospasm that causes Raynaud's attacks
4. Starting dose 5 mg daily, can increase to 10 mg
5. Trial of 4-6 weeks to assess efficacy
6. Reduces frequency and severity of attacks by 30-50%
7. Alternative: nifedipine 30 mg extended-release daily

Propranolol (A) is CONTRAINDICATED — beta-blockers worsen Raynaud's by blocking beta-2 mediated vasodilation, leaving unopposed alpha-mediated vasoconstriction. Warfarin (C) is not indicated. Aspirin (D) has limited evidence. Prednisone (E) is not first-line. Methotrexate (F) is for autoimmune diseases.

356 **Answers:** A B C

Non-pharmacological advice for Raynaud's:

- **Minimize cold exposure, dress warmly (A)**: Cold is the primary trigger for Raynaud's attacks. Wear insulated gloves, warm socks, and layered clothing. Keep the core body warm — core cooling triggers peripheral vasoconstriction. Use hand warmers in cold weather.
- **Avoid vasoconstricting drugs and smoking (B)**: Smoking causes vasoconstriction through nicotine's effect on alpha-adrenergic receptors and endothelial damage. Vasoconstricting drugs include decongestants (pseudoephedrine), beta-blockers, ergotamine, and amphetamines. All worsen Raynaud's.
- **Avoid vibrating tools (C)**: Vibration causes direct vascular damage and triggers vasospasm. Occupational use of vibrating tools (jackhammers, chainsaws, drills) is a recognized cause of secondary Raynaud's (vibration white finger/hand-arm vibration syndrome).

Exposing hands to cold water (D) triggers attacks. Increasing caffeine (E) causes vasoconstriction. Tight gloves (F) restrict blood flow — loose, insulated gloves are preferred.

357 **Answers:** A B C D

Risk factor modifications for pressure ulcers:

- **Pressure minimization — repositioning (A)**: The most important intervention. Reposition every 2 hours to redistribute pressure and allow tissue reperfusion. Use pressure-relieving surfaces (alternating pressure mattresses, foam overlays). Avoid positioning directly on the ulcer.
- **Prevent excessive moisture (B)**: Moisture from incontinence, perspiration, or wound drainage macerates the skin, weakening the epidermis and increasing friction. Use moisture barriers (zinc oxide, dimethicone), absorbent pads, and manage incontinence with catheterization if necessary.
- **Improve nutritional status (C)**: Malnutrition impairs wound healing. Ensure adequate protein intake (1.25-1.5 g/kg/day), calories (30-35 kcal/kg/day), and micronutrients (vitamin C, zinc). Nutritional supplementation accelerates healing.
- **Early mobilization and physiotherapy (D)**: Immobility is the primary risk factor for pressure ulcers. Progressive mobilization — bed exercises, sitting, standing, walking — reduces pressure exposure time and improves circulation.

Keeping patient in same position (E) worsens pressure damage. Restricting oral intake (F) worsens malnutrition. Applying pressure to the wound (G) is harmful.

358 **Answer:** A

Meningitis close contact prophylaxis:

1. Rifampin is the standard chemoprophylaxis for Neisseria meningitidis contacts

2. Dose: 600 mg PO every 12 hours for 2 days (4 doses total)

3. Children: 10 mg/kg (max 600 mg) every 12 hours for 2 days

4. Rifampin eradicates nasopharyngeal carriage of N. meningitidis

5. Close contacts include: household members, daycare contacts, anyone with direct exposure to oral secretions (kissing, sharing utensils, mouth-to-mouth resuscitation)

6. Prophylaxis should be given within 24 hours of diagnosis for maximum benefit

Amoxicillin (B) does not reliably eradicate meningococcal carriage. Doxycycline (C) is not standard prophylaxis. Metronidazole (D) is for anaerobes. Fluconazole (E) is antifungal.

359 **Answers:** A B

Alternative meningitis prophylaxis:

- **Ciprofloxacin 500 mg PO single dose (A)**: A fluoroquinolone that achieves high nasopharyngeal concentrations, effectively eradicating N. meningitidis carriage with a single oral dose. Convenient, well-tolerated, and does not require multiple doses like rifampin. Preferred for adult contacts. Not recommended for pregnant women or children.

- **Ceftriaxone 250 mg IM single dose (B)**: A third-generation cephalosporin that eradicates nasopharyngeal carriage with a single IM injection. Preferred for pregnant women (safe in pregnancy) and children. Also effective when oral medication is not feasible.

Azithromycin (C) is not standard meningococcal prophylaxis. Vancomycin (D) is not effective for nasopharyngeal carriage eradication. Metronidazole (E) covers anaerobes only.

360 **Answers:** A B

Medications to stop in a newly confirmed pregnancy:

- **ACE inhibitor (A)**: Causes renal tubular dysgenesis, oligohydramnios, fetal hypotension, skull ossification defects, and limb contractures. Contraindicated in all trimesters but especially dangerous in 2nd/3rd trimesters. Switch to labetalol or nifedipine.

- **TMP-SMX (B)**: Trimethoprim is a folate antagonist — it inhibits dihydrofolate reductase, potentially causing neural tube defects in the first trimester. Sulfamethoxazole in the third trimester can cause neonatal kernicterus by displacing bilirubin from albumin. Should be replaced with a pregnancy-safe alternative.

Prenatal vitamins (C) should be CONTINUED. Folic acid (D) should be CONTINUED — essential for preventing neural tube defects. Iron supplements (E) should be CONTINUED for preventing anemia.

361 **Answer:** B

UTI prophylaxis in pregnancy (replacing TMP-SMX):

1. Cephalexin is a first-generation cephalosporin safe in all trimesters of pregnancy

2. It covers common uropathogens: E. coli, Klebsiella, Proteus

3. Low-dose nightly administration (250 mg qhs) provides effective suppressive prophylaxis

4. It does not cross the placenta in significant amounts

5. Well-tolerated with minimal side effects

6. Category B in pregnancy — no evidence of fetal harm

Ciprofloxacin (A) is contraindicated in pregnancy — causes cartilage damage in developing fetus. Nitrofurantoin (C) is generally safe but avoided in the first trimester (possible association with birth defects) and at term (risk of neonatal hemolytic anemia). Doxycycline (D) is contraindicated — teeth discoloration and bone growth inhibition. Metronidazole (E) is not standard UTI prophylaxis.

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362 **Answer:** B

A woman who took 20 lorazepam tablets after a fight — likely a chronic benzodiazepine user given she has a prescription:

1. Flumazenil is a competitive benzodiazepine receptor antagonist

2. In CHRONIC benzodiazepine users, flumazenil precipitates acute withdrawal

3. Acute benzodiazepine withdrawal causes life-threatening SEIZURES

4. The seizure risk from flumazenil far outweighs the benefit in this scenario

5. The patient's vitals are relatively stable — RR 8 but SpO2 99%, BP 110/60

6. Management: supportive care with airway management, bag-valve-mask ventilation if needed, and monitoring

7. Flumazenil is only safe in benzodiazepine-naive patients (e.g., procedural sedation reversal)

Giving flumazenil immediately (A) risks fatal seizures. Naloxone (C) is for opioid overdose — lorazepam is a benzodiazepine. Activated charcoal (D) may be considered if within 1 hour of ingestion and airway is protected. Immediate intubation (E) is not necessary if SpO2 is maintained with supportive measures.

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363 **Answer:** **A**

Medical treatment for ectopic pregnancy:

1. Methotrexate is a folate antagonist that inhibits dihydrofolate reductase

2. This blocks DNA synthesis in rapidly dividing trophoblastic cells

3. The trophoblast stops growing and is reabsorbed by the body

4. Single-dose IM injection (50 mg/m² body surface area)

5. Criteria for medical management:

- Hemodynamically stable
- Unruptured ectopic
- hCG <5000 mIU/mL (some guidelines say <10,000)
- No fetal cardiac activity
- Ectopic mass <3.5-4 cm
- Patient reliable for follow-up

6. Monitor serial hCG levels — should decline by ≥15% between days 4 and 7

7. Success rate: 85-90% for single dose

Misoprostol (B) is for medical abortion of intrauterine pregnancy — not effective for ectopic. Mifepristone (C) is an antiprogestogen for intrauterine pregnancy termination. Oxytocin (D) stimulates uterine contractions — not effective for ectopic. Progesterone (E) would support the ectopic pregnancy — contraindicated.

These case-based questions are designed to strengthen your clinical reasoning and therapeutic decision-making skills. Continue reviewing, practicing, and always verify with current clinical guidelines and evidence-based resources.

Dr Hamza Abu Nawara

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